New

101 Days
Self-Preparation module

2nd edition

ENGLISH Medium

SSC MATHS

CHAPTER WISE TYPE WISE

6500+ TCS MCQ

- Complete Coverage for 2025 Exams
- Smart Approach & Short Tricks
- With Best Explanations

Useful for

SSC CGL (Tier-I & II), CHSL (Tier-I & II), CPO, Selection Post, GD, MTS & other Govt. Exams

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(Excise Inspector)

SELECTED हैं SELECTION दिलाएँगे



MASTER SSC MATHS LIKE A TOPPER

Your Ultimate Guide to Using This Book Effectively

Welcome to a power-packed Maths resource designed exclusively for SSC aspirants! This book isn't just a collection of past year questions-it's a strategic tool crafted to help you master every concept in a smart, structured, and exam-oriented way.

Why This Book is a Game-Changer?

- ✅ Complete Coverage of SSC Exams - Chapter-wise compilation of all questions asked in the last 5 years across SSC CGL, CHSL, CPO, MTS, and more!
- ✅ Type-wise & Level-wise Arrangement - Focus on one concept at a time and progress from basics to advanced effortlessly.
- ✅ Smart & Time-Saving Solutions - Master multiple solving techniques (traditional, shortcut, and conceptual) for every question.
- ✅ Designed for Targeted Practice - Learn a concept in class? Jump straight to practice only that type from this book!

How to Use This Book for Maximum Success?

📖 **STEP 1: Learn the Concept First**
Before diving into questions, understand the fundamentals from your teacher, notes, or video lectures.

🎯 **STEP 2: Targeted Concept Practice**
Thanks to the type-wise arrangement, you can instantly practice only the relevant question types, reinforcing your learning.

📊 **STEP 3: Level-Up Gradually**
Move step by step from basic to high-difficulty questions. This systematic progression boosts confidence and accuracy.

⚡ **STEP 4: Master Shortcuts & Multiple Methods**
For every question, compare different solving techniques-pick the fastest one to optimize your speed in exams.

🔄 **STEP 5: Revise & Track Your Progress**
Use the previous year questions (PYQs) as a self-evaluation tool. Reattempt difficult questions and refine your approach.

Pro-Tips to Ace SSC Maths Using This Book

- 🔥 Smart Work > Hard Work - Don't just practice, analyze solutions to find the best method.
- 🔥 Time Yourself - Try to solve questions within exam-time constraints for real-time improvement.
- 🔥 Make Notes of Tricks & Formulas - A quick revision sheet will be your best friend before the exam.
- 🔥 Solve Questions Daily - Even 15-20 quality questions per day can skyrocket your preparation.

🚀 Let's Ace SSC Maths Together!

With this book as your trusted guide, be assured of concept clarity, speed, and accuracy-the three pillars of SSC exam success!

🔷 Stay Consistent. Stay Focused. Stay Ahead. 🔷
🎯 All the Best! Your Success Story Starts Here. 🎯

Reverse Calendar Technique

101 days Self-Preparation module

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			Total	162		
30.	Data Interpretation	TYPE-I	Bar Graph	105	645-645	DAY 98-101
		TYPE-II	Tabular Chart	206		
		TYPE-III	Pie Chart	91		
		TYPE-IV	Line Graph	21		
			Answer Key	–		
			Solutions	–		
			Total	423		
			Grand Total	6459		

New

SSC MATHS

2nd edition

ENGLISH Medium

TYPE WISE

6500+ TCS MCQ

FREE

TRIGONOMETRY

22 Chapter TRIGONOMETRY

◆ TYPE – I ◆
BASIC TRIGONOMETRIC RATIOS

1. If $\tan A = \frac{4}{3}, 0 \le A \le 90°$, then find the value of sin A.
SSC CPO 03/10/2023 (Shift-01)
(a) $\frac{3}{5}$ (b) $\frac{7}{5}$
(c) $\frac{4}{5}$ (d) $\frac{4}{7}$

2. If $\tan(t) = \frac{3}{7}$, t is an acute angle, then what is the value of cosec(t)?
SSC CHSL 08/07/2024 (Shift-02)
(a) $\frac{\sqrt{53}}{3}$ (b) $\frac{\sqrt{58}}{7}$
(c) $\frac{\sqrt{53}}{7}$ (d) $\frac{\sqrt{58}}{3}$

3. If θ is an acute angle and $\cos\theta = \frac{11}{17}$, what is the value of $\tan\theta$?
SSC CHSL 08/08/2023 (Shift-02)
(a) $\frac{4\sqrt{10}}{11}$ (b) $\frac{13}{11}$
(c) $\frac{2\sqrt{42}}{11}$ (d) $\frac{2\sqrt{42}}{17}$

4. If θ is an acute angle and $\sin\theta = \frac{13}{19}$, what is the value of $\cos\theta$?
SSC CHSL 08/08/2023 (Shift-03)
(a) $\frac{6}{19}$ (b) $\frac{10\sqrt{2}}{19}$
(c) $\frac{14}{19}$ (d) $\frac{8\sqrt{3}}{19}$

5. If $\tan(A) = \frac{3}{5}$, then cos(A) is equal to:
SSC CGL 17/09/2024 (Shift-03)
(a) $\frac{5}{\sqrt{34}}$ (b) $\frac{9}{\sqrt{34}}$
(c) $\frac{3}{\sqrt{34}}$ (d) $\frac{25}{\sqrt{34}}$

6. If θ is an acute angle and $\sin\theta = \frac{21}{25}$, then what is the value of $\tan\theta$?
SSC CHSL 11/08/2023 (Shift-04)
(a) $\frac{2\sqrt{46}}{21}$ (b) $\frac{25}{2\sqrt{46}}$
(c) $\frac{21}{2\sqrt{46}}$ (d) $\frac{2\sqrt{46}}{25}$

7. If θ is an acute angle and $\sin\theta = \frac{43}{47}$, what is the value of $\cot\theta$?
SSC CHSL 14/08/2023 (Shift-03)
(a) $\frac{6\sqrt{10}}{47}$ (b) $\frac{43}{6\sqrt{10}}$
(c) $\frac{47}{6\sqrt{10}}$ (d) $\frac{6\sqrt{10}}{43}$

8. In a right triangle for an acute angle x, find $\sin x$. It is given that $\tan x = 5$.
SSC CPO 09/11/2022 (Shift-02)
(a) $\frac{1}{3}$ (b) $\frac{5}{\sqrt{26}}$
(c) $\frac{2}{5}$ (d) $\frac{5}{\sqrt{28}}$

9. If $\sec(t) = \frac{x}{y}$, then cot(t) is equal to:
SSC CGL 18/09/2024 (Shift-01)
(a) $\frac{y}{\sqrt{x^2-y^2}}$ (b) $\frac{x}{\sqrt{x^2+y^2}}$
(c) $\frac{x}{\sqrt{x^2+y^2}}$ (d) $\frac{y}{\sqrt{x^2+y^2}}$

10. If θ be acute angle and $\cos\theta = \frac{24}{25}$, then the value of $\cot(90-\theta)$ is equal to:
SSC CHSL 05/07/2024 (Shift-03)
(a) $\frac{6}{25}$ (b) $\frac{7}{24}$
(c) $\frac{7}{25}$ (d) $\frac{5}{24}$

11. If $\cos A = \frac{15}{17}, 0 \le A \le 90°$, then the value of cot (90° – A) is:
SSC CPO 03/10/2023 (Shift-02)
(a) $\frac{8}{15}$ (b) $\frac{7}{15}$
(c) $\frac{\sqrt{2}}{15}$ (d) $\frac{2\sqrt{2}}{15}$

12. What is the value of sec(t), if $\tan(t) = \frac{1}{3}$?
SSC CGL 09/09/2024 (Shift-02)
(a) $\frac{2\sqrt{2}}{3}$ (b) $\frac{\sqrt{10}}{9}$
(c) $\frac{\sqrt{10}}{3}$ (d) $\frac{\sqrt{3}}{3}$

13. If $\sin\theta = \frac{7}{25}$, then $\cos\theta$ and $\tan\theta$ are respectively _____ .
(where θ is in the first quadrant)
SSC CHSL 09/07/2024 (Shift-03)
(a) $\frac{25}{24}$ and $\frac{24}{7}$ (b) $\frac{25}{7}$ and $\frac{7}{24}$
(c) $\frac{24}{25}$ and $\frac{24}{7}$ (d) $\frac{24}{25}$ and $\frac{7}{24}$

14. If $\tan A = \frac{1}{\sqrt{10}}$, A is an acute angle, then the value of sinA + cosecA is.
SSC CGL 23/09/2024 (Shift-03)
(a) $\frac{11}{\sqrt{10}}$ (b) $\frac{1}{\sqrt{10}}$
(c) $\frac{12}{\sqrt{11}}$ (d) $\frac{10}{\sqrt{11}}$

15. If $\sin A = \frac{2}{3}$, then find the value of (7 – tanA) (3 + cosA).
SSC CPO 05/10/2023 (Shift-01)
(a) $\frac{61}{3}+\frac{17}{\sqrt{5}}$ (b) $\frac{61}{3}-\frac{17}{3\sqrt{5}}$
(c) $\frac{61}{3}+\frac{17}{3\sqrt{5}}$ (d) $\frac{61}{3\sqrt{5}}+\frac{17}{3}$

16. If $\sin\beta = \frac{1}{3}$, $(\sec\beta - \tan\beta)^2$ is equal to:
SSC CPO 05/10/2023 (Shift-01)
(a) $\frac{1}{3}$ (b) $\frac{1}{2}$
(c) $\frac{2}{3}$ (d) $\frac{3}{4}$

17. If $\cos A = \frac{63}{65}$, then find the value of tan A + cot A (up to two places of decimal).
SSC CHSL 02/08/2023 (Shift-04)
(a) 4.19 (b) 2.76
(c) 3.19 (d) 5.23

18. If $\sin\alpha = \frac{5}{13}$, then the value of $\cos\alpha.\operatorname{cosec}\alpha.\cot\alpha$ is _____ ?
SSC CPO 27/06/2024 (Shift-02)
(a) $\frac{12}{5}$ (b) $\frac{144}{25}$
(c) $\frac{5}{12}$ (d) $\frac{25}{144}$

19. If $\tan\theta = \frac{7}{24}$, then find the value of $(\cos^2\theta - \sin^2\theta)$.
SSC CPO 27/06/2024 (Shift-03)
(a) $\frac{576}{550}$ (b) $\frac{576}{550}$
(c) $\frac{527}{625}$ (d) $\frac{520}{625}$

20. If $\cot\theta = \frac{4}{3}, 0 < \theta < \frac{\pi}{2}$ and $5p\cos^2\theta \sin\theta = \cot^2\theta$, then the value of p is:

SSC CPO 03/10/2023 (Shift-02)

(a) $\frac{7}{27}$ (b) $\frac{5}{27}$

(c) $\frac{25}{27}$ (d) $\frac{125}{27}$

21. If $\sin\theta = \frac{5}{6}$, the value of $\cot\theta \cdot \sin\theta \cdot \cos\theta$ is .....

SSC CHSL 08/08/2023 (Shift-03)

(a) $\frac{6}{5}$ (b) $\frac{25}{36}$

(c) $\frac{5}{6}$ (d) $\frac{11}{36}$

22. If $\sin\theta = \frac{4}{5}$, find the value of $\tan\theta - \cot\theta$.

SSC CGL 17/07/2023 (Shift-02)

(a) $\frac{5}{12}$ (b) $\frac{7}{9}$

(c) $\frac{7}{12}$ (d) $\frac{7}{8}$

23. If $\cot B = \frac{15}{8}$, where B is an acute angle, what is the value of sec B + tan B?

SSC CHSL 03/08/2023 (Shift-02)

(a) $\frac{4}{5}$ (b) $\frac{5}{3}$

(c) $\frac{5}{4}$ (d) $\frac{3}{5}$

24. If $\sin A = \frac{8}{17}$ then what is the value cot A + sec A?

SSC CHSL TIER-II 26/06/2023

(a) $4\frac{1}{120}$ (b) $2\frac{1}{120}$

(c) $5\frac{1}{120}$ (d) $3\frac{1}{120}$

25. If 4cotA = 5, then what is the value of 6 secA tanA?

SSC CHSL 10/03/2023 (Shift-01)

(a) $\frac{20\sqrt{41}}{21}$ (b) $\frac{25\sqrt{41}}{24}$

(c) $\frac{24\sqrt{41}}{25}$ (d) $\frac{3}{2}\sqrt{41}$

26. If $\cot\theta = \frac{28}{45}$, then find the value of $(\sin\theta - \cos\theta)$.

SSC CPO 27/06/2024 (Shift-01)

(a) $\frac{45}{28}$ (b) $\frac{7}{53}$

(c) $\frac{28}{53}$ (d) $\frac{17}{53}$

27. If 3 tanA = 4 and A is an acute angle, then the value of 4sinA + 3 cosA is:

SSC CGL 24/09/2024 (Shift-02)

(a) 1 (b) 3

(c) 4 (d) 5

28. If $\cos A = \frac{4}{5}$, then find the value of (8 + sinA) (3 – tanA).

SSC CPO 29/06/2024 (Shift-03)

(a) $\frac{373}{20}$ (b) $\frac{367}{20}$

(c) $\frac{387}{20}$ (d) $\frac{381}{20}$

29. If $\sec\theta = \frac{29}{20}$ where $0 < \theta < 90°$ then what is the value of $3\operatorname{cosec}\theta + 3\cot\theta$?

SSC CGL 17/09/2024 (Shift-01)

(a) $\frac{7}{3}$ (b) 7

(c) 14 (d) 49

30. If 4 tan A = 3, then $\cos^2 A - \sin^2 A$ equals:

SSC Phase-XI 27/06/2023 (Shift-03)

(a) $\frac{1}{5}$ (b) $\frac{7}{25}$

(c) $\frac{9}{25}$ (d) $\frac{3}{5}$

31. If $\sin B = \frac{15}{17}$, What is the value of cosB(secB – tanB)? Given that $0 < B < \frac{\pi}{2}$

SSC CGL 13/09/2024 (Shift-02)

(a) $\frac{1}{16}$ (b) $\frac{2}{15}$

(c) $\frac{8}{15}$ (d) $\frac{2}{17}$

32. If $\tan A = \sqrt{2} - 1$ then find the value of sinA cosA + tanA.

SSC CHSL 10/07/2024 (Shift-01)

(a) $\frac{24\sqrt{2}+15}{4}$ (b) $\frac{5\sqrt{2}-4}{4}$

(c) $\frac{24\sqrt{2}-15}{4}$ (d) $\frac{5\sqrt{2}+4}{4}$

33. If $\sqrt{3}\tan\theta = 3\sin\theta$, then what is the value of $\sin^2\theta - \cos^2\theta$?

SSC CGL TIER-II 08/08/2022

(a) $\frac{1}{2}$ (b) $\frac{2}{3}$

(c) $\frac{1}{3}$ (d) $\frac{1}{4}$

34. If $\sin A = \frac{5}{13}$ and 7 cot B = 24, then the value of (secA cosB)(cosecBtan A) is:

SSC CGL TIER-II 29/01/2022

(a) $\frac{13}{7}$ (b) $\frac{65}{42}$

(c) $\frac{15}{13}$ (d) $\frac{13}{14}$

35. If $\operatorname{cosec}\theta = \frac{5}{3}$, then evaluate $(\sec^2\theta - 1) \times \cot^2\theta \times (1 + \cot^2\theta)$.

SSC CGL 09/09/2024 (Shift-01)

(a) $\frac{25}{16}$ (b) $\frac{9}{16}$

(c) $\frac{25}{9}$ (d) $\frac{4}{5}$

36. If $\sin A = \frac{5}{13}$, then find the value of $\frac{\cos A - 2\tan A}{\sin A + 3\tan A}$.

SSC CGL TIER-II (20/01/2025)

(a) $\frac{11}{255}$ (b) $\frac{12}{255}$

(c) $\frac{14}{255}$ (d) $\frac{13}{255}$

37. If $\cot\theta = \frac{3}{4}, \theta$ is an acute angle, then $\sin\theta + \cos\theta - \tan\theta$ is equal to:

SSC CHSL 04/07/2024 (Shift-04)

(a) $\frac{1}{11}$ (b) $\frac{1}{13}$

(c) $\frac{1}{18}$ (d) $\frac{1}{15}$

38. If cot A = 7, then the value of $\frac{5\cos A + 4\sin A}{\cos^3 A + 7\sin^3 A + 6\sin A}$ is:

SSC CHSL 01/07/2024 (Shift-03)

(a) 3 (b) 2

(c) 0 (d) 1

39. If $\cot A = \frac{5}{12}$, then find the value of $\frac{\sqrt{\sec A - \operatorname{cosec} A}}{\sqrt{\sec A + \operatorname{cosec} A}} - \sqrt{\frac{17}{7}}$.

SSC CHSL 08/07/2024 (Shift-01)

(a) $\frac{24}{\sqrt{119}}$ (b) 0

(c) $-\frac{10}{\sqrt{119}}$ (d) $\frac{10}{\sqrt{119}}$

40. If $8\cot\theta = 7$, then the value of $\frac{1+\sin\theta}{\cos\theta}$ is:

SSC CGL 26/09/2024 (Shift-02)

(a) $\frac{8+\sqrt{113}}{7}$ (b) $\frac{1+\sqrt{113}}{8}$

(c) $\frac{7+\sqrt{113}}{8}$ (d) $\frac{1+\sqrt{113}}{7}$

41. If $\sqrt{13}\sin\theta = 2$, then the value of $\frac{3\tan\theta + \sqrt{13}\sin\theta}{\sqrt{13}\cos\theta - 3\tan\theta}$ is:

SSC CHSL 02/07/2024 (Shift-04)

(a) 4 (b) 3

(c) 2 (d) 0

42. Given that $\sin\theta = \frac{15}{17}$. Evaluate $\frac{15\cot\theta + 17\sin\theta}{8\tan\theta + 16\sec\theta}$, where $0° < \theta < 90°$.

SSC CHSL 02/07/2024 (Shift-02)

(a) $\frac{23}{49}$ (b) $\frac{49}{26}$

(c) $\frac{49}{23}$ (d) $\frac{26}{23}$

43. If $\tan\theta = \frac{7}{4}$, then the value of $\frac{4\sin\theta - 3\cos\theta}{4\sin\theta + 3\cos\theta}$ is equal to:

SSC CHSL 09/07/2024 (Shift-02)

(a) $\frac{2}{5}$ (b) $\frac{1}{3}$

(c) $\frac{4}{5}$ (d) $\frac{3}{5}$

44. If $\tan x = \frac{7}{5}$, then the value of $\frac{9\sin x - \frac{42}{5}\cos x}{15\sin x + 21\cos x}$ is

SSC CGL 24/07/2023 (Shift-03)

(a) 0 (b) 1

(c) 0.1 (d) 0.5

45. If $3\cot A = \frac{7}{3}$, what is the value of given expression $\frac{3\cos A + 2\sin A}{3\cos A - 2\sin A}$ = ?

SSC CGL 27/07/2023 (Shift-01)

(a) $\frac{2}{3}$ (b) $\frac{1}{3}$

(c) 13 (d) 1

46. Evaluate $\frac{(9\sin\theta + 2\cos\theta)}{(9\sin\theta - 2\cos\theta)}$, if 3 tanθ = 4.

SSC CHSL 10/07/2024 (Shift-02)

(a) $\frac{7}{5}$ (b) $\frac{9}{7}$

(c) $\frac{5}{7}$ (d) $\frac{6}{5}$

47. If 8 tanA = 5, what is the value of $\frac{8\sin A - 7\cos A}{8\sin A + 11\cos A}$?

SSC Phase-XII 21/06/2024 (Shift-02)

(a) $\frac{1}{19}$ (b) $\frac{-1}{6}$

(c) $\frac{-1}{8}$ (d) $\frac{1}{8}$

48. If $\cot A = \frac{5}{12}$, find the value of the following expression. $\frac{5(1 - \cos^2 A)}{6(1 - \sin^2 A)}$

SSC CHSL 09/08/2023 (Shift-02)

(a) $\frac{144}{5}$ (b) $\frac{24}{25}$

(c) $\frac{144}{25}$ (d) $\frac{24}{5}$

49. If 2 cotθ = 3, then find the value of $\frac{\sqrt{13}\sin\theta - 3\tan\theta}{3\tan\theta + \sqrt{13}\cos\theta}$

SSC CGL 24/07/2023 (Shift-01)

(a) $\frac{1}{\sqrt{13}}$ (b) $\frac{2}{\sqrt{13}}$

(c) 0 (d) $\frac{2}{3}$

50. If $\cos\theta = \frac{12}{13}$, then the value of $\frac{\sin\theta(1 - \tan\theta)}{\tan\theta(1 + \text{cosec}\,\theta)}$ is:

SSC CGL TIER-II 03/02/2022

(a) $\frac{25}{78}$ (b) $\frac{35}{234}$

(c) $\frac{35}{108}$ (d) $\frac{25}{156}$

51. If sin A = $\frac{1}{2}$, A is an acute angle, then find the value of $\frac{\tan A - \cot A}{\sqrt{3}(1 + \text{cosec}A)}$.

SSC CGL 17/08/2021 (Shift-02)

(a) $-\frac{2}{9}$ (b) $-\frac{4\sqrt{3}}{9}$

(c) $\frac{4\sqrt{3}}{9}$ (d) $\frac{2}{9}$

52. If 2 tanθ = 3, then $\frac{3\sin\theta - 2\cos\theta}{3\sin\theta + 2\cos\theta}$ is equal to:

SSC CGL 10/09/2024 (Shift-02)

(a) $\frac{6}{13}$ (b) $\frac{6}{7}$

(c) $\frac{5}{7}$ (d) $\frac{5}{13}$

53. If 7tanθ = 3, and θ is an acute angle, then $\frac{5\sin\theta - \cos\theta}{5\sin\theta + 2\cos\theta}$ is equal to

SSC CGL 10/09/2024 (Shift-03)

(a) $\frac{8}{29}$ (b) $\frac{11}{29}$

(c) $\frac{7}{29}$ (d) $\frac{9}{29}$

54. If 7 tanθ = 3, then $\frac{7\sin\theta + 5\cos\theta}{7\sin\theta - 2\cos\theta}$ is:

SSC CGL 12/09/2024 (Shift-02)

(a) 7 (b) 5

(c) 8 (d) 6

55. If 7 cosA = 6. then the numerical value of $\frac{\text{cosec}A + \cot A}{\text{cosec}A - \cot A}$ is:

SSC CGL 12/09/2024 (Shift-03)

(a) 1 (b) 0

(c) 13 (d) –13

56. If tanA = $\frac{5}{7}$, then find the value of $\frac{7\sin A - 3\cos A}{7\sin A + 3\cos A}$ + 4.

SSC CGL 13/09/2024 (Shift-02)

(a) $4\frac{1}{4}$ (b) $3\frac{5}{7}$

(c) $3\frac{1}{5}$ (d) $7\frac{1}{3}$

57. If sinθ = $\frac{5}{13}$, then what is the value of $\frac{\cos^2\theta - \sin^2\theta}{2\cos\theta.\sin\theta}$?

SSC CHSL 17/03/2023 (Shift-03)

(a) $\frac{15}{126}$ (b) $\frac{13}{120}$

(c) $\frac{17}{136}$ (d) $\frac{119}{120}$

58. If $\frac{7\sin\theta + 4\cos\theta}{9\sin\theta - 2\cos\theta}$ = $\frac{5}{4}$, then the value of $\frac{\tan^2\theta + 5}{\tan^2\theta - 5}$ is:

SSC CGL 09/09/2024 (Shift-03)

(a) $\frac{2121}{769}$ (b) $-\frac{2121}{769}$

(c) $\frac{769}{2121}$ (d) $-\frac{769}{2121}$

59. If $\frac{\sec\alpha + \tan\alpha}{\sec\alpha - \tan\alpha}$ = $\frac{7}{4}$, then find the value of cosecα.

SSC CGL 09/09/2024 (Shift-03)

(a) $\frac{1}{3}$ (b) $\frac{1}{11}$

(c) $\frac{11}{3}$ (d) $\frac{3}{11}$

60. If cosec$x$ = $\frac{5}{4}$, then the value of $\frac{5\sin^2 x + 8\cot^2 x}{27\tan^2 x - 125\cos^2 x}$ is:

SSC CGL 17/09/2024 (Shift-03)

(a) $\frac{33}{91}$ (b) $\frac{77}{30}$

(c) $\frac{55}{87}$ (d) $\frac{99}{47}$

61. If cotθ = $\sqrt{11}$, then the value of $\frac{\text{cosec}^2\theta - \sec^2\theta}{\text{cosec}^2\theta + \sec^2\theta}$

SSC CGL 26/09/2024 (Shift-03)

(a) $\frac{3}{5}$ (b) $\frac{5}{6}$

(c) $\frac{4}{5}$ (d) $\frac{6}{7}$

62. If $\frac{2\sin A - \cos A}{\sin A + \cos A}$ = 1, then find the value of cot A.

SSC CPO 05/10/2023 (Shift-03)

(a) 1 (b) 2

(c) $\frac{1}{2}$ (d) $\frac{1}{3}$

63. If $\frac{\sin\theta-\cos\theta}{\sin\theta+\cos\theta}=\frac{4}{5}$, then the value of $\frac{\text{cosec}^2\theta}{2-\text{cosec}^2\theta}$ is:

SSC CHSL 09/08/2023 (Shift-01)

(a) $\frac{16}{25}$ (b) $\frac{40}{41}$

(c) $\frac{41}{40}$ (d) $\frac{31}{30}$

64. If $\frac{\cos\beta}{\sec\alpha}=15$ and $\frac{\sin\beta}{\sec\alpha}=16$, then the value of $\sin^2\beta$ is ____.

SSC CGL 14/07/2023 (Shift-03)

(a) $\frac{256}{481}$ (b) $-\frac{256}{481}$

(c) $\frac{481}{256}$ (d) $-\frac{481}{256}$

65. If $\frac{\sin\theta+\cos\theta}{\sin\theta-\cos\theta}=5$, then the value of $\frac{4\sin^2\theta+3}{2\cos^2\theta+2}$ is:

SSC CGL 24/08/2021 (Shift-03)

(a) $\frac{75}{34}$ (b) $\frac{75}{17}$

(c) $\frac{3}{2}$ (d) $\frac{1}{2}$

66. If $\frac{(\sec A+\tan A)}{\sec A-\tan A}=2\frac{51}{79}$, then the value of sinA is equal to:

SSC CPO 28/06/2024 (Shift-01)

(a) $\frac{77}{144}$ (b) $\frac{61}{169}$

(c) $\frac{87}{169}$ (d) $\frac{65}{144}$

67. If $\frac{21\cos A+3\sin A}{3\cos A+4\sin A}=2$, then find the value of cot A

SSC CGL 26/07/2023 (Shift-02)

(a) $\frac{9}{11}$ (b) $\frac{11}{9}$

(c) $\frac{1}{3}$ (d) $\frac{11}{10}$

68. If $\frac{\sec\theta-\tan\theta}{\sec\theta+\tan\theta}=\frac{1}{7}$, θ lies in first quadrant, then the value of $\frac{\text{cosec}\,\theta+\cot^2\theta}{\text{cosec}\,\theta-\cot^2\theta}$ is:

SSC CGL TIER-II 03/02/2022

(a) $\frac{19}{5}$ (b) $\frac{22}{3}$

(c) $\frac{37}{12}$ (d) $\frac{37}{19}$

69. If $\frac{\sec\theta+\tan\theta}{\sec\theta-\tan\theta}=2\frac{51}{79}$, then the value of $\sin\theta$ is equal to :

SSC CGL TIER-II 18/11/2020

(a) $\frac{65}{144}$ (b) $\frac{35}{72}$

(c) $\frac{91}{144}$ (d) $\frac{39}{72}$

70. If $\tan\theta=\frac{3}{10}$, then find the value of $\frac{(1+\sin\theta)(1-\sin\theta)}{(1+\cos\theta)(1-\cos\theta)}$.

SSC CPO 27/06/2024 (Shift-03)

(a) $\frac{100}{7}$ (b) $\frac{100}{3}$

(c) $\frac{100}{9}$ (d) $\frac{50}{3}$

71. If $\cot\theta=\frac{4}{3}$, then evaluate $\frac{(\sec\theta(1+\cot^2\theta)(\text{cosec}^2\theta-\cot^2\theta))}{\text{cosec}^3\theta}$.

SSC CGL 10/09/2024 (Shift-02)

(a) $\frac{3}{4}$ (b) $\frac{4}{5}$

(c) $\frac{4}{3}$ (d) $\frac{3}{5}$

72. If $\frac{\sec\theta-\tan\theta}{\sec\theta+\tan\theta}=\frac{1}{9}$, θ lies in the firs quadrant, then the value of $\frac{\sin\theta-\tan^2\theta}{\sin\theta+\tan^2\theta}$ is:

SSC CGL TIER-II (18/01/2025)

(a) $-\frac{13}{27}$ (b) $-\frac{11}{29}$

(c) $\frac{13}{27}$ (d) $\frac{11}{29}$

73. ΔRMS is right-angled at M. The length of the base, RM = 4cm and the length of perpendicular MS = 3cm. Find the value of sec R.

SSC CHSL 10/08/2023 (Shift-03)

(a) $\frac{3}{4}$ (b) $\frac{5}{4}$

(c) $\frac{4}{5}$ (d) $\frac{2}{5}$

74. In ΔABC, AB = 20 cm, BC = 21 cm and AC = 29 cm. What is the value of cot C + cosec C – 2 tanA?

SSC CGL 16/08/2021 (Shift-02)

(a) $\frac{9}{20}$ (b) $\frac{7}{20}$

(c) $\frac{2}{5}$ (d) $\frac{3}{5}$

75. In ΔABC, right angled at B, if cot A = $\frac{1}{2}$, then the value of $\frac{\text{sinA}(\text{cosC}+\text{cosA})}{\cos C(\text{sinC}-\text{sinA})}$ is:

SSC CGL 24/08/2021 (Shift-01)

(a) –3 (b) 2

(c) 3 (d) –2

76. In ΔPQR, $\angle$Q = 90°. If tan R = $\frac{1}{3}$, then what is the value of $\frac{\sec P(\cos R+\sin P)}{\text{cosec}R(\sin R-\text{cosec}P)}$?

SSC CGL 18/08/2021 (Shift-02)

(a) $-\frac{2}{7}$ (b) $\frac{18}{7}$

(c) $\frac{2}{7}$ (d) $-\frac{18}{7}$

77. In a ΔABC right - angled at C, if tan A = $\sqrt{3}$ then the value of sin A cos B cot (A + B) is:

SSC CGL 11/09/2024 (Shift-01)

(a) $\sqrt{3}$ (b) 1

(c) 0 (d) 2

78. In a triangle ABC, the $\angle$ABC = 90°. If sin(A) = $\frac{1}{2}$, then cos(C) is equal to:

SSC CGL 11/09/2024 (Shift-02)

(a) $\frac{1}{\sqrt{2}}$ (b) 1

(c) $\frac{\sqrt{3}}{2}$ (d) $\frac{1}{2}$

79. In a right angle ΔABC, $\angle$B = 90°, if tanA = 1, then 4sinA cosA =_____.

SSC CGL 17/09/2024 (Shift-01)

(a) 1 (b) 2

(c) $\frac{1}{2}$ (d) 4

80. In a ΔABC, right angled at B if tanC = $\sqrt{3}$, then find $\frac{\sin^2 C+\cos^2 C}{1+\cot^2 C}$.

SSC CGL 26/09/2024 (Shift-02)

(a) $\frac{3}{4}$ (b) $\frac{16}{3}$

(c) $\frac{4}{15}$ (d) $\frac{3}{16}$

81. In right angled triangle ABC, $\angle$B = 90° and angle C are acute angles. If cosesA = $2\sqrt{2}$, then find the value of sinA. cosC + cosA. sinC.

SSC CHSL 01/07/2024 (Shift-01)

(a) 0 (b) $2\sqrt{2}$

(c) 1 (d) $\sqrt{2}$

82. In ΔABC, $\angle$B = 90° and AB : BC = 1 : 2. The value of cos A + tan C is.

SSC CPO 04/10/2023 (Shift-01)

(a) $\frac{1+\sqrt{5}}{2\sqrt{5}}$ (b) $\frac{2\sqrt{5}}{2+\sqrt{5}}$

(c) $\frac{5+\sqrt{5}}{2\sqrt{5}}$ (d) $\frac{2+\sqrt{5}}{2\sqrt{5}}$

83. ΔPQR is right angled at Q. If PQ = 12 cm, PR = 13 cm, find tanP + cotR.

SSC CHSL 09/08/2023 (Shift-01)

(a) $\frac{9}{10}$ (b) 0

(c) $\frac{10}{12}$ (d) $\frac{12}{10}$

84. ABC is a right trianlge. If $\angle B = 90°$ and $\tan A = \frac{1}{\sqrt{3}}$, then the value of sinA cosC + cosA sinC is:

SSC CGL 26/07/2023 (Shift-02)

(a) 1 (b) 2

(c) $\frac{2}{3}$ (d) $\frac{1}{3}$

85. Consider a triangle PQR, right angled at R, in which PQ = 29 units, QR = 21 units and $\angle PQR = \theta$. Find the value of $\cos^2\theta - \sin^2\theta$.

SSC CGL 11/09/2024 (Shift-03)

(a) $\frac{21}{841}$ (b) 1

(c) $\frac{20}{841}$ (d) $\frac{41}{841}$

86. If $\sec 31° = x$, then $\sin^2 59° + \frac{1}{\sec^2 31°} - \frac{1}{\sin^2 59° \operatorname{cosec}^2 59°}$ is equal to:

SSC CGL 24/08/2021 (Shift-03)

(a) $\frac{x^2-2}{x}$ (b) $\frac{2-x^2}{x^2}$

(c) $\frac{x^2-2}{x^2}$ (d) $\frac{2-x^2}{x}$

87. If $\cos 27° = \frac{p}{q}$, then find the value of $\operatorname{cosec} 27° - \cos 63°$.

SSC CGL TIER-II (20/01/2025)

(a) $\frac{q^2}{p\sqrt{q^2-p^2}}$ (b) $\frac{p^2}{q\sqrt{q^2-p^2}}$

(c) $\frac{q}{p\sqrt{q^2-p^2}}$ (d) $\frac{p}{q\sqrt{q^2-p^2}}$

88. If $\cos 24° = \frac{m}{n}$, then the value of $(\operatorname{cosec} 24° - \cos 66°)$ is:

SSC CGL 17/09/2024 (Shift-01)

(a) $\frac{m^2}{n\sqrt{m^2-n^2}}$ (b) $\frac{n^2}{m\sqrt{m^2-n^2}}$

(c) $\frac{n^2}{m\sqrt{n^2-m^2}}$ (d) $\frac{m^2}{n\sqrt{n^2-m^2}}$

89. If $\sin\theta = \frac{2xy}{x^2+y^2}$, then the value of $\tan\theta$ is:

SSC CHSL 01/07/2024 (Shift-02)

(a) $\frac{2xy}{x^2-y^2}$ (b) $\frac{x^2+y^2}{x^2-y^2}$

(c) $\frac{x^2-y^2}{2xy}$ (d) $\frac{x^2-y^2}{x^2+y^2}$

90. If A is an acute angle, what is the value of cosec A, if $\cot A = \frac{2p}{p^2-1}$?

SSC Phase-XII 26/06/2024 (Shift-02)

(a) $\frac{2p}{p^2+1}$ (b) $\frac{2^2}{p^2-1}$

(c) $\frac{p^2+1}{p^2-1}$ (d) $\frac{p^2-1}{p^2+1}$

91. If $\sin 23° = \frac{a}{b}$, then the value of $\sec 23° - \sin 67°$ is _____.

SSC CGL 14/07/2023 (Shift-03)

(a) $\frac{a^2}{\sqrt{b^2-a^2}}$ (b) $\frac{b^2-a^2}{ab}$

(c) $\frac{a^2}{b\sqrt{b^2+a^2}}$ (d) $\frac{a^2}{b\sqrt{b^2-a^2}}$

92. If $\cos 48° = \frac{m}{n}$ then, $\sec 48° - \cot 42°$ is equal to:

SSC CPO 10/11/2022 (Shift-01)

(a) $\frac{m-\sqrt{n^2-m^2}}{m}$ (b) $\frac{m-\sqrt{n^2-m^2}}{m}$

(c) $\frac{m-\sqrt{n^2-m^2}}{m}$ (d) $\frac{n-\sqrt{n^2-m^2}}{m}$

93. If $\cos 53° = \frac{x}{y}$, then $\sec 53° + \cot 37°$ is equal to:

SSC CGL 19/04/2022 (Shift-02)

(a) $\frac{y+\sqrt{y^2-x^2}}{x}$ (b) $\frac{x+\sqrt{y^2-x^2}}{y}$

(c) $\frac{y+\sqrt{y^2-x^2}}{y}$ (d) $\frac{y-\sqrt{y^2-x^2}}{x}$

94. If $\sin\theta = \frac{2\sqrt{ab}}{a+b}$, a > b > 0, then the value of $\frac{\cos\theta+1}{\cos\theta-1}$ will be:

SSC CHSL 19/04/2021 (Shift-02)

(a) $-\frac{a}{b}$ (b) $\frac{a}{b}$

(c) $-\frac{b}{a}$ (d) $\frac{b}{a}$

95. If $\cos\theta = \frac{P^2-1}{P^2+1}$, $0° < \theta < 90°$, then $\operatorname{cosec}\theta$ is equal to:

SSC CHSL 05/08/2021 (Shift-02)

(a) $\frac{2P}{1-P^2}$ (b) $\frac{1+P^2}{2P}$

(c) $\frac{1-P^2}{2P}$ (d) $\frac{2P}{1+P^2}$

96. If $\sec\theta = \frac{a}{b}$, $b \neq 0$, then $\frac{1-\tan^2\theta}{2-\sin^2\theta} = ?$

SSC CGL TIER-II 15/11/2020

(a) $\frac{a^2(2b^2+a^2)}{b^2(a^2-b^2)}$ (b) $\frac{a^2(2b^2+a^2)}{b^2(a^2+b^2)}$

(c) $\frac{a^2(2b^2-a^2)}{b^2(a^2+b^2)}$ (d) $\frac{a^2(2b^2-a^2)}{a^2(a^2+b^2)}$

97. If $\operatorname{cosec}\theta = \frac{b}{a}$, then $\frac{\sqrt{3}\cot\theta+1}{\tan\theta+\sqrt{3}}$ is equal to:

SSC CGL TIER-II 16/11/2020

(a) $\frac{\sqrt{b^2-a^2}}{a}$ (b) $\frac{\sqrt{a^2+b^2}}{a}$

(c) $\frac{\sqrt{a^2+b^2}}{b}$ (d) $\frac{\sqrt{b^2-a^2}}{b}$

◆ TYPE – II ◆
VALUES OF TRIGONOMETRIC RATIOS

98. What is the value of $\left(\frac{5}{\sqrt{3}} - \operatorname{cosec} 60°\right)$?

SSC CGL 25/09/2024 (Shift-03)

(a) $\sqrt{3}$ (b) $\frac{2}{\sqrt{3}}$

(c) $3\sqrt{3}$ (d) $\frac{1}{\sqrt{3}}$

99. Simplify $\frac{\sin 60° - \cos 60°}{\sin 60° + \cos 60°}$

SSC CHSL 11/07/2024 (Shift-03)

(a) $2+\sqrt{3}$ (b) $\sqrt{3}-1$

(c) $2-\sqrt{3}$ (d) $\sqrt{3}-2$

100. Find the value of the following expression.

$\sin(30 + \theta) - \cos(60 - \theta)$

SSC CHSL 03/07/2024 (Shift-02)

(a) 0 (b) $2\sin(30 - \theta)$

(c) $\frac{\sqrt{3}}{2}$ (d) 1

101. Find the value of the given expression.

$\frac{4}{3}\tan^2 45° + 3\cos^2 30° - 2\sec^2 30° - \frac{3}{4}\cot^2 60°$

SSC CGL 18/07/2023 (Shift-04)

(a) $\frac{2}{3}$ (b) $\frac{3}{2}$

(c) $\frac{\sqrt{2}}{3}$ (d) $\frac{3}{\sqrt{2}}$

102. Find the value of $\frac{2}{3}\tan^2 60° + 3\cos^2 30° - 2\sec^2 30° - \frac{3}{4}\cot^2 60°$.

SSC CGL 20/07/2023 (Shift-03)

(a) 2.33 (b) 2.01

(c) 1.33 (d) 3.33

103. If $\sec\theta = \sqrt{2}$, then $\operatorname{cosec}\theta$ is equal to:

SSC CGL 24/09/2024 (Shift-01)

(a) 0 (b) $\sqrt{2}$

(c) –1 (d) 0

104. $16\cos^3\frac{\pi}{6} - 12\cos\frac{\pi}{6}$ is equal to:

SSC CGL 01/12/2022 (Shift-04)

(a) 0 (b) 2
(c) 1 (d) –1

105. Evaluate the following expression: $3\tan^2 45° + 2\cos 60° - 2\sin 30°$

SSC CHSL 26/05/2022 (Shift-03)

(a) 3 (b) 4
(c) 0 (d) 2

106. What is the value of $8\sqrt{3}\sin30°\tan60° - 3\cos 0° + 3\sin^2 45° + 2\cos^2 30°$?

SSC CGL 11/04/2022 (Shift-02)

(a) 15 (b) 12
(c) 9 (d) 18

107. If $2k\sin30°\cos30°\cot60° = \frac{\cot^2 30°\sec60°\tan45°}{\text{cosec}^2 45°\text{cosec}30°}$, then find the value of k.

SSC CGL 12/04/2022 (Shift-03)

(a) $\frac{3}{2}$ (b) 3
(c) 1 (d) 6

108. Find the value of

$[\frac{4}{3}\tan^2 60° + 3\cos^2 30° - 2\sec^2 30° - \frac{3}{4}\cot^2 60°] \div [\sin60°\cos30° - \cos60°\sin30°]$

SSC CGL 11/09/2024 (Shift-02)

(a) $6\frac{2}{3}$ (b) $3\frac{1}{2}$
(c) $4\frac{1}{3}$ (d) $6\frac{3}{2}$

109. If $p\left(\sqrt{3}+\cot30°\right) = \tan^3 60° - 2\sin60°$, then the value of p is:

SSC CGL 13/09/2024 (Shift-01)

(a) –1 (b) $2\sqrt{3}$
(c) 1 (d) $\sqrt{3}$

110. Given $\theta_1 + \theta_2 = \frac{\pi}{2}$ & $\sin\theta_1 = \frac{1}{2}$ find the value of θ_2.

SSC CGL 27/07/2023 (Shift-01)

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$
(c) $\frac{\pi}{3}$ (d) π

111. If $k(\tan 45° \sin 60°) = \cos 60° \cot 30°$, then the value of k is:

SSC CPO 04/10/2023 (Shift-01)

(a) 1 (b) $\frac{1}{\sqrt{3}}$
(c) $\sqrt{3}$ (d) 3

112. Find the value of $\cos0° + \cos30° - \tan 45° + \text{cosec } 60° + \cot 90°$.

SSC CHSL 03/08/2023 (Shift-01)

(a) $\frac{7}{6\sqrt{3}}$ (b) $\frac{\sqrt{3}}{6}$
(c) $\frac{7}{6}$ (d) $\frac{7}{2\sqrt{3}}$

113. Find the value of $\sin60° \sec30° + \tan45° \cos0° + \sec60° \sin30°$.

SSC CHSL 07/08/2023 (Shift-02)

(a) 0 (b) 3
(c) 1 (d) 2

114. At what angle θ, the value of sin (θ) and cos (θ) will have the same value?

SSC CGL 18/09/2024 (Shift-01)

(a) 45° (b) 30°
(c) 60° (d) 75°

115. The value of $\frac{\sin^2\theta - 2\sin^4\theta}{2\cos^4\theta + \cos^2\theta}$ at θ = 45° is:

SSC CGL 23/09/2024 (Shift-03)

(a) 1 (b) 0
(c) 3 (d) 2

116. If A = 15°, then what is the value of $\frac{11\sqrt{2}\cos3A + 10\sin2A}{7\sqrt{3}\sin4A}$?

SSC CGL 24/09/2024 (Shift-03)

(a) $\frac{11}{7}$ (b) $\frac{21}{32}$
(c) $\frac{32}{21}$ (d) $\frac{7}{11}$

117. If $\tan\frac{\pi}{6} + \sec\frac{\pi}{6} = x$, then find x

SSC CGL 17/07/2023 (Shift-02)

(a) $\sqrt{3}$ (b) $\frac{1}{\sqrt{3}}$
(c) $\frac{-1}{\sqrt{3}}$ (d) $\frac{2}{\sqrt{3}}$

118. If $\frac{k - k\cot^2 30°}{1+\cot^2 30°} = \sin^2 60° + 4\tan^2 45° - \text{cosec}^2 60°$, then the value of k (correct to two decimal places) is:

SSC CGL 03/12/2022 (Shift-03)

(a) 5.55 (b) –6.83
(c) –5.58 (d) 6.83

119. If $\frac{x - x\,\text{cosec}^2 30°}{1+\text{cosec}^2 30°} = \cos^2 60° + 4\cot^2 45° - \sec^2 60°$, then the value of x is _____.

SSC CHSL 25/05/2022 (Shift-01)

(a) $\frac{5}{12}$ (b) $-\frac{5}{12}$
(c) $\frac{1}{5}$ (d) $\frac{-1}{12}$

120. If $\cot\theta = \frac{1}{\sqrt{3}}$ 0° < θ < 90°, then the value of $\frac{2-\sin^2\theta}{1-\cos^2\theta} + (\text{cosec}^2\theta - \sec\theta)$ is:

SSC CPO 25/11/2020 (Shift-02)

(a) 5 (b) 0
(c) 1 (d) 2

121. If $3\tan\theta = 2\sqrt{3}\sin\theta$, 0°<θ<90°, then the value of $\frac{\text{cosec}^2 2\theta + \cot^2 2\theta}{\sin^2\theta + \tan^2 2\theta}$ is:

SSC CGL TIER-II 29/01/2022

(a) $\frac{4}{13}$ (b) $\frac{20}{39}$
(c) $\frac{4}{3}$ (d) $\frac{20}{27}$

122. If $4\sin^2(2x-10)° = 3$, $0° \le (2x-10) \le 90°$, then find the value of $\frac{\sin^4(x-5)° + \cos^4(x-5)°}{1-2\sin^2(3x-15)°\cos^2(3x-15)°}$.

SSC CGL 17/08/2021 (Shift-02)

(a) 1 (b) $\frac{5}{8}$
(c) $-\frac{5}{8}$ (d) –1

123. If $3(\cot^2\phi - \cos^2\phi) = \cos^2\phi$, 0° < ϕ < 90°, then the value of $(\tan^2\phi + \text{cosec}^2\phi + \sin^2\phi)$ is :

SSC CGL TIER-II 12/09/2019

(a) $\frac{13}{3}$ (b) $\frac{61}{12}$
(c) $\frac{25}{12}$ (d) $\frac{15}{4}$

124. If $\frac{\cos^2\theta}{\cot^2\theta - \cos^2\theta} = 3$, 0° < θ < 90°, then the value of cotθ + cosecθ is :

SSC CGL 06/06/2019 (Shift-02)

(a) $\sqrt{3}$ (b) $\frac{\sqrt{3}}{2}$
(c) $2\sqrt{3}$ (d) $\frac{3\sqrt{3}}{4}$

125. $\frac{\tan(x)}{\sec(x)} = \frac{1}{2}$, then the value of $(\sin x + \cos x)^2$ is _____.

SSC CGL 13/09/2024 (Shift-02)

(a) $\frac{2+\sqrt{3}}{3}$ (b) $\frac{2+\sqrt{3}}{2}$
(c) $\frac{2-\sqrt{3}}{2}$ (d) $\frac{2-\sqrt{3}}{3}$

126. If $\sin\theta + \cos\theta = \sqrt{3}\cos\theta$, then the value of cotθ is:

SSC CGL 17/09/2024 (Shift-03)

(a) $\sqrt{3} - 1$ (b) $\frac{(\sqrt{3}+1)}{2}$
(c) $\frac{(\sqrt{3}-1)}{2}$ (d) $\sqrt{3} + 1$

127. If $\cot^2\theta - 2\cos^2\theta = 0$, $(0° < \theta < 90°)$, then find the value of θ ?

SSC CGL 25/09/2024 (Shift-02)

(a) 45° (b) 60°
(c) 90° (d) 30°

128. If $\sin^4\theta + \cos^4\theta = 2\sin^2\theta\cos^2\theta$, $0°<\theta<90°$, then find the value of $\cot\theta$.

SSC CPO 29/06/2024 (Shift-01)

(a) $\frac{1}{2}$ (b) $\frac{1}{\sqrt{2}}$
(c) $\frac{1}{2}$ (d) 1

129. If $(\sin\theta - \cos\theta) = 0$, then the value of $\sin(\pi - \theta) + \sin\left(\frac{\pi}{2} - \theta\right)$ is:

SSC CHSL 02/08/2023 (Shift-02)

(a) 1 (b) 0
(c) $\sqrt{3}$ (d) $\sqrt{2}$

130. If $4\sin^2A - 3 = 0$ and $0 \le A \le 90°$, then $3\sin A - 4\sin^3 A$ is:

SSC CHSL 10/08/2023 (Shift-01)

(a) $\frac{1}{2}$ (b) $\sqrt{\frac{3}{2}}$
(c) 0 (d) 1

131. If $4\sin^2\theta = 3(1 + \cos\theta)$, $0° < \theta < 90°$, then what is the value of $(2\tan\theta + 4\sin\theta - \sec\theta)$?

SSC CGL 11/04/2022 (Shift-01)

(a) $3\sqrt{15} - 4$ (b) $15\sqrt{3} - 4$
(c) $15\sqrt{3} + 3$ (d) $4\sqrt{15} - 3$

132. If $3\tan\theta = 2\sqrt{3}\sin\theta$, $0° < \theta < 90°$, then find the value of $2\sin^2 2\theta - 3\cos^2 3\theta$.

SSC CGL 13/08/2021 (Shift-03)

(a) 1 (b) $\frac{3}{2}$
(c) $\frac{1}{2}$ (d) $\frac{-3}{2}$

133. If $\sin\left(\frac{2A+B}{2}\right) = \cos\left(\frac{2A-B}{2}\right) = \frac{\sqrt{3}}{2}, 0° < \frac{2A+B}{2} < 90°$ and $0° < \frac{2A+B}{2} < 90°$ then find the value of $\sin[3(A - B)]$.

SSC CGL 16/08/2021 (Shift-03)

(a) 1 (b) $\frac{1}{\sqrt{2}}$
(c) $\frac{1}{2}$ (d) $\frac{\sqrt{3}}{2}$

134. If $\sin(x + y) = 1$ and $\cos(x - y) = \frac{\sqrt{3}}{2}$, find the value of 'y'.

SSC CPO 28/06/2024 (Shift-03)

(a) 90° (b) 60°
(c) 30° (d) 20°

135. If $\sin(p + q) = 1$ and $\cos(p - q) = \frac{\sqrt{3}}{2}$, find p.

SSC CHSL 03/08/2023 (Shift-04)

(a) 90° (b) 80°
(c) 60° (d) 120°

136. If $\sin(A - B) = \frac{1}{2}$, and $\sin(A + B) = 1$, where A and B are positive acute angles and $A \ge B$, then A and B are respectively:

SSC CHSL 04/08/2023 (Shift-04)

(a) 80°, 10° (b) 60°, 30°
(c) 70°, 20° (d) 50°, 40°

137. If A, B, C are the acute angles, such that $\tan(A + B - C) = \frac{1}{\sqrt{3}}$, $\cos(B + C - A) = \frac{1}{2}$ and $\sin(C + A - B) = \frac{1}{\sqrt{2}}$. The value of A + B + C will be equal to:

SSC CHSL 10/08/2023 (Shift-04)

(a) 90° (b) 110°
(c) 170° (d) 135°

138. If $\sin(x - y) = \frac{1}{2}$ and $\cos(x + y) = \frac{1}{2}$, then what is the value of $\sin x\cos x + 2\sin^2 x + \cos^3 x\sec x$?

SSC CHSL 15/03/2023 (Shift-03)

(a) 2 (b) $\frac{3}{4}$
(c) $\sqrt{2} + 1$ (d) 1

139. If $\tan(x + y) = 1$ and $\cos(x - y) = \frac{\sqrt{3}}{2}$, then what is the value of x and y?

SSC CHSL 16/03/2023 (Shift-04)

(a) $x = 3$, $y = 4.5$
(b) $x = 37.5$, $y = 7.5$
(c) $x = 7.5$, $y = 37.5$
(d) $x = 4.5$, $y = 3$

140. If $\tan(A + B) = \sqrt{3}$ and $\tan(A - B) = \frac{1}{\sqrt{3}}$; $0° < (A + B) < 90°$; $A > B$, then the values of A and B are _____.respectively.

SSC CGL TIER-II 02/03/2023

(a) 45° & 15° (b) 15° & 45°
(c) 30° & 30° (d) 60° & 30°

141. If $\cos 2\theta = 0$, where θ is an acute angle, then find the value of $\sin(75° - \theta)$.

SSC Phase-XII 26/06/2024 (Shift-03)

(a) $\frac{1}{2}$ (b) $\frac{1}{\sqrt{2}}$
(c) $\sqrt{2} - 1$ (d) $\sqrt{2} + 1$

142. If $P\sin60° = Q\operatorname{cosec}45°$, then the value of $\frac{P^2 + Q^2}{P^2 - Q^2}$ is equal to:

SSC Phase-XII 26/06/2024 (Shift-04)

(a) $\frac{11}{7}$ (b) $\frac{11}{5}$
(c) $\frac{5}{11}$ (d) $\frac{7}{11}$

143. If $\alpha\sin45° = \beta\operatorname{cosec}30°$, then $\frac{\alpha^4}{\beta^4}$ is:

SSC CHSL 07/08/2023 (Shift-01)

(a) 4^4 (b) 3^3
(c) 2^3 (d) 4^3

144. If $\sin x = \frac{1}{2}$, then find the value of $(\sec^4 x + \operatorname{cosec}^4 x) - (\tan^4 x + \cot^4 x)$.

SSC CHSL 10/03/2023 (Shift-02)

(a) $\frac{26}{3}$ (b) $\frac{26}{33}$
(c) $\frac{26}{33}$ (d) 2

145. If $\cos\theta - \sin\theta = \sqrt{3}\cos(90° - \theta)$, $0° < \theta < 90°$ then find the value of $\tan\theta - \cot\theta$.

SSC CGL 17/08/2021 (Shift-02)

(a) $\frac{3 + 2\sqrt{3}}{(1 + \sqrt{3})}$ (b) $-\frac{3 + 2\sqrt{3}}{(1 + \sqrt{3})}$
(c) $-\frac{3 + 2\sqrt{3}}{(1 - \sqrt{3})}$ (d) $\frac{3 - 2\sqrt{3}}{(1 + \sqrt{3})}$

◆ TYPE – III ◆
BASED ON QUADRANTS

146. If $\mu = 60°$, then $\sin\mu + \cos(90° - \mu)$ = _____.

SSC Phase-XII 25/06/2024 (Shift-03)

(a) $\sqrt{3}$ (b) $\frac{\sqrt{3}}{2}$
(c) $2\sqrt{3}$ (d) $\frac{1}{2}$

147. If $\tan(90° - \theta) = \frac{2}{\sqrt{3}}$, then the value of $2\sqrt{3}\tan\theta + 1$ is:

SSC CHSL 11/08/2023 (Shift-02)

(a) 4 (b) 5
(c) 3 (d) 6

148. The value of $\sin^2\frac{2\pi}{3} + \cos^2\frac{5\pi}{6} - \tan^2\frac{3\pi}{4}$ is:

SSC Phase-X 03/08/2022 (Shift-02)

(a) $\frac{1}{2}$ (b) $\frac{1}{4}$
(c) 4 (d) 2

149. What is the simplified value of $\cos^2(90° - \theta) - \left[\frac{\{\cos(90° - \theta)\cos\theta\}}{\cot\theta}\right]$?

SSC CHSL 31/05/2022 (Shift- 3)

(a) 4 (b) 2
(c) 0 (d) 1

150. Express sin 74° + tan 74° in terms of trigonometric ratios of angles between 0° and 45°.

SSC CGL 11/09/2024 (Shift-03)

(a) sec16° + cot16°
(b) cos 16° + cot 16°
(c) cosec 16° + cot 16°
(d) cosec 16° + sec 16°

151. Find the value of

$$\left[\frac{4\cos(90-\theta)\sin^3(90+\theta)-4\sin(90+\theta)\cos^3(90-\theta)}{\cos\left(\frac{180+8\theta}{2}\right)}\right]$$

SSC CGL 17/09/2024 (Shift-02)

(a) −1 (b) 1
(c) 2 (d) 0

152. Evaluate the following.

$$\frac{5\cos^2 120° + 4\sec^2 30° - \tan^2 135°}{\sin^2 30° + \cos^2 30°}$$

SSC Phase-XII 25/06/2024 (Shift-02)

(a) $\frac{67}{12}$ (b) $\frac{57}{5}$
(c) $\frac{27}{5}$ (d) $\frac{37}{12}$

153. Find the exact value of sin150°.

SSC CGL 12/09/2024 (Shift-03)

(a) 1.5 (b) 0.5
(c) 0.75 (d) –0.5

154. What is the value of tan 570°

SSC CHSL 04/07/2024 (Shift-02)

(a) $-\frac{1}{\sqrt{3}}$ (b) $-\sqrt{3}$
(c) $\frac{1}{\sqrt{3}}$ (d) $\sqrt{3}$

155. If $\cos x = -\frac{1}{2}$, x lies in third quadrant, then $\tan x$ = ?

SSC CGL 17/07/2023 (Shift-03)

(a) $\sqrt{3}$ (b) $\frac{\sqrt{3}}{2}$
(c) $\frac{2}{\sqrt{3}}$ (d) $\frac{1}{\sqrt{3}}$

156. Find $\cos\left(-\frac{7\pi}{2}\right)$.

SSC CPO 10/11/2022 (Shift-03)

(a) $\frac{1}{2}$ (b) 1
(c) −1 (d) 0

157. cosec 2910° + sec 4260° + tan 2565° + cot 1755° = ?

SSC CGL 20/07/2023 (Shift-01)

(a) 3 (b) 1
(c) 4 (d) 0

158. Simplify

$$\frac{1}{\cos x}\sqrt{\frac{\cos(\pi+x)\cos(-x)}{\sin(\pi-x)\cos\left(\frac{\pi}{2}+x\right)}}$$

SSC Phase-X 04/08/2022 (Shift-03)

(a) tan x (b) cot x
(c) sec x (d) cosec x

159. If A is an acute angle, the simplified form of

$$\frac{\cos(\pi-A).\cot\left(\frac{\pi}{2}+A\right)\cos(-A)}{\tan(\pi+A)\tan\left(\frac{3\pi}{2}+A\right)\sin(2\pi-A)}$$ is :

SSC CGL 13/12/2022 (Shift-04)

(a) $\cos^2 A$ (b) $\sin A$
(c) $\sin^2 A$ (d) $\cos A$

◆ TYPE – IV (A) ◆
IF $\alpha + \beta = 90° \Rightarrow \alpha$ & β ARE COMPLEMENTARY

160. Evaluate the value of
(cosec56° cos34° – cos59°cosec31°)

SSC CGL 11/09/2024 (Shift-02)

(a) 1 (b) 2
(c) −1 (d) 0

161. Find the value of cos47°sec133° + sin44°cosec136°.

SSC CGL 17/07/2023 (Shift-02)

(a) $\frac{1}{2}$ (b) 1
(c) 0 (d) −1

162. 2(sin 1° × sec 89°) + 3 (cos 11° × cosec 79°) + 5 (tan 21° × tan 69°) = ?

SSC CPO 04/10/2023 (Shift-02)

(a) 20 (b) 12
(c) 11 (d) 10

163. The value of cot 13° cot 27° cot 60° cot 63° cot 77° is:

SSC CHSL 02/08/2023 (Shift-02)

(a) $\frac{1}{\sqrt{3}}$ (b) 0
(c) $\sqrt{3}$ (d) 1

164. Simplify the following 7(cot 11° cot13° cot 19° cot 29° cot 61° cot 71° cot 77° cot 79°)

SSC CHSL 09/08/2023 (Shift-03)

(a) 7 (b) –7
(c) – 1 (d) 1

165. Find the value of $\frac{\tan 55°}{\cot 35°}$.

SSC CGL 18/09/2024 (Shift-02)

(a) cos35° (b) 1
(c) sec55° (d) 0

166. $\frac{3\sin 58°}{5\cos 32°} + \frac{2\cos 62°}{5\sin 28°}$ is equal to:

SSC CGL TIER-II 15/11/2020

(a) 0 (b) 2
(c) 3 (d) 1

167. Find the value of

$$\frac{\sec 35°}{\text{cosec}55°} - \frac{\cos 25°}{\sin 65°}$$

SSC CGL 11/09/2024 (Shift-01)

(a) 0 (b) 1
(c) $\frac{1}{2}$ (d) 2

168. Find the value of $2\frac{\tan 54°}{\cot 36°} - \frac{\cot 41°}{\tan 49°}$.

SSC CGL 25/09/2024 (Shift-01)

(a) −1 (b) 1
(c) 0 (d) 2

169. Find the value of $\frac{\cos 25° - \sin 65°}{\cos 25° + \sin 65°}$

SSC Phase-XII 24/06/2024 (Shift-03)

(a) 1 (b) $\frac{1}{2}$
(c) −1 (d) 0

170. Evaluate:

$$7\left(\frac{\text{cosec}\,24^\circ}{\sec 66^\circ}\right)^3 + 8\left(\frac{\cot 37^\circ}{\tan 53^\circ}\right)^4 - \left(2\frac{\sec 14^\circ}{\text{cosec}\,76^\circ}\right)^2 + \left(-3\frac{\tan 82^\circ}{\cot 8^\circ}\right)^3$$

SSC Phase-XII 20/06/2024 (Shift-02)

(a) 40 (b) 2
(c) −16 (d) 10

171. Find the value of

$$\frac{\cos 65°}{\sin 25°} + \frac{5\sin 19°}{\cos 71°} - \frac{3\cos 28°}{\sin 62°}.$$

SSC CPO 03/10/2023 (Shift-02)

(a) 3 (b) 2
(c) 1 (d) 0

172. Find the value of $\frac{\cos 37°}{\sin 53°}$ – cos 47° cosec 43°.

SSC CPO 04/10/2023 (Shift-01)

(a) −1 (b) 1
(c) 2 (d) 0

173. The value of $\frac{\cos 65°}{\sin 25°} + \frac{\cos 55°}{\sin 35°}$ – tan50° cot50° is equal to:

SSC CHSL 08/08/2023 (Shift-01)

(a) 2 (b) 0
(c) – 1 (d) 1

174. What is the value of

$$\frac{\cos 50°}{\sin 40°} + \frac{3\,\text{cosec}80°}{\sec 10°} -$$

2 cos50°. cosec40° ?

SSC CGL TIER-II 08/08/2022

(a) 3 (b) 4
(c) 2 (d) 5

175. The value of $\frac{2\cos 15°\sin 15°}{\cos^2 15° - \sin^2 15°}$ is:

SSC CPO 29/06/2024 (Shift-02)

(a) $\frac{1}{\sqrt{3}}$ (b) ∞
(c) $\frac{\sqrt{5}-1}{\sqrt{5}+1}$ (d) $\sqrt{3}$

176. If tan(x + y)tan(x – y) = 1, then find the value of tan$x$.

SSC CGL TIER-II (18/01/2025)

(a) $\sqrt{3}$ (b) $\frac{1}{\sqrt{3}}$
(c) 1 (d) $\frac{1}{2\sqrt{3}}$

177. If $\cos(2\theta+54°) = \sin\theta$, $0° < (2\theta + 54°) < 90°$, then what is the value of $\frac{1}{\cot 5\theta + \sec\frac{5\theta}{2}}$?

SSC CGL 20/08/2021 (Shift- 02)

(a) $\frac{\sqrt{3}}{2}$ (b) $\frac{1}{3}$

(c) $\frac{\sqrt{3}}{3}$ (d) $2\sqrt{3}$

178. If $\sin3A = \cos(A + 10°)$, where 3A is an acute angle then what is the value of $2\,\text{cosec}\frac{3A}{2} + 6\tan^2 3A - \frac{3}{2}$ $\tan^2 3A$?

SSC CGL TIER-II 15/11/2020

(a) $\frac{7}{4}$ (b) $\frac{35}{2}$

(c) $\frac{17}{2}$ (d) 5

179. If $\sin 3A = \cos(A - 26°)$ then A = ___ where 3A is an acute angle.

SSC CGL 24/09/2024 (Shift-01)

(a) 29° (b) 52°

(c) 26° (d) 78°

180. If $\sec6A = \text{cosec}(A - 29°)$, where 2A is an acute angle, then the measure of $\angle A$ = ___ °.

SSC CPO 27/06/2024 (Shift-01)

(a) 18° (b) 19°

(c) 21° (d) 17°

181. If $\sin(x + y) = \cos\{3(x + y)\}$, then, the value of $\cot\{2(x + y)\}$ is _____.

SSC CPO 29/06/2024 (Shift-02)

(a) –1 (b) 1

(c) –2 (d) 2

182. If $\sin(5x - 40°) = \cos(5y + 40°)$, then the value of $x + y$ is:

SSC Phase-XII 24/06/2024 (Shift-03)

(a) 18° (b) 40°

(c) 15° (d) 20°

183. If $\cos(3\alpha) = \sin(\alpha - 22°)$, where $3\alpha < 90°$, then what is the value of α?

SSC CHSL 02/08/2023 (Shift-03)

(a) 29° (b) 26°

(c) 27° (d) 28°

184. If $\tan(5\theta - 10°) = \cot(5\phi + 20°)$, then the value of $\theta + \phi$ is:

SSC CHSL 10/08/2023 (Shift-02)

(a) 16° (b) 18°

(c) 15° (d) 20°

185. If $\sin(5x - 25°) = \cos(5y + 25°)$, where $5x - 25°$ and $5y + 25°$ are acute angles, then the value of $(x + y)$ is:

SSC CGL 18/07/2023 (Shift-03)

(a) 50° (b) 40°

(c) 18° (d) 16°

186. If $\cos\theta = \sin(2\theta) \neq 0$, what is the value of $\cos^4\theta + \sin^4\theta + \cos^3\theta + \sin^3\theta + \sin^2\theta + \cos^2\theta + \sin\theta + \cos\theta$?

SSC Phase-X 05/08/2022 (Shift-03)

(a) $\frac{18+8\sqrt{3}}{7}$ (b) $\frac{8+7\sqrt{3}}{18}$

(c) $\frac{7+18\sqrt{3}}{8}$ (d) $\frac{18+7\sqrt{3}}{8}$

187. If $\sec(5\alpha - 15°) = \text{cosec}(15° - 2\alpha)$, then the value of $\cos3\alpha + \sin2\alpha + \cot\alpha$ is:

SSC PHASE-IX 2022

(a) $\frac{\sqrt{3}}{2}$ (b) $\frac{3\sqrt{3}}{2}$

(c) $\frac{\sqrt{3}(\sqrt{3}-1)}{2}$ (d) $\frac{\sqrt{3}(\sqrt{3}+1)}{2}$

◆ TYPE – IV (B) ◆
IF $\alpha + \beta = 90° \Rightarrow \alpha$ & β ARE COMPLEMENTARY

188. Find the value of $\sin^2 35° + \sin^2 55°$ = _____

SSC CHSL 03/08/2023 (Shift-03)

(a) 0 (b) 1

(c) $2\sin35°$ (d) $2\sin55°$

189. The value of $\cos^2 29° + \cos^2 61°$ is:

SSC CGL 09/09/2024 (Shift-01)

(a) $\frac{\sqrt{3}}{2}$ (b) 2

(c) 1 (d) 0

190. The value of $\sin^2 30° - \sin^2 40° + \sin^2 45° - \sin^2 55° - \sin^2 35° + \sin^2 45° - \sin^2 50° + \sin^2 60°$ is:

SSC CGL 09/09/2024 (Shift-01)

(a) 1 (b) 2

(c) 4 (d) 0

191. The value of $\sin^2 15° + \sin^2 25° + \sin^2 35° + \sin^2 45° + \sin^2 55° + \sin^2 65° + \sin^2 75°$ is:

SSC CPO 29/06/2024 (Shift-03)

(a) 4 (b) 7

(c) $\frac{7}{2}$ (d) $\frac{7}{3}$

192. What is the value of $\sec^2 54° - \cot^2 36° + \frac{3}{2}\sin^2 37° \times \sec^2 53° + \frac{2}{\sqrt{3}}\tan60°$?

SSC CGL 14/07/2023 (Shift-02)

(a) $\frac{1}{3}$ (b) $\frac{9}{2}$

(c) $\frac{3}{2}$ (d) $\frac{7}{2}$

193. The value of $\frac{\sin^2 30° + \cos^2 60° - \sec35°.\sin55°}{\sec60° + \text{cosec}30°}$

SSC CPO 23/11/2020 (Shift-02)

(a) $-\frac{1}{4}$ (b) $\frac{1}{4}$

(c) $-\frac{1}{8}$ (d) $\frac{1}{8}$

194. The value of $\frac{\sin23°\cos67° + \sec52°\sin38° + \cos23°\sin67° + \text{cosec}52°\cos38°}{\text{cosec}^2 20° - \tan^2 70°}$

SSC CGL 11/04/2022 (Shift- 01)

(a) 3 (b) 4

(c) 2 (d) 0

195. The value of $\frac{3(\cot^2 47° - \sec^2 43°) - 2}{(\tan^2 23° - \text{cosec}^2 67°)}$ $\frac{}{\cos ec^2(68° + \theta) - \tan(\theta + 61°) - \tan^2(22° - \theta) + \cot(29° - \theta)}$

SSC CGL TIER-II 29/01/2022

(a) 1 (b) 0

(c) 5 (d) –1

196. The value of $\frac{2\sin^2 38°\sec^2 52° + \cos64°\sin26° + \sin^2 64°}{\tan^2 23° + \cot^2 23° - \sec^2 67° - \text{cosec}^2 67°}$ is:

SSC CGL TIER-II 15/10/2020

(a) – 2 (b) $\frac{3}{2}$

(c) 2 (d) $-\frac{3}{2}$

197. What is the value of $\sqrt{\frac{9\sin^2 26° + 9\sin^2 64° + 36}{32 - 7\cos^2 32° - 7\cos^2 58°}}$

SSC CGL 18/09/2024 (Shift-03)

(a) $\frac{3\sqrt{5}}{5}$ (b) $\frac{5\sqrt{3}}{3}$

(c) $\frac{5\sqrt{5}}{3}$ (d) $\frac{3\sqrt{3}}{5}$

198. Evaluate: $\left(\frac{\sin23°\cos67° + \cos23°\sin67°}{\text{cosec}^2 15° - \tan^2 75°}\right)^3$

SSC Phase-XII 25/06/2024 (Shift-04)

(a) 1 (b) 0

(c) –1 (d) 8

199. If A + B = 90°, then the expression $\frac{\cot A}{\cot B} + \cos^2 A + \cos^2 B$ is equal to:

SSC CPO 03/10/2023 (Shift-03)

(a) $\cot^2 B$ (b) $\text{cosec}^2 A$

(c) $\text{cosec}^2 B$ (d) $\cot^2 A$

200. The value of $[\tan^2(22° - \theta) - \tan^2(\theta + 68°) - \cos ec^2(68° + \theta) + \cot^2(22° - \theta)]$ $\frac{}{[3(\cot^2 52° - \sec^2 38°) + 2(\cos ec^2 28° - \tan^2 62°)]}$

SSC PHASE-IX 2022

(a) 0 (b) -1

(c) 1 (d) $\frac{1}{5}$

201. The value of $\frac{4\tan^2 30° + \sin^2 30°\cos^2 45° + \sec^2 48° - \cot^2 42°}{\cos37°\sin53° + \sin37°\cos53° + \tan18°\tan72°}$

SSC CGL TIER-II 29/01/2022

(a) $\frac{35}{48}$ (b) $\frac{59}{48}$

(c) $\frac{49}{24}$ (d) $\frac{35}{24}$

202. The value of

$$\frac{3(\cos ec^2 26^\circ - \tan^2 64^\circ) + (\cot^2 42^\circ - \sec^2 48^\circ)}{\cot(22^\circ - \theta) - \cos ec^2(62^\circ + \theta) - \tan(\theta + 68^\circ) + \tan^2(28^\circ - \theta)}$$

SSC CGL TIER-II 03/02/2022

(a) 3 (b) 4
(c) –1 (d) –2

203. The value of

$$\frac{(\cos 9^\circ + \sin 81^\circ)(\sec 9^\circ + \cos ec\, 81^\circ)}{\cos ec^2 71^\circ + \cos^2 15^\circ - \tan^2 19^\circ + \cos^2 75^\circ}$$

is:

SSC CGL TIER-II 03/02/2022

(a) 1 (b) 4
(c) – 3 (d) 2

◆ TYPE – V (A) ◆
TRIGONOMETRIC IDENTITIES (BASIC QUESTIONS.)

204. Select the INCORRECT formula from the following options.

SSC CGL 17/07/2023 (Shift-01)

(a) $\sec^2\theta - \tan^2\theta = 1$
(b) $\sin^2\theta + \cos^2\theta = 1$
(c) $\text{cosec}^2\theta - \cot^2\theta = 1$
(d) $\sec^2\theta + \cos^2\theta = 1$

205. Select the correct identity from the following options.

SSC CHSL 07/06/2022 (Shift-002)

(a) $1 + \cos^2 A = \sin^2 A$
(b) $1 + \sin^2 A = \cos^2 A$
(c) $1 + \tan^2 A = \sec^2 A$
(d) $1 + \sec^2 A = \tan^2 A$

206. Select the incorrect formula from the following options.

SSC CGL TIER-II 03/03/2023

(a) $\text{cosec}^2\theta - \cot^2\theta = 1$
(b) $\sin^2\theta + \cos^2\theta = 1$
(c) $\sec^2\theta + \cos^2\theta = 1$
(d) $\sec^2\theta - \tan^2\theta = 1$

207. Which of the following is FALSE statement?

SSC CGL 18/07/2023 (Shift-03)

(a) $\tan^2 A = 1 - \sec^2 A$
(b) $\sin A = \tan A \times \cos A$
(c) $\text{cosec}^2 A - 1 = \cot^2 A$
(d) $\cos A \times \sec A = 1$

208. $\sec\theta\sqrt{1-\sin^2\theta} = ?$

SSC CGL 25/07/2023 (Shift-02)

(a) –1 (b) 1
(c) ∞ (d) 0

209. Solve the following equation.

$\sec^2\theta(\sqrt{1-\sin^2\theta})$

SSC CPO 10/11/2022 (Shift-01)

(a) $\tan\theta$ (b) $\text{cosec}\theta$
(c) $\sec\theta$ (d) 1

210. Which of the following is equal to $\frac{1}{\tan\theta} + \tan\theta$?

SSC CGL TIER-II 08/08/2022

(a) $\frac{\cos ec\,\theta}{\sec\theta}$ (b) $\tan^2\theta$
(c) $\sec\theta.\ \text{cosec}$ (d) 1

211. The given expression $\frac{\cos^2 A}{(1+\sin A)}$ is equal to.

SSC CHSL 07/08/2023 (Shift-03)

(a) cos A (b) 1 + cos A
(c) sin A (d) 1 – sin A

212. What is the value of $\frac{1+\sin A}{\cos^2 A}$?

SSC CHSL 13/03/2023 (Shift-01)

(a) $\frac{1}{1-\sin A}$ (b) $\frac{1}{1+\sin A}$
(c) $\frac{1}{1+\cos A}$ (d) $\frac{1}{1-\cos A}$

213. Which of the following is equal to $\left[\frac{\cos\theta}{\sin\theta} + \frac{\sin\theta}{\cos\theta}\right]$?

SSC CGL TIER-II 08/08/2022

(a) $\cot\theta.\sec\theta$ (b) $\text{cosec}\theta.\sec\theta$
(c) $\sec\theta.\tan\theta$ (d) $\text{cosec}\theta.\ \tan\theta$

214. The value of $\frac{32}{\sec^2\theta} - \frac{20}{1+\tan^2\theta} + 12\sin^2\theta$ is:

SSC Phase-XI 27/06/2023 (Shift-04)

(a) 20 (b) 32
(c) 24 (d) 12

215. Find the value of $\frac{2\sin\theta}{\sqrt{4-4\sin^2\theta}}$

SSC Phase-X 04/08/2022 (Shift-03)

(a) $2\tan\theta$ (b) $\tan\theta$
(c) $2\cot\theta$ (d) $\cot\theta$

216. Simplify the following:

$(1 + \cot^2\theta)(1 - \cos\theta)(1 + \cos\theta)$

SSC CPO 10/11/2022 (Shift-03)

(a) 1 (b) –5
(c) 3 (d) –3

217. The value of (cosec A + cot A) (1 – cosA) is:

SSC CPO 10/11/2022 (Shift-03)

(a) cos A (b) tan A
(c) cot A (d) sin A

218. Which of the following is equal to secA – cosA?

SSC CGL TIER-II 08/08/2022

(a) sinA. cotA (b) cotA. cosA
(c) tanA. sinA (d) cosA sinA

219. The given expression is equal to

$1 - \frac{\tan^2\phi}{\sec^2\phi}$

SSC CHSL 07/08/2023 (Shift-01)

(a) $\sin^2\phi\cos^2\phi$ (b) $\sin^2\phi\cot^2\phi$
(c) $\cot^2\phi\cos^2\phi$ (d) $\tan^2\phi\cos^2\phi$

220. The value of $(\text{cosec}^2 B - 1)(\sec^2 B - 1)$ is:

SSC CHSL 03/08/2023 (Shift-02)

(a) 4 (b) 3
(c) 2 (d) 1

221. $\cos^4 A - \sin^4 A$ is equal to:

SSC CHSL 03/08/2023 (Shift-03)

(a) $2\cos^2 A + 1$ (b) $1 - 2\sin^2 A$
(c) $2\sin^2 A - 1$ (d) $-(2\sin^2 A + 1)$

222. Find the value of

$\text{cosec}\theta\,(1 - \cos\theta)\,(\text{cosec}\theta + \cot\theta)$.

SSC CPO 27/06/2024 (Shift-02)

(a) –1 (b) 2
(c) 0 (d) 1

223. $(1 - \cos^2 A)\sec^2 A$ is equal to

SSC CPO 28/06/2024 (Shift-02)

(a) $\cos^2 A \cot^2 A$ (b) $\tan^2 A$
(c) $\cot^2 A$ (d) $\text{cosec}^2 A$

224. The value of $\sec x - \cos x$ = ?

SSC CGL 21/07/2023 (Shift-02)

(a) $\tan x \sin x$ (b) $\sec x \tan x$
(c) $\tan x \cos x$ (d) $\sec x \cos x$

225. Solve the following.

$$\frac{1}{1+\cos\theta} + \frac{1}{1-\cos\theta} = ?$$

SSC CGL 12/09/2024 (Shift-02)

(a) $2\text{cosec}^2\theta$ (b) $2\sin^2\theta$
(c) $2\sec^2\theta$ (d) $2\tan^2\theta$

226. If $\frac{1}{1-\sin\theta} + \frac{1}{1+\sin\theta} = 4\sec\theta$, $(0 < \theta < 90^\circ)$, then the value of $(\cot\theta + \text{cosec}\theta)$ is:

SSC CGL 13/09/2024 (Shift-01)

(a) $\sqrt{3}$ (b) $\sqrt{6}$
(c) $\sqrt{2}$ (d) $\sqrt{5}$

227. Simplify $\left[1 - \sin^2 32^\circ + \frac{1}{1+\tan^2 58^\circ}\right]$.

SSC CGL 19/09/2024 (Shift-02)

(a) $\frac{1}{2}$ (b) $\sin 32^\circ$
(c) 1 (d) $\sqrt{3}$

228. The value of $\left(\frac{\sin\theta}{1+\cos\theta} + \frac{1+\cos\theta}{\sin\theta}\right)\left(\frac{1}{\tan\theta+\cot\theta}\right)$ is equal to:

SSC CHSL 11/07/2024 (Shift-01)

(a) $\tan\theta$ (b) 1
(c) $2\cos\theta$ (d) $2\sin\theta$

229. Evaluate the given expression.

$$\frac{5}{1+\cot^2\theta} + \frac{3}{1+\tan^2\theta} + 2\cos^2\theta$$

SSC CGL 26/09/2024 (Shift-01)

(a) 0 (b) 3
(c) 5 (d) 2

230. The value of $\frac{1+\sin A}{\cos A} + \frac{\cos A}{1+\sin A}$ is :

SSC CHSL 14/08/2023 (Shift-01)

(a) 2 sec A (b) 2 cosecA
(c) sec A (d) cosec A

231. What is the value of following in the term of trigonometric ratios?

$\frac{\sin A}{1+\cos A}+\frac{1+\cos A}{\sin A}$

SSC CPO 10/11/2022 (Shift-02)

(a) 2 cosec A (b) 2 cos A
(c) 2 sec A (d) 2 sin A

232. Find the value of

$\frac{\cos A}{1+\sin A}+\frac{1+\sin A}{\cos A}+2$

SSC CHSL 08/08/2023 (Shift-03)

(a) 1 + 2 secA (b) 2 – secA
(c) 2 + secA (d) 2(1 + secA)

233. Find the value of $3(\sec^2A - 1)(\text{cosec}A - 1)(\text{cosec}A + 1) + 4(\text{cosec}^2A - \cot^2A)$.

SSC CHSL TIER-II (18/11/2024)

(a) 7 (b) 3
(c) 10 (d) 12

234. If $(1 + \tan^2\theta) + \left[1+\frac{1}{\tan^2\theta}\right] = k$, then $\sqrt{k}$ = ?

SSC CGL 06/06/2019 (Shift-03)

(a) $\sec\theta.\text{cosec}\theta$
(b) $\cos\theta.\text{cosec}\theta$
(c) $\sin\theta.\cos\theta$
(d) $\sec\theta.\sin\theta$

235. The value of $\left(\frac{1}{\sin\theta}+\frac{1}{\tan\theta}\right)\left(\frac{1}{\sin\theta}-\frac{1}{\tan\theta}\right)$

SSC CGL 18/07/2023 (Shift-02)

(a) 0 (b) 2
(c) 3 (d) 1

236. $(\sin\theta + \text{cosec}\theta)^2 + (\cos\theta + \sec\theta)^2$ is equal to ?

SSC CGL TIER-II 07/03/2023

(a) $5 + \tan^2\theta + \cot^2\theta$
(b) $7 + \tan^2\theta - \cot^2\theta$
(c) $7 + \tan^2\theta + \cot^2\theta$
(d) $5 + \tan^2\theta - \cot^2\theta$

237. If $\sin^2\theta\cos^2\theta = \frac{2}{9}$, then what will be the value of $\text{cosec}^2\theta + \sec^2\theta$?

SSC CGL 09/12/2022 (Shift-03)

(a) $\frac{7}{2}$ (b) $\frac{5}{2}$
(c) $\frac{9}{2}$ (d) $9\sqrt{2}$

238. If $\sec\theta - \cos\theta = 14$ and $14\sec\theta = x$, then the value of x is _____.

SSC CHSL 25/05/2022 (Shift-02)

(a) $\tan^2\theta$ (b) $\sec^2\theta$
(c) $2\sec\theta$ (d) $2\tan\theta$

239. Simplify: $\frac{\cos^4\theta - \sin^4\theta}{\sin^2\theta}$

SSC CPO 11/11/2022 (Shift-02)

(a) $1 - \tan^2\theta$ (b) $\tan^2\theta - 1$
(c) $\cot^2\theta - 1$ (d) $1 - \cot^2\theta$

240. If $\cos^4\alpha - \sin^4\alpha = \frac{5}{6}$, then the value of $2\cos^2\alpha - 1$ _____.

SSC CPO 11/11/2022 (Shift-03)

(a) $\frac{11}{6}$ (b) $\frac{5}{6}$
(c) $\frac{6}{11}$ (d) $\frac{6}{5}$

241. The given expression is equal to:

$\frac{(1+\tan^2 A)}{\text{cosec}^2A.\tan A}$

SSC CHSL 03/08/2023 (Shift-03)

(a) $\sec^2A$ (b) secA
(c) tanA (d) $\tan^2A$

242. $\frac{\cos\theta}{\sec\theta-1}+\frac{\cos\theta}{\sec\theta+1}$ is equal to:

SSC CHSL 02/08/2023 (Shift-01)

(a) $2\sec^2\theta$ (b) $2\cot^2\theta$
(c) $2\cos^2\theta$ (d) $2\sin^2\theta$

243. If $\frac{\cos\theta}{(1+\sin\theta)}+\frac{\cos\theta}{(1-\sin\theta)} = 4$ and θ is acute, then the value of θ is:

SSC CHSL 03/08/2023 (Shift-01)

(a) 60° (b) 15°
(c) 45° (d) 30°

244. Simplify the expression

$\frac{3-\sin^2 A+\cos^2 A}{2+2\cos^2 A}$

SSC CHSL 04/08/2023 (Shift-01)

(a) 1 (b) 0
(c) –1 (d) 2

245. The given expression is equal to:

$\frac{\sin^4 A+\cos^4 A}{1-2\sin^2 A\cos^2 A}$

SSC CHSL 09/08/2023 (Shift-02)

(a) 1 (b) –1
(c) 0 (d) 2

246. If $\sin\theta\cos\theta = \frac{1}{\sqrt{3}}$ then the value of $(\sin^4\theta + \cos^4\theta)$ is:

SSC CHSL 17/08/2023 (Shift-03)

(a) 1 (b) $\frac{5}{3}$
(c) $\frac{2}{3}$ (d) $\frac{1}{3}$

247. Simplify the given expression

$\frac{1+\sin^4\theta+\cos^4\theta}{\cos^2\theta+\sin^4\theta}$

SSC CGL 24/07/2023 (Shift-04)

(a) 3 (b) 1
(c) 2 (d) 4

248. $\sin^4\theta + \cos^4\theta$ in terms of $\sin\theta$ can be written as:

SSC CGL 03/12/2022 (Shift-02)

(a) $2\sin^4\theta + 2\sin^2\theta - 1$
(b) $2\sin^4\theta - 2\sin^2\theta$
(c) $2\sin^4\theta - 2\sin^2\theta - 1$
(d) $2\sin^4\theta - 2\sin^2\theta + 1$

249. Find the value of the following expression. $2(\tan^2A\cos^2A + \cos^2A)$

SSC CHSL 10/08/2023 (Shift-04)

(a) 4 (b) 1
(c) 3 (d) 2

250. If $\tan\theta = \frac{8}{19}$, find the value of $\sec^2\theta$.

SSC CHSL 02/08/2023 (Shift-03)

(a) $\frac{11}{19}$ (b) $1\frac{8}{19}$
(c) $\frac{297}{361}$ (d) $1\frac{64}{361}$

251. If $\text{cosec}\theta = 1\frac{7}{22}$, find the value of $\cot^2\theta$.

SSC CHSL 03/08/2023 (Shift-04)

(a) $\frac{49}{484}$ (b) $\frac{357}{484}$
(c) $\frac{225}{484}$ (d) $\frac{7}{22}$

252. If $\cot\theta = \frac{9}{17}$, find the value of $\text{cosec}^2\theta$.

SSC CHSL 02/08/2023 (Shift-01)

(a) $1\frac{9}{17}$ (b) $\frac{208}{289}$
(c) $\frac{8}{17}$ (d) $1\frac{81}{289}$

253. Simplify the following.

$\frac{\sin^3 A-\cos^3 A}{\sin A-\cos A}$, where A is an acute angle.

SSC CGL 09/09/2024 (Shift-02)

(a) 1 + sinA cosA (b) 1 – 3sinA
(c) 3 cosA – 1 (d) sinA + cosA

254. Simplify: $\frac{\sin^3 A+\cos^3 A}{\sin A+\cos A}$

SSC CPO 28/06/2024 (Shift-03)

(a) 1 – tanA secA
(b) 1 – cosA cotA
(c) 1 – sinA tanA
(d) 1 – sinA cosA

255. Simplify the given equation. $\frac{\cot^3 A-1}{\cot A-1}$ = ?

SSC CHSL 08/08/2023 (Shift-02)

(a) cosec^2 A – cot A
(b) cosec^2 A + cot A
(c) $\cot^2$ A + cosec A
(d) $\cot^2$ A – cosec A

256. Simplify the following.

$\frac{\sin^3\alpha+\cos^3\alpha}{\sin\alpha+\cos\alpha}$

SSC CGL 14/07/2023 (Shift-04)

(a) $1 + \sin\alpha\cos\alpha$
(b) $\tan\alpha$
(c) $1 - \sin\alpha\cos\alpha$
(d) $\sec\alpha$

257. Solve the following to find its value in terms of trigonometric ratios.

(sin A + cos A) (1 – sin A cosA)

SSC CGL 06/12/2022 (Shift-03)

(a) $\sin^3A + \cos^3A$
(b) $\sin^2A - \cos^2A$
(c) [cosA – sinA] [$\sin^2A + \cos^2A$]
(d) $\sin^3A - \cos^3A$

258. If $\sin^6\theta + \cos^6\theta = \frac{1}{3}$, $0° < \theta < 90°$, then what is the value of sinθcosθ?

SSC CGL 18/08/2021 (Shift-03)

(a) $\frac{\sqrt{6}}{6}$ (b) $\frac{\sqrt{2}}{3}$
(c) $\frac{\sqrt{2}}{\sqrt{3}}$ (d) $\frac{2}{2}$

259. The value of $\frac{1}{\sin\theta} - \frac{\cot^2\theta}{1+\operatorname{cosec}\theta}$ is:

SSC CHSL 07/08/2023 (Shift-04)

(a) 2 (b) 1
(c) –1 (d) 0

260. Simplify the given equation

$(1 + \tan^2 A)(1 + \cot^2 A) = ?$

SSC CHSL 14/08/2023 (Shift-01)

(a) $\frac{1}{\cos^2A(1-\sin^2A)}$
(b) $\frac{1}{\sin^2A(1-\sin^2A)}$
(c) $\frac{1}{\sin^2A + \operatorname{cosec}^2A}$
(d) $\frac{1}{\sin^2A(1+\cos^2A)}$

261. The value of

$\frac{17}{\sec^2\theta} + 10\sin^2\theta + \frac{7}{1+\cot^2\theta}$ is:

SSC CPO 29/06/2024 (Shift-03)

(a) 34 (b) 20
(c) 17 (d) 19

262. Which of the following is NOT a correct formula?

SSC Phase-XII 24/06/2024 (Shift-02)

(a) $\frac{1+\tan^2A}{\tan^2A} = \sin^2A$
(b) secAsinA = tanA
(c) $\operatorname{cosec}^2A - \cot^2A = 1$
(d) $\cos^2A\cot^2A = \cot^2A - \cos^2A$

263. Find the value of $(\tan^2\theta + \tan^4\theta)$.

SSC CPO 03/10/2023 (Shift-01)

(a) $\operatorname{Sec}^2\theta - \sec^4\theta$
(b) $\operatorname{Cot}^2\theta - \tan^2\theta$
(c) $\operatorname{Sec}^4\theta + \sec^2\theta$
(d) $\operatorname{Sec}^4\theta - \sec^2\theta$

264. If $\left(\frac{\cos A}{1-\sin A}\right) + \left(\frac{\cos A}{1+\sin A}\right) = 4$, then what will be the value of A? ($0° < \theta < 90°$)

SSC CGL 19/07/2023 (Shift-01)

(a) 90° (b) 45°
(c) 60° (d) 30°

265. If $3\sin^2\theta + 5\cos^2\theta = 4$ and θ is an acute angle, then the value of tanθ is:

SSC CGL 19/09/2024 (Shift-01)

(a) $\frac{1}{\sqrt{2}}$ (b) $\frac{1}{\sqrt{3}}$
(c) $\frac{1}{2}$ (d) 1

266. If $a = x\cos\theta + y\sin\theta$ and $b = x\sin\theta - y\cos\theta$, then $a^2 + b^2$ is equal to:

SSC CGL 17/09/2024 (Shift-02)

(a) $x + y$ (b) $x^2 - y^2$
(c) $x^2 + y^2$ (d) $x - y$

267. $(\sec\phi - \tan\phi)^2 (1 + \sin\phi)^2 \div \cos^2\phi = ?$

SSC CGL 13/04/2022 (Shift-03)

(a) 1 (b) – 1
(c) cot2 ϕ (d) cos2 ϕ

268. If pcosA = 2qsinA and 2pcosecA – qsecA = 3, then the value of $p^2 + 4q^2$ is:

SSC CGL 10/09/2024 (Shift-02)

(a) 2 (b) 4
(c) 1 (d) 3

269. Let x = r cos (t), y = r sin (t) cos (u), z = r sin (t)sin (u). Then the value of $x^2 + y^2 + z^2$ is ___.

SSC CGL 23/09/2024 (Shift-01)

(a) r^2 cos (u) (b) r^2
(c) 2r (d) r^2 sin (t)

270. The value of $\frac{1}{\operatorname{cosec}\theta - \cot\theta} - \frac{1}{\sin\theta}$ is equal to:

SSC CGL 23/09/2024 (Shift-02)

(a) tanθ (b) cosecθ
(c) cotθ (d) secθ

271. The value of $(\sin A + \operatorname{cosec} A)^2 + (\cos A + \sec A)^2$ is:

SSC CGL 24/09/2024 (Shift-02)

(a) $3 + \tan^2A + \cot^2A$
(b) $7 + \tan^2A + \cot^2A$
(c) $5 + \tan^2A + \cot^2A$
(d) $1 + \tan^2A + \cot^2A$

272. Find the value of $\frac{1-\cot^2\theta}{\tan^2\theta - 1}$.

SSC CPO 28/06/2024 (Shift-01)

(a) tanθ (b) cotθ
(c) $\tan^2\theta$ (d) $\cot^2\theta$

273. If $\left\{\frac{(3\sin\theta - \cos\theta)}{(\cos\theta + \sin\theta)}\right\} = 1$, then the value of cotθ is:

SSC CGL 14/07/2023 (Shift-01)

(a) 3 (b) 0
(c) 1 (d) 2

274. If $\frac{\sin x - \cos x}{\sin x + \cos x} = \frac{2}{5}$, then the value of $\frac{1+\cot^2 x}{1-\cot^2 x}$ is:

SSC CGL 19/07/2023 (Shift-01)

(a) 2.25 (b) 1.45
(c) 3.75 (d) 5.25

275. If $(\sin\theta + \operatorname{cosec}\theta)^2 + (\cos\theta + \sec\theta)^2 = k + \tan^2\theta + \cot^2\theta$, then the value of k is equal to :

SSC CGL TIER-II 18/11/2020

(a) 7 (b) 2
(c) 9 (d) 5

276. If $1 + \sin^2\theta = a\cos^2\theta$, then find $\sec^2\theta + \tan^2\theta$.

SSC Phase-XI 30/06/2023 (Shift-01)

(a) a^{-2} (b) a^{-1}
(c) a (d) a^2

277. If $m + 5 = \sec^2A\operatorname{cosec}^2A(1 - \cos^2A)(1 - \sin^2A) + 5$, then find the value of m:

SSC CGL 25/09/2024 (Shift-01)

(a) 10 (b) 0
(c) 1 (d) 5

278. If x = p secA cosB, y = q secA sinB and z = r tanA, what is the value of the following expression?

$\frac{x^2}{p^2} + \frac{y^2}{q^2} - \frac{z^2}{r^2}$

SSC CGL 25/09/2024 (Shift-02)

(a) 1 (b) $p^2 + q^2 - r^2$
(c) $p^2 - q^2 + r^2$ (d) 0

279. If $x = a\sec\theta\cos\phi$, $y = b\sec\theta\sin\phi$ and $z = c\tan\theta$, then the value of $\left(\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2}\right)$ is equal to:

SSC CGL 26/09/2024 (Shift-03)

(a) 2 (b) 1
(c) 3 (d) 0

280. If $\tan\theta = \frac{5}{8}$, then find the value of $\frac{(1+\cos\theta)(1-\cos\theta)}{(1+\sin\theta)(1-\sin\theta)}$

SSC CGL 19/09/2024 (Shift-03)

(a) $\frac{5}{64}$ (b) $\frac{2}{25}$
(c) $\frac{25}{64}$ (d) $\frac{64}{25}$

281. $\frac{3\cos^3\theta - 2\cos\theta}{\sin\theta - 3\sin^3\theta}$ is equal to

SSC CHSL 03/07/2024 (Shift-03)

(a) tanθ (b) $\cot^2\theta$
(c) cotθ (d) $\tan^2\theta$

282. Simplify the given expression

$\frac{3\left(\sin^4 z - \cos^4 z + 1\right)}{\sin^2 z}$

SSC CHSL 11/08/2023 (Shift-04)

(a) 9 (b) 2
(c) 4 (d) 6

283. The value of $\left(\frac{\operatorname{cosec}A}{\operatorname{cosec}A - 1} + \frac{\operatorname{cosec}A}{\operatorname{cosec}A + 1}\right)\left(\frac{\sec A}{\sec A - 1} + \frac{\sec A}{\sec A + 1}\right)^{-1}$, when A = 60° is:

SSC CGL 13/09/2024 (Shift-03)

(a) 4 (b) 3
(c) 1 (d) 2

284. If tan(t) + cot(t) = 1, then one of the values of the expression $\frac{1}{\sin(t)+\cos(t)}$ is _____.

SSC CPO 27/06/2024 (Shift-01)

(a) $\frac{1}{\sqrt{3}}$ (b) $\frac{1}{\sqrt{2}}$

(c) $\frac{\sqrt{3}}{2}$ (d) $\sqrt{3}$

285. The value of (cosecθ – sinθ)(secθ – cosθ)(tanθ + cotθ) is:

SSC CPO 29/06/2024 (Shift-01)

(a) –3 (b) 3

(c) secθ cosecθ (d) 1

286. If sinθ + tanθ = p and tanθ – sinθ = q, then which of the relation satisfies for given values of p and q?

SSC CHSL 11/08/2023 (Shift-04)

(a) $p^2 - q^2 = 4\sqrt{pq}$

(b) $p^2 - q^2 = 8\sqrt{pq}$

(c) $p^2 + q^2 = 4\sqrt{pq}$

(d) $p^2 + q^2 = 4pq$

287. If P cos α = 3 and 4 tan α = Q, then what is the relation between P and Q, which is independent of α?

SSC CGL 19/07/2023 (Shift-02)

(a) $\frac{9}{P^2}+\frac{16}{Q^2}=1$ (b) $\frac{9}{P^2}-\frac{16}{Q^2}=1$

(c) $\frac{P^2}{9}-\frac{Q^2}{16}=1$ (d) $\frac{P^2}{9}+\frac{Q^2}{16}=1$

288. If $\cos x + \sin x = \sqrt{2}\cos x$ what is the value of $(\cos x - \sin x)^2 + (\cos x + \sin x)^2$?

SSC CGL 19/07/2023 (Shift-04)

(a) 2 (b) 1

(c) 0 (d) $\frac{1}{\sqrt{2}}$

289. Simplify: $\frac{1+\sin t}{4-4\sin t}-\frac{1-\sin t}{4+4\sin t}$

SSC CGL 21/07/2023 (Shift-04)

(a) 4tant.sint (b) tant.sect

(c) tant – sint (d) tant + sint

290. If r = 15(sinθ + cosθ) and s = 16(sinθ – cosθ), then the value of $\frac{r^2}{15^2}+\frac{s^2}{16^2}$ is _____.

SSC CGL 25/07/2023 (Shift-03)

(a) 8 (b) 6

(c) 4 (d) 2

291. If sin m + sin n = p, cos m + cos n = q, then find the value of sin m × sin n + cos m × cos n.

SSC CHSL 20/03/2023 (Shift-02)

(a) $p^2 - q^2 - 2$ (b) $\frac{(p^2+q^2-2)}{2}$

(c) p + q – pq (d) p + q + pq

292. $\frac{(1+\cos\theta)^2+\sin^2\theta}{(\operatorname{cosec}^2\theta-1)\times\sin^2\theta}$ = ?

SSC CGL TIER-II 11/09/2019

(a) cosθ(1 + sinθ)

(b) 2 cosθ(1 + secθ)

(c) secθ(1 + sinθ)

(d) 2 secθ(1 + secθ)

293. The value of $\sqrt{\sec^2\theta+\operatorname{cosec}^2\theta}\times\sqrt{\tan^2\theta-\sin^2\theta}$ is equal to :

SSC CGL 06/06/2019 (Shift-01)

(a) cosecθ. $\sec^2\theta$

(b) sinθ. $\sec^2\theta$

(c) sinθ. $\cos^2\theta$

(d) cosecθ. $\cos^2\theta$

294. What is the value of $\frac{1}{\sin^4(90^\circ-2\alpha)}+\frac{1}{\cos^2(90^\circ-2\alpha)-1}$

SSC CHSL 04/07/2024 (Shift-02)

(a) $\sin^2 2\alpha \times \tan^2 2\alpha$

(b) $\sec^2 2\alpha \times \tan^2 2\alpha$

(c) $\sec^2 2\alpha \times \cot^2 2\alpha$

(d) $\cos^2 2\alpha \times \cot^2 2\alpha$

295. Simplify:$(1 + \tan^2 A) \times \left(1+\frac{1}{\tan^2 A}\right)$

SSC CPO 27/06/2024 (Shift-01)

(a) $\frac{1}{\sin^2 A-\sin^4 A}$

(b) $\frac{1}{\sin^2 A-\cos^4 A}$

(c) $\frac{1}{\cos^2 A-\sin^4 A}$

(d) $\frac{1}{\cos^2 A-\tan^4 A}$

296. If A, B and C are the interior angles of ΔABC, then what is the value of ?

$$\left\{\tan\frac{A}{2}+\operatorname{cosec}\frac{B+C}{2}\right\}\left\{\tan\frac{A}{2}-\operatorname{cosec}\frac{B+C}{2}\right\}$$

SSC CPO 28/06/2024 (Shift-03)

(a) 2 (b) –1

(c) –2 (d) 1

297. $\tan^4 A + \tan^2 A$ is equal to:

SSC CPO 29/06/2024 (Shift-01)

(a) $\sec^2 A - \sec^4 A$

(b) $-(\sec^2 A + \sec^4 A)$

(c) $\sec^4 A - \sec^2 A$

(d) $\sec^4 A + \sec^2 A$

298. The value of.....

$$\operatorname{cosec}\theta\left[\frac{1+\cos\theta}{\sin\theta}+\frac{\sin\theta}{1+\cos\theta}\right]-2\cot^2\theta$$

SSC Phase-XII 26/06/2024 (Shift-02)

(a) 0 (b) 1

(c) 3 (d) 2

299. What is value of the expression

$$\sin A\left(1+\frac{\sin A}{\cos A}\right)+\cos A\left(1+\frac{\cos A}{\sin A}\right)$$

SSC CHSL 04/08/2023 (Shift-01)

(a) secA + cosecA

(b) sinA + cosA

(c) sinA – cosA

(d) secA – cosecA

◆ TYPE – V (B) ◆
BASED ON ALGEBRAIC FORMULAE

300. If tanθ + cotθ = 2 and θ is an acute angle, then the value of θ is:

SSC CGL 19/09/2024 (Shift-03)

(a) 30° (b) 15°

(c) 60° (d) 45°

301. If tanA + cotA = 2, then the value of $\tan^2 A + \cot^2 A$ is:

SSC CGL 24/09/2024 (Shift-02)

(a) 8 (b) 2

(c) 6 (d) 4

302. If A is an acute angle and tanA + cotA = 2, find the value of $7\tan^3 A - 6\cot^3 A + 8\sec^2 A$.

SSC CGL 26/09/2024 (Shift-01)

(a) 6 (b) 17

(c) 16 (d) 7

303. If θ is an acute angle and tanθ + cotθ = 2, then find the value of $\tan^2\theta + \cot^3\theta + 6\tan^3\theta\cot^2\theta$.

SSC CHSL 02/07/2024 (Shift-03)

(a) 8 (b) 6

(c) 10 (d) 12

304. If θ is an acute angle and cotθ + tanθ = 2, then find the value of $\tan^{12}\theta + \cot^{12}\theta + 2\tan^5\theta\cot^7\theta$.

SSC CHSL 04/07/2024 (Shift-03)

(a) 2 (b) 3

(c) 4 (d) 1

305. If tanθ + cotθ = 2, then find the value of $\tan^{53}\theta + \cot^{38}\theta$.

SSC Phase-XII 20/06/2024 (Shift-03)

(a) – 1 (b) 2

(c) 0 (d) 1

306. If θ is an angle and sin θ + cosec θ = 2, then the value of $\sin^5\theta + \operatorname{cosec}^5\theta$ is:

SSC CPO 03/10/2023 (Shift-01)

(a) 2 (b) 10

(c) 4 (d) 5

307. If cosθ + secθ = 2, then the value of $\sin^6\theta + \cos^6\theta$ is:

SSC CPO 10/11/2022 (Shift-03)

(a) $\frac{1}{3}$ (b) 0

(c) 1 (d) $\frac{1}{2}$

308. If tanθ + cotθ = 6, then the value of $\tan^2\theta + \cot^2\theta$ is ____ .

SSC CPO 28/06/2024 (Shift-02)

(a) 24 (b) 34

(c) 20 (d) 22

309. If sinθ + cosecθ = $\sqrt{5}$, then the value of $\sin^3\theta + \operatorname{cosec}^3\theta$ is:

SSC Phase-XII 26/06/2024 (Shift-03)

(a) $\sqrt{5}$ (b) 0

(c) $\frac{1}{\sqrt{5}}$ (d) $2\sqrt{5}$

310. If $\cos\theta + \sec\theta = \sqrt{3}$, then the value of $\cos^3\theta + \sec^3\theta$ is:

SSC CHSL 02/08/2023 (Shift-03)

(a) 0 (b) $\frac{1}{\sqrt{3}}$

(c) $\sqrt{3}$ (d) $2\sqrt{3}$

311. If $\sec\theta + \cos\theta = 32$, then $\sec^2\theta + \cos^2\theta$ is ____.

SSC CGL 14/07/2023 (Shift-04)

(a) 1000 (b) 1022

(c) 1024 (d) 1020

312. If $3x$ = secA and $\frac{3}{x}$ = tanA then $9\left(x^2 - \frac{1}{x^2}\right)$ is ——.

SSC CGL 27/07/2023 (Shift-02)

(a) 3 (b) 9

(c) 1 (d) $\frac{1}{9}$

313. If $\sin\theta - \operatorname{cosec}\theta = 6$, then what is the value of $\sin\theta + \operatorname{cosec}\theta$?

SSC CHSL 14/03/2023 (Shift-02)

(a) 6 (b) $\sqrt{40}$

(c) $\sqrt{34}$ (d) $\sqrt{38}$

314. If $\sec x - \cos x = 4$, then what will be the value of $\frac{(1+\cos^2 x)}{\cos x}$?

SSC CGL 08/12/2022 (Shift-03)

(a) $\frac{9}{4}$ (b) $\frac{1}{4}$

(c) $2\sqrt{5}$ (d) $\sqrt{5}$

315. The value of the expression $\sin x + \operatorname{cosec} x = 2$, then the value of $\sin^7 x + \operatorname{cosec}^7 x$ is:

SSC CGL 19/09/2024 (Shift-01)

(a) 0 (b) 1

(c) 2 (d) 4

◆ TYPE – V (C) ◆

If $\sec\theta + \tan\theta = K$, Then $\sec\theta - \tan\theta = 1/K$, If $\operatorname{cosec}\theta - \cot\theta = K$, Then $\operatorname{cosec}\theta + \cot\theta = 1/K$

316. If $\sec\theta + \tan\theta = x$, then find $\sin\theta$.

SSC CGL 25/09/2024 (Shift-03)

(a) $\frac{x^2+1}{1-x^2}$ (b) $\frac{x^2-1}{1+2x^2}$

(c) $\frac{x^2-1}{1+x^2}$ (d) $\frac{1-x^2}{1+x^2}$

317. If tanA + secA = K, then what is the value of sinA.

SSC CHSL 10/03/2023 (Shift-01)

(a) $\frac{2K}{K^2+1}$ (b) $\frac{K^2-1}{2K}$

(c) $\frac{K^2+1}{K^2-1}$ (d) $\frac{K^2-1}{K^2+1}$

318. If sec A – tan A = p, then find the value of secA.

SSC CPO 04/10/2023 (Shift-02)

(a) $\frac{p^2-1}{p^2+1}$ (b) $\frac{p^2+1}{p^2-1}$

(c) $\frac{p^2+1}{2p}$ (d) $\frac{p^2+1}{p}$

319. If $\tan\theta + \sec\theta = a$, then what is $\sec\theta$?

SSC Phase-XI 28/06/2023 (Shift-03)

(a) $\frac{a-1}{2a}$ (b) $\frac{a^2+1}{2a}$

(c) $\frac{1}{2a}$ (d) $\frac{a^2-1}{2a}$

320. If $\sec\theta + \tan\theta = 5$, $(\theta \neq 0)$ then $\sec\theta$ is equal to:

SSC CGL 14/07/2023 (Shift-04)

(a) $\left(5+\frac{1}{5}\right)$ (b) $\frac{1}{2}\left(3+\frac{1}{3}\right)$

(c) $\frac{1}{2}\left(5+\frac{1}{5}\right)$ (d) $\left(3+\frac{1}{3}\right)$

321. If $\sec\theta + \tan\theta = 5$, then $\sin\theta$?

SSC CHSL 11/08/2023 (Shift-02)

(a) 0 (b) $\frac{12}{13}$

(c) $\frac{13}{12}$ (d) $\frac{1}{5}$

322. If $\sec\theta + \tan\theta = 5$, then what is the value of $\sec\theta - \tan\theta$?

SSC CHSL 09/03/2023 (Shift-01)

(a) $\frac{1}{5}$ (b) $\frac{1}{3}$

(c) $\frac{1}{6}$ (d) $\frac{1}{2}$

323. If $\sec\theta - \tan\theta = m$, then $2\tan\theta$ = ?

SSC CHSL 02/08/2023 (Shift-03)

(a) $\frac{1-m}{m}$ (b) $\frac{1-m^2}{m}$

(c) $\frac{1+m^2}{m}$ (d) $\frac{m^2+1}{m}$

324. If $\sec\theta \tan\theta = 3\sqrt{}$ then find the positive value of $\sin\theta + \cos\theta$.

SSC CHSL 04/08/2023 (Shift-03)

(a) $\frac{\sqrt{3}}{2}$ (b) $\frac{2}{\sqrt{3}}$

(c) $\frac{2}{(1+\sqrt{3})}$ (d) $\frac{(1+\sqrt{3})}{2}$

325. If $\sec^2\theta + \tan^2\theta = \frac{25}{18}$, then the value of $\sec^4\theta - \tan^4\theta$ is:

SSC CGL 14/07/2023 (Shift-03)

(a) $\frac{18}{25}$ (b) $\frac{25}{12}$

(c) $\frac{25}{9}$ (d) $\frac{25}{18}$

326. If $\operatorname{cosec}\theta + \cot\theta = \frac{3}{2}$, then what is the value of $\operatorname{cosec}\theta$?

SSC CGL 24/07/2023 (Shift-03)

(a) $\frac{13}{12}$ (b) $\frac{3}{2}$

(c) $\frac{11}{12}$ (d) $\frac{9}{13}$

327. If the value of cosecA + cotA = m, then the value of cosecA – cotA is:

SSC CPO 10/11/2022 (Shift-02)

(a) $\frac{1}{m}$ (b) m

(c) $\sqrt{m}$ (d) m^2

328. If $\operatorname{cosec}\theta + \cot\theta = m$, find the value of $\frac{m^2-1}{m^2+1}$.

SSC CGL 24/07/2023 (Shift-02)

(a) 1.1 (b) 2.0

(c) $\cos\theta$ (d) $-\sin\theta$

329. If cosecA + cotA = $a\sqrt{b}$, then find the value of $\frac{(a^2b-1)}{(a^2b+1)}$.

SSC CHSL 17/08/2023 (Shift-02)

(a) cosA (b) tanA

(c) $\frac{1}{\sin A}$ (d) $\frac{1}{\cot A}$

330. If $\sec x + \tan x = 5$ and $\operatorname{cosec} y - \cot y = \frac{1}{3}$, then find the value of $(\sec x + \operatorname{cosec} y) - (\tan x - \cot y)$.

SSC CHSL 15/03/2023 (Shift-01)

(a) 2.2 (b) 3.2

(c) 4.2 (d) 3.1

◆ TYPE – V (D) ◆

BASED ON QUADRATIC EQUATIONS

331. If $2\sin^2\theta + 3\sin\theta - 2 = 0$, $(0 < q < 90°)$ then the value of θ is:

SSC CGL 24/09/2024 (Shift-01)

(a) 15° (b) 30°

(c) 60° (d) 45°

332. If $\tan^2 A + 2\tan A - 63 = 0$ Given that $0 < A < \frac{\pi}{4}$ what is the value of $(2\sin A + 5\cos A)$?

SSC CGL 11/04/2022 (Shift-03)

(a) $19\sqrt{50}$ (b) $15\sqrt{50}$

(c) $\frac{19}{\sqrt{50}}$ (d) $\frac{15}{\sqrt{50}}$

333. If $10\sin^2\theta + 6\cos^2\theta = 7$, $0 < \theta < 90°$, then find the value of $\tan\theta$.

SSC CGL 24/09/2024 (Shift-03)

(a) $\sqrt{3}$ (b) 1

(c) $\frac{1}{\sqrt{3}}$ (d) $\frac{1}{2}$

334. $12\sin^2A + 2\cos^2 A = 7$, Where $0 < A < 180°$. Then cot A is:

SSC CHSL 01/07/2024 (Shift-04)

(a) ± 1 (b) $\pm\frac{1}{2}$

(c) $\pm\frac{1}{\sqrt{2}}$ (d) $\pm\frac{1}{\sqrt{3}}$

335. If $4\sin^2\theta - 4\sqrt{3}\sin\theta + 3 = 0$, then find the value of θ. $(0 < \theta < 90°)$

SSC CHSL 10/07/2024 (Shift-03)

(a) 30° (b) 75°

(c) 60° (d) 45°

336. Solve the equation for A (in degrees). $2\cos^2 A + 3\cos A - 2 = 0$, $0 < A < 90°$

SSC CPO 28/06/2024 (Shift-01)

(a) 30° (b) 80°

(c) 60° (d) 45°

337. If $4\sin^2\theta = 3(1 + \cos\theta)$, $0° < \theta < 90°$, then what is the value of $\sqrt{15}\tan\theta + \frac{4}{\sqrt{15}}\sin\theta + 2\sec\theta$?

SSC CPO 28/06/2024 (Shift-02)

(a) 24 (b) $4\sqrt{15}$

(c) $\frac{8}{\sqrt{15}}$ (d) 8

338. If $3\sec^4\theta + 8 = 10\sec^2\theta$, then the value of $\tan\theta$ can be:

SSC Phase-XII 26/06/2024 (Shift-04)

(a) $1, \frac{1}{\sqrt{3}}$ (b) $1, \sqrt{3}$

(c) $2, \frac{1}{\sqrt{3}}$ (d) $2, \sqrt{3}$

339. If $7\sin^2 A + 3\cos^2 A = 4$, then find cot A:

SSC CPO 04/10/2023 (Shift-02)

(a) $\frac{1}{\sqrt{3}}$ (b) $\frac{1}{\sqrt{2}}$

(c) $\sqrt{3}$ (d) $\sqrt{2}$

340. If $2\text{cosec}^2\theta + 3\cot^2\theta = 17$, then the value of '$\theta$' when $0° \le \theta \le 90°$ is:

SSC CGL 12/09/2024 (Shift-01)

(a) 45° (b) 90°

(c) 30° (d) 75°

341. If $\sec^2\alpha + 4\cos^2\alpha = 4$ and $0° \le \alpha \le 90°$, then find the value of α.

SSC CGL 21/04/2022 (Shift-03)

(a) 30° (b) 60°

(c) 0° (d) 45°

342. If $7\cos^2\theta + 5\sin^2\theta - 6 = 0$, $(0° < \theta < 90°)$, then what is the value of $1+\sqrt{\frac{\sec\theta+\tan\theta}{\sec\theta-\tan\theta}}$?

SSC CGL 10/09/2024 (Shift-03)

(a) $\sqrt{2} + 2$ (b) $\sqrt{2} + 1$

(c) $\sqrt{2} - 1$ (d) $\sqrt{2} - 2$

343. If $\sqrt{2}\sec^2\theta - 4\sec\theta + 2\sqrt{2} = 0$, then what is the value of $\sin^2\theta + \tan^2\theta$?

SSC CHSL 14/03/2023 (Shift-03)

(a) $\frac{1}{2}$ (b) $\frac{2}{3}$

(c) $\frac{5}{2}$ (d) $\frac{3}{2}$

344. If $\sin^2\theta - \cos^2\theta - 3\sin\theta + 2 = 0$, $0° < \theta < 90°$, Then what is the value of $1 + \sec\theta + \tan\theta$?

SSC CGL 13/04/2022 (Shift-03)

(a) $-1-\sqrt{3}$ (b) $-1+\sqrt{3}$

(c) $1+\sqrt{3}$ (d) $1-\sqrt{3}$

345. If $6\cos^2\theta + \cos\theta = 2, 0° < \theta < 90°$ then the value of $(\text{cosec}\theta + \cot\theta + \tan\theta)$ is

SSC PHASE-IX 2022

(a) $2\sqrt{3}$ (b) $4\sqrt{3}$

(c) $3\sqrt{3}$ (d) $6\sqrt{3}$

346. If $8\cos^2\theta - 2\cos\theta - 3 = 0$, $0° < \theta < 90°$, then what is the value of $3\tan\theta + \text{cosec}\theta + \cot\theta$?

SSC PHASE-IX 2022

(a) $\frac{3}{4}\sqrt{7}$ (b) $4\sqrt{7}$

(c) $2\sqrt{7}$ (d) $\frac{4}{3}\sqrt{7}$

347. If $7\sin^2\theta + 4\cos^2\theta = 5$ and θ lies in the first quadrant, then what is the value of $\frac{\sqrt{3}\sec\theta + \tan\theta}{\sqrt{2}\cot\theta - \sqrt{3}\cos\theta}$?

SSC CGL TIER-II 29/01/2022

(a) $2(1+\sqrt{2})$ (b) $3\sqrt{2}$

(c) $2(\sqrt{2}-1)$ (d) $4\sqrt{2}$

348. If $\sin^2\theta - \cos^2\theta - 3\sin\theta + 2 = 0$, $0° < \theta < 90°$, then what is the value of $\frac{1}{\sqrt{\sec\theta - \tan\theta}}$ is:

SSC CGL 23/08/2021 (Shift-01)

(a) $\sqrt[4]{3}$ (b) $\sqrt[2]{2}$

(c) $\sqrt[2]{3}$ (d) $\sqrt[4]{2}$

349. Find the value of θ, if $\sec^2\theta + (1-\sqrt{3})\tan\theta - (1+\sqrt{3}) = 0$, where θ is an acute angle.

SSC CHSL 05/08/2021 (Shift-03)

(a) 30° (b) 15°

(c) 45° (d) 60°

350. If $4 - 2\sin^2\theta - 5\cos\theta = 0, 0° < \theta < 90°$, then the value of $\cos\theta + \tan\theta$ is:

SSC CPO 23/11/2020 (Shift-01)

(a) $\frac{2+\sqrt{3}}{2}$ (b) $\frac{2-\sqrt{3}}{2}$

(c) $\frac{1+2\sqrt{3}}{2}$ (d) $\frac{1-2\sqrt{3}}{2}$

351. If $\frac{\sin^2\theta - 3\sin\theta + 2}{\cos^2\theta} = 1$, where $0° < \theta < 90°$, then what is the value of $(\cos 2\theta - \sin 3\theta + \text{cosec}2\theta)$

SSC CGL 21/04/2022 (Shift-02)

(a) $\frac{3+4\sqrt{3}}{6}$ (b) $\frac{-3-4\sqrt{3}}{6}$

(c) $\frac{3-4\sqrt{3}}{6}$ (d) $\frac{-3+4\sqrt{3}}{6}$

352. If $\frac{\sin^2\theta}{\cos^2\theta - 3\cos\theta + 2} = 1$, θ lies in the first quadrant, then the value of $\frac{\tan^2\frac{\theta}{2} + \sin^2\frac{\theta}{2}}{\tan\theta + \sin\theta}$ is:

SSC CGL TIER-II 03/02/2022

(a) $\frac{2\sqrt{3}}{27}$ (b) $\frac{5\sqrt{3}}{27}$

(c) $\frac{2\sqrt{3}}{9}$ (d) $\frac{7\sqrt{3}}{54}$

353. If θ is an acute angle and $\sin\theta\cos\theta = 2\cos^3\theta - \frac{1}{4}\cos\theta$, then the value of $\sin\theta$ is:

SSC CHSL 10/08/2023 (Shift-02)

(a) $\frac{\sqrt{15}-1}{8}$ (b) $\frac{\sqrt{15}-1}{4}$

(c) $\frac{\sqrt{15}+1}{4}$ (d) $\frac{\sqrt{15}-1}{2}$

◆ TYPE – V (E) ◆
TRIGONOMETRIC IDENTITIES (ADVANCE QUESTIONS.)

354. $[(\sin\theta\tan\theta + \cos\theta)^2 - 1]$ is equal to:

SSC Phase-X 02/08/2022 (Shift-02)

(a) $\sec^2\theta$ (b) $\sec\theta$

(c) $\text{cosec}\,\theta$ (d) $\tan^2\theta$

355. Find the value of $(\text{cosec}\theta - \sin\theta)(\sec\theta - \cos\theta)(\tan\theta + \cot\theta)$.

SSC CGL 25/09/2024 (Shift-01)

(a) 0 (b) $\frac{1}{2}$

(c) 1 (d) -1

356. The expression $(\tan\theta + \cot\theta)(\sec\theta + \tan\theta)(1 - \sin\theta)$, $0° < A < 90°$, is equal to:

SSC CGL TIER-II 03/02/2022

(a) $\sec\theta$ (b) $\text{cosec}\theta$

(c) $\cot\theta$ (d) $\sin\theta$

357. Find the value of $(\sin^6A + \cos^6A + 3\sin^2A\cos^2A) - (\sin^4A - \cos^4A + 1)\text{cosec}^2A$.

SSC CHSL TIER-II (18/11/2024)

(a) 3 (b) 0

(c) -3 (d) -1

358. The value of the expression $4(\sin^6A + \cos^6A) - 6(\sin^4A + \cos^4A) + 8$ is:

SSC CHSL 02/08/2023 (Shift-04)

(a) 4 (b) 8
(c) 7 (d) 6

359. Find the value of the following expression.

$12(\sin^4\theta + \cos^4\theta) + 18(\sin^6\theta + \cos^6\theta) + 78\sin^2\theta\cos^2\theta$

SSC CGL 18/07/2023 (Shift-03)

(a) 30 (b) 40
(c) 10 (d) 20

360. Find the value of the following expression $5(\sin^4\theta + \cos^4\theta) + 3(\sin^6\theta + \cos^6\theta) + 19\sin^2\theta\cos^2\theta$.

SSC CGL 07/12/2022 (Shift-03)

(a) 8 (b) 5
(c) 6 (d) 7

361. The expression $(\cos^6\theta + \sin^6\theta - 1)(\tan^2\theta + \cot^2\theta + 2) + 3$ is equal to:

SSC CGL 13/04/2022 (Shift-02)

(a) 0 (b) – 1
(c) 2 (d) 1

362. If $p\sin A - \cos A = 1$, then $p^2 - (1 + p^2)\cos A$ is equal to:

SSC CGL 11/09/2024 (Shift-03)

(a) 1 (b) –1
(c) 2 (d) 0

363. If $\sin A + \cos A = \frac{1}{2\sqrt{2}}$, then find the value of $\sin^4A + \cos^4A$.

SSC CGL 17/09/2024 (Shift-02)

(a) $\frac{79}{128}$ (b) $\frac{7}{13}$
(c) $\frac{79}{126}$ (d) $\frac{5}{13}$

364. If $\sin A + \cos A = \sqrt{3}$, then find the value of $\tan A + \cot A + 2\sin A\cos A$.

SSC CGL 19/09/2024 (Shift-01)

(a) 6 (b) 2
(c) 1 (d) 3

365. If $\sin A + \cos A = \frac{4}{3}$, then find the value of $\tan A + \cot A$.

SSC Phase-XII 20/06/2024 (Shift-03)

(a) $\frac{4}{3}$ (b) $\frac{3}{4}$
(c) $\frac{18}{7}$ (d) $\frac{7}{18}$

366. If $\sin A - \cos A = \frac{\sqrt{3}-\sqrt{2}}{2}$ then the value of $\sin A \cdot \cos A$ is:

SSC CGL 23/09/2024 (Shift-02)

(a) $\frac{2\sqrt{3}-1}{4}$ (b) $\frac{2\sqrt{2}-1}{8}$
(c) $\frac{2\sqrt{6}-1}{8}$ (d) $\frac{3\sqrt{2}-1}{4}$

367. If $\sin\alpha + \cos\alpha = \frac{2}{\sqrt{3}}$, then what is $(\tan\alpha + \cot\alpha)$ equal to?

SSC CHSL 03/08/2023 (Shift-01)

(a) 1 (b) 2
(c) 6 (d) 0

368. If $\sin\theta + \cos\theta = \frac{\sqrt{11}}{3}$, then what is $\tan\theta + \cot\theta$ equal to?

SSC Phase-XI 27/06/2023 (Shift-04)

(a) 5 (b) 9
(c) 7 (d) 11

369. If $\sin\theta + \cos\theta = \frac{\sqrt{3}-1}{2\sqrt{2}}$, then what is the value of $\tan\theta + \cot\theta$?

SSC CGL 05/12/2022 (Shift-01)

(a) $8(\sqrt{3}-2)$ (b) $12(\sqrt{3}-2)$
(c) $12(\sqrt{3}+2)$ (d) $8(\sqrt{3}+2)$

370. If $\sin t + \cos t = \frac{4}{5}$, then find $\sin t \cdot \cos t$.

SSC CGL 17/07/2023 (Shift-01)

(a) $\frac{9}{50}$ (b) $\frac{-9}{50}$
(c) $\frac{9}{25}$ (d) $\frac{-9}{25}$

371. If $f(p) = \sin^{p+2}x + \cos^{p+2}x$, then the value of $6f(2) - 4f(4) + 10f(0)$

SSC CHSL 10/07/2024 (Shift-02)

(a) 14 (b) 10
(c) 12 (d) 11

372. If $\tan A + \sin A = x$, $\tan A - \sin A = y$, what is the value of $x^2 - y^2$?

SSC CHSL 11/07/2024 (Shift-04)

(a) $2\sqrt{\frac{x}{y}}$ (b) $4\sqrt{\frac{x}{y}}$
(c) $\sqrt{xy}$ (d) $4\sqrt{xy}$

373. If $(\sin x + \sin y) = a$ and $(\cos x + \cos y) = b$, then find the value of $(\sin x \sin y + \cos x \cos y)$.

SSC CPO 27/06/2024 (Shift-02)

(a) $\frac{a^2+b^2-1}{2}$ (b) $\frac{a^2+b^2-1}{2}$
(c) $\frac{a^2+b^2-2}{2}$ (d) $\frac{a^2+b^2+2}{2}$

374. If $\cot^2\theta = 1 - e^2$, then the value of $\text{cosec}\,\theta + \cot^3\theta\sec\theta$ is:

SSC CPO 05/10/2023 (Shift-03)

(a) $(2-e^2)^{\frac{3}{2}}$ (b) $(1-e^2)$
(c) $(1-e^2)^{\frac{3}{2}}$ (d) $(2-e^2)^{\frac{1}{2}}$

375. If $\tan^2\theta = 1 - a^2$, then the value of $\sec\theta + \tan^3\theta\,\text{cosec}\,\theta$ is:

SSC CGL 01/12/2022 (Shift-04)

(a) $(2-a)^{\frac{3}{2}}$ (b) $(a^2-1)^{\frac{3}{2}}$
(c) $(2-a^2)^{\frac{3}{2}}$ (d) $a^{\frac{3}{2}}$

376. If $\tan^2\alpha = 3 + Q^2$, then $\sec\alpha + \tan^3\alpha\,\text{cosec}\alpha$ = ?

SSC CGL 07/12/2022 (Shift-01)

(a) $(3+Q^2)^{\frac{3}{2}}$ (b) $(7+Q^2)^{\frac{3}{2}}$
(c) $(5-Q^2)^{\frac{3}{2}}$ (d) $(4+Q^2)^{\frac{3}{2}}$

377. Let $0° < \theta < 90°$, $(1 + \cot^2\theta)(1 + \tan^2\theta) \times (\sin\theta - \text{cosec}\theta)(\cos\theta - \sec\theta)$ is equal to:

SSC CGL TIER-II 29/01/2022

(a) $\sec\theta\,\text{cosec}\theta$ (b) $\sec\theta + \text{cosec}\theta$
(c) $\sin\theta + \cos\theta$ (d) $\sin\theta\cos\theta$

378. If $x\sin^3\theta + y\cos^3\theta = \sin\theta.\cos\theta$ and $x\sin\theta = y\cos\theta$ then the value of $x^2 + y^2$ is:

SSC CHSL 20/10/2020 (Shift-01)

(a) 0 (b) 4
(c) 1 (d) 2

379. If $a\sin^3 X + b\cos^3 X = \sin X\cos X$ and $a\sin X = b\cos X$, then find the value of $a^2 + b^2$, provided that X is neither 0° nor 90°.

SSC CGL 11/09/2024 (Shift-01)

(a) 0 (b) 1
(c) $a^2 - b^2$ (d) $a^2 + b^2$

380. The value of

$\left(\frac{\sin A}{1-\cos A} + \frac{1-\cos A}{\sin A}\right) \div \left(\frac{\cot^2 A}{1+\text{cosec}\,A} + 1\right)$ is:

SSC CGL TIER-II 12/09/2019

(a) $\frac{3}{2}$ (b) $\frac{1}{2}$
(c) 1 (d) 2

381. The value of

$\frac{\sec^2\theta(2+\tan^2\theta+\cot^2\theta) \div (\sin^2\theta - \tan^2\theta)}{(\text{cosec}^2\theta + \sec^2\theta)(1+\cot^2\theta)^2}$ is:

SSC CGL TIER-II 15/10/2020

(a) –1 (b) 1
(c) – 2 (d) 2

382. $\frac{2 + \tan^2 2A + \cot^2 2A}{\sec 2A.\ \text{cosec}\,2A}$ is equal to

SSC CGL 10/09/2024 (Shift-01)

(a) sec2A + tan2A
(b) sec2A + cosec2A
(c) cot2A – cosec2A
(d) sec2A cosec2A

383. The value of

$\left(\frac{\sin\theta - 2\sin^3\theta}{2\cos^3\theta - \cos\theta}\right)^3 \frac{1}{\tan\theta} - \sec^2\theta$ is:

SSC CGL 12/09/2024 (Shift-01)

(a) 2 (b) 1
(c) 0 (d) –1

384. The value of

$\frac{\sin A}{\cot A + \text{cosec}\,A} - \frac{\sin A}{\cot A - \text{cosec}\,A} - 1$

SSC CPO 11/11/2022 (Shift-02)

(a) $\frac{1}{2}$ (b) 1
(c) 0 (d) 2

385. $\left(\frac{\tan^3\theta}{\sec^2\theta}+\frac{\cot^3\theta}{\text{cosec}^2\theta}+2\sin\theta\cos\theta\right) \div (1 + \text{cosec}^2\theta + \tan^2\theta)$, $0° < \theta < 90°$ is equal to:

SSC CGL TIER-II 29/01/2022

(a) $\sin\theta\cos\theta$ (b) $\text{cosec}\theta$
(c) $\sec\theta$ (d) $\text{cosec}\theta\sec\theta$

386. $\frac{(1+\sec\theta\,\text{cosec}\,\theta)^2(\sec\theta-\tan\theta)^2(1+\sin\theta)}{(\sin\theta+\sec\theta)^2+(\cos\theta+\text{cosec}\,\theta)^2}$, $0°, < \theta < 90°$, is equal to:

SSC CGL TIER-II 29/01/2022

(a) $1 - \cos\theta$ (b) $1 - \sin\theta$
(c) $\cos\theta$ (d) $\sin\theta$

387. $\frac{\cot^3\theta}{\text{cosec}^2\theta}+\frac{\tan^3\theta}{\sec^2\theta} + 2\sin\theta\cos\theta = ?$

SSC CGL 16/08/2021 (Shift-02)

(a) $\text{cosec}\theta\sec\theta$ (b) $\text{cosec}^2\theta\sec^2\theta$
(c) $\sin\theta\cos\theta$ (d) $\sin\theta$

388. $\left(\sqrt{\sec^2\theta+\text{cosec}^2\theta}\right)\left(\frac{\sin\theta(1+\cos\theta)}{1+\cos\theta-\sin^2\theta}\right)$, $0° < \theta < 90°$ is equal to:

SSC CGL 17/08/2021 (Shift-01)

(a) $\cot\theta$ (b) $\text{cosec}^2\theta$
(c) $\sec^2\theta$ (d) $\tan\theta$

389. $\frac{(2\sin A)(1+\sin A)}{1+\sin A+\cos A}$ is equal to :

SSC CGL TIER-II 11/09/2019

(a) 1 + sinA – cosA
(b) 1 – sinA.cosA
(c) 1 + cosA – sinA
(d) 1 + sinA.cosA

390. The value of

$\frac{\sin\phi(1-\sin\phi)(\sin\phi+\cos\phi)(\sec\phi+\tan\phi)}{\sin\phi(1+\tan\phi)+\cos\phi(1+\cot\phi)}$ is:

SSC CGL TIER-II 12/09/2019

(a) $2\cos\phi$ (b) $\text{cosec}\phi.\sec\phi$
(c) $2\sin\phi$ (d) $\sin^2\phi.\cos^2\phi$

391. $\frac{1+\cos\theta-\sin^2\theta}{\sin\theta(1+\cos\theta)}\times\frac{\sqrt{\sec^2\theta+\text{cosec}^2\theta}}{\tan\theta+\cot\theta}$, $0° < \theta < 90°$

SSC CGL TIER-II 29/01/2022

(a) $\tan\theta$ (b) $\sec\theta$
(c) $\text{cosec}\theta$ (d) $\cot\theta$

392. The expression

$\frac{\tan^6\theta-\sec^6\theta+3\sec^2\theta\tan^2\theta}{\tan^2\theta+\cot^2\theta+2}$

$0° < \theta < 90°$, is equal to:

SSC CGL TIER-II 03/02/2022

(a) $\sec^2\theta\text{cosec}^2\theta$ (b) $-\sec^2\theta\text{cosec}^2\theta$
(c) $\cos^2\theta\sin^2\theta$ (d) $-\cos^2\theta\sin^2\theta$

393. The value of

$\frac{\sec^6\theta-\tan^6\theta-3\sec^2\theta\tan^2\theta+1}{\cos^4\theta-\sin^4\theta+2\sin^2\theta+2}$ is:

SSC CGL 05/03/2020 (Shift-01)

(a) $\frac{2}{3}$ (b) 1
(c) $\frac{3}{4}$ (d) $\frac{1}{2}$

394. The expression

$\frac{\cos^4\theta-\sin^4\theta+2\sin^2\theta+3}{(\text{cosec}\,\theta+\cot\theta+1)(\text{cosec}\,\theta-\cot\theta+1)-2}$

$0° < \theta < 90°$, is equal to

SSC CGL TIER-II 03/02/2022

(a) $\frac{1}{2}\sin\theta$ (b) $2\sin\theta$
(c) $\sec\theta$ (d) $2\,\text{cosec}\theta$

395. The expression

$\frac{(1-\sin\theta+\cos\theta)^2(1-\cos\theta)\sec^3\theta\,\text{cosec}^2\theta}{(\sec\theta-\tan\theta)(\tan\theta+\cot\theta)}$,

$0° < \theta < 90°$, is equal to

SSC CGL TIER-II 03/02/2022

(a) $\sin\theta$ (b) $2\cos\theta$
(c) $\cot\theta$ (d) $2\tan\theta$

396. The expression

$\frac{(1-2\sin^2\theta\cos^2\theta)(\cot\theta+1)\cos\theta}{(\sin^4\theta+\cos^4\theta)(1+\tan\theta)\text{cosec}\,\theta}-1$,

$0° < \theta < 90°$, is equal to:

SSC CGL TIER-II 03/02/2022

(a) $\cos^2\theta$ (b) $-\sin^2\theta$
(c) $\sec^2\theta$ (d) $-\sec^2\theta$

◆ TYPE – V (F) ◆
BAHUBALI CONCEPT

397. If $\cot^4\theta - \cot^2\theta = 1$, then the value of $\cos^4\theta + \cos^2\theta$ is:

SSC CGL 12/09/2024 (Shift-02)

(a) 2 (b) 1
(c) $\frac{1}{\sqrt{3}}$ (d) $\frac{3}{2}$

398. If $\cot^4A + \cot^2A = 1$ then find the value of $\cos^4A - 3\cos^2A$.

SSC CHSL TIER-II 02/11/2023

(a) 2 (b) –1
(c) 1 (d) –2

399. If $\sin A + \sin^2 A = 1$, then the value of $\cos^4 A + \cos^6 A$ is:

SSC CPO 04/10/2023 (Shift-01)

(a) 1 (b) cos A
(c) sin A (d) 0

400. If $\sin A + \sin^2A = 1$, then the value of the expression $(\cos^2A + \cos^4A)$ is .....

SSC CHSL 02/08/2023 (Shift-01)

(a) $\frac{3}{2}$ (b) 1
(c) 2 (d) $\frac{1}{2}$

401. If $\sin x + \sin^2 x = 1$, then what is the value of $\cos^8 x + 2\cos^6 x + \cos^4 x$?

SSC CHSL TIER-II 26/06/2023

(a) 1 (b) 0
(c) 4 (d) 2

402. If $\sin^4 x + \sin^2 x = 1$, then what is the value of $\cot^4 x + \cot^2 x$?

SSC CHSL 17/03/2023 (Shift-02)

(a) – 2 (b) 2
(c) – 1 (d) 1

403. If $\cos A + \cos^2A = 1$ then $\sin^2A + \sin^4A$ is equal to:

SSC CGL 23/09/2024 (Shift-02)

(a) $\frac{\cos^2 A}{\cos A}$ (b) 0
(c) $\frac{\cos A}{\cos^2 A}$ (d) 1

404. If $\cos A + \cos^2A = 1$, then the value of $\sin^4A + \sin^6A$ is:

SSC CGL 26/07/2023 (Shift-01)

(a) 2 (b) cosA
(c) 1 (d) sinA

405. If $\tan^4\theta + \tan^2\theta = 1$, what is the value of $11(\cos^4\theta + \cos^2\theta)$?

SSC CGL 27/07/2023 (Shift-03)

(a) –11 (b) 8
(c) 0 (d) 11

406. If $\cos A = \sin^2A$, and $a\sin^{12}A + b\sin^{10}A + c\sin^8A + \sin^6A = 1$, then $a + b + c = ?$

SSC CPO 11/11/2022 (Shift-01)

(a) 7 (b) 8
(c) 9 (d) 6

407. If $\sin^2\theta = \cos^3\theta$, then the value of $\cot^2\theta - \cot^6\theta$ is:

SSC CPO 11/11/2022 (Shift-02)

(a) –1 (b) 0
(c) 2 (d) 1

◆ TYPE – V (G) ◆
SOME SPECIAL CONCEPTS

408. Find the value of the following expression.

$\sqrt{\frac{1+\sin\theta}{1-\sin\theta}}$

SSC CGL 05/12/2022 (Shift-04)

(a) $\sec\theta + \tan\theta$
(b) $\text{cosec}\theta + \tan\theta$
(c) $\text{cosec}\theta + \cot\theta$
(d) $\sec\theta + \cot\theta$

409. What is $\frac{\sqrt{1-\sin A}}{\sqrt{1+\sin A}}$?

SSC CHSL 09/03/2023 (Shift-02)

(a) $\sec^2A$ (b) secA + tanA
(c) $\tan^2A$ (d) secA – tanA

410. The value of

$\frac{\sin\theta+\cos\theta-1}{\sin\theta-\cos\theta+1}\times\sqrt{\frac{1+\sin\theta}{1-\sin\theta}}$

SSC CGL TIER-II 16/11/2020

(a) 1 (b) – 1
(c) – 2 (d) 2

411. What is the value of

$\sqrt{\frac{\text{cosec}A+1}{\text{cosec}A-1}}+\sqrt{\frac{\text{cosec}A-1}{\text{cosec}A+1}} = ?$

SSC CHSL 13/03/2023 (Shift-01)

(a) 2 cosA (b) secA
(c) cosA (d) 2 secA

412. Simplify the given expression

$\sqrt{\frac{1+\cos P}{1-\cos P}}$

SSC CGL 25/07/2023 (Shift-03)
SSC CGL 03/12/2022 (Shift-04)
SSC CHSL 17/08/2023 (Shift-03)

(a) cosecP–cotP (b) secP–tanP
(c) secP+tanP (d) cosecP+cotP

413. If A is an acute angle, then $\sqrt{\frac{1-\cos A}{1+\cos A}}+\sqrt{\frac{1+\cos A}{1-\cos A}}$ is equal to:

SSC CGL 05/12/2022 (Shift-02)
SSC CGL 19/09/2024 (Shift-03)

(a) 2cosecA (b) 2cosA
(c) 2secA (d) 2sinA

414. The given expression is equivalent to:

$\sqrt{\frac{\sec\theta+1}{\sec\theta-1}}+\sqrt{\frac{\sec\theta-1}{\sec\theta+1}}$

SSC CGL 18/09/2024 (Shift-03)

(a) 2sinθ (b) 2cosecθ
(c) 2tanθ (d) 2tanθ secθ

415. If $\left\{\left(\frac{\sec\theta-1}{\sec\theta+1}\right)\right\}^{n}=\text{cosec}\,\theta-\cot\theta$ then n =?:

SSC CPO 09/11/2022 (Shift-01)

(a) 1 (b) 0.5
(c) –1 (d) – 0.5

416. Find the value of $\sqrt{\frac{1-\tan A}{1+\tan A}}$

SSC CPO 04/10/2023 (Shift-02)

(a) $\sqrt{\frac{1+\sin A}{\cos A}}$ (b) $\sqrt{\frac{1+\sin 2A}{\cos 2A}}$
(c) $\sqrt{\frac{1-\sin A}{\cos A}}$ (d) $\sqrt{\frac{1-\sin 2A}{\cos 2A}}$

417. Which of the following is equal to $\left[\frac{\tan\theta+\sec\theta-1}{\tan\theta-\sec\theta+1}\right]$?

SSC CGL TIER-II 08/08/2022

(a) $\frac{1+\cos\theta}{\sin\theta}$ (b) $\frac{1+\cot\theta}{\tan\theta}$
(c) $\frac{1+\sin\theta}{\cos\theta}$ (d) $\frac{1+\tan\theta}{\cot\theta}$

418. What is the value of $\frac{\tan\theta-\sec\theta+1}{\tan\theta+\sec\theta-1}$?

SSC CHSL 21/03/2023 (Shift-03)

(a) secθ + tanθ (b) $\frac{(1+\sin\theta)}{\cos\theta}$
(c) 2secθ (d) $\frac{\cos\theta}{(1+\sin\theta)}$

419. What is the value of $\frac{\cot\theta+\text{cosec}\,\theta-1}{\cot\theta-\text{cosec}\,\theta+1}$?

SSC CHSL 20/03/2023 (Shift-02)

(a) 2 secθ (b) 2 cosecθ
(c) 2 cotθ (d) (cosecθ + cotθ)

420. If A is an acute angle, which of the following is equal to $\frac{\sin A}{1+\cos A}$?

SSC CGL 09/09/2024 (Shift-03)

(a) $\frac{1-\cos A}{\sin A}$ (b) $\frac{1+\cos A}{\sin A}$
(c) $\frac{1-\sin A}{\cos A}$ (d) $\frac{1+\sin A}{\cos A}$

421. If $p=\frac{\sin A}{1+\cos A}$ then $\frac{\sin A}{1-\cos A}$ is equal to:

SSC CGL 26/09/2024 (Shift-02)

(a) $\frac{1}{p}$ (b) $\frac{1}{1-p}$
(c) $\frac{1}{p+1}$ (d) p – 1

422. Simplify. $\frac{\sin A}{1+\cos A}$ + cotA, where A is an acute angle.

SSC CGL 18/09/2024 (Shift-02)

(a) SinA (b) CosA
(c) sinA + cosA (d) cosecA

423. Simplify. $\frac{1+\sin\theta-\cos\theta}{1+\sin\theta+\cos\theta}$.

SSC CGL 24/09/2024 (Shift-03)

(a) $\cos\frac{\theta}{2}$ (b) $\sin\frac{\theta}{2}$
(c) $\cot\frac{\theta}{2}$ (d) $\tan\frac{\theta}{2}$

424. Find the value of

$\frac{\sin\theta+\cos\theta-1}{\sin\theta-\cos\theta+1}\times\frac{\tan^2\theta(\text{cosec}^2\theta-1)}{\sec\theta-\tan\theta}$.

SSC CGL TIER-II 13/09/2019

(a) 0 (b) – 1
(c) 1 (d) $\frac{1}{2}$

425. $\frac{1+\sin\theta}{\cos\theta}$ is equal to which of the following $\left(\text{where }\theta\neq\frac{\pi}{2}\right)$?

SSC CGL 12/12/2022 (Shift-02)

(a) $\frac{1+\cos\theta}{\sin\theta}$ (b) $\frac{\tan\theta+1}{\tan\theta-1}$
(c) $\frac{\tan\theta-1}{\tan\theta+1}$ (d) $\frac{\cos\theta}{1-\sin\theta}$

426. Find the value of

(1 + cotA – cosecA) (1 + tanA + secA) –3(sin²A + cos²A).

SSC CPO 27/06/2024 (Shift-02)

(a) –1 (b) 1
(c) 2 (d) –2

427. Solve the following:

(tanθ + secθ + 1)(cotθ – cosecθ + 1) =?

SSC CPO 29/06/2024 (Shift-01)

(a) 2 (b) 1
(c) 0 (d) –1

428. What is the value of (1 + tanθ – secθ)(1 + cotθ + cosecθ)?

SSC CHSL 09/03/2023 (Shift-03)

(a) – 1 (b) 0
(c) 1 (d) 2

429. The value of (1 + cot A – cosec A)(1 + tanA + sec A)–1 is:

SSC CPO 09/11/2022 (Shift-02)

(a) 1 (b) 2
(c) 3 (d) 0

430. The value of (cosecA + cotA+1)(cosecA–cotA + 1) – 2cosecA is

SSC CGL 09/03/2020 (Shift-02)

(a) 4cosecA (b) 2
(c) 2cosecA (d) 0

431. What is the value of (1 + cotA + tanA) (sinA – cosA) $\left(\frac{\sin A\cos A}{\sin^3 A-\cos^3 A}\right)$?

SSC CHSL TIER-II 26/06/2023

(a) 2 (b) – 1
(c) 1 (d) 0

432. If $\tan\frac{A}{2}=x$, then find *x*.

SSC CGL 18/07/2023 (Shift-01)

(a) $\frac{\sqrt{1+\cos A}}{\sqrt{1-\cos A}}$ (b) $\frac{\sqrt{1-\sin A}}{\sqrt{1+\cos A}}$
(c) $\frac{\sqrt{1-\cos A}}{\sqrt{1+\cos A}}$ (d) $\frac{\sqrt{\cos A-1}}{\sqrt{1+\cos A}}$

433. Find the value of $\sqrt{\frac{1-\sin 3\theta}{1+\sin 3\theta}}$

SSC CGL 21/07/2023 (Shift-02)

(a) sec3θ – tan3θ
(b) $(\sec 3\theta-\tan 3\theta)^2$
(c) $(\sec 3\theta-\tan 3\theta)^3$
(d) sec3θ + tan3θ

434. What is the value of $\tan\frac{\theta}{2}$?

SSC CGL 26/07/2023 (Shift-04)

(a) $\frac{\cos\theta}{1-\sin\theta}$ (b) $\frac{\sin\theta}{1-\cos\theta}$
(c) $\frac{\cos\theta}{1-\cos\theta}$ (d) $\frac{\sin\theta}{1+\cos\theta}$

435. What is the value of $\frac{\tan A-\sin A}{\tan A+\sin A}$?

SSC CHSL 09/03/2023 (Shift-02)

(a) 1 (b) $\frac{\sec A+1}{\sec A-1}$
(c) $\frac{\sec A-1}{\sec A+1}$ (d) secA

436. If $\frac{(1-\cos\theta)}{\sin\theta}=\frac{1}{5}$, then what will be the value of $\frac{(1+\cos\theta)}{\sin\theta}$?

SSC CGL 02/12/2022 (Shift-04)

(a) 5 (b) $\frac{2}{5}$
(c) $\frac{4}{5}$ (d) $\frac{1}{5}$

437. Simplify: $\frac{\cos A}{1+\tan A}-\frac{\sin A}{1+\cot A}$

SSC CGL 03/12/2022 (Shift-01)

(a) tanA (b) cosA – sinA
(c) cosA. sinA (d) cosA + sinA

438. Simplify: $\frac{\cos A}{1-\tan A}+\frac{\sin A}{1-\cot A}$

SSC CGL 13/12/2022 (Shift-02)

(a) tanA – cotA (b) tanA + cotA
(c) sinA – cosA (d) sinA + cosA

439. Simplify the following expression:

$\frac{\cos A}{1-\tan A}+\frac{\sin A}{1-\cot A}-\sin A$

SSC CGL 19/04/2022 (Shift-03)

(a) 1 + sinA (b) (1 + sinA) cosA
(c) cosA (d) 1 + cosA

440. If 1 + sinθ = mcosθ, then what is the value of sinθ?

SSC CGL 07/12/2022 (Shift-02)

(a) $\frac{2m^2-1}{m^2+1}$ (b) $\frac{m^2-1}{m^2+1}$
(c) $\frac{m^2+1}{2m^2-1}$ (d) $\frac{m^2+1}{m^2-1}$

441. The value of $1+\sqrt{\frac{\cot\theta+\cos\theta}{\cot\theta-\cos\theta}}$, if 0° <θ<90°, is equal to:

SSC CGL 12/04/2022 (Shift-02)

(a) 1 – secθ + tanθ
(b) 1 – secθ – tanθ
(c) 1 + secθ – tanθ
(d) 1 + secθ + tanθ

442. The value of $2-\sqrt{\frac{\cot\theta+\cos\theta}{\cot\theta-\cos\theta}}$, when 0° < θ < 90° is equal to:

SSC CGL 18/04/2022 (Shift-02)

(a) 2 – sec θ + tan θ
(b) 2 + sec θ – tan θ
(c) 2 – sec θ – tan θ
(d) 2 + sec θ + tan θ

443. $1 + 2\tan^2\theta + 2\sin\theta\sec^2\theta$, 0°< θ <90°, is equal to :

SSC CGL 16/08/2021 (Shift-01)

(a) $\frac{1-\sin\theta}{1+\sin\theta}$ (b) $\frac{1+\cos\theta}{1-\cos\theta}$
(c) $\frac{1-\cos\theta}{1+\cos\theta}$ (d) $\frac{1+\sin\theta}{1-\sin\theta}$

444. If $1 + 2\tan^2\theta + 2\sin\theta\sec^2\theta = \frac{a}{b}$, 0°<θ< 90°, then $\frac{a+b}{a-b}$ = ?

SSC CGL TIER-II 29/01/2022

(a) sinθ (b) cosθ
(c) cosecθ (d) secθ

445. What is the value of

$\left(\frac{1+\sec^2 A}{1+\cos^2 A}\right)\left(\frac{1+\sin^2 A}{1+\operatorname{cosec}^2 A}\right)$?

SSC CHSL 09/03/2023 (Shift-01)

(a) $\cos^2 A$ (b) $\operatorname{cosec}^2 A$
(c) $\sec^2 A$ (d) $\tan^2 A$

446. What is the value of

$\frac{\sin\theta+\cos\theta}{\sin\theta-\cos\theta}+\frac{\sin\theta-\cos\theta}{\sin\theta+\cos\theta}$?

SSC CHSL 14/03/2023 (Shift-04)

(a) $\frac{1}{(\sin^2\theta-\cos^2\theta)}$
(b) $2(\sin^2\theta-\cos^2\theta)$
(c) $\frac{2}{(\sin^2\theta-\cos^2\theta)}$
(d) $\sin^2\theta-\cos^2\theta$

447. The value of $(2\cos^2\theta - 1)\left[\frac{1+\tan\theta}{1-\tan\theta}+\frac{1-\tan\theta}{1+\tan\theta}\right]$ is:

SSC CPO 09/11/2022 (Shift-01)

(a) 2 (b) 0
(c) $\frac{\sqrt{3}}{2}$ (d) 1

448. What is the value of $\frac{1+\tan A}{\operatorname{cosec} A}+\frac{1+\cot A}{\sec A}$?

SSC CHSL 16/03/2023 (Shift-01)

(a) $2\sec^2 A$
(b) secA + cosecA
(c) secA – cosecA
(d) $2\operatorname{cosec}^2 A$

449. Simplify the following expression:

$\operatorname{cosec}^4 A(1-\cos^4 A) - 2\cot^2 A - 1$

SSC CGL 20/04/2022 (Shift-01)

(a) 1 (b) $\sin^2 A$
(c) 0 (d) $\operatorname{cosec}^2 A$

◆ TYPE – VI ◆
asinθ + bcosθ – x,
acosθ – bsinθ = y,
Then, $a^2 + b^2 = x^2 + y^2$

450. If $\sin\theta-\cos\theta=\frac{1}{5}$, then find the value of $\sin\theta+\cos\theta$.

SSC CHSL 02/08/2023 (Shift-04)

(a) $\frac{5}{7}$ (b) $\frac{7}{5}$
(c) $\frac{5}{3}$ (d) $\frac{3}{5}$

451. If cosθ – sinθ = $\sqrt{2}$ sinθ, then (cosθ + sinθ) is:

SSC CPO 03/10/2023 (Shift-01)

(a) $-\sqrt{2}\sin\theta$ (b) $\sqrt{2}\tan\theta$
(c) $\sqrt{2}\cos\theta$ (d) $-\sqrt{2}\cos\theta$

452. If 4 cos θ + 3 sin θ = x and 4 sin θ – 3 cos θ = y, find the value of $x^2 + y^2$.

SSC CPO 11/11/2022 (Shift-01)

(a) 16 (b) 9
(c) 25 (d) 1

453. If sin$x$ + cos$x$ = $\sqrt{2}$ sin$x$, then the value of sin$x$ – cos$x$ is:

SSC CPO 28/06/2024 (Shift-02)

(a) sin$x$ (b) 2sin$x$
(c) $-\sqrt{2}$ cos$x$ (d) $\sqrt{2}$ cos$x$

454. If $\sin\theta+\cos\theta=\frac{\sqrt{11}}{3}$, then the value of (cosθ – sinθ) is:

SSC CPO 03/10/2023 (Shift-03)

(a) $\frac{\sqrt{5}}{3}$ (b) $\frac{7}{3}$
(c) $\frac{5}{3}$ (d) $\frac{\sqrt{7}}{3}$

455. If $\sin\theta+\cos\theta=\frac{1}{29}$, find the value of $\frac{\sin\theta+\cos\theta}{\sin\theta-\cos\theta}$.

SSC CHSL 03/08/2023 (Shift-02)

(a) $\frac{1}{41}$ (b) $\frac{43}{29}$
(c) $\frac{41}{29}$ (d) $\frac{1}{43}$

456. If 3 sinA + 4cosA = 5, then the value of tan A is:

SSC CGL 18/09/2024 (Shift-01)

(a) $\frac{3}{4}$ (b) $\frac{3}{5}$
(c) $\frac{5}{4}$ (d) $\frac{4}{5}$

457. If a cotθ + b cosecθ = p and b cotθ + a cosecθ = q then $p^2 - q^2$ is equal to _____.

SSC CGL TIER-II 07/03/2023

(a) $b^2 - a^2$ (b) $a^2 - b^2$
(c) b – a (d) $a^2 + b^2$

◆ TYPE – VII ◆
CONCEPT OF VALUE PUTTING

458. The value of θ, when $\sqrt{3}$ cosθ + sinθ = 1 ($1 \le \theta \le 90°$), is:

SSC CGL 23/09/2024 (Shift-01)

(a) 20° (b) 60°
(c) 90° (d) 30°

459. For what value of t is the value of $\sin^2$ (t), half of the value of tan (t)?

SSC CGL 23/09/2024 (Shift-03)

(a) 60° (b) 45°
(c) 30° (d) 22.5°

460. The value of $(1+\sin^4 A - \cos^4 A)\operatorname{cosec}^2 A$ is:

SSC CPO 03/10/2023 (Shift-02)

(a) –1 (b) 1
(c) –2 (d) 2

461. (cot θ + tan θ)(cosec θ – sin θ)(cos θ – sec θ) = -------.

SSC CHSL 03/07/2024 (Shift-01)

(a) 0 (b) 1
(c) 2 (d) –1

462. If $2\tan^2 A + 4\cos^4 A = 3$, then the possible value of A is:

SSC CHSL 04/08/2023 (Shift-04)

(a) 45° (b) 60°
(c) 0° (d) 30°

463. If m = asecA and y = btanA, then find the value of $b^2m^2 - a^2y^2 + \frac{a^2y^2}{b^2m^2} + \cos^2A$.

SSC CGL 13/09/2024 (Shift-03)

(a) a^2b^2 (b) $1-a^2b^2$
(c) $a^2b^2 + 2$ (d) $a^2b^2 + 1$

464. If m = secθ – tanθ and n = cosecθ + cotθ, then what is the value of m + n(m – 1)?

SSC CHSL 13/03/2023 (Shift-02)

(a) 2 (b) 1
(c) 0 (d) –1

465. The value of the expression $\frac{1-\sin(2t)}{1+\sin(2t)} \times \frac{\cos(t)+\sin(t)}{\cos(t)-\sin(t)}$ is:

SSC CGL 19/09/2024 (Shift-02)

(a) $\frac{1-2\tan(t)}{1+2\tan(t)}$ (b) $\frac{1-\tan(t)}{1+\tan(t)}$
(c) $\frac{1+2\tan(t)}{1-2\tan(t)}$ (d) $\frac{1+\tan(t)}{1-\tan(t)}$

466. If tanA = a tanB and sinA = b sinB, then the value of $\cos^2A$ is:

SSC CHSL 05/07/2024 (Shift-01)

(a) $\frac{b^2-1}{a^2-1}$ (b) $\frac{a^2-1}{b^2-1}$
(c) $\frac{b^2+1}{a^2+1}$ (d) $\frac{a^2+1}{b^2+1}$

467. If $\cot\alpha = \frac{P}{Q}$, then find the value of $\frac{P\cos\alpha - Q\sin\alpha}{P\cos\alpha + Q\sin\alpha} - \frac{P^2-Q^2}{P^2+Q^2} + 3$.

SSC CHSL 09/07/2024 (Shift-01)

(a) $\frac{P_2}{P_2+Q_2}$ (b) $\frac{P}{Q}$
(c) 0 (d) 3

468. Which of the following is true when x = sinA + cosA; y = secA + cosecA?

SSC CPO 27/06/2024 (Shift-03)

(a) $y(1 + x^2) = 2x$ (b) $y - 2x = x^2y$
(c) $y + 2x = x^2y$ (d) $y(1 - 2x^2) = x$

469. What is the value of $\frac{[(\sin x + \sin y)(\sin y - \sin x)]}{[(\cos x + \cos y)(\cos y - \cos x)]}$?

SSC CPO 29/06/2024 (Shift-03)

(a) 0 (b) 1
(c) –1 (d) 2

470. If cotA + cosA = p and cot A – cosA = q. then which of the following relation is correct.?

SSC CHSL 03/08/2023 (Shift-04)

(a) $\sqrt{\frac{1}{16pq}} = p^2 + q^2$
(b) $\frac{1}{4pq} = p^2 + q^2$
(c) $\sqrt{16pq} = p^2 - q^2$
(d) $4pq = p^2 + q^2$

471. Using trigonometric formulas, find the value of $\left(\frac{\sin(x-y)}{\sin(x+y)}\right)\left(\frac{\tan x + \tan y}{\tan x - \tan y}\right)$

SSC CGL 18/07/2023 (Shift-01)

(a) –2 (b) 2
(c) 0 (d) 1

472. Using $\operatorname{cosec}(\alpha+\beta) = \frac{\sec\alpha \times \sec\beta \times \operatorname{cosec}\alpha \times \operatorname{cosec}\beta}{\sec\alpha \times \operatorname{cosec}\beta + \operatorname{cosec}\alpha \times \sec\beta}$, find the value of cosec 75°.

SSC CGL 25/07/2023 (Shift-01)

(a) $\frac{\sqrt{6}+\sqrt{2}}{4}$ (b) $\frac{\sqrt{6}-\sqrt{2}}{4}$
(c) $\sqrt{6}-\sqrt{2}$ (d) $\sqrt{6}+\sqrt{2}$

◆ TYPE – VIII ◆
MISCELLANEOUS QUESTIONS

473. If α + β = 45° and (tanα + 1)(tanβ + 1) = 2x, then x is:

SSC CHSL 03/07/2024 (Shift-04)

(a) 2 (b) –1
(c) 0 (d) 1

474. The value of (1 + tan 10°) (1 + tan 35°) is:

SSC CHSL 25/05/2022 (Shift-03)

(a) $\frac{1}{2}$ (b) $\frac{3}{4}$
(c) 1 (d) 2

475. The expression $\sin^2\theta + \cos^2\theta - 1 = 0$ is satisfied by how many value of θ?

SSC Phase-XII 21/06/2024 (Shift-03)

(a) No value
(b) Only one value
(c) Infinitely many values
(d) Two values

476. If a = cotA + cosA and b = cotA – cosA, then find the value of $a^2 - b^2 - 4\sqrt{ab}$.

SSC CHSL 04/08/2023 (Shift-04)

(a) 0 (b) –1
(c) 1 (d) –4

477. The value of the equation $\cos^2\theta + \sin\theta + 1 = 0$ lies between:

SSC Phase-XI 30/06/2023 (Shift-01)

(a) $\left(\frac{3\pi}{4}, \frac{5\pi}{4}\right)$ (b) $\left(\frac{5\pi}{4}, \frac{7\pi}{4}\right)$
(c) $\left(\frac{\pi}{4}, \frac{3\pi}{4}\right)$ (d) $\left(-\frac{\pi}{4}, \frac{\pi}{4}\right)$

478. Let, 0° < t < 90°. Then which of the followings is true?

SSC CGL 10/09/2024 (Shift-01)

(a) sin(t) ≠ cos(t), when t = 45°
(b) sin(t) > cos(t), when t < 45°
(c) sin(t) < cos(t), when t < 45°
(d) sin(t) < cos(t), when t > 45°

479. If $2\left(\frac{\cos^2x - \sec^2x}{\tan^2x}\right) = a + b\cos2x$ then a, b = ?

SSC CGL 09/12/2022 (Shift-04)

(a) $\frac{-3}{2}, \frac{-1}{2}$ (b) $\frac{3}{2}, \frac{1}{2}$
(c) –3, –1 (d) 3, 1

480. If $x = \frac{2\sin\theta}{(1+\cos\theta+\sin\theta)}$, then the value of $\frac{1-\cos\theta+\sin\theta}{(1+\sin\theta)}$ is:

SSC CPO 09/11/2022 (Shift-01)

(a) $\frac{x}{(1+x)}$ (b) x
(c) $\frac{1}{x}$ (d) $\frac{(1+x)}{x}$

481. Let a = $\frac{2\sin x}{1+\sin x+\cos x}$ and b = $\frac{c}{1+\sin x}$. If a = b then c = ?

SSC CGL 06/06/2019 (Shift-02)

(a) 1 – sin$x$.cos$x$
(b) 1 + sin$x$ – cos$x$
(c) 1 + sin$x$.cos$x$
(d) 1 + cos$x$.sin$x$

482. If $\frac{\cos\alpha}{\sin\beta} = 10$ and $\frac{\cos\alpha}{\cos\beta} = 11$, then the value of $\cos^2\beta$ is:

SSC CGL 21/07/2023 (Shift-03)

(a) $\frac{121}{132}$ (b) $\frac{100}{221}$
(c) $\frac{88}{108}$ (d) $\frac{221}{121}$

483. The value of sin18° is given as $\frac{\sqrt{5}-1}{4}$. Using this value, find the value of cosec18°.

SSC CHSL 30/05/2022 (Shift-02)

(a) $\frac{\sqrt{10+2\sqrt{5}}}{4}$ (b) $\sqrt{5}+1$
(c) $\frac{\sqrt{6}+\sqrt{2}}{4}$ (d) $\frac{\sqrt{10-2\sqrt{5}}}{4}$

484. What is the value of $\cos^{-1}\left(\frac{7}{25}\right) - \cos^{-1}\left(\frac{15}{17}\right)$

SSC CHSL 11/07/2024 (Shift-02)

(a) $\sin^{-1}\left(\frac{304}{425}\right)$
(b) $\cot^{-1}\left(\frac{297}{425}\right)$
(c) $\tan^{-1}\left(\frac{297}{425}\right)$
(d) $\cos^{-1}\left(\frac{304}{425}\right)$

◆ TYPE – IX (A) ◆
TRIGONOMETRIC FUNCTIONS OF 2A & 3A

485. If $\cos^4\theta - \sin^4\theta = \frac{4}{5}$, then find the value of $\sin 4\theta$.

SSC CHSL 08/07/2024 (Shift-04)

(a) $\frac{24}{25}$ (b) $\frac{21}{25}$

(c) $\frac{16}{25}$ (d) $\frac{18}{25}$

486. What is the value of $\cos^2 15°$?

SSC CGL 06/12/2022 (Shift-04)

(a) $(2+\sqrt{3})$ (b) $\frac{(2+\sqrt{3})}{4}$

(c) $\frac{(2+\sqrt{3})}{2}$ (d) $\frac{(1+\sqrt{3})}{2}$

487. Solve for θ : $\cos^2\theta - \sin^2\theta = \frac{1}{2}$, $0 < \theta < 90°$.

SSC CPO 25/11/2020 (Shift-2)

(a) 45° (b) 60°

(c) 40° (d) 30°

488. If $\sin\theta = \frac{6}{10}$ and $\cos\theta = \frac{8}{10}$, then $\sin 2\theta =$ _____ .

SSC CHSL 11/08/2023 (Shift-03)

(a) $\frac{10}{8}$ (b) $\frac{12}{10}$

(c) $\frac{16}{10}$ (d) $\frac{24}{25}$

489. Using the formula $\tan\frac{x}{2} = \frac{1-\cos x}{\sin x}$, find the value of tan 22.5°.

SSC CHSL 10/06/2022 (Shift- 01)

(a) $\sqrt{2}-1$ (b) $\sqrt{2}+1$

(c) $\frac{\sqrt{3}+\sqrt{2}}{2}$ (d) $\frac{\sqrt{3}-\sqrt{2}}{2}$

490. If $4\tan\theta - 3 = 0$, then the value of $\frac{1-\cos 2\theta}{1+\cos 2\theta}$ is?

SSC CGL 23/09/2024 (Shift-01)

(a) 1 (b) $\frac{7}{15}$

(c) $\frac{4}{3}$ (d) $\frac{9}{16}$

491. The value of $\left(\frac{2\cos^3\theta - \cos\theta}{\sin\theta - 2\sin^3\theta}\right)^2 + 1, \theta \neq 45°$ is:

SSC CPO 28/06/2024 (Shift-03)

(a) $\sin^2\theta$ (b) $\sec^2\theta$

(c) $\text{cosec}^2\theta$ (d) $\cot^2\theta$

492. $\frac{2\tan A}{1+\tan^2 A} = ?$

SSC CGL 20/07/2023 (Shift-02)

(a) cos2A (b) sinA

(c) cosA (d) sin2A

493. Simplify the following expression

$\frac{\sin\theta - 2\sin^3\theta}{2\cos^3\theta - \cos\theta}$

SSC CGL 21/07/2023 (Shift-03)

(a) $\tan\theta$ (b) $\sin\theta$

(c) $\sec\theta$ (d) $\cos\theta$

494. The value of $\frac{2\cos^3\theta - \cos\theta}{\sin\theta - 2\sin^3\theta}$ is:

SSC CPO 10/11/2022 (Shift-01)

(a) $\sec\theta$ (b) $\sin\theta$

(c) $\cot\theta$ (d) $\tan\theta$

495. Evaluate the following:

$\sqrt{2+\sqrt{2+\sqrt{2+2\cos 8\theta}}}$

SSC CGL 27/07/2023 (Shift-03)

(a) $2\cos\theta$ (b) $2\cos 2\theta$

(c) $\sin 2\theta$ (d) $\cos 2\theta$

496. If $(2\cos^3 A - \cos A) / (\sin A - 2\sin^3 A) = 1$, then what is the value of A?

SSC CHSL 21/03/2023 (Shift-02)

(a) 90° (b) 30°

(c) 45° (d) 60°

497. If $0° < A, B < 45°$, $\cos(A+B) = \frac{24}{25}$ and $\sin(A-B) = \frac{15}{17}$, then tan2A = ?

SSC CGL 06/03/2020 (Shift-02)

(a) 0 (b) 1

(c) $\frac{416}{87}$ (d) $\frac{213}{4}$

498. $3\sin A - 4\sin^3 A$ = ?

SSC CHSL 02/06/2022 (Shift- 01)

(a) cot3A (b) sin3A

(c) cos3A (d) tan3A

499. Find the value of $3\sin 15° - 4\sin^3 15°$.

SSC CGL 25/09/2024 (Shift-03)

(a) 2 (b) $\frac{1}{2}$

(c) $\frac{1}{\sqrt{2}}$ (d) $\sqrt{2}$

◆ TYPE – IX (B) ◆
SUM & DIFFERENCE OF ANGLES OF TRIGONOMETRIC RATIOS

500. If A = 60°, B = 30°, then find the value of sinA cosB + cosA sinB.

SSC CHSL 02/08/2023 (Shift-02)

(a) 0 (b) 12

(c) 13 (d) 1

501. using cos (A + B) = cosA cosB – sin A sin B, find the value of cos75°

SSC CGL 17/07/2023 (Shift-03)

(a) $\frac{\sqrt{5}-1}{4}$ (b) $\frac{\sqrt{5}+1}{4}$

(c) $\frac{\sqrt{6}-\sqrt{2}}{4}$ (d) $\frac{\sqrt{6}+\sqrt{2}}{4}$

502. Let ABC be a right angled triangle with a right angle at B. If $\tan A = \sqrt{3}$, then find the value of sinA cosC + cosA sinC and cosA cosC – sinA sinC.

SSC CGL TIER-II (18/01/2025)

(a) $\frac{\sqrt{3}}{2}, \frac{1}{2}$ (b) $\frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}$

(c) $\frac{1}{2}, \frac{1}{2}$ (d) 1, 0

503. What is the value of cos (*x* + y)–sin (*x* – y) + tan (2z)?

SSC CHSL 02/07/2024 (Shift-01)

(a) cos *x*. cos y – sin *x*. sin y – sin *x*. cos y – cos *x* . sin y + $\frac{2\tan z}{1-\tan^2 z}$

(b) cos *x*. cos y – sin *x*. sin y – sin *x*. cos y + cos *x*. sin y + $\frac{2\tan z}{1-\tan^2 z}$

(c) cos x. cos y – sin x. sin y – sin *x*. cos y + cos *x*. sin y + $\frac{\tan z}{1-\tan^z z}$

(d) sin *x*. sin y + $\frac{2\tan z}{1-\tan^z z}$

504. Find the value of tan 72° – tan 27° – tan 72° tan 27°.

SSC CGL 25/09/2024 (Shift-02)

(a) –1 (b) 1

(c) –2 (d) 0

505. If sin(X – Y) = sinXcosY – cosXsinY, then the value of $\frac{\sin(A-B)}{\cos A\cos B} + \frac{\sin(B-C)}{\cos B\cos C}$ is:

SSC CHSL 05/07/2024 (Shift-02)

(a) $\frac{\sin(C-A)}{\cos A\cos C}$ (b) $\frac{\sin(A-C)}{\cos A\cos C}$

(c) $\frac{\cos(A+C)}{\cos A\cos C}$ (d) $\frac{\cos(A+C)}{\cos A\sin C}$

506. $\left[\frac{\text{cosec}31° - \sec 59°}{(\sin 18° \div \cos 72°)} \times \frac{\tan 26°}{\cot 64°}\right] +$

$\left[\frac{\tan 30° + \tan 15°}{(1 - \tan 30°\tan 15°)}\right]$ = _____.

SSC CHSL 05/07/2024 (Shift-04)

(a) 0 (b) –1

(c) 1 (d) 2

507. Simplify: $\frac{\cos 13° + \sin 13°}{\cos 13° - \sin 13°}$

SSC CPO 28/06/2024 (Shift-01)

(a) cot 58° (b) tan 32°

(c) tan 58° (d) cos 26°

508. Simplify: $\frac{\tan 71° + \tan 19°}{1 - \tan 71°\tan 19°}$

SSC CPO 29/06/2024 (Shift-02)

(a) Not defined (b) 1

(c) $\sqrt{3}$ (d) 0

509. Using $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$, find the value of tan 15°.

SSC CGL 18/07/2023 (Shift-04)

(a) $\sqrt{3}+1$ (b) $\sqrt{3}-1$

(c) $2-\sqrt{3}$ (d) $2+\sqrt{3}$

510. If sin(A + B) = cos(A + B), what is the value of tanA?

SSC CGL 02/12/2022 (Shift-01)

(a) $\frac{1-\tan B}{1+\tan B}$ (b) $\frac{1+\tan B}{1-\tan B}$

(c) $\frac{1+\sec B}{1-\sec B}$ (d) $\frac{1-\operatorname{cosec} B}{1+\operatorname{cosec} B}$

511. If $\sin A = \frac{4}{5}$ and $\sin B = \frac{15}{17}$, what is the value of sin(A – B)?

SSC CGL 02/12/2022 (Shift-04)

(a) $-\frac{32}{45}$ (b) $-\frac{13}{85}$

(c) $\frac{13}{85}$ (d) $\frac{32}{45}$

512. Simplify the following:

$\frac{\cos x - \sqrt{3}\sin x}{2}$

SSC CGL 03/12/2022 (Shift-03)

(a) $\cot\left(\frac{\pi}{3}-x\right)$ (b) $\sin\left(\frac{\pi}{3}+x\right)$

(c) $\cos\left(\frac{\pi}{3}+x\right)$ (d) $\sin\left(\frac{\pi}{3}-x\right)$

513. If tan A – tan B – tan C = tan A tan B tan C, what is the value of A in terms of B and C?

SSC CGL 07/12/2022 (Shift-01)

(a) A = B + C (b) A = 2B – 2C

(c) A = B – C (d) $A = \frac{B-C}{2}$

514. If tan(α + β) = a, tan(α – β) = b, then the value of tan2α is:

SSC CGL 13/12/2022 (Shift-04)

(a) $\frac{a+b}{1-ab}$ (b) $\frac{a+b}{1+ab}$

(c) $\frac{a-b}{1+ab}$ (d) $\frac{a-b}{1-ab}$

515. If $\tan A \tan B + \frac{\cos x}{\cos A \cos B} = 1$, then x = ?

SSC CGL 08/12/2022 (Shift-04)

(a) A (b) B

(c) A + B (d) A – B

516. Given that A and B are second quadrant angles, $\sin A = \frac{1}{3}$ and $\sin B = \frac{1}{5}$, then find the value of cos (A – B).

SSC CPO 09/11/2022 (Shift-03)

(a) $\frac{4\sqrt{3}+1}{15}$ (b) $\frac{8\sqrt{3}-1}{15}$

(c) $\frac{8\sqrt{3}+1}{15}$ (d) $\frac{4\sqrt{3}-1}{15}$

517. What is the value of $\frac{\sin(A+B)}{\sin A \cos B}$?

SSC CPO 09/11/2022 (Shift-03)

(a) 1 + cot A tan B

(b) 1 + tan A cot B

(c) 1 – sin A cos B

(d) 1 – cot A tan B

◆ TYPE – IX (C) ◆
TRANSFORMATION OF SUM OR DIFFERENCE INTO PRODUCT

518. Find the value of:

(sin 75° + sin 15°).

SSC Phase-XII 21/06/2024 (Shift-03)

(a) $\frac{\sqrt{6}}{2}$ (b) $\sqrt{6}$

(c) $\frac{\sqrt{3}}{2}$ (d) $\sqrt{3}$

519. Simplify the following:

sin 2x + 2sin4x + sin 6x

SSC CGL 18/07/2023 (Shift-02)

(a) 4 $\cos^2 x \sin 4x$

(b) 4 $\cos^2 x \sin x$

(c) 2 $\cos^2 x \sin 4x$

(d) 4 $\sin^2 x \sin 4x$

520. What is sinα – sinβ ?

SSC CGL 01/12/2022 (Shift-02)

(a) $2\cos\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2}$

(b) $2\sin\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2}$

(c) $2\cos\frac{\alpha-\beta}{2}\sin\frac{\alpha+\beta}{2}$

(d) $2\cos\frac{\alpha+\beta}{2}\cos\frac{\alpha-\beta}{2}$

521. The value of $\frac{\sin 4\theta + \sin 2\theta}{\cos 4\theta + \cos 2\theta}$ is:

SSC CHSL 27/05/2022 (Shift-01)

(a) tan3θ (b) cot3θ

(c) tanθ (d) cotθ

◆ TYPE – IX (D) ◆
TRANSFORMATION OF PRODUCT INTO SUM OR DIFFERENCE

522. Evaluate the following:

cos(36° + A).cos(36° – A) + cos(54° + A).cos(54° – A)

SSC CGL 14/07/2023 (Shift-02)

(a) sin 2A (b) cos A

(c) sin A (d) cos 2A

523. Using 2cosA cosB = cos(A + B) + cos(A – B), find the value of cos75°cos15°.

SSC CHSL 01/06/2022 (Shift-02)

(a) $\frac{1}{2}$ (b) $\frac{1}{4}$

(c) $\frac{\sqrt{3}}{2}$ (d) $\frac{\sqrt{3}}{4}$

524. sinA sinB = _____.

SSC CGL 18/09/2024 (Shift-03)

(a) $\frac{1}{2}\{\sin(A+B) + \sin(A-B)\}$

(b) $\frac{1}{2}\{\sin(A+B) - \sin(A-B)\}$

(c) $\frac{1}{2}\{\cos(A+B) + \cos(A-B)\}$

(d) $\frac{1}{2}\{\cos(A-B) - \cos(A+B)\}$

525. Simplify: $\frac{\sin 8\theta \cos\theta - \sin 6\theta \cos 3\theta}{\cos 2\theta \cos\theta - \sin 3\theta \sin 4\theta}$

SSC CPO 09/11/2022 (Shift-02)

(a) cotθ (b) cot2θ

(c) tanθ (d) tan2θ

◆ TYPE – IX (E) ◆
MAXIMUM/MINIMUM VALUES OF TRIGONOMETRIC FUNCTIONS

526. The maximum value of (2sinθ + 3cosθ)

SSC CGL 19/09/2024 (Shift-02)

(a) $\sqrt{17}$ (b) $\sqrt{11}$

(c) 9 (d) $\sqrt{13}$

527. What is the maximum value of 7cosA + 24sinA + 32?

SSC CHSL 09/07/2024 (Shift-04)

(a) 57 (b) 32

(c) 25 (d) 39

528. Find the maximum value of (19sinθ + 6cotθ sinθ)

SSC CPO 27/06/2024 (Shift-03)

(a) $\sqrt{397}$ (b) $\sqrt{197}$

(c) $\sqrt{297}$ (d) $\sqrt{497}$

529. Find the least value of $16\operatorname{cosec}^2\theta + 25\sin^2\theta$.

SSC CPO 04/10/2023 (Shift-03)

(a) 35 (b) 38

(c) 42 (d) 40

530. The least value of cosθ sinθ is:

SSC CGL TIER-II 16/10/2020

(a) $\left(\frac{1}{2}\right)$ (b) 0

(c) (– 1) (d) $\left(-\frac{1}{2}\right)$

531. The maximum value of $(\sin^{12}\theta + \cos^{20}\theta)$ for all the real values of θ is:

SSC CGL 20/07/2023 (Shift-03)

(a) 2 (b) 3

(c) 1 (d) 0

532. The greatest value of $\sin^4\theta + \cos^4\theta$ is:

SSC CGL 26/09/2024 (Shift-03)

(a) 2 (b) 1

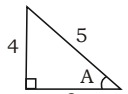
(c) 3 (d) 4

ANSWER KEY

1.	(c)	2.	(d)	3.	(c)	4.	(d)	5.	(a)	6.	(c)	7.	(d)	8.	(b)	9.	(a)	10.	(b)
11.	(a)	12.	(c)	13.	(d)	14.	(c)	15.	(c)	16.	(b)	17.	(a)	18.	(b)	19.	(c)	20.	(c)
21.	(d)	22.	(c)	23.	(b)	24.	(d)	25.	(c)	26.	(d)	27.	(d)	28.	(c)	29.	(b)	30.	(b)
31.	(d)	32.	(b)	33.	(c)	34.	(b)	35.	(c)	36.	(c)	37.	(d)	38.	(a)	39.	(c)	40.	(a)
41.	(a)	42.	(a)	43.	(a)	44.	(c)	45.	(c)	46.	(a)	47.	(c)	48.	(d)	49.	(c)	50.	(b)
51.	(a)	52.	(d)	53.	(a)	54.	(c)	55.	(c)	56.	(a)	57.	(d)	58.	(b)	59.	(c)	60.	(b)
61.	(b)	62.	(c)	63.	(c)	64.	(a)	65.	(a)	66.	(d)	67.	(c)	68.	(a)	69.	(a)	70.	(c)
71.	(a)	72.	(b)	73.	(b)	74.	(c)	75.	(a)	76.	(d)	77.	(c)	78.	(d)	79.	(b)	80.	(a)
81.	(c)	82.	(d)	83.	(c)	84.	(a)	85.	(d)	86.	(b)	87.	(b)	88.	(d)	89.	(a)	90.	(c)
91.	(d)	92.	(d)	93.	(a)	94.	(a)	95.	(b)	96.	(c)	97.	(a)	98.	(a)	99.	(c)	100.	(a)
101.	(a)	102.	(c)	103.	(b)	104.	(a)	105.	(a)	106.	(b)	107.	(b)	108.	(a)	109.	(c)	110.	(c)
111.	(a)	112.	(d)	113.	(b)	114.	(a)	115.	(b)	116.	(c)	117.	(a)	118.	(b)	119.	(b)	120.	(c)
121.	(b)	122.	(b)	123.	(b)	124.	(a)	125.	(b)	126.	(b)	127.	(a)	128.	(d)	129.	(d)	130.	(c)
131.	(a)	132.	(b)	133.	(b)	134.	(c)	135.	(c)	136.	(b)	137.	(d)	138.	(a)	139.	(b)	140.	(a)
141.	(a)	142.	(b)	143.	(d)	144.	(a)	145.	(b)	146.	(a)	147.	(a)	148.	(a)	149.	(c)	150.	(b)
151.	(a)	152.	(a)	153.	(b)	154.	(c)	155.	(a)	156.	(d)	157.	(c)	158.	(d)	159.	(d)	160.	(d)
161.	(c)	162.	(d)	163.	(a)	164.	(a)	165.	(b)	166.	(d)	167.	(a)	168.	(b)	169.	(d)	170.	(c)
171.	(a)	172.	(d)	173.	(d)	174.	(c)	175.	(a)	176.	(c)	177.	(c)	178.	(b)	179.	(a)	180.	(d)
181.	(b)	182.	(a)	183.	(d)	184.	(a)	185.	(c)	186.	(d)	187.	(b)	188.	(b)	189.	(c)	190.	(d)
191.	(c)	192.	(b)	193.	(c)	194.	(a)	195.	(d)	196.	(d)	197.	(a)	198.	(a)	199.	(b)	200.	(c)
201.	(b)	202.	(d)	203.	(d)	204.	(d)	205.	(c)	206.	(c)	207.	(a)	208.	(b)	209.	(c)	210.	(c)
211.	(d)	212.	(a)	213.	(b)	214.	(d)	215.	(b)	216.	(a)	217.	(d)	218.	(c)	219.	(b)	220.	(d)
221.	(b)	222.	(d)	223.	(b)	224.	(a)	225.	(a)	226.	(a)	227.	(c)	228.	(c)	229.	(c)	230.	(a)
231.	(a)	232.	(d)	233.	(a)	234.	(a)	235.	(d)	236.	(c)	237.	(c)	238.	(a)	239.	(c)	240.	(b)
241.	(c)	242.	(b)	243.	(a)	244.	(a)	245.	(a)	246.	(d)	247.	(c)	248.	(d)	249.	(d)	250.	(d)
251.	(b)	252.	(d)	253.	(a)	254.	(d)	255.	(b)	256.	(c)	257.	(a)	258.	(b)	259.	(b)	260.	(b)
261.	(c)	262.	(a)	263.	(d)	264.	(c)	265.	(d)	266.	(c)	267.	(a)	268.	(b)	269.	(b)	270.	(c)
271.	(b)	272.	(d)	273.	(c)	274.	(b)	275.	(a)	276.	(c)	277.	(c)	278.	(a)	279.	(b)	280.	(c)
281.	(c)	282.	(d)	283.	(b)	284.	(a)	285.	(d)	286.	(a)	287.	(c)	288.	(a)	289.	(b)	290.	(d)
291.	(b)	292.	(d)	293.	(b)	294.	(b)	295.	(a)	296.	(b)	297.	(c)	298.	(d)	299.	(a)	300.	(d)
301.	(b)	302.	(b)	303.	(a)	304.	(c)	305.	(b)	306.	(a)	307.	(c)	308.	(b)	309.	(d)	310.	(a)
311.	(b)	312.	(c)	313.	(b)	314.	(c)	315.	(c)	316.	(c)	317.	(c)	318.	(c)	319.	(b)	320.	(c)
321.	(b)	322.	(a)	323.	(b)	324.	(d)	325.	(d)	326.	(a)	327.	(a)	328.	(c)	329.	(a)	330.	(b)
331.	(b)	332.	(c)	333.	(c)	334.	(a)	335.	(c)	336.	(c)	337.	(a)	338.	(a)	339.	(c)	340.	(c)
341.	(d)	342.	(a)	343.	(d)	344.	(c)	345.	(a)	346.	(c)	347.	(a)	348.	(a)	349.	(d)	350.	(c)
351.	(d)	352.	(d)	353.	(b)	354.	(d)	355.	(c)	356.	(b)	357.	(d)	358.	(d)	359.	(a)	360.	(a)
361.	(a)	362.	(a)	363.	(a)	364.	(d)	365.	(c)	366.	(c)	367.	(c)	368.	(b)	369.	(a)	370.	(b)
371.	(c)	372.	(d)	373.	(c)	374.	(a)	375.	(c)	376.	(d)	377.	(a)	378.	(c)	379.	(b)	380.	(d)
381.	(b)	382.	(d)	383.	(d)	384.	(b)	385.	(a)	386.	(b)	387.	(a)	388.	(c)	389.	(a)	390.	(d)
391.	(d)	392.	(d)	393.	(a)	394.	(b)	395.	(d)	396.	(b)	397.	(b)	398.	(b)	399.	(c)	400.	(b)
401.	(a)	402.	(d)	403.	(d)	404.	(b)	405.	(d)	406.	(a)	407.	(a)	408.	(a)	409.	(d)	410.	(a)
411.	(d)	412.	(d)	413.	(a)	414.	(b)	415.	(b)	416.	(d)	417.	(c)	418.	(d)	419.	(d)	420.	(a)
421.	(a)	422.	(d)	423.	(d)	424.	(c)	425.	(d)	426.	(a)	427.	(a)	428.	(d)	429.	(a)	430.	(b)
431.	(c)	432.	(c)	433.	(a)	434.	(d)	435.	(c)	436.	(a)	437.	(b)	438.	(d)	439.	(c)	440.	(b)
441.	(d)	442.	(c)	443.	(d)	444.	(c)	445.	(d)	446.	(c)	447.	(a)	448.	(b)	449.	(c)	450.	(b)
451.	(c)	452.	(c)	453.	(d)	454.	(d)	455.	(a)	456.	(a)	457.	(a)	458.	(c)	459.	(b)	460.	(d)
461.	(d)	462.	(a)	463.	(d)	464.	(d)	465.	(b)	466.	(a)	467.	(d)	468.	(c)	469.	(c)	470.	(c)
471.	(d)	472.	(c)	473.	(d)	474.	(d)	475.	(c)	476.	(a)	477.	(b)	478.	(c)	479.	(c)	480.	(b)
481.	(b)	482.	(b)	483.	(b)	484.	(a)	485.	(a)	486.	(b)	487.	(d)	488.	(d)	489.	(a)	490.	(d)
491.	(c)	492.	(d)	493.	(a)	494.	(c)	495.	(a)	496.	(c)	497.	(c)	498.	(b)	499.	(c)	500.	(d)
501.	(c)	502.	(d)	503.	(b)	504.	(b)	505.	(b)	506.	(c)	507.	(c)	508.	(a)	509.	(c)	510.	(a)
511.	(b)	512.	(c)	513.	(a)	514.	(a)	515.	(c)	516.	(c)	517.	(a)	518.	(a)	519.	(a)	520.	(a)
521.	(a)	522.	(d)	523.	(b)	524.	(d)	525.	(d)	526.	(d)	527.	(a)	528.	(a)	529.	(d)	530.	(d)
531.	(c)	532.	(b)																

SOLUTIONS

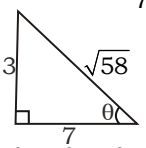
1. (c) $\tan A = \frac{4}{3}$



With the help of triplet (3, 4, 5),
Hypotenuse = 5

$\therefore \sin A = \frac{4}{5}$

2. (d) $\tan(t) = \frac{3}{7}$



$h^2 = p^2 + b^2$

$\Rightarrow h^2 = 3^2 + 7^2 \Rightarrow h = \sqrt{58}$

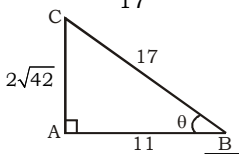
$\therefore \text{cosec}(t) = \frac{h}{p} = \frac{\sqrt{58}}{3}$

Alternate Method:

$\text{cosec}(t) = \sqrt{1+\cot^2 t}$

$= \sqrt{1+\frac{49}{9}} = \frac{\sqrt{58}}{3}$

3. (c) $\cos\theta = \frac{11}{17}$



Perpendicular $= \sqrt{17^2 - 11^2} = 2\sqrt{42}$

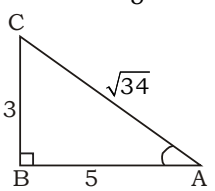
$\therefore \tan\theta = \frac{2\sqrt{42}}{11}$

4. (d) $\sin\theta = \frac{13}{19}$

$\cos\theta = \sqrt{1-\sin^2\theta} = \sqrt{1-\frac{169}{361}}$

$= \frac{\sqrt{192}}{19} = \frac{8\sqrt{3}}{19}$

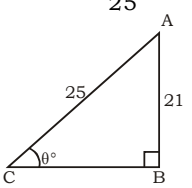
5. (a) $\tan A = \frac{3}{5}$



Hypotenuse (AC) $= \sqrt{5^2+3^2}$

$= \sqrt{25+9} = \sqrt{34} \quad \therefore \cos A = \frac{5}{\sqrt{34}}$

6. (c) $\sin\theta = \frac{21}{25}$

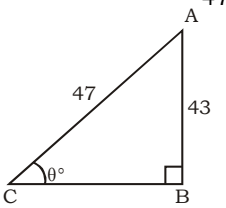


BC $= \sqrt{25^2 - 21^2} = \sqrt{625-441}$

$= \sqrt{184} = 2\sqrt{46}$

$\tan\theta = \frac{21}{2\sqrt{46}}$

7. (d) Given, $\sin\theta = \frac{43}{47}$



BC $= \sqrt{47^2 - 43^2} = \sqrt{90\times 4} = 6\sqrt{10}$

$\therefore \cot\theta = \frac{6\sqrt{10}}{43}$

8. (b) We know that,

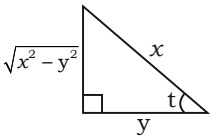
$\tan x = \frac{5}{1} = \frac{P}{B}$

By Pythagoras Theorm,

$H = \sqrt{P^2+B^2} = \sqrt{5^2+1^2} = \sqrt{26}$

$\text{Sin}x = \frac{P}{B} = \frac{5}{\sqrt{26}}$

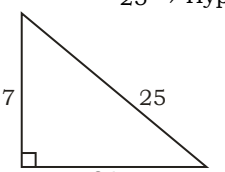
9. (a) $\sec(t) = \frac{x \to \text{Hypotenous}}{y \to \text{base}}$



Perpendicular $= \sqrt{x^2-y^2}$

$\therefore \cot(t) = \frac{y}{\sqrt{x^2-y^2}}$

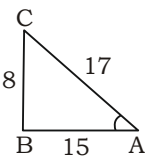
10. (b) $\cos\theta = \frac{24 \to \text{Base}}{25 \to \text{Hypotenuse}}$



Triplet (7, 24, 25)

$\therefore \cot(90° - \theta) = \tan\theta = \frac{7}{24}$

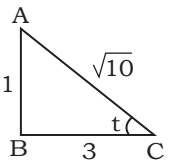
11. (a) $\cos A = \frac{15 \to \text{Base}}{17 \to \text{Hypotenuse}}$



With the help of triplet (8, 15, 17),
Perpendicular = 8
$\cot(90° - A) = \tan A$

$\therefore \tan A = \frac{8}{15}$

12. (c)



Given,

$\text{tant} = \frac{1 \to P}{3 \to B}$

$AC = \sqrt{3^2+1^2} = \sqrt{10}$

$\therefore \text{sect} = \frac{\sqrt{10}}{3}$

Alternate Method:

$\text{tant} = \frac{1}{3}$

We know that,

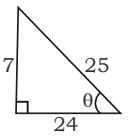
$\sec^2 t - \tan^2 t = 1$

$\Rightarrow \sec^2 t - \frac{1}{9} = 1$

$\Rightarrow \sec^2 t = \frac{10}{9}$

$\Rightarrow \text{sect} = \frac{\sqrt{10}}{3}$

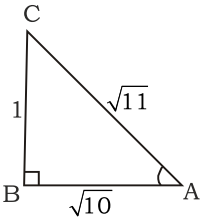
13. (d) $\sin\theta = \frac{7}{25}$



By triplet of 7, 24, 25
Base = 24

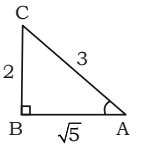
$\cos\theta = \frac{24}{25}, \tan\theta = \frac{7}{24}$

14. (c) $\tan A = \frac{1}{\sqrt{10}}$



$\therefore \sin A + \text{cosec}A = \frac{1}{\sqrt{11}} + \frac{\sqrt{11}}{1} = \frac{12}{\sqrt{11}}$

15. (c) $\sin A = \frac{2}{3}$



$\therefore (7 - \tan A)(3 + \cos A)$

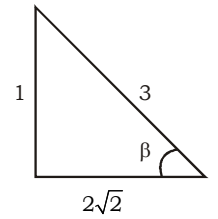
$= \left(7 - \frac{2}{\sqrt{5}}\right)\left(3 + \frac{\sqrt{5}}{3}\right)$

$= \left(\frac{7\sqrt{5}-2}{\sqrt{5}}\right)\left(\frac{9+\sqrt{5}}{3}\right)$

$= \frac{63\sqrt{5}+35-18-2\sqrt{5}}{3\sqrt{5}}$

$= \frac{61\sqrt{5}+17}{3\sqrt{5}} = \frac{61}{3} + \frac{17}{3\sqrt{5}}$

16. (b) $\sin\beta = \frac{1}{3}$

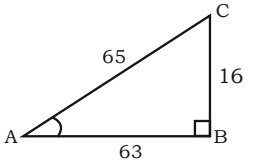


$\therefore (\sec\beta - \tan\beta)^2 = \left(\frac{3}{2\sqrt{2}} - \frac{1}{2\sqrt{2}}\right)^2 = \frac{1}{2}$

17. (a) Given,

$\cos A = \frac{63}{65}$

With the help of triplet (16, 63, 65), perpendicular = 16



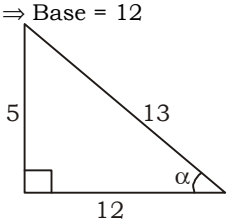
$\therefore \tan A + \cot A$

$= \frac{16}{63} + \frac{63}{16} = \frac{256 + 3969}{1008} = \frac{4225}{1008} = 4.19$

18. (b) $\sin\alpha = \frac{5}{13}$

[5, 12, 13 → Triplet]

$\Rightarrow$ Base = 12

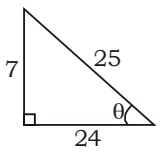


$\therefore \cos\alpha \operatorname{cosec}\alpha \times \cot\alpha = \frac{\cos\alpha}{\sin\alpha} \times \cot\alpha$

$= \cot^2\alpha = \left(\frac{12}{5}\right)^2 = \frac{144}{25}$

19. (c) $\tan\theta = \frac{7}{24}$

[7, 24, 25, → Triplet]



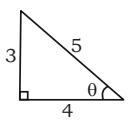
$\therefore \cos^2\theta - \sin^2\theta$

$= \left(\frac{24}{25}\right)^2 - \left(\frac{7}{25}\right)^2 = \frac{576}{625} - \frac{49}{625} = \frac{527}{625}$

20. (c) Given,

$\cot\theta = \frac{4}{3}$

With the help of triplet (3, 4, 5), Hypotenuse = 5



Consider,

$5p \cos^2\theta \sin\theta = \cot^2\theta$

$\Rightarrow 5p\left(\frac{4}{5}\right)^2 \times \frac{3}{5} = \left(\frac{4}{3}\right)^2 \Rightarrow p = \frac{25}{27}$

21. (d) $\cot\theta.\sin\theta.\cos\theta$

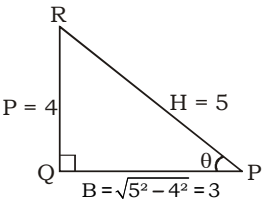
$= \frac{\cos\theta}{\sin\theta} \times \sin\theta \times \cos\theta$

$= \cos^2\theta = 1 - \sin^2\theta$

Put $\sin\theta = \frac{5}{6}$

$\Rightarrow 1 - \frac{25}{36} = \frac{11}{36}$

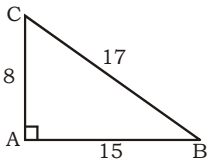
22. (c) $\sin\theta = \frac{4}{5}$



$\therefore \tan\theta - \cot\theta$

$= \frac{P}{B} - \frac{B}{P} = \frac{4}{3} - \frac{3}{4} = \frac{7}{12}$

23. (b) Given, $\cot B = \frac{15 \rightarrow \text{Base}}{8 \rightarrow P}$



$BC = \sqrt{15^2 + 8^2} = 17$

$\therefore$ secB + tanB

$= \frac{17}{15} + \frac{8}{15} = \frac{25}{15} = \frac{5}{3}$

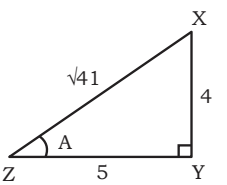
24. (d) $\sin A = \frac{8}{17}$

using triplet (8, 15, 17),

Base = 15

$\therefore \cot A + \sec A = \frac{15}{8} + \frac{17}{15} = \frac{361}{120} = 3\frac{1}{120}$

25. (c) $4\cot A = 5 \Rightarrow \cot A = \frac{5}{4}$



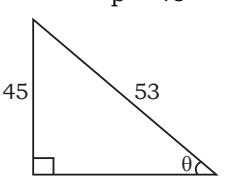
$XZ = \sqrt{4^2 + 5^2} = \sqrt{41}$

$\therefore 6\sec A \times \tan A$

$= 6 \times \frac{\sqrt{41}}{5} \times \frac{4}{5} = \frac{24\sqrt{41}}{25}$

26. (d) $\cot\theta = \frac{28}{45}$

$\Rightarrow \cot\theta = \frac{b}{p} = \frac{28}{45}$



(45, 28, 53 Triplet)

$\therefore \sin\theta - \cos\theta = \frac{45}{53} - \frac{28}{53} = \frac{17}{53}$

27. (d) $3\tan A = 4$

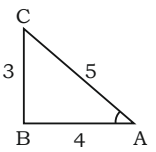
$\Rightarrow \tan A = \frac{4}{3}$, (Triplet of 3, 4, 5)

$\therefore 4\sin A + 3\cos A$

$= 4 \times \frac{4}{5} + 3 \times \frac{3}{5}$

$= \frac{16}{5} + \frac{9}{5} = \frac{25}{5} = 5$

28. (c) $\cos A = \frac{4}{5} = \frac{B}{H}$



[3, 4, 5 triplet]

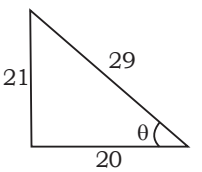
$\sin A = \frac{3}{5}$, $\tan A = \frac{3}{4}$

$\therefore (8 + \sin A)(3 - \tan A)$

$= \left(8 + \frac{3}{5}\right)\left(3 - \frac{3}{4}\right) = \frac{43}{5} \times \frac{9}{4}$

$= \frac{387}{20}$

29. (b) $\sec\theta = \frac{29}{20}$



By triplet of 20, 21, 29

perpendicular = 21

$\therefore 3\operatorname{cosec}\theta + 3\cot\theta = 3(\operatorname{cosec}\theta + \cot\theta)$

$= 3\left(\frac{29}{21} + \frac{20}{21}\right) = 3 \times \frac{49}{21} = 7$

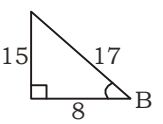
30. (b) Given, $4\tan A = 3$

$\Rightarrow \tan A = \frac{3 \rightarrow P}{4 \rightarrow B}$

H = 5

$\therefore \cos^2 A - \sin^2 A = \frac{16}{25} - \frac{9}{25} = \frac{7}{25}$

31. (d) $\sin B = \frac{15}{17}$



Triplet(8, 15, 17)

Base = 8

$\therefore \cos B(\sec B - \tan B)$

$= \frac{8}{17}\left(\frac{17}{8} - \frac{15}{8}\right)$

$= \frac{8}{17} \times \frac{2}{8} = \frac{2}{17}$

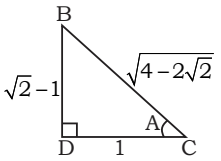
Alternate Method:-

$\cos B(\sec B - \tan B)$

$= 1 - \sin B$

$= 1 - \frac{15}{17} = \frac{2}{17}$

32. (b) $\tan A = \sqrt{2} - 1$



$BC = \sqrt{2+1-2\sqrt{2}+1} = \sqrt{4-2\sqrt{2}}$

$\therefore \sin A \cos A + \tan A =$

$$\frac{\sqrt{2}-1}{\left(\sqrt{4-2\sqrt{2}}\right)^2} + \sqrt{2} - 1 = \frac{\sqrt{2}-1}{4-2\sqrt{2}} + \sqrt{2} - 1$$

$$= \frac{1}{2\sqrt{2}} + \sqrt{2} - 1 = \frac{1+4-2\sqrt{2}}{2\sqrt{2}}$$

$$= \frac{5-2\sqrt{2}}{2\sqrt{2}} = \frac{5\sqrt{2}-4}{4}$$

33. (c) $\sqrt{3}\tan\theta = 3\sin\theta$

$$\Rightarrow \sqrt{3}\frac{\sin\theta}{\cos\theta} = 3\sin\theta$$

$$\Rightarrow \frac{1}{\cos\theta} = \sqrt{3}$$

$$\Rightarrow \cos\theta = \frac{1 \to B}{\sqrt{3} \to H}$$

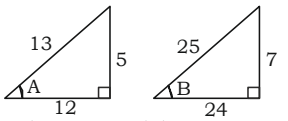
Perpendicular = $\sqrt{2}$

$\therefore \sin^2\theta - \cos^2\theta$

$$= \left(\frac{\sqrt{2}}{\sqrt{3}}\right)^2 - \left(\frac{1}{\sqrt{3}}\right)^2 = \frac{1}{3}$$

34. (b) $\sin A = \frac{5}{13}$

$7\cot B = 24 \Rightarrow \cot B = \frac{24}{7}$



$\therefore$ (secA cosB) (cosec B tan A)

$$= \left(\frac{13}{12}\times\frac{24}{25}\right)\left(\frac{25}{7}\times\frac{5}{12}\right) = \frac{65}{42}$$

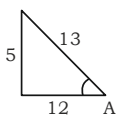
35. (c) $\text{cosec}\theta = \frac{5}{3}$

$\therefore (\sec^2\theta - 1) \times \cot^2\theta \times (1 + \cot^2\theta)$

$= \tan^2\theta \times \cot^2\theta \times \text{cosec}^2\theta$

$= \text{cosec}^2\theta = \frac{25}{9}$

36. (c) $\sin A = \frac{5}{13}$ (5, 12, 13)

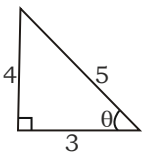


$$\frac{\cos A - 2\tan A}{\sin A + 3\tan A}$$

$$= \frac{\frac{12}{13} - 2\times\frac{5}{12}}{\frac{5}{13} + 3\times\frac{5}{12}} = \frac{\frac{12}{13} - \frac{5}{6}}{\frac{5}{13} + \frac{5}{4}}$$

$$= \frac{\frac{7}{78}}{\frac{85}{52}} = \frac{7\times 52}{78\times 85} = \frac{14}{255}$$

37. (d) $\cot\theta = \frac{3}{4} \to$ Base $\to$ Perpendicular

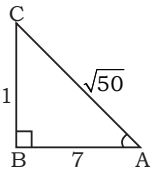


Triplet $\to$ (3, 4, 5)

$\therefore \sin\theta + \cos\theta - \tan\theta$

$$= \frac{4}{5} + \frac{3}{5} - \frac{4}{3} = \frac{12+9-20}{15} = \frac{1}{15}$$

38. (a) $\cot A = 7$



By pythagoras theorem,

$AC = \sqrt{7^2+1^2} = \sqrt{50}$

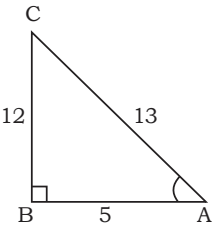
$$\therefore \frac{5\cos A + 4\sin A}{\cos^3 A + 7\sin^3 A + 6\sin A}$$

$$= \frac{5\times\frac{7}{\sqrt{50}} + 4\times\frac{1}{\sqrt{50}}}{\left(\frac{7}{\sqrt{50}}\right)^3 + 7\left(\frac{1}{\sqrt{50}}\right)^3 + 6\times\frac{1}{\sqrt{50}}}$$

$$= \frac{\frac{39}{\sqrt{50}}}{\left(\frac{343}{50\sqrt{50}}\right) + \left(\frac{7}{50\sqrt{50}}\right) + \frac{6}{\sqrt{50}}}$$

$$= \frac{\frac{39}{\sqrt{50}}}{\frac{650}{50\sqrt{50}}} = 3$$

39. (c) $\cot A = \frac{5}{12}$



AC = 13 (triplet of 5, 12, 13)

$$\therefore \frac{\sqrt{\sec A - \text{cosec}A}}{\sqrt{\sec A + \text{cosec}A}} - \sqrt{\frac{17}{7}}$$

$$= \frac{\sqrt{\frac{13}{5} - \frac{13}{12}}}{\sqrt{\frac{13}{5} + \frac{13}{12}}} - \sqrt{\frac{17}{7}} = \frac{\sqrt{\frac{1}{5} - \frac{1}{12}}}{\sqrt{\frac{1}{5} + \frac{1}{12}}} - \sqrt{\frac{17}{7}}$$

$$= \frac{\sqrt{\frac{12-5}{60}}}{\sqrt{\frac{12+5}{60}}} - \sqrt{\frac{17}{7}} = \frac{\sqrt{7}}{\sqrt{17}} - \frac{\sqrt{17}}{\sqrt{7}}$$

$$= \frac{7-17}{\sqrt{119}} = \frac{-10}{\sqrt{119}}$$

Alternate Method:

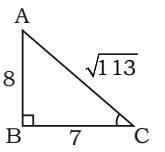
$$\cot A = \frac{\text{cosec}A}{\sec A} = \frac{5}{12}$$

$$\therefore \frac{\sqrt{\sec A - \text{cosec}A}}{\sqrt{\sec A + \text{cosec}A}} - \sqrt{\frac{17}{7}}$$

$$= \frac{\sqrt{12-5}}{\sqrt{12+5}} - \frac{\sqrt{17}}{\sqrt{7}} = \frac{\sqrt{7}}{\sqrt{17}} - \frac{\sqrt{17}}{\sqrt{7}}$$

$$= \frac{-10}{\sqrt{119}}$$

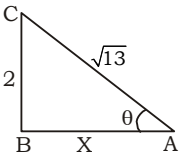
40. (a) $\cot\theta = \frac{7}{8}$



$AC = \sqrt{8^2+7^2} = \sqrt{113}$

$$\therefore \frac{1+\sin\theta}{\cos\theta} = \frac{1+\frac{8}{\sqrt{113}}}{\frac{7}{\sqrt{113}}} = \frac{8+\sqrt{113}}{7}$$

41. (a) $\sqrt{13}\sin\theta = 2 \Rightarrow \sin\theta = \frac{2}{\sqrt{13}}$

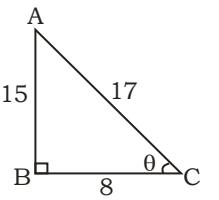


Base (X) = $\sqrt{13-4} = 3$

$$\therefore \frac{3\tan\theta + \sqrt{13}\sin\theta}{\sqrt{13}\cos\theta - 3\tan\theta}$$

$$= \frac{3\times\frac{2}{3} + \sqrt{13}\times\frac{2}{\sqrt{13}}}{\sqrt{13}\times\frac{3}{\sqrt{13}} - 3\times\frac{2}{3}} = \frac{2+2}{3-2} = 4$$

42. (a)



$\sin\theta = \frac{15 \to \text{Perpendicular}}{17 \to \text{Hypotenuse}}$

Base (BC) = 8 (with triplet of 8,15,17)

$$\therefore \frac{15\cot\theta + 17\sin\theta}{8\tan\theta + 16\sec\theta}$$

$$= \frac{15\times\frac{8}{15} + 17\times\frac{15}{17}}{8\times\frac{15}{8} + 16\times\frac{17}{8}} = \frac{8+15}{15+34} = \frac{23}{49}$$

43. (a) $\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{7}{4}$

$$\therefore \frac{4\sin\theta - 3\cos\theta}{4\sin\theta + 3\cos\theta} = \frac{4\times 7 - 3\times 4}{4\times 7 + 3\times 4}$$

$$= \frac{16}{40} = \frac{2}{5}$$

44. (c) Given, $\tan x = \frac{7}{5}$

$\Rightarrow \frac{\sin x}{\cos x} = \frac{7}{5}$

$\therefore \frac{9\sin x - \frac{42}{5}\cos x}{15\sin x + 21\cos x}$

$= \frac{9\times 7 - \frac{42}{5}\times 5}{15\times 7 + 21\times 5} = \frac{63-42}{105+105}$

$= \frac{21}{210} = 0.1$

45. (c) Given, $3\cot A = \frac{7}{3}$

$\Rightarrow \cot A = \frac{7}{9} \Rightarrow \frac{\cos A}{\sin A} = \frac{7}{9}$

$\therefore \frac{3\cos A + 2\sin A}{3\cos A - 2\sin A}$

$= \frac{3\times 7 + 2\times 9}{3\times 7 - 2\times 9} = \frac{39}{3} = 13$

46. (a) $3\tan\theta = 4 \Rightarrow \tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{4}{3}$

$\therefore \frac{9\sin\theta + 2\cos\theta}{9\sin\theta - 2\cos\theta} = \frac{9\times 4 + 2\times 3}{9\times 4 - 2\times 3}$

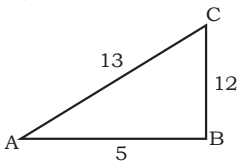
$= \frac{42}{30} = \frac{7}{5}$

47. (c) $\tan A = \frac{5}{8} \Rightarrow \frac{\sin A}{\cos A} = \frac{5}{8}$

$\therefore \frac{8\sin A - 7\cos A}{8\sin A + 11\cos A}$

$= \frac{8\times 5 - 7\times 8}{8\times 5 + 11\times 8} = \frac{40-56}{40+88} = \frac{-16}{128} = \frac{-1}{8}$

48. (d) We know triplet (5, 12, 13)



$\cot A = \frac{5}{12}$, $\tan A = \frac{12}{5}$

$\therefore \frac{5\times \sin^2 A}{6\times \cos^2 A} = \frac{5}{6}\times \tan^2 A$

$= \frac{5}{6}\times\frac{12}{5}\times\frac{12}{5} = \frac{24}{5}$

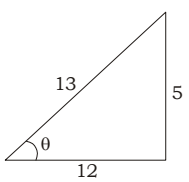
49. (c) $2\cot\theta = 3$

$\Rightarrow \cot\theta = \frac{3}{2} \to \frac{B}{P}, H = \sqrt{2^2+3^2} = \sqrt{13}$

$\therefore \frac{\sqrt{13}\sin\theta - 3\tan\theta}{3\tan\theta + \sqrt{13}\cos\theta}$

$= \frac{\sqrt{13}\times\frac{2}{\sqrt{13}} - 3\times\frac{2}{3}}{3\times\frac{2}{3} - \sqrt{13}\times\frac{3}{13}} = 0$

50. (b) $\cos\theta = \frac{12}{13}$



$\therefore \frac{\sin\theta(1-\tan\theta)}{\tan\theta(1+\cos ec\theta)} = \frac{\frac{5}{13}\left(1-\frac{5}{12}\right)}{\frac{5}{12}\left(1+\frac{13}{5}\right)}$

$= \frac{\frac{5}{13}\times\left(\frac{7}{12}\right)}{\frac{5}{12}\times\frac{18}{5}} = \frac{35}{234}$

51. (a) $\sin A = \frac{1}{2} \Rightarrow A = 30°$

$\therefore \frac{\tan A - \cot A}{\sqrt{3}(1+\cos ecA)}$

$= \frac{\frac{1}{\sqrt{3}} - \sqrt{3}}{\sqrt{3}(1+2)} = \frac{-2}{\sqrt{3}}\times\frac{1}{3\sqrt{3}} = \frac{-2}{3(3)} = \frac{-2}{9}$

52. (d) Given,

$2\tan\theta = 3 \Rightarrow \tan\theta = \frac{3}{2}$

$\Rightarrow \frac{\sin\theta}{\cos\theta} = \frac{3}{2}$

$\therefore \frac{3\sin\theta - 2\cos\theta}{3\sin\theta + 2\cos\theta} = \frac{3\times 3 - 2\times 2}{3\times 3 + 2\times 2}$

$= \frac{9-4}{9+4} = \frac{5}{13}$

53. (a) $7\tan\theta = 3 \Rightarrow \tan\theta = \frac{3}{7}$

$\Rightarrow \frac{\sin\theta}{\cos\theta} = \frac{3}{7}$

$\therefore \frac{5\sin\theta - \cos\theta}{5\sin\theta + 2\cos\theta} = \frac{5\times 3 - 7}{5\times 3 + 2\times 7}$

$= \frac{8}{29}$

54. (c) $7\tan\theta = 3$

$\Rightarrow \tan\theta = \frac{3}{7} \Rightarrow \frac{\sin\theta}{\cos\theta} = \frac{3}{7}$

$\therefore \frac{7\sin\theta + 5\cos\theta}{7\sin\theta - 2\cos\theta} = \frac{7\times 3 + 5\times 7}{7\times 3 - 7\times 2}$

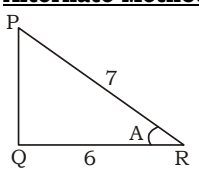
$= \frac{56}{7} = 8$

55. (c) $7\cos A = 6 \Rightarrow \cos A = \frac{6}{7}$

$\therefore \frac{\cos ecA + \cot A}{\cos ecA - \cot A} = \frac{\frac{1}{\sin A} + \frac{\cos A}{\sin A}}{\frac{1}{\sin A} - \frac{\cos A}{\sin A}}$

$= \frac{1+\cos A}{1-\cos A} = \frac{1+\frac{6}{7}}{1-\frac{6}{7}} = \frac{13}{1} = 13$

Alternate Method:



$PQ = \sqrt{7^2 - 6^2} = \sqrt{13}$

$\therefore \frac{\cos ecA + \cot A}{\cos ecA - \cot A}$

$= \frac{7+6}{7-6} = \frac{13}{1} = 13$

56. (a) Given,

$\tan A = \frac{5}{7} \Rightarrow \frac{\sin A}{\cos A} = \frac{5}{7}$

$\therefore \frac{7\sin A - 3\cos A}{7\sin A + 3\cos A} + 4$

$= \frac{7\times 5 - 3\times 7}{7\times 5 + 3\times 7} + 4 = \frac{14}{56} + 4$

$= \frac{1}{4} + 4 = 4\frac{1}{4}$

57. (d) $\sin\theta = \frac{5}{13} = \frac{P}{h}$

We have triplet (5, 12, 13)

b = 12

$\therefore \frac{\cos^2\theta - \sin^2\theta}{2\cos\theta\sin\theta}$

$= \frac{\left(\frac{12}{13}\right)^2 - \left(\frac{5}{13}\right)^2}{2\times\frac{12}{13}\times\frac{5}{13}} = \frac{\frac{144}{169} - \frac{25}{169}}{\frac{120}{169}}$

$= \frac{\frac{119}{169}}{\frac{120}{169}} = \frac{119}{120}$

58. (b) $\frac{7\sin\theta + 4\cos\theta}{9\sin\theta - 2\cos\theta} = \frac{5}{4}$

$\Rightarrow 28\sin\theta + 16\cos\theta = 45\sin\theta - 10\cos\theta$

$\Rightarrow 26\cos\theta = 17\sin\theta$

$\Rightarrow \tan\theta = \frac{26}{17}$

$\therefore \frac{\tan^2\theta + 5}{\tan^2\theta - 5} = \frac{\left(\frac{26}{17}\right)^2 + 5}{\left(\frac{26}{17}\right)^2 - 5} = -\frac{2121}{769}$

59. (c) $\frac{\sec\alpha + \tan\alpha}{\sec\alpha - \tan\alpha} = \frac{7}{4}$

$\Rightarrow 4\sec\alpha + 4\tan\alpha = 7\sec\alpha - 7\tan\alpha$

$\Rightarrow 11\tan\alpha = 3\sec\alpha$

$\Rightarrow \frac{\tan\alpha}{\sec\alpha} = \frac{3}{11}$

$\Rightarrow \sin\alpha = \frac{3}{11}$

$\therefore \operatorname{cosec}\alpha = \frac{11}{3}$

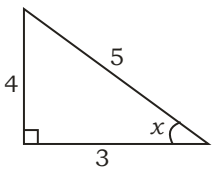
Alternate Method:-

By using componendo and dividendo

$\frac{\sec\alpha}{\tan\alpha} = \frac{11}{3}$

$\Rightarrow \operatorname{cosec}\alpha = \frac{11}{3}$

60. (b) $\text{cosec}x = \begin{matrix} 5 \to H \\ 4 \to P \end{matrix}$



$\Rightarrow$ Base = 3

(3, 4, 5 are triplet)

$$\therefore \frac{5\sin^2 x + 8\cot^2 x}{27\tan^2 x - 125\cos^2 x}$$

$$= \frac{5\times\frac{16}{25} + 8\times\frac{9}{16}}{27\times\frac{16}{9} - 125\times\frac{9}{25}} = \frac{\frac{16}{5}+\frac{9}{2}}{48-45}$$

$$= \frac{3.2+4.5}{3} = \frac{7.7}{3} = \frac{77}{30}$$

61. (b) $\cot\theta = \frac{\sqrt{11}}{1} \Rightarrow \frac{\cos\theta}{\sin\theta} = \frac{\sqrt{11}}{1}$

$$\therefore \frac{\text{cosec}^2\theta - \sec^2\theta}{\text{cosec}^2 + \sec^2\theta}$$

$$= \frac{\frac{1}{\sin^2\theta} - \frac{1}{\cos^2\theta}}{\frac{1}{\sin^2\theta} + \frac{1}{\cos^2\theta}} = \frac{\cos^2\theta - \sin^2\theta}{\cos^2\theta + \sin^2\theta}$$

$$= \frac{11-1}{11+1} = \frac{10}{12} = \frac{5}{6}$$

62. (c) $\frac{2\sin A - \cos A}{\sin A + \cos A} = 1$

$\Rightarrow 2\sin A - \cos A = \sin A + \cos A$

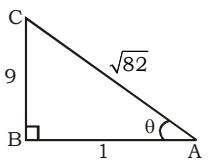
$\Rightarrow \sin A = 2\cos A \Rightarrow \tan A = 2$

$$\therefore \cot A = \frac{1}{2}$$

63. (c) $\frac{\sin\theta - \cos\theta}{\sin\theta + \cos\theta} = \frac{4}{5}$,

$\Rightarrow 5\sin\theta - 5\cos\theta = 4\sin\theta + 4\cos\theta$

$\Rightarrow \sin\theta = 9\cos\theta \Rightarrow \tan\theta = \frac{9}{1}$



$$\text{cosec}\theta = \frac{\sqrt{82}}{9}$$

$$\therefore \frac{\text{cosec}^2\theta}{2 - \text{cosec}^2\theta} = \frac{\frac{82}{81}}{2 - \frac{82}{81}} = \frac{\frac{82}{81}}{\frac{80}{81}} = \frac{82}{80} = \frac{41}{40}$$

64. (a) $\frac{\cos\beta}{\sec\alpha} = 15, \frac{\sin\beta}{\sec\alpha} = 16$

$$\Rightarrow \frac{\frac{\cos\beta}{\sec\alpha}}{\frac{\sin\beta}{\sec\alpha}} = \frac{15}{16}$$

$$\cot\beta = \frac{15 \to B}{16 \to P}$$

$$\therefore H = \sqrt{16^2 + 15^2} = \sqrt{481}$$

$$\sin^2\beta = \left(\frac{P}{H}\right)^2 = \left(\frac{16}{\sqrt{481}}\right)^2 = \frac{256}{481}$$

65. (a) $\frac{\sin\theta + \cos\theta}{\sin\theta - \cos\theta} = \frac{5}{1}$

By componendo and dividendo

$$\frac{\sin\theta}{\cos\theta} = \frac{6}{4} = \frac{3}{2} \Rightarrow \tan\theta = \frac{3}{2}$$

$$H = \sqrt{9+4} = \sqrt{13}$$

$$\therefore \sin\theta = \frac{3}{\sqrt{13}}, \cos\theta = \frac{2}{\sqrt{13}}$$

$$\therefore \frac{4\sin^2\theta + 3}{2\cos^2\theta + 2} = \frac{4\times\frac{9}{13} + 3}{2\times\frac{4}{13} + 2} = \frac{\frac{36}{13}+3}{\frac{8}{13}+2} = \frac{75}{34}$$

66. (d) $\frac{\sec A + \tan A}{\sec A - \tan A} = \frac{209}{79}$

using C & D rule

$$\Rightarrow \frac{\sec A}{\tan A} = \frac{288}{130}$$

$$\Rightarrow \frac{\cos A}{\cos A \times \sin A} = \frac{144}{65}$$

$$\Rightarrow \frac{1}{\sin A} = \frac{144}{65} \Rightarrow \sin A = \frac{65}{144}$$

67. (c) $\frac{21\cos A + 3\sin A}{3\cos A + 4\sin A} = 2$

Divide N^r & D^r by sinA

$$\Rightarrow \frac{21\cot A + 3}{3\cot A + 4} = 2$$

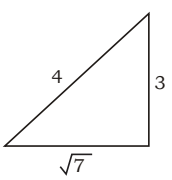
$\Rightarrow 21\cot A + 3 = 6\cot A + 8$

$$\Rightarrow 15\cot A = 5 \Rightarrow \cot A = \frac{1}{3}$$

68. (a) $\frac{\sec\theta - \tan\theta}{\sec\theta + \tan\theta} = \frac{1}{7}$

By C and D rule,

$$\Rightarrow \frac{\sec\theta}{\tan\theta} = \frac{8}{6} = \frac{4}{3} \Rightarrow \frac{\frac{1}{\cos\theta}}{\frac{\sin\theta}{\cos\theta}} = \frac{4}{3} \Rightarrow \text{Sin}\theta = \frac{3}{4}$$



$$\therefore \frac{\text{cosec}\theta + \cot^2\theta}{\text{cosec}\theta - \cot^2\theta} = \frac{\frac{4}{3} + \left(\frac{\sqrt{7}}{3}\right)^2}{\frac{4}{3} - \left(\frac{\sqrt{7}}{3}\right)^2}$$

$$= \frac{\frac{4}{3}+\frac{7}{9}}{\frac{4}{3}-\frac{7}{9}} = \frac{\frac{19}{9}}{\frac{5}{9}} = \frac{19}{5}$$

69. (a) $\frac{\sec\theta + \tan\theta}{\sec\theta - \tan\theta} = 2\frac{51}{79} = \frac{209}{79}$

By C & D,

$$\Rightarrow \frac{\sec\theta}{\tan\theta} = \frac{288}{130} = \frac{144}{65}$$

$$\Rightarrow \frac{\frac{1}{\cos\theta}}{\frac{\sin\theta}{\cos\theta}} = \frac{144}{65} \Rightarrow \sin\theta = \frac{65}{144}$$

70. (c) $\tan\theta = \frac{3}{10} \Rightarrow \cot\theta = \frac{10}{3}$

$$\therefore \frac{(1+\sin\theta)(1-\sin\theta)}{(1+\cos\theta)(1-\cos\theta)}$$

$$= \frac{(1-\sin^2\theta)}{(1-\cos^2\theta)} = \frac{\cos^2\theta}{\sin^2\theta} = \cot^2\theta = \left(\frac{10}{3}\right)^2 = \frac{100}{9}$$

71. (a) Given, $\cot\theta = \frac{4}{3}$

$$\therefore \frac{\sec\theta(1+\cot^2\theta)(\text{cosec}^2\theta - \cot^2\theta)}{\text{cosec}^3\theta}$$

$$= \frac{\sec\theta \times \text{cosec}^2\theta \times 1}{\text{cosec}^3\theta} = \frac{\sec\theta}{\text{cosec}\theta}$$

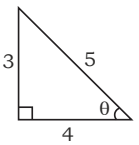
$$= \tan\theta = \frac{1}{\cot\theta} = \frac{3}{4}$$

Note: $1 + \cot^2\theta = \text{cosec}^2\theta$

Alternate Method:-

Given,

$\cot\theta = \frac{4}{3} \Rightarrow$



$$\therefore \frac{\sec\theta(1+\cot^2\theta)(\text{cosec}^2\theta - \cot^2\theta)}{\text{cosec}^3\theta}$$

$$= \frac{5}{4} \times \frac{\left(1+\frac{16}{9}\right)\left(\frac{25}{9}-\frac{16}{9}\right)}{\frac{25}{9}\times\frac{5}{3}}$$

$$= \frac{5}{4} \times \frac{25}{9} \times 1 \times \frac{9}{25} \times \frac{3}{5}$$

$$= \frac{5}{4} \times \frac{3}{5} = \frac{3}{4}$$

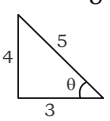
72. (b) $\frac{\sec\theta - \tan\theta}{\sec\theta + \tan\theta} = \frac{1}{9}$

$$\frac{1-\sin\theta}{1+\sin\theta} = \frac{1}{9}$$

$9 - 9\sin\theta = 1 + \sin\theta$

$10\sin\theta = 8$

$$\sin\theta = \frac{4}{5}$$



$$\frac{\sin\theta - \tan^2\theta}{\sin\theta + \tan^2\theta} = \frac{\frac{4}{5} - \left(\frac{4}{3}\right)^2}{\frac{4}{5} + \left(\frac{4}{3}\right)^2} = \frac{\frac{4}{5}-\frac{16}{9}}{\frac{4}{5}+\frac{16}{9}}$$

$$= -\frac{11}{29}$$

Alternate Method:

$$\frac{\sec\theta - \tan\theta}{\sec\theta + \tan\theta} = \frac{1}{9}$$

$$\frac{\sec\theta + \tan\theta}{\sec\theta - \tan\theta} = \frac{9}{1}$$

by C & D Rule

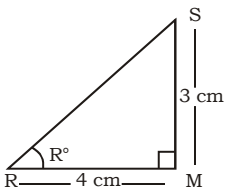
$$\frac{\sec\theta}{\tan\theta} = \frac{10}{8} = \frac{5}{4}$$

$\text{cosec}\theta = \frac{5}{4}$ than, $\sin\theta = \frac{4}{5}$

$$\frac{\sin\theta - \tan^2\theta}{\sin\theta + \tan^2\theta} = \frac{\frac{4}{5} - \left(\frac{4}{3}\right)^2}{\frac{4}{5} + \left(\frac{4}{3}\right)^2} = \frac{\frac{4}{5}-\frac{16}{9}}{\frac{4}{5}+\frac{16}{9}}$$

$$= -\frac{11}{29}$$

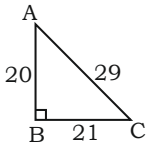
73. (b)



$\because$ SR = $\sqrt{4^2+3^2}$ = 5 cm

$\sec R = \frac{\text{Hypotenuse}}{\text{Base}} = \frac{5}{4}$

74. (c)



We have triplet (20, 21, 29)

cotcC + cosecC – 2tanA

$= \frac{21}{20} + \frac{29}{20} - 2 \times \frac{21}{20} = \frac{8}{20} = \frac{2}{5}$

75. (a) Given, $\angle B = 90°$

Then, remaining angles $\angle A + \angle C = 90°$

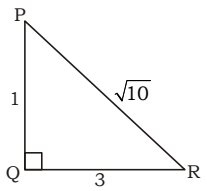
So, $\frac{\sin A}{\cos C} = 1$ & $\cot A = \frac{\cos A \to 1}{\sin A \to \frac{1}{2}}$

$\therefore \frac{\sin A(\cos C + \cos A)}{\cos C(\sin C - \sin A)}$

$= \frac{\cos(90-A)+\cos A}{\sin(90-A)-\sin A} = \frac{\sin A + \cos A}{\cos A - \sin A}$

$= \frac{1+2}{1-2} = \frac{3}{-1} = -3$

76. (d)



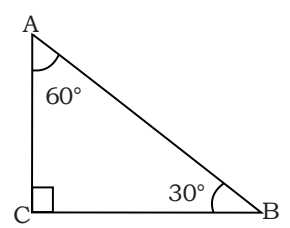
$PR = \sqrt{3^2+1} = \sqrt{10}$

$\frac{\sec P(\cos R + \sin P)}{\text{cosec}R(\sin R - \text{cosec}P)}$

$= \frac{\frac{\sqrt{10}}{1}\left(\frac{3}{\sqrt{10}} + \frac{3}{\sqrt{10}}\right)}{\frac{\sqrt{10}}{1}\left(\frac{1}{\sqrt{10}} - \frac{\sqrt{10}}{3}\right)}$

$= \frac{\frac{6}{\sqrt{10}}}{\frac{-7}{3\sqrt{10}}} = \frac{6}{\sqrt{10}} \times \frac{3\sqrt{10}}{-7} = \frac{-18}{7}$

77. (c)



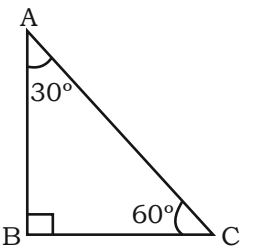
$\tan A = \sqrt{3}$

$\Rightarrow A = 60°$

$\because \cot(A + B) = \cot 90° = 0$

$\therefore \sin A \cos B \cot(A + B) = 0$

78. (d)



$\sin A = \frac{1}{2} \Rightarrow A = 30°$

So, $\angle C = 60°$, then $\cos 60° = \frac{1}{2}$

Alternate Method:

Here, $\angle B = 90°$

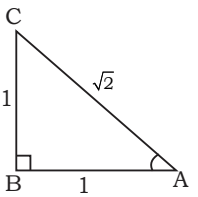
So, $\angle A + \angle C = 90°$

Then, $\sin A = \cos C = \frac{1}{2}$

Note: $\sin\alpha = \cos\beta$

When, $\alpha + \beta = 90°$

79. (b)



$\tan A = 1 \Rightarrow AC = \sqrt{2}$

$\therefore 4 \sin A \cos A = 4 \times \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} = 2$

Alternate Method:

$\tan A = 1 \Rightarrow A = 45°$

$\therefore 4\sin A \cos A = 4\sin 45° \cos 45°$

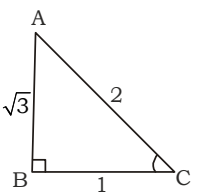
$= 4 \times \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} = 2$

Alternate Method:

$A = 45°$

$4 \sin A.\cos A = 2\sin 2A = 2\sin 90° = 2$

80. (a)



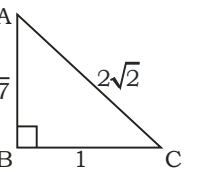
$\tan C = \frac{\sqrt{3}}{1}$

$\therefore \frac{\sin^2 C + \cos^2 C}{1+\cot^2 C} = \frac{\frac{3}{4}+\frac{1}{4}}{1+\frac{1}{3}} \Rightarrow \frac{1}{\frac{4}{3}} = \frac{3}{4}$

Alternate Method:

$\frac{\sin^2 C + \cos^2 C}{1+\cot^2 C} = \frac{1}{\text{cosec}^2 C} \Rightarrow \sin^2 C = \frac{3}{4}$

81. (c) $\text{cosec}A = \frac{2\sqrt{2} \to H}{1 \to P}$



$\therefore \sin A \cos C + \cos A \sin C$

$= \frac{1}{2\sqrt{2}} \times \frac{1}{2\sqrt{2}} + \frac{\sqrt{7}}{2\sqrt{2}} \times \frac{\sqrt{7}}{2\sqrt{2}}$

$= \frac{1}{8} + \frac{7}{8} = \frac{8}{8} = 1$

Alternate Method:

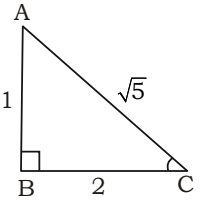
$\angle B = 90°$, $\angle A + \angle C = 180° - 90°$

$= 90°$

$\therefore \sin A \cos C + \cos A \sin C$

$= \sin(A + C) = \sin 90° = 1$

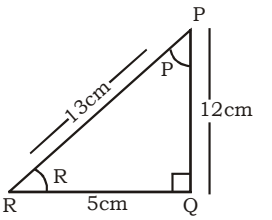
82. (d)



$\therefore \cos A + \tan C$

$= \frac{1}{\sqrt{5}} + \frac{1}{2} = \frac{2+\sqrt{5}}{2\sqrt{5}}$

83. (c)

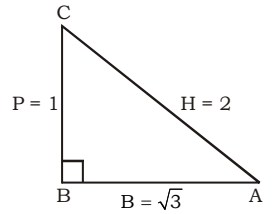


We know triplet (5, 12, 13)

RQ = 5 cm

$\therefore \tan P + \cot R = \frac{5}{12} + \frac{5}{12} = \frac{10}{12}$

84. (a)



$\therefore \sin A \cos C + \cos A \sin C$

$= \frac{1}{2} \times \frac{1}{2} + \frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2}$

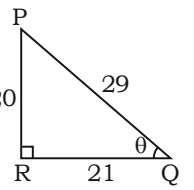
$= \frac{1+3}{4} = 1$

Alternate Method:

$\sin A \cos C + \cos A \sin C = \sin(A + C)$

$= \sin 90° = 1$

85. (d)



Triplet $\Rightarrow$ (20, 21, 29)

Then,

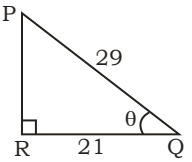
$\cos^2\theta - \sin^2\theta$

$= (\cos\theta + \sin)(\cos\theta - \sin\theta)$

$= \left(\frac{21}{29} + \frac{20}{29}\right)\left(\frac{21}{29} - \frac{20}{29}\right)$

$= \frac{41}{29} \times \frac{1}{29} = \frac{41}{841}$

Alternate Method:



$\therefore \cos^2\theta - \sin^2\theta$
$= \cos^2\theta - (1 - \cos^2\theta)$
$= 2\cos^2\theta - 1$

$= 2 \times \left(\frac{21}{29}\right)^2 - 1 = 2 \times \frac{441}{841} - 1$

$= \frac{882 - 841}{841} = \frac{41}{841}$

86. (b) $\sec 31° = x$

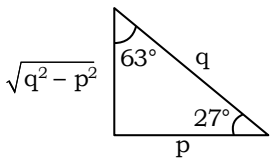
$\therefore \sin^2 59° + \frac{1}{\sec^2 31°} - \frac{1}{\sin^2 59° \operatorname{cosec}^2 59°}$

$= \sin^2 59° + \cos^2 31° - 1$

$= 2\cos^2 31° - 1$

$= \frac{2}{\sec^2 31°} - 1 = \frac{2}{x^2} - 1 = \frac{2 - x^2}{x^2}$

87. (b)

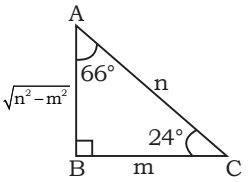


cosec 27° – cos 63°
= cosec 27° – sin27°

$= \frac{q}{\sqrt{q^2 - p^2}} - \frac{\sqrt{q^2 - p^2}}{q}$

$= \frac{q^2 - q^2 + p^2}{q\sqrt{q^2 - p^2}} = \frac{p^2}{q\sqrt{q^2 - p^2}}$

88. (d)



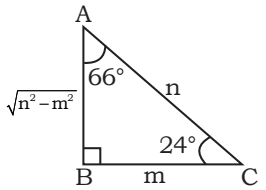
$\text{Cos}24° = \frac{m}{n} \Rightarrow AB = \sqrt{n^2 - m^2}$

$\therefore$ Cosec24° – Cos66°

$= \frac{n}{\sqrt{n^2 - m^2}} - \frac{\sqrt{n^2 - m^2}}{n}$

$= \frac{n^2 - n^2 + m^2}{n\sqrt{n^2 - m^2}} = \frac{m^2}{n\sqrt{n^2 - m^2}}$

Alternate Method:



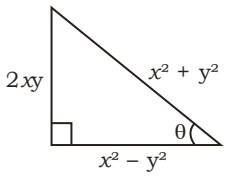
$\cos 24° = \frac{m}{n}$

$\therefore$ cosec24° – cos66° = cosec24° – sin24°

$= \frac{1}{\sin 24°} - \sin 24° = \frac{\cos^2 24°}{\sin 24°}$

$= \frac{\frac{m^2}{n^2}}{\frac{\sqrt{n^2 - m^2}}{n}} = \frac{m^2}{n\sqrt{n^2 - m^2}}$

89. (a)



$\sin\theta = \frac{2xy}{x^2 + y^2}$

Base = $x^2 - y^2$

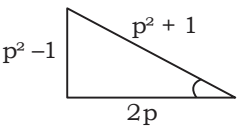
$\therefore \tan\theta = \frac{\text{Perpendicular}}{\text{Base}} = \frac{2xy}{x^2 - y^2}$

90. (c) $\cot A = \frac{B}{P} = \frac{2p}{p^2 - 1}$

Hypotenuse = $\sqrt{(2p)^2 + (p^2 - 1)^2}$

$= \sqrt{4p^2 + p^4 + 1 - 2p^2} = \sqrt{2p^2 + p^4 + 1}$

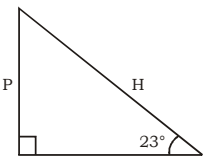
$= \sqrt{(p^2 + 1)^2} = p^2 + 1$



$\text{cosec}A = \frac{p^2 + 1}{p^2 - 1}$

Note:- 2p, $p^2 - 1$ & $p^2 + 1$ are triplet.

91. (d) $\sin 23° = \frac{a \to P}{b \to H}$



$B = \sqrt{b^2 - a^2}$

We know, [sin (90° – θ) = cosθ]

$\therefore \sec 23° - \sin 67° = \sec 23° - \cos 23°$

$= \frac{b}{\sqrt{b^2 - a^2}} - \frac{\sqrt{b^2 - a^2}}{b} = \frac{a^2}{b\sqrt{b^2 - a^2}}$

92. (d) $\cos 48° = \frac{m \to B}{n \to H}$

$P = \sqrt{n^2 - m^2}$

Now,
sec48° – cot42°
= sec48° – cot(90° – 48°)
= sec48° – tan48°

$= \frac{n}{m} - \frac{\sqrt{n^2 - m^2}}{m}$

$= \frac{n - \sqrt{n^2 - m^2}}{m}$

Alternate Method:

Let $\cos 48° = \frac{4 \to m}{5 \to n}$

tiplet (3, 4, 5)
↓ ↓ ↓
P B H

sec48° – cot42° = sec48° – tan48°

$= \frac{5}{4} - \frac{3}{4} = \frac{2}{4} = \frac{1}{2}$

Go through option - (d)

$\frac{n - \sqrt{n^2 - m^2}}{m} = \frac{5 - \sqrt{25 - 16}}{4}$

$= \frac{5 - 3}{4} = \frac{1}{2}$ (satisfied)

93. (a) $\cos 53° = \frac{x}{y}$

So, $\sec 53° = \frac{y}{x}$

sec53° = sec (90° – 37°)
= cosec37°

$\text{Cot}37° = \sqrt{\text{cosec}^2 37° - 1} = \sqrt{\left(\frac{y}{x}\right)^2 - 1}$

$= \sqrt{\frac{y^2 - x^2}{x^2}} = \frac{\sqrt{y^2 - x^2}}{x}$

$\therefore$ sec53° + cot37°

$= \frac{y}{x} + \frac{\sqrt{y^2 - x^2}}{x} = \frac{y + \sqrt{y^2 - x^2}}{x}$

SMART APPROACH:-

$\cos 53° = \frac{x \text{—B}}{y \text{—H}}$

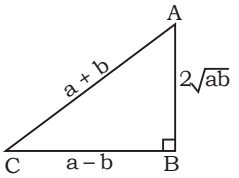
$P = \sqrt{y^2 - x^2}$

$\therefore$ sec53° + cot37°
= sec53° + tan53°

$= \frac{y}{x} + \frac{\sqrt{y^2 - x^2}}{x}$

$= \frac{y + \sqrt{y^2 - x^2}}{x}$

94. (a) $\sin\theta = \frac{2\sqrt{ab}}{a + b}$



$BC^2 = (a + b)^2 - 4ab$
$\Rightarrow BC^2 = a^2 + b^2 + 2ab - 4ab$
$\Rightarrow BC = a - b$

$\therefore \frac{\cos\theta + 1}{\cos\theta - 1} = \frac{\frac{a - b}{a + b} + 1}{\frac{a - b}{a + b} - 1} = -\frac{a}{b}$

Alternate Method:

Let, a = 4, b = 1

Then $\sin\theta = \frac{4 \to P}{5 \to H}$

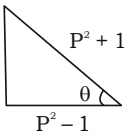
by triplet (3, 4, 5)
↓ ↓ ↓
B P H

$\cos\theta = \frac{3}{5}$

$\frac{\cos\theta + 1}{\cos\theta - 1} = \frac{\frac{3}{5} + 1}{\frac{3}{5} - 1} = \frac{\frac{8}{5}}{\frac{-2}{5}} = -4$

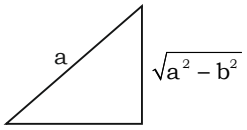
by option, $\frac{-a}{b} = \frac{-4}{1} = -4$ (satisfied)

95. (b) $\cos\theta = \frac{P^2 - 1}{P^2 + 1}$



$\therefore \text{cosec}\,\theta = \frac{P^2 + 1}{2P}$

96. (c)



$\sec\theta = \frac{a}{b}$

$$\frac{1-\tan^2\theta}{2-\sin^2\theta} = \frac{1-\left(\frac{\sqrt{a^2-b^2}}{b}\right)^2}{2-\left(\frac{\sqrt{a^2-b^2}}{a}\right)^2}$$

$$= \frac{\frac{b^2-a^2+b^2}{b^2}}{\frac{2a^2-a^2+b^2}{a^2}} = \frac{\frac{2b^2-a^2}{b^2}}{\frac{a^2+b^2}{a^2}}$$

$$= \frac{a^2\left(2b^2-a^2\right)}{b^2\left(a^2+b^2\right)}$$

Alternate Method:-

I. $\sec\theta = \frac{a}{b}$

Put $\theta = 60°$

$\sec 60° = \frac{2}{1} = \frac{a\to H}{b\to B}$

$P = \sqrt{2^2-1^2} = \sqrt{3}$

$\frac{1-\tan^2\theta}{2-\sin^2\theta} = \frac{1-3}{2-\frac{3}{4}} = \frac{-2}{5} \times 4 = \frac{-8}{5}$

Go through option,

$\frac{a^2(2b^2-a^2)}{b^2(a^2+b^2)} = \frac{4(2-4)}{1(4+1)} = \frac{4\times(-2)}{5} = \frac{-8}{5}$

II. $\text{Sec}\theta = \frac{a}{b}$

with the help of triplet 3, 4, 5 (P, B, H)

$\sec\theta = \frac{H}{B} = \frac{5}{4} = \frac{a}{b}$

Let a = 5, b = 4

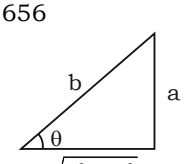
$\therefore \frac{1-\tan^2\theta}{2-\sin^2\theta} = \frac{1-\frac{9}{16}}{2-\frac{9}{25}} = \frac{7}{16}\times\frac{25}{41} = \frac{175}{656}$

Go through option,

$\frac{a^2(2b^2-a^2)}{b^2(a^2+b^2)} = \frac{25(32-25)}{16(25+11)} = \frac{25\times7}{16\times41}$

$= \frac{175}{656}$

97. (a)



$\text{cosec}\,\theta = \frac{b}{a}$

$$\frac{\sqrt{3}\cot\theta+1}{\tan\theta+\sqrt{3}} = \frac{\sqrt{3}\left(\frac{\sqrt{b^2-a^2}}{a}\right)+1}{\frac{a}{\sqrt{b^2-a^2}}+\sqrt{3}}$$

$$= \frac{\frac{\sqrt{3}\left(\sqrt{b^2-a^2}\right)+a}{a}}{\frac{a+\sqrt{3}\left(\sqrt{b^2-a^2}\right)}{\sqrt{b^2-a^2}}} = \frac{\sqrt{b^2-a^2}}{a}$$

Alternate Method:-

Put $\theta = 45°$

$\text{cosec}\,45° = \frac{b}{a} \Rightarrow \frac{\sqrt{2}\to H}{1\to P}$

$B = \sqrt{(\sqrt{2})^2-1^2} = 1$

$\therefore \frac{\sqrt{3}\cot\theta+1}{\tan\theta+\sqrt{3}} = \frac{\sqrt{3}+1}{\sqrt{3}+1} = 1$

Go through option,

$\frac{\sqrt{b^2-a^2}}{a} = \frac{\sqrt{(\sqrt{2})^2-1}}{1} = 1$

98. (a) $\frac{5}{\sqrt{3}} - \text{cosec}60°$

$= \frac{5}{\sqrt{3}} - \frac{2}{\sqrt{3}} = \frac{3}{\sqrt{3}} = \sqrt{3}$

99. (c) $\frac{\sin60°-\cos60°}{\sin60°+\cos60°} = \frac{\frac{\sqrt{3}}{2}-\frac{1}{2}}{\frac{\sqrt{3}}{2}+\frac{1}{2}} = \frac{\sqrt{3}-1}{\sqrt{3}+1}$

$= 2 - \sqrt{3}$

100. (a) $\sin(30° + \theta) - \cos(60° - \theta)$

Put, $\theta = 0°$

$\Rightarrow \sin30° - \cos60° = \frac{1}{2} - \frac{1}{2} = 0$

101. (a) $\frac{4}{3}\tan^2 45° + 3\cos^2 30° - 2\sec^2 30° - \frac{3}{4}\cot^2 60°$

$= \frac{4}{3}\times1 + 3\times\frac{3}{4} - 2\times\frac{4}{3} - \frac{3}{4}\times\frac{1}{3}$

$= \frac{4}{3} + \frac{9}{4} - \frac{8}{3} - \frac{1}{4}$

$= \frac{16+27-35}{12} = \frac{8}{12} = \frac{2}{3}$

102. (c) $\frac{2}{3}\tan^2 60° + 3\cos^2 30° - 2\sec^2 30° - \frac{3}{4}\cot^2 60°$

$= \frac{2}{3}\times\left(\sqrt{3}\right)^2 + 3\times\left(\frac{\sqrt{3}}{2}\right)^2 - 2\times\left(\frac{2}{\sqrt{3}}\right)^2 - \frac{3}{4}\times\left(\frac{1}{\sqrt{3}}\right)^2$

$= \frac{2}{3}\times3 + 3\times\frac{3}{4} - 2\times\frac{4}{3} - \frac{3}{4}\times\frac{1}{3}$

$= 2 + \frac{9}{4} - \frac{8}{3} - \frac{1}{4}$

$= 2 - \frac{8}{3} + 2 = 4 - \frac{8}{3} = \frac{4}{3} = 1.33$

103. (b) $\sec\theta = \sqrt{2} \Rightarrow \theta = 45°$

$\therefore \text{cosec}\,45° = \sqrt{2}$

104. (a) $16\cos^3\frac{\pi}{6} - 12\cos\frac{\pi}{6}$ ($\pi = 180°$)

$\Rightarrow 16\times\left(\frac{\sqrt{3}}{2}\right)^3 - 12\times\frac{\sqrt{3}}{2}$

$\Rightarrow 6\sqrt{3} - 6\sqrt{3} = 0$

105. (a) $3\tan^2 45° + 2\cos60° - 2\sin30°$

$= 3 \times 1^2 + 2\times\frac{1}{2} - 2\times\frac{1}{2}$

$= 3 + 1 - 1 = 3$

106. (b) $8\sqrt{3}\sin30°\tan60° - 3\cos0° + 3\sin^2 45° + 2\cos^2 30°$

$= 8\sqrt{3} \times\frac{1}{2}\times\sqrt{3} - 3\times1 + 3\times\left(\frac{1}{\sqrt{2}}\right)^2 + 2\times\left(\frac{\sqrt{3}}{2}\right)^2$

$= 12 - 3 + \frac{3}{2} + \frac{3}{2}$

$= 12 - 3 + 3 = 12$

107. (b) $2k\sin30°\cos30°\cot60°$

$= \frac{\cot^2 30°\sec60°\tan45°}{\text{cosec}^2 45°\text{cosec}30°}$

$\Rightarrow 2k\times\left(\frac{1}{2}\right)\times\left(\frac{\sqrt{3}}{2}\right)\times\frac{1}{\sqrt{3}} = \frac{(\sqrt{3})^2\times2\times1}{(\sqrt{2})^2\times2}$

$\Rightarrow 2k\times\left(\frac{1}{4}\right) = \frac{3\times2}{2\times2}$

$\Rightarrow k = \frac{(6\times2)}{4} = \frac{12}{4} = 3$

108. (a)

$$\frac{\frac{4}{3}\tan^2 60 + 3\cos^2 30° - 2\sec^2 30 - \frac{3}{4}\cot^2 60}{\sin60°.\cos30° - \cos60°.\sin30}$$

$$= \frac{\frac{4}{3}(\sqrt{3})^2 + 3\times\left(\frac{\sqrt{3}}{2}\right)^2 - 2\times\left(\frac{2}{\sqrt{3}}\right)^2 - \frac{3}{4}\times\left(\frac{1}{\sqrt{3}}\right)^2}{\sin(60°-30°)}$$

$$= \frac{4 + 3\times\frac{3}{4} - 2\times\frac{4}{3} - \frac{3}{4}\times\frac{1}{3}}{\sin30°}$$

$$= \frac{\left(4 + \frac{9}{4} - \frac{8}{3} - \frac{1}{4}\right)}{\frac{1}{2}}$$

$$= \left(\frac{48+27-32-3}{12}\right)\times2 = 6\frac{2}{3}$$

109. (c) $p\left(\sqrt{3} + \cot30°\right)$

$= \tan^3 60° - 2\sin60°$

$\Rightarrow p\left(\sqrt{3}+\sqrt{3}\right) = 3\sqrt{3} - 2\times\frac{\sqrt{3}}{2}$

$\Rightarrow 2\sqrt{3}\,p = \sqrt{3}\,(3-1)$

$\Rightarrow p = 1$

110. (c) $\theta_1 + \theta_2 = \frac{\pi}{2}$

$\because \sin\theta_1 = \sin30°$

$\Rightarrow \theta_1 = 30° = \frac{\pi}{6} \Rightarrow \theta_2 = 60° = \frac{\pi}{3}$

111. (a) $k(\tan45°\sin60°)$

$= \cos60°\cot30°$

$\Rightarrow k\left(1\times\frac{\sqrt{3}}{2}\right) = \frac{1}{2}\times\sqrt{3} \Rightarrow k = 1$

112. (d) A.T.Q,

$\cos0° + \cos30° - \tan45° + \text{cosec}60° + \cot90°$

$= 1 + \frac{\sqrt{3}}{2} - 1 + \frac{2}{\sqrt{3}} + 0 = \frac{2}{\sqrt{3}} + \frac{\sqrt{3}}{2} = \frac{7}{2\sqrt{3}}$

113. (b) $\sin60°\sec30° + \tan45°.\cos0° + \sec60°\sin30°$

$= \frac{\sqrt{3}}{2}\times\frac{2}{\sqrt{3}} + 1\times1 + 2\times\frac{1}{2}$

$= 1 + 1 + 1 = 3$

114. (a) If $\theta = 45°$

Then, $\sin45° = \cos45° = \frac{1}{\sqrt{2}}$

115. (b) $\frac{\sin^2\theta - 2\sin^4\theta}{2\cos^4\theta + \cos^2\theta}$

Put, $\theta = 45°$

$\frac{\sin^2 45° - 2\sin^4 45°}{2\cos^4 45° + \cos^2 45°}$

$= \frac{\left(\frac{1}{\sqrt{2}}\right)^2 - 2\left(\frac{1}{\sqrt{2}}\right)^4}{2\times\left(\frac{1}{\sqrt{2}}\right)^4 + \left(\frac{1}{\sqrt{2}}\right)^2} = \frac{\frac{1}{2} - \frac{1}{2}}{\frac{1}{2} + \frac{1}{2}} = 0$

116. (c) $\frac{11\sqrt{2}\cos3A + 10\sin2A}{7\sqrt{3}\sin4A}$

$A = 15°$

$\therefore \frac{11\sqrt{2}\cos45° + 10°\sin30°}{7\sqrt{3}\sin60°}$

$= \frac{11\sqrt{2}\times\frac{1}{\sqrt{2}} + 10\times\frac{1}{2}}{7\sqrt{3}\times\frac{\sqrt{3}}{2}}$

$= \frac{11+5}{\frac{21}{2}} = \frac{32}{21}$

117. (a) $\tan\frac{\pi}{6} + \sec\frac{\pi}{6} = x$

$\Rightarrow x = \tan30° + \sec30°$

$\Rightarrow x = \frac{1}{\sqrt{3}} + \frac{2}{\sqrt{3}} = \frac{3}{\sqrt{3}} = \sqrt{3}$

118. (b) $\frac{k - k\cot^2 30°}{1 + \cot^2 30°} = \sin^2 60° + 4\tan^2 45° - \text{cosec}^2 60°$

$\Rightarrow \frac{k(1 - \cot^2 30°)}{1 + \cot^2 30°} = \sin^2 60° + 4\tan^2 45° - \text{cosec}^2 60°$

$\Rightarrow \frac{k(-2)}{4} = \frac{3}{4} + 4 - \frac{4}{3}$

$\Rightarrow k = \left[\frac{9 + 48 - 16}{12}\right]\times(-2)$

$\Rightarrow k = -6.83$

119. (b) $\frac{x - x\,\text{cosec}^2 30°}{1 + \text{cosec}^2 30°}$

$= \cos^2 60° + 4\cot^2 45° - \sec^2 60°$

$\Rightarrow \frac{x - 4x}{1 + 4} = \frac{1}{4} + 4 - 4$

$\Rightarrow \frac{x - 4x}{1 + 4} = \frac{1}{4} \Rightarrow x - 4x = \frac{5}{4}$

$\Rightarrow -3x = \frac{5}{4} \Rightarrow x = \frac{-5}{12}$

120. (c) $\cot\theta = \frac{1}{\sqrt{3}} \Rightarrow \theta = 60°$

$\therefore \frac{2 - \sin^2\theta}{1 - \cos^2\theta} + (\text{cosec}^2\theta - \sec\theta)$

Put $\theta = 60°$

$= \frac{2 - \frac{3}{4}}{\frac{3}{4}} + \left(\frac{4}{3} - 2\right)$

$= \frac{8-3}{\frac{4}{3}} + \frac{(4-6)}{3} = \frac{5}{3} - \frac{2}{3} = 1$

121. (b) $3\tan\theta = 2\sqrt{3}\sin\theta$

$\Rightarrow \frac{3\sin\theta}{\cos\theta} = 2\sqrt{3}\sin\theta \Rightarrow \cos\theta = \frac{\sqrt{3}}{2}$

$\Rightarrow \theta = 30°$

$\therefore \frac{\text{cosec}^2 2\theta + \cot^2 2\theta}{\sin^2\theta + \tan^2 2\theta}$

$= \frac{\left(\frac{2}{\sqrt{3}}\right)^2 + \left(\frac{1}{\sqrt{3}}\right)^2}{\left(\frac{1}{2}\right)^2 + \left(\sqrt{3}\right)^2} = \frac{\frac{5}{3}}{\frac{13}{4}} = \frac{20}{39}$

122. (b) $4\sin^2(2x - 10)° = 3$

$\Rightarrow \sin(2x - 10)° = \frac{\sqrt{3}}{2}$

$\Rightarrow \sin(2x - 10)° = \sin60°$

$\Rightarrow 2x - 10° = 60°$

$\Rightarrow x = 35°$

$\therefore \frac{\sin^4 30° + \cos^4 30°}{1 - 2\sin^2 90°\cos^2 90°}$

$= \frac{1}{16} + \frac{9}{16} = \frac{5}{8}$

123. (b) $3(\cot^2\phi - \cos^2\phi) = \cos^2\phi$

$\Rightarrow 3\cot^2\phi - 3\cos^2\phi = \cos^2\phi$

$\Rightarrow 3\cot^2\phi = 4\cos^2\phi$

$\Rightarrow \frac{3\cos^2\phi}{\sin^2\phi} = 4\cos^2\phi$

$\Rightarrow \frac{3}{4} = \sin^2\phi$

$\Rightarrow \sin\phi = \frac{\sqrt{3}}{2} = \sin60°$

$\Rightarrow \phi = 60°$

Now, $(\tan^2\phi + \text{cosec}^2\phi + \sin^2\phi)$

$= \tan^2 60° + \text{cosec}^2 60° + \sin^2 60°$

$= \left(\sqrt{3}\right)^2 + \left(\frac{2}{\sqrt{3}}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2$

$= 3 + \frac{4}{3} + \frac{3}{4} = \frac{61}{12}$

124. (a) $\frac{\cos^2\theta}{\cot^2\theta - \cos^2\theta} = 3$

$\Rightarrow \cos^2\theta = 3\cot^2\theta - 3\cos^2\theta$

$\Rightarrow 4\cos^2\theta = \frac{3\cos^2\theta}{\sin^2\theta}$

$\Rightarrow \sin^2\theta = \frac{3}{4} \Rightarrow \sin\theta$

$= \frac{\sqrt{3}}{2} = \sin60°$

$\Rightarrow \theta = 60°$

Now, $\cot\theta + \text{cosec}\theta = \cot60° + \text{cosec}60°$

$= \frac{1}{\sqrt{3}} + \frac{2}{\sqrt{3}} = \frac{3}{\sqrt{3}} = \sqrt{3}$

125. (b) $\frac{\tan x}{\sec x} = \frac{1}{2} \Rightarrow \frac{\sin x}{\cos x \times \sec x} = \frac{1}{2}$

$\Rightarrow \sin x = \frac{1}{2} = \sin30° \Rightarrow x = 30°$

$\therefore (\sin x + \cos x)^2 = \left(\frac{1}{2} + \frac{\sqrt{3}}{2}\right)^2$

$= \left(\frac{\sqrt{3}+1}{2}\right)^2 = \frac{4 + 2\sqrt{3}}{4} = \frac{2 + \sqrt{3}}{2}$

126. (b) $\sin\theta + \cos\theta = \sqrt{3}\cos\theta$

divided by $\cos\theta$

$\Rightarrow \frac{\sin\theta}{\cos\theta} + 1 = \sqrt{3}$

$\Rightarrow \tan\theta = \sqrt{3} - 1$

$\therefore \cot\theta = \frac{1}{\sqrt{3} - 1}$

$= \frac{1}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1} = \frac{\sqrt{3} + 1}{2}$

127. (a) $\cot^2\theta - 2\cos^2\theta = 0$

$\Rightarrow \cot^2\theta = 2\cos^2\theta$

$\Rightarrow \frac{\cos^2\theta}{\sin^2\theta} = 2\cos^2\theta$

$\Rightarrow \frac{1}{\sin^2\theta} = 2 \Rightarrow \sin^2\theta = \frac{1}{2}$

$\Rightarrow \sin\theta = \frac{1}{\sqrt{2}} \Rightarrow \theta = 45°$

Alternate Method:-

$\cot^2\theta - 2\cos^2\theta = 0$

Put $\theta = 45°$

$\Rightarrow \cot^2 45 - 2\cos^2 45° = 0$

$\Rightarrow 1 - 2 \times \left(\frac{1}{\sqrt{2}}\right)^2 = 0$

$\Rightarrow 1 - 2 \times \frac{1}{2} = 0 \Rightarrow 1 - 1 = 0$

$\therefore \theta = 45°$

128. (d) $\sin^4\theta + \cos^4\theta = 2\sin^2\theta\cos^2\theta$

$\Rightarrow \sin^4\theta + \cos^4\theta - 2\sin^2\theta\cos^2\theta = 0$

$\Rightarrow \left(\sin^2\theta - \cos^2\theta\right)^2 = 0$

$\Rightarrow \sin^2\theta - \cos^2\theta = 0$

$\Rightarrow \sin^2\theta = \cos^2\theta$

$\therefore [\sin\alpha = \cos\beta \text{ when } \alpha + \beta = 90°]$

$\Rightarrow \theta = 45°$

$\therefore \cot45° = 1$

Alternate Method:

Put $\theta = 45°$

$\therefore \sin^4 45° + \cos^4 45° = 2\sin^2 45°.\cos^2 45°$

$\Rightarrow \left(\frac{1}{\sqrt{2}}\right)^4 + \left(\frac{1}{\sqrt{2}}\right)^4 = 2\times\left(\frac{1}{\sqrt{2}}\right)^2 \cdot \left(\frac{1}{\sqrt{2}}\right)^2$

$\Rightarrow \frac{1}{4} + \frac{1}{4} = 2\times\frac{1}{2}\times\frac{1}{2} \Rightarrow \frac{2}{4} = \frac{2}{4}$

LHS = RHS

So, $\cot45° = 1$

SMART APPROACH:-

We know, $\cot\theta$ possible value only and only could be ∞, $\sqrt{3}, 1, \frac{1}{\sqrt{3}}$ and 0

In given option, only option 'd' has 1 value which shows $\cot45° = 1$

129. (d) Given, $\sin\theta - \cos\theta = 0$
$\Rightarrow \sin\theta = \cos\theta$
$\Rightarrow \theta = 45°$
$\therefore \sin(\pi - \theta) + \sin\left(\frac{\pi}{2} - \theta\right)$
$= \sin(\pi - 45) + \sin\left(\frac{\pi}{2} - 45\right)$
$= \sin45° + \cos45° = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \sqrt{2}$

130. (c) $4\sin^2 A - 3 = 0$
$\Rightarrow 4\sin^2 A = 3 \Rightarrow \sin^2 A = \frac{3}{4}$
$\Rightarrow \sin A = \frac{\sqrt{3}}{2} \Rightarrow A = 60°$
$\because 3\sin A - 4\sin^3 A$
$= \sin3A = \sin3 \times 60° = \sin180° = 0$

131. (a) $4\sin^2\theta = 3(1 + \cos\theta)$
$\Rightarrow 4(1 - \cos^2\theta) = 3(1 + \cos\theta)$
$\Rightarrow 4\cos\theta = 1$
$\Rightarrow \cos\theta = \frac{1 \to B}{4 \to H}$
$P = \sqrt{4^2 - 1^2} = \sqrt{15}$
$\therefore 2\tan\theta + 4\sin\theta - \sec\theta$
$= \frac{2\sqrt{15}}{1} + 4\frac{\sqrt{15}}{4} - 4 = 3\sqrt{15} - 4$

132. (b) $3\tan\theta = 2\sqrt{3}\sin\theta$
$\Rightarrow \frac{3\sin\theta}{\cos\theta} = 2\sqrt{3}\sin\theta$
$\Rightarrow \cos\theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = 30°$
$\therefore 2\sin^2 2\theta - 3\cos^2 3\theta$
$= 2\sin^2 60° - 3\cos^2 90°$
$= 2 \times \left(\frac{3}{4}\right) - 3 \times 0 = \frac{3}{2}$

133. (b) $\sin\frac{(2A+B)}{2} = \frac{\sqrt{3}}{2}$
$\Rightarrow \sin\frac{(2A+B)}{2} = \sin60°$
$\Rightarrow 2A + B = 120°$ _____(i)
Now,
$\cos\frac{(2A-B)}{2} = \frac{\sqrt{3}}{2}$
$\Rightarrow \cos\frac{(2A-B)}{2} = \cos30°$
$\Rightarrow 2A - B = 60°$ _____(ii)
On solving eq. (i) and (ii)
$A = 45°, B = 30°$
$\therefore \sin[3(A - B)] = \sin[3(45° - 30°)]$
$= \sin 45° = \frac{1}{\sqrt{2}}$

134. (c) $\sin(x + y) = 1 \Rightarrow x + y = 90°$
$\cos(x - y) = \frac{\sqrt{3}}{2} \Rightarrow x - y = 30°$
$\therefore y = \frac{90° - 30°}{2} = 30°$

135. (c) Given, $\sin(p + q) = 1$
$\Rightarrow \sin(p + q) = \sin90°$
$\Rightarrow p + q = 90°$ .....(i)
$\cos(p - q) = \frac{\sqrt{3}}{2} = 30°$
$\Rightarrow p - q = 30°$ .....(ii)
Solve Eq^n (i) and (ii)
$p = \frac{90° + 30°}{2} = 60°$

136. (b) Given, $\sin(A - B) = \frac{1}{2} = \sin30°$
$\Rightarrow A - B = 30°$ .....(i)
$\sin(A + B) = 1 = 90°$
$\Rightarrow A + B = 90°$ .....(ii)
From Eq^n (i) and (ii)
$A = \frac{90 + 30}{2} = 60°,$
$B = \frac{90 - 30}{2} = 30°$

137. (d) $\tan(A + B - C) = \frac{1}{\sqrt{3}} = \tan30°$
$\cos(B + C - A) = \frac{1}{2} = \cos60°$
$\sin(C + A - B) = \frac{1}{\sqrt{2}} = \sin45°$
Now,
$\Rightarrow A + B - C = 30°$ ....(i)
$\Rightarrow B + C - A = 60°$ ....(ii)
$\Rightarrow C + A - B = 45°$
.....(iii)
Add Eq^n (i) + (ii) + (iii)
$\Rightarrow A + B - C + B + C - A + C + A - B$
$= 30° + 60° + 45°$
$\Rightarrow A + B + C = 135°$

138. (a) $\sin(x - y) = \frac{1}{2}$ and $\cos(x + y) = \frac{1}{2}$
$x - y = 30°$
$x + y = 60$
$2x = 90°$
$x = 45°$
$\sin x\cos x + 2\sin^2 x + \cos^3 x\sec x$
$= \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} + 2 \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{\sqrt{2}} \times \sqrt{2}$
$= \frac{1}{2} + \frac{1}{1} + \frac{1}{2} = \frac{4}{2} = 2$

139. (b) $\tan(x + y) = 1$ and
$\cos(x - y) = \frac{\sqrt{3}}{2}$ (given)
$x + y = 45°$
$x - y = 30°$
$2x = 75° \Rightarrow x = 37.5 \Rightarrow y = 7.5$

140. (a) $\tan(A + B) = \sqrt{3}$
$\Rightarrow \tan(A + B) = \tan60°$
$\Rightarrow A + B = 60°$ .....(i)
$\tan(A - B) = \frac{1}{\sqrt{3}}$
$\Rightarrow \tan(A - B) = \tan30°$
$\Rightarrow A - B = 30°$ .....(ii)
Adding (i) and (ii) we get
$A = 45°, B = 15°$

141. (a) $\because \cos2\theta = 0$
$\Rightarrow 2\theta = 90° \Rightarrow \theta = 45°$
$\therefore \sin(75° - 45°) = \sin30° = \frac{1}{2}$

142. (b) $P\sin60° = Q\operatorname{cosec}45°$
$\Rightarrow \frac{P}{Q} = \frac{\operatorname{cosec}45°}{\sin60°} = \frac{\sqrt{2}}{\frac{\sqrt{3}}{2}}$
$\Rightarrow \frac{P}{Q} = \frac{2\sqrt{2}}{\sqrt{3}} \Rightarrow \frac{P^2}{Q^2} = \frac{8}{3}$
Use C and D rule
$\therefore \frac{P^2 + Q^2}{P^2 - Q^2} = \frac{8 + 3}{8 - 3} = \frac{11}{5}$

143. (d) Given,
$\alpha\sin45° = \beta.\operatorname{cosec}30°$
$\Rightarrow \frac{\alpha}{\beta} = 2\sqrt{2} \Rightarrow \left(\frac{\alpha}{\beta}\right)^4 = (2\sqrt{2})^4$
$\therefore \frac{\alpha^4}{\beta^4} = 4^3$

144. (a) $\sin x = \frac{1}{2} \Rightarrow x = 30°$
$\therefore \sec^4 x + \operatorname{cosec}^4 x - \tan^4 x - \cot^4 x$
$= \left(\frac{2}{\sqrt{3}}\right)^4 + (2)^4 - \left(\frac{1}{\sqrt{3}}\right)^4 - \left(\sqrt{3}\right)^4$
$= \frac{16}{9} + 16 - \frac{1}{9} - 9 = \frac{15}{9} + 7$
$= \frac{15 + 63}{9} = \frac{78}{9} = \frac{26}{3}$

145. (b) $\cos\theta - \sin\theta = \sqrt{3}\cos(90° - \theta)$
$\Rightarrow \cos\theta - \sin\theta = \sqrt{3}\sin\theta$
$\Rightarrow \cot\theta - 1 = \sqrt{3}$
$\Rightarrow \cot\theta = \sqrt{3} + 1$
$\therefore \tan\theta - \cot\theta = \frac{1 - \operatorname{Cot}^2\theta}{\operatorname{Cot}\theta}$
$= \frac{1 - (\sqrt{3} + 1)^2}{\sqrt{3} + 1}$
$= \frac{1 - 3 - 1 - 2\sqrt{3}}{\sqrt{3} + 1} = -\frac{(3 + 2\sqrt{3})}{\sqrt{3} + 1}$

146. (a) $\mu = 60°$
$\therefore \sin\mu + \cos(90° - \mu)$
$= \sin60° + \cos(90° - 60°)$
$= \sin60° + \sin60° = \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} = \sqrt{3}$

147. (a) $\tan(90 - \theta) = \frac{2}{\sqrt{3}}$
$\Rightarrow \cot\theta = \frac{2}{\sqrt{3}} \Rightarrow \tan\theta = \frac{\sqrt{3}}{2}$
$\therefore 2\sqrt{3}\tan\theta + 1 = 2\sqrt{3} \times \frac{\sqrt{3}}{2} + 1 = 4$

148. (a) $\sin^2\frac{2\pi}{3} + \cos^2\frac{5\pi}{6} - \tan^2\frac{3\pi}{4}$
$= \sin^2 120° + \cos^2 150° - \tan^2 135°$
$= \cos^2 30° + \sin^2 60° - \cot^2 45°$
$= \frac{3}{4} + \frac{3}{4} - 1 = \frac{1}{2}$

149. (c) $\cos^2(90°-\theta)-\left[\frac{\cos(90°-\theta)\cos\theta}{\cot\theta}\right]$

$= \sin^2\theta - \frac{\sin\theta\cos\theta}{\cot\theta}$

$= \sin^2\theta - \sin^2\theta = 0$

150. (b) $\sin74° + \tan74°$

$= \sin(90° - 16°) + \tan(90° - 16°)$

$= \cos16° + \cot16°$

151. (a) $\frac{4\cos(90-\theta)\sin^3(90+\theta) - 4\sin(90+\theta)\cos^3(90-\theta)}{\cos\left(\frac{180+8\theta}{2}\right)}$

$= \frac{4\sin\theta\cos^3\theta - 4\cos\theta\sin^3\theta}{-\sin4\theta}$

$= \frac{4\sin\theta\cos\theta[\cos^2\theta - \sin^2\theta]}{-\sin4\theta}$

$= \frac{2\sin2\theta\cos2\theta}{-\sin4\theta} = \frac{\sin4\theta}{-\sin4\theta} = -1$

152. (a) $\frac{5\cos^2120° + 4\sec^230° - \tan^2135°}{\sin^230° + \cos^230°}$

$= \frac{5\cos^2(90°+30°) + 4\sec^230° - \tan^2(90°+45°)}{\sin^230° + \cos^230°}$

$= \frac{5\sin^230° + 4\sec^230° - \cot^245°}{\sin^230° + \cos^230°}$

[$\because \sin^2\theta + \cos^2\theta = 1$]

$= \frac{5\times\frac{1}{4} + 4\times\frac{4}{3} - (+1)}{1} \Rightarrow \frac{5}{4} + \frac{16}{3} - \frac{1}{1}$

$= \frac{15 + 64 - 12}{12} = \frac{67}{12}$

$\cos(90 + \theta) = \sin\theta$ and in 2nd Quadrant cos(–ve), $\tan(90 + \theta) = \cot\theta$ in 2nd quadrant tan (–ve) and $\sin^2\theta + \cos^2\theta = 1$

153. (b) $\sin150° = \sin(90° + 60°)$

$= \cos60° = \frac{1}{2} = 0.5$

154. (c) $\tan570° = \tan(360° + 210°)$

$= \tan210° = \tan(180° + 30°)$

$= \tan30° = \frac{1}{\sqrt{3}}$

155. (a) $\cos x = \frac{-1}{2}$

$\Rightarrow \cos x = -(\cos60°)$

$\Rightarrow \cos x = \cos60°$

x lies in IIIrd quad, $x = 60°$

in third quadrant, $\tan\theta$ is positive

So, $\tan x = \tan60°$

$\Rightarrow x = \sqrt{3}$

156. (d) $\cos\left(\frac{-7\pi}{2}\right) = \cos\left(4\pi - \frac{\pi}{2}\right)$

$= \cos\frac{\pi}{2} = 0$

157. (c) $\operatorname{cosec}\ 2910° + \sec\ 4260° + \tan\ 2565° + \cot\ 1755°$

$= \operatorname{cosec}(360°\times 8 + 30°) + \sec\ (360°\times 12 - 60°] + \tan\ (360°\times 7 + 45°) + \cot(360°\times 5 - 45°)$

$= \operatorname{cosec}30° + \sec60° + \tan45° - \cot45°$

$= 2 + 2 + 1 - 1 = 4$

158. (d) $\frac{1}{\cos x}\sqrt{\frac{\cos(\pi+x)\cos(-x)}{\sin(\pi-x)\cos\left(\frac{\pi}{2}+x\right)}}$

$= \frac{1}{\cos x}\sqrt{\frac{(-\cos x)\cos x}{\sin x(-\sin x)}}$

$= \frac{1}{\cos x}\sqrt{\frac{\cos^2 x}{\sin^2 x}} = \operatorname{cosec} x$

159. (d) $\frac{\cos(\pi - A)\cot\left(\frac{\pi}{2}+A\right)\cos(-A)}{\tan(\pi+A)\tan\left(\frac{3\pi}{2}+A\right)\sin(2\pi - A)}$

$= \frac{(-\cos A)(-\tan A)\cos A}{\tan A(-\cot A)(-\sin A)}$

$= \frac{\sin A\cos A}{\sin A} = \cos A$

160. (d) $\operatorname{cosec}56°.\cos34° - \cos59°.\operatorname{cosec}31°$

$= 1 - 1 = 0$

Note: $\operatorname{cosec}\alpha \times \cos\beta = 1$

When, $\alpha + \beta = 90°$

161. (c) $\cos47°.\sec133° + \sin44°.\operatorname{cosec}136°$

[We know, $\sec(180° - \theta) = -\sec\theta$

$\therefore \sec133° = \sec(180° - 74) = -\sec47°$]

$\cos47°(-\sec47°) + \sin44°.(\operatorname{cosec}44°) = -1 + 1 = 0$

162. (d) $2(\sin1° \times \sec 89°) + 3(\cos 11° \times \operatorname{cosec} 79°) + 5(\tan 21° \times \tan 69°)$

$= 2 + 3 + 5 = 10$

[If A + B = 90° Then, sinA secB = 1, cosA cosecB = 1, tanA cotB = 1]

163. (a) $\cot13°\cot27°\cot60°\cot63°\cot77°$

[If A + B = 90° Then, cotA cotB = 1]

$= \cot13.\cot77.\cot27.\cot63.\cot60$

$= \cot 60° = \frac{1}{\sqrt{3}}$

164. (a) $7(\cot11°\cot13°\cot19°\cot29°\cot61°\cot71°\cot77°\cot79°)$

[If (A + B) = 90° Then, cotA.cotB = 1]

$\Rightarrow 7(\cot11.\cot79.\cot13.\cot77.\cot19.\cot71.\cot29.\cot61°)$

$= 7 \times 1 = 7$

165. (b) $\frac{\tan55°}{\cot35°} = 1$

Note: $\tan\alpha = \cot\beta$ When, $\alpha + \beta = 90°$

166. (d) $\frac{3\sin58°}{5\cos32°} + \frac{2\cos62°}{5\sin28°} = \frac{3}{5} + \frac{2}{5} = 1$

Note: $\sin\alpha = \cos\beta$

When, $\alpha + \beta = 90°$

167. (a) $\frac{\sec35°}{\operatorname{cosec}55°} - \frac{\cos25°}{\sin65°}$

$= 1 - 1 = 0$

Note: $\sec\alpha = \operatorname{cosec}\beta$

When, $\alpha + \beta = 90°$

168. (b) $2\frac{\tan54°}{\cot36°} - \frac{\cot41°}{\tan49°}$

$= 2 \times 1 - 1 = 1$

Note: $\tan\alpha = \cot\beta$

When, $\alpha + \beta = 90°$

169. (d) $\frac{\cos25° - \sin65°}{\cos25° + \sin65°}$

{$\because \cos(90° - 65°) = \sin65°$}

$= \frac{\sin65° - \sin65°}{\cos25° + \sin65°} = 0$

170. (c) $7\left(\frac{\operatorname{cosec}24°}{\sec66°}\right)^3 + 8\left(\frac{\cot37°}{\tan53°}\right)^4 - \left(\frac{2\sec14°}{\operatorname{cosec}76°}\right)^2 + \left(\frac{-3\tan82°}{\cot8°}\right)^3$

[$\operatorname{cosec}\alpha = \sec\beta$, $\tan\alpha = \cot\beta$, If $\alpha + \beta = 90°$]

$= 7(1)^3 + 8(1)^4 - (2)^2 + (-3)^3$

$= 7 + 8 - 4 - 27$

$= -16$

171. (a) If (A + B) = 90°

Then,

$\frac{\sin A}{\cos B} = 1$ and $\frac{\cos A}{\sin B} = 1$

$\frac{\cos65°}{\sin25°} + \frac{5\sin19°}{\cos71°} - \frac{3\cos28°}{\sin62°}$

$= 1 + 5 - 3 = 3$

172. (d) $\frac{\cos37°}{\sin53°} - \cos47°\operatorname{cosec}43°$

$= 1 - 1 = 0$

[If A + B = 90° Then, $\frac{\sin A}{\cos B} = 1$ and cosA cosecB = 1]

173. (d) $\frac{\cos65°}{\sin25°} + \frac{\cos55°}{\sin35°} - \tan50°\cot50°$

$= 1 + 1 - 1 = 1$

[If A + B = 90° Then, $\frac{\sin A}{\cos B} = 1$]

174. (c) $\frac{\cos50°}{\sin40°} + \frac{3\operatorname{cosec}80°}{\sec10°} - 2\cos50°\operatorname{cosec}40°$

$= \frac{\cos(90° - 40°)}{\sin40°} + \frac{3\operatorname{cosec}(90° - 10°)}{\sec10°} - 2\cos50°\operatorname{cosec}(90° - 50°)$

$= \frac{\sin40°}{\sin40°} + \frac{3\sec10°}{\sec10°} - 2\cos50°\sec50°$

$= 1 + 3 - 2 = 4 - 2 = 2$

Alternate Method:

If A + B = 90°

Then, $\frac{\cos A}{\sin B} = 1$

$\frac{\cos50°}{\sin40°} + \frac{3\operatorname{cosec}80°}{\sec10°} - 2\cos50°\operatorname{cosec}40°$

$= 1 + 3 - 2 = 2$

175. (a) $\frac{2\cos15°\sin15°}{\cos^215° - \sin^215°}$

$= \frac{\sin30°}{\cos30°} = \tan30° = \frac{1}{\sqrt{3}}$

$\left[\begin{array}{l}2\sin\theta\cos\theta = \sin2\theta \\ \cos^2\theta - \sin^2\theta = \cos2\theta\end{array}\right]$

176. (c) Given,

$\tan(x + y)\tan(x - y) = 1$

If $\tan A \times \tan B = 1$

then, A + B = 90°

Now,

$x + y + x - y = 90°$

then, $2x = 90°$, $x = 45°$

Hence, $\tan45° = 1$

Alternate Method:

$\tan(x + y)\tan(x - y) = 1$

Put y = 0

$\tan^2x = 1 \Rightarrow \tan x = 1$

$x = 45° \Rightarrow \tan45° = 1$

177. (c) $\cos(2\theta + 54°) = \sin\theta$
$\Rightarrow 2\theta + 54° + \theta = 90°$
$\Rightarrow 3\theta = 36° \Rightarrow \theta = 12°$
$\therefore \dfrac{1}{\cot 5\theta + \sec\dfrac{5\theta}{2}} = \dfrac{1}{\cot 60° + \sec 30°}$
$= \dfrac{1}{\dfrac{1}{\sqrt{3}} + \dfrac{2}{\sqrt{3}}} = \dfrac{1}{\sqrt{3}} = \dfrac{\sqrt{3}}{3}$

178. (b) $\sin 3A = \cos(A + 10°)$
$\Rightarrow 3A + A + 10 = 90° \Rightarrow 4A = 80°$
$\Rightarrow A = 20°$
$\therefore 2\operatorname{cosec}\dfrac{3A}{2} + 6\tan^2 3A - \dfrac{3}{2}\tan^2 3A$
$= 2\operatorname{cosec} 30° + 6\tan^2 60° - \dfrac{3}{2}\tan^2 60°$
$= 2 \times 2 + 6 \times 3 - \dfrac{3}{2} \times 3$
$= 22 - \dfrac{9}{2} = \dfrac{44 - 9}{2} = \dfrac{35}{2}$

179. (a) $\sin 3A = \cos(A - 26°)$
If $A + B = 90°$, then, $\sin A = \cos B$
So, $3A + A - 26° = 90°$
$\Rightarrow 4A = 116° \quad \therefore A = 29°$

180. (d) $\sec 6A = \operatorname{cosec}(A - 29°)$
If $\sec A = \operatorname{cosec} B$
Then, $A + B = 90°$
$\Rightarrow 6A + A - 29° = 90°$
$\Rightarrow 7A = 90° + 29° = 119°$
$\Rightarrow A = \dfrac{119°}{7} = 17°$

181. (b) If $\sin\alpha = \cos\beta$ then $\alpha + \beta = 90°$
$\sin(x + y) = \cos\{3(x + y)\}$
$\Rightarrow x + y + 3x + 3y = 90°$
$\Rightarrow 4x + 4y = 90°$
$\Rightarrow 2(x + y) = 45°$
$\therefore \cot\{2(x + y)\} = \cot 45° = 1$

182. (a) $\sin(5x - 40°) = \cos(5y + 40°)$
$\Rightarrow \sin(5x - 40°) = \sin(90° - 5y - 40°)$
$\Rightarrow 5x - 40° = 50° - 5y$
$\Rightarrow x + y = \dfrac{90°}{5} = 18°$

183. (d) Given, $\cos(3\alpha) = \sin(\alpha - 22°)$
(If $\sin\theta = \cos\phi$, then $\theta + \phi = 90°$)
$\Rightarrow 3\alpha + \alpha - 22° = 90°$
$\Rightarrow 4\alpha = 112° \Rightarrow \alpha = 28°$

184. (a) $\tan(5\theta - 10°) = \cot(5\phi + 20°)$
$\Rightarrow \tan(5\theta - 10°) = \tan(90° - 5\phi - 20°)$
$\Rightarrow 5\theta - 10° = 70° - 5\phi$
$\Rightarrow 5\theta + 5\phi = 80 \Rightarrow \theta + \phi = 16°$

185. (c) If $A + B = 90°$, then $\sin\theta = \cos\theta$
$\Rightarrow \sin(5x - 25°) = \cos(5y + 25°)$
$\Rightarrow 5x - 25° + 5y + 25° = 90°$
$\Rightarrow 5x + 5y = 90°$
$\Rightarrow x + y = 18°$

186. (d) $\cos\theta = \sin 2\theta$
[$\because \sin 2\theta = 2\sin\theta \times \cos\theta$]
$\Rightarrow \cos\theta = 2\sin\theta\cos\theta \Rightarrow \dfrac{1}{2} = \sin\theta$
$\Rightarrow \sin 30° = \sin\theta \Rightarrow \theta = 30°$
$\therefore \cos^4 30° + \sin^4 30° + \cos^3 30° + \sin^3 30° + \sin^2 30° + \cos^2 30° + \sin 30° + \cos 30°$
$= \dfrac{9}{16} + \dfrac{1}{16} + \dfrac{3\sqrt{3}}{8} + \dfrac{1}{8} + \dfrac{1}{4} + \dfrac{3}{4} + \dfrac{1}{2} + \dfrac{\sqrt{3}}{2}$
$= \dfrac{36 + 14\sqrt{3}}{16} = \dfrac{18 + 7\sqrt{3}}{8}$

187. (b) $\sec(5\alpha - 15°) = \operatorname{cosec}(15° - 2\alpha)$
$\Rightarrow (5\alpha - 15°) + (15° - 2\alpha) = 90°$
$\Rightarrow 3\alpha = 90°$
$\Rightarrow \alpha = 30°$
$\therefore \cos 3\alpha + \sin 2\alpha + \cot\alpha$
$= 0 + \dfrac{\sqrt{3}}{2} + \sqrt{3} = \dfrac{3\sqrt{3}}{2}$

188. (b) $\sin^2 35° + \sin^2 55°$
$= \sin^2 35° + \sin^2(90° - 35°)$
$= \sin^2 35 + \cos^2 35° = 1$

189. (c) $\cos^2 29° + \cos^2 61° = 1$
Note: $\cos^2\alpha + \cos^2\beta = 1$
When, $\alpha + \beta = 90°$

190. (d) $\sin^2 30° - \sin^2 40° + \sin^2 45° - \sin^2 55° - \sin^2 35° + \sin^2 45° - \sin^2 50° + \sin^2 60°$
$= \sin^2 30° + 2\sin^2 45° + \sin^2 60° - (\sin^2 40° + \sin^2 50° + \sin^2 55° + \sin^2 35°)$
$= \dfrac{1}{4} + 2 \times \dfrac{1}{2} + \dfrac{3}{4} - (1 + 1)$
$= \dfrac{1}{4} + 1 + \dfrac{3}{4} - 2$
$= 2 - 2 = 0$
Note: $\sin^2\alpha + \sin^2\beta = 1$
When, $\alpha + \beta = 90°$

191. (c)
$\sin^2 15° + \sin^2 25° + \sin^2 35° + \sin^2 45° + \sin^2 55° + \sin^2 65° + \sin^2 75°$
$\therefore \left[\begin{matrix}\sin^2 A + \sin^2 B = 1 \\ \text{When } A + B = 90°\end{matrix}\right]$
$= 1 + 1 + 1 + \dfrac{1}{2}$
$= 3 + \dfrac{1}{2} = \dfrac{7}{2}$

192. (b) $\sec^2 54° - \cot^2 36° + \dfrac{3}{2}\sin^2 37° \times \sec^2 53° + \dfrac{2}{\sqrt{3}}\tan 60°$
We know, $\boxed{\begin{matrix}\sec(90° - x) = \operatorname{cosec} x \\ \operatorname{cosec}(90° - x) = \sec x\end{matrix}}$
$= \operatorname{cosec}^2 36° - \cot^2 36° + \dfrac{3}{2}\sin^2 37° \times \operatorname{cosec}^2 37° + \dfrac{2}{\sqrt{3}}\tan 60°$
$= 1 + \dfrac{3}{2} \times 1 + \dfrac{2}{\sqrt{3}} \times \sqrt{3}$
$= 1 + \dfrac{3}{2} + 2 = \dfrac{9}{2}$

193. (c)
$\dfrac{\sin^2 30° + \cos^2 60° - \sec 35° . \sin 55°}{\sec 60° + \operatorname{cosec} 30°}$
$= \dfrac{\dfrac{1}{4} + \dfrac{1}{4} - \operatorname{cosec} 55° \sin 55°}{2 + 2}$
$= \dfrac{\dfrac{1}{2} - 1}{4} = \dfrac{-1}{8}$

194. (a) $\dfrac{\sin 23°\cos 67° + \sec 52°\sin 38° + \cos 23°\sin 67° + \operatorname{cosec} 52°\cos 38°}{\operatorname{cosec}^2 20° - \tan^2 70°}$
$= \dfrac{\sin 23°\cos(90° - 23°) + \sec(90 - 38°)\sin 38° + \cos 23°\sin(90 - 23°) + \operatorname{cosec}(90 - 38°)\cos 38°}{\operatorname{cosec}^2(90 - 70°) - \tan^2 70°}$
$= \dfrac{\sin^2 23° + \operatorname{cosec} 38°\sin 38° + \cos^2 23° + \sec 38°\cos 38°}{\sec^2 70° - \tan^2 70°}$
[$\because \sin^2\theta + \cos^2\theta = 1$, $\sec^2\theta - \tan^2\theta = 1$]
$= \dfrac{1 + 1 + 1}{1} = 3$

195. (d) $\dfrac{3(\cot^2 47° - \sec^2 43°) - 2(\tan^2 23° - \operatorname{cosec}^2 67°)}{\operatorname{cosec}^2(68° + \theta) - \tan(\theta + 61°) - \tan^2(22° - \theta) + \cot(29° - \theta)}$
$= \dfrac{3[\cot^2(90° - 43°) - \sec^2 43°] - 2[\tan^2(90° - 67°) - \operatorname{cosec}^2 67°]}{\operatorname{cosec}^2(68° + \theta) - \tan^2[90° - (68° + \theta)] - \tan(\theta + 61°) + \cot[90° - (61° + \theta)]}$
$= \dfrac{3(\tan^2 43° - \sec^2 43°) - 2(\cot^2 67° - \operatorname{cosec}^2 67°)}{\operatorname{cosec}^2(68° + \theta) - \cot^2(68° + \theta) - \tan(\theta + 61°) + \tan(\theta + 61°)}$
$= \dfrac{3(-1) - 2(-1)}{1} = -1$

196. (d)
$\dfrac{2\sin^2 38°\sec^2 52° + \cos 64°\sin 26° + \sin^2 64°}{\tan^2 23° + \cot^2 23° - \sec^2 67° - \operatorname{cosec}^2 67°}$
$= \dfrac{2\sin^2 38°\sec^2(90 - 38°) + \cos 64°}{\dfrac{\sin(90 - 26°) + \sin^2 64°}{\tan^2(90 - 67°) + \cot^2(90 - 67°) - \sec^2 67° - \operatorname{cosec}^2 67°}}$
$= \dfrac{2\sin^2 38°\operatorname{cosec}^2 38° + \cos^2 64° + \sin^2 64°}{\tan^2 67° + \cot^2 67° - \sec^2 67° - \operatorname{cosec}^2 67°}$
$= \dfrac{2\sin^2 38° \times \dfrac{1}{\sin^2 38°} + \sin^2 64° + \cos^2 64°}{\tan^2 67° - \sec^2 67° + \cot^2 67° - \operatorname{cosec}^2 67°}$
[$\because \sin^2\theta + \cos^2\theta = 1$, $\sec^2\theta - \tan^2\theta = 1$
$\operatorname{cosec}^2\theta - \cot^2\theta = 1$]
$= \dfrac{2 + 1}{-1 - 1} = \dfrac{-3}{2}$

197. (a) $\sqrt{\dfrac{9\sin^2 26° + 9\sin^2 64° + 36}{32 - 7\cos^2 32° - 7\cos^2 58°}}$
$= \sqrt{\dfrac{9\sin^2 26° + 9\cos^2 26° + 36}{32 - 7\sin^2 58° - 7\cos^2 58°}}$
$= \sqrt{\dfrac{9 + 36}{32 - 7}} = \sqrt{\dfrac{45}{25}} = \sqrt{\dfrac{9}{5}}$
$= \dfrac{3}{\sqrt{5}} = \dfrac{3\sqrt{5}}{5}$

198. (a) $\left(\dfrac{\sin 23°\cos 67° + \cos 23°\sin 67°}{\operatorname{cosec}^2 15° - \tan^2 75°}\right)^3$
$= \left(\dfrac{\sin 23°\cos(90 - 23°) + \cos 23°\sin(90 - 23°)}{\operatorname{cosec}^2 15° - \tan^2(90° - 15°)}\right)^3$
$= \left(\dfrac{\sin^2 23° + \cos^2 23°}{\operatorname{cosec}^2 15° - \cot^2 15°}\right)^3$
$\therefore [\sin^2\theta + \cos^2\theta = 1, \operatorname{cosec}^2\theta - \cot^2\theta = 1]$
$= (1)^3 = 1$

199. (b) If A + B = 90° then

$$\frac{\cot A}{\cot B} + \cos^2 A + \cos^2 B$$

$$= \frac{\cot(90° - B)}{\cot B} + \cos^2(90° - B) + \cos^2 B$$

$$= \frac{\tan B}{\cot B} + \sin^2 B + \cos^2 B$$

$$= \tan^2 B + 1 = \sec^2 B = \sec^2(90° - A)$$

$$= \text{cosec}^2 A$$

Alternate Method:

Let A = 60°, B = 30°

$$\frac{\cot 60°}{\cot 30°} + \cos^2 60° + \cos^2 30°$$

$$= \frac{1}{\sqrt{3} \times \sqrt{3}} + \frac{1}{4} + \frac{3}{4} = \frac{1}{3} + 1 = \frac{4}{3}$$

Go through options,

$$\text{cosec}^2 A = \text{cosec}^2 60° = \left(\frac{2}{\sqrt{3}}\right)^2 = \frac{4}{3}$$

200. (c)

$$\frac{[\tan^2(22° - \theta) - \tan^2(\theta + 68°) - \text{cosec}^2(68° + \theta) + \cot^2(22° - \theta)]}{\left[3\left(\cot^2 52° - \sec^2 38°\right) + 2\left(\text{cosec}^2 28° - \tan^2 62°\right)\right]}$$

Put θ = 0°

$$= \tan^2 22° - \tan^2 68° - \text{cosec}^2 68° + \cot^2 22°$$

$$= \tan^2(90° - 68°) - \tan^2 68° - \text{cosec}^2 68° + \cot^2(90° - 68°)$$

$$= \cot^2 68° - \text{cosec}^2 68° = -1$$

$$\Rightarrow \frac{-1}{-3 + 2} = 1$$

Using
$\tan(90° - \theta) = \cot\theta$
$\text{cosec}(90° - \theta) = \sec\theta$
$\sec^2\theta - \tan^2\theta = 1$
$\text{cosec}^2\theta - \cot^2\theta = 1$

201. (b)

$$\frac{4\tan^2 30° + \sin^2 30° \cos^2 45° + \sec^2 48° - \cot^2 42°}{\cos 37° \sin 53° + \sin 37° \cos 53° + \tan 18° \tan 72°}$$

$$4 \times \left(\frac{1}{\sqrt{3}}\right)^2 + \left(\frac{1}{2}\right)^2 \times \left(\frac{1}{\sqrt{2}}\right)^2$$

$$= \frac{+\sec^2 48° - \cot^2(90° - 48°)}{\cos 37° \sin 53° + \sin 37° \cos 53° + \tan 18° \tan(90° - 18°)}$$

$$= \frac{\frac{4}{3} + \frac{1}{8} + \sec^2 48° - \tan^2 48°}{\sin 90° + 1}$$

$$= \frac{\frac{4}{3} + \frac{1}{8} + 1}{1 + 1} = \frac{32 + 3 + 24}{24/2}$$

$$= \frac{59}{24 \times 2} = \frac{59}{48}$$

202. (d)

$$\frac{3(\text{cosec}^2 26° - \tan^2 64°) + (\cot^2 42° - \sec^2 48°)}{\cot(22° - \theta) - \text{cosec}^2(62° + \theta) - \tan(\theta + 68°) + \tan^2(28° - \theta)}$$

$$= \frac{3[(\text{cosec}^2 26° - \tan^2(90° - 26°)] + (\cot^2 42° - \sec^2(90° - 42°))}{\cot(22° - \theta) - \tan[90° - (22° - \theta)] - \text{cosec}^2(62° + \theta) + \tan^2[90° - (62° + \theta)]}$$

$$= \frac{3[(\text{cosec}^2 26° - (\cot^2 26°)] + (\cot^2 42° - \text{cosec}^2 42°)}{\cot(22° - \theta) - \cot(22° - \theta) - \text{cosec}^2(62° + \theta) + \cot^2(62° + \theta)}$$

$$= \frac{3(1) + (-1)}{-1} = \frac{2}{-1} = -2$$

Alternate Method:-

Put θ = 0°

$$\frac{3(\text{cosec}^2 26° - \tan^2 64°) + \cot^2 42 - \sec^2 48°}{\cot 22° - \text{cosec}^2 62° - \tan 68° + \tan^2 28°}$$

$$= \frac{3(\sec^2 64° - \tan^2 64°) + (\tan^2 48° - \sec^2 48°)}{-\text{cosec}^2 62° + \cot^2 62°}$$

$$= \frac{3 - 1}{-1} = -2$$

203. (d)

$$\frac{(\cos 9° + \sin 81°)(\sec 9° + \text{cosec} 81°)}{\text{cosec}^2 71° + \cos^2 15° - \tan^2 19° + \cos^2 75°}$$

$$= \frac{[\cos 9° + \sin(90° - 9°)][\sec 9° + \text{cosec}(90° - 9°)]}{\text{cosec}^2 71° - \tan^2(90° - 71°) + \cos^2(90° - 15°) + \cos^2 75°}$$

$$= \frac{(\cos 9° + \cos 9°)(\sec 9° + \sec 9°)}{\text{cosec}^2 71° - \cot^2 71° + \cos^2 15° + \sin^2 15°}$$

$$= \frac{2\cos 9° \times 2\sec 9°}{1 + 1} = \frac{4}{2} = 2$$

204. (d) Incorrect formula in the given options is, $\sec^2\theta + \cos^2\theta = 1$

205. (c) we know that.

$\sec^2\theta - \tan^2\theta = 1$

$\Rightarrow \sec^2\theta = 1 + \tan^2\theta$

so, option (c) satisfied

206. (c) We know,

$\sec^2\theta + \cos^2\theta \neq 1$

207. (a) $\tan^2 A = 1 - \sec^2 A$ (False)

208. (b) $\sec\theta\sqrt{1 - \sin^2\theta}$

$= \sec\theta\sqrt{\cos^2\theta}$

$= \sec\theta \times \cos\theta = 1$

209. (c) $\therefore \sec^2\theta\left(\sqrt{1 - \sin^2\theta}\right)$

$= \sec^2\theta \times \cos\theta = \sec\theta$

210. (c) $\frac{1}{\tan\theta} + \tan\theta$

$= \frac{1 + \tan^2\theta}{\tan\theta}$

$= \frac{\sec^2\theta}{\tan\theta} = \frac{\cos\theta}{\cos^2\theta \times \sin\theta}$

$= \frac{1}{\cos\theta \sin\theta} = \sec\theta\ \text{cosec}\theta$

Alternate Method:

Put, θ = 45°

$\frac{1}{\tan 45°} + \tan 45° = 1 + 1 = 2$

By options,

$\sec\theta \times \text{cosec}\theta = \sec 45° \text{cosec} 45°$

$= \sqrt{2} \times \sqrt{2} = 2$

211. (d) $\frac{\cos^2 A}{1 + \sin A} = \frac{1 - \sin^2 A}{1 + \sin A}$

$= \frac{(1 - \sin A)(1 + \sin A)}{1 + \sin A} = 1 - \sin A$

212. (a) $\frac{1 + \sin A}{\cos^2 A} = \frac{1 + \sin A}{1 - \sin^2 A}$

$= \frac{(1 + \sin A)}{(1 + \sin A)(1 - \sin A)} = \frac{1}{(1 - \sin A)}$

213. (b) $\left[\frac{\cos\theta}{\sin\theta} + \frac{\sin\theta}{\cos\theta}\right]$

$= \frac{\cos^2\theta + \sin^2\theta}{\sin\theta\cos\theta} = \frac{1}{\sin\theta\cos\theta}$

$= \text{cosec}\theta\ \sec\theta$

Alternate Method:

Put θ = 45°,

$\frac{\cos 45°}{\sin 45°} + \frac{\sin 45°}{\cos 45°} = 1 + 1 = 2$

Go through options,

$\sec\theta\ \text{cosec}\theta = \sec 45° \times \text{cosec} 45°$

$= \sqrt{2} \times \sqrt{2} = 2$

214. (d) $\frac{32}{\sec^2\theta} - \frac{20}{1 + \tan^2\theta} + 12\sin^2\theta$

$= 32\cos^2\theta - 20\cos^2\theta + 12\sin^2\theta$

$= 12(\cos^2\theta + \sin^2\theta) = 12$

SMART APPROACH:-

Triplet = (3, 4, 5) → (P, B, H)

$\left(\frac{1}{\sin\theta} + \frac{1}{\tan\theta}\right)\left(\frac{1}{\sin\theta} - \frac{1}{\tan\theta}\right)$

$= \left(\frac{5}{3} + \frac{4}{3}\right)\left(\frac{5}{3} - \frac{4}{3}\right) = \frac{9}{3} \times \frac{1}{3} = 1$

215. (b) $\frac{2\sin\theta}{2\sqrt{1 - \text{Sin}^2\theta}} = \frac{\sin\theta}{\cos\theta} = \tan\theta$

216. (a) $(1 + \cot^2\theta)(1 - \cos\theta)(1 + \cos\theta)$

$= (1 + \cot^2\theta)(1 - \cos^2\theta)$

$= \text{cosec}^2\theta \times \sin^2\theta = 1$

Alternate Method:

Put θ = 45°

$(1 + \cot^2 45°)\ (1 - \cos^2 45°) = (1 + 1)\left(1 - \frac{1}{2}\right) = 2 \times \frac{1}{2} = 1$

217. (d) $(\text{cosec}\ A + \cot A)(1 - \cos A)$

$= \left(\frac{1}{\sin A} + \frac{\cos A}{\sin A}\right)(1 - \cos A)$

$= \frac{(1 + \cos A)}{\sin A} \times (1 - \cos A)$

$= \frac{1 - \cos^2 A}{\sin A} = \frac{\sin^2 A}{\sin A} = \sin A$

Alternate Method:

Put A = 30°

$(\text{cosec}30° + \cot 30°)\ (1 - \cos 30°)$

$= (2 + \sqrt{3})\left(1 - \frac{\sqrt{3}}{2}\right) = (2 + \sqrt{3})\ \frac{(2 - \sqrt{3})}{2} = \frac{1}{2}$

Go through options,

$\sin 30° = \frac{1}{2}$

218. (c) $\sec A - \cos A = \frac{1}{\cos A} - \cos A$

$= \frac{1-\cos^2 A}{\cos A} = \frac{\sin^2 A}{\cos A}$

$= \frac{\sin A}{\cos A}\sin A$

$= \tan A \sin A$

Alternate Method:

Put A = 60°,

$\sec 60° - \cos 60° = 2 - \frac{1}{2} = \frac{3}{2}$

Go through options,

$\tan A \sin A = \tan 60° \times \sin 60°$

$= \sqrt{3} \times \frac{\sqrt{3}}{2} = \frac{3}{2}$

219. (b) $1 - \frac{\tan^2\phi}{\sec^2\phi} = \frac{\sec^2\phi - \tan^2\phi}{\sec^2\phi} = \cos^2\phi$

Go through option,

$\cot^2\phi \sin^2\phi = \frac{\cos^2\phi}{\sin^2\phi} \times \sin^2\phi = \cos^2\phi$

220. (d) ATQ,

$(\text{cosec}^2 B - 1)(\sec^2 B - 1)$

$= \cot^2 B \times \tan^2 B = 1$

221. (b) $\cos^4 A - \sin^4 A$

$= (\cos^2 A - \sin^2 A)(\cos^2 A + \sin^2 A)$

$= (\cos^2 A - \sin^2 A)$

$= (1 - \sin^2 A - \sin^2 A) = (1 - 2\sin^2 A)$

Alternate Method:

Put θ = 0°

$\Rightarrow \cos^4 0° - \sin^4 0° = 1 - 0 = 1$

Go through options,

$1 - 2\sin^2 0° = 1 - 0 = 1$

222. (d) $\text{cosec}\theta\,(1 - \cos\theta)(\text{cosec}\theta + \cot\theta)$

$= \text{cosec}\theta\,(1 - \cos\theta)\left(\frac{1}{\sin\theta} + \frac{\cos\theta}{\sin\theta}\right)$

$= \text{cosec}\theta\,(1 - \cos\theta)\left(\frac{1+\cos\theta}{\sin\theta}\right)$

$= \text{cosec}\theta \times \frac{1-\cos^2\theta}{\sin\theta}$

$= \text{cosec}\theta \times \frac{\sin^2\theta}{\sin\theta} = \text{cosec}\theta \times \sin\theta = 1$

223. (b) $(1 - \cos^2 A)\sec^2 A$

$= \sin^2 A \times \sec^2 A = \tan^2 A$

224. (a) $\sec x - \cos x$

$= \frac{1}{\cos x} - \cos x$

$= \frac{1-\cos^2 x}{\cos x} = \frac{\sin^2 x}{\cos x} = \frac{\sin x \times \sin x}{\cos x}$

$= \tan x \sin x$

SMART APPROACH:-

We know, Triplet

P B H

↓ ↓ ↓

3 4 5

$\sec x - \cos x$

$= \frac{5}{4} - \frac{4}{5} = \frac{9}{20}$

option (a) = $\tan x \sin x$

$= \frac{3}{4} \times \frac{3}{5} = \frac{9}{20}$ (Satisfied)

225. (a) $\frac{1}{1+\cos\theta} + \frac{1}{1-\cos\theta}$

$= \frac{1-\cos\theta+1+\cos\theta}{1-\cos^2\theta} = \frac{2}{\sin^2\theta}$

$= 2\,\text{cosec}^2\theta$

226. (a)

$\frac{1}{1-\sin\theta} + \frac{1}{1+\sin\theta} = 4\sec\theta$

$\Rightarrow \frac{2}{1-\sin^2\theta} = \frac{4}{\cos\theta}$

$\Rightarrow \frac{2}{\cos^2\theta} = \frac{4}{\cos\theta}$

$\Rightarrow 2\cos\theta = 1$

$\Rightarrow \cos\theta = \frac{1}{2} \Rightarrow \theta = 60°$

$\therefore \cot 60° + \text{cosec}\,60° = \frac{1}{\sqrt{3}} + \frac{2}{\sqrt{3}}$

$= \frac{3}{\sqrt{3}} = \sqrt{3}$

227. (c) $1 - \sin^2 32° + \frac{1}{1+\tan^2 58}$

$= \cos^2 32° + \frac{1}{\sec^2 58}$

$= \cos^2 32° + \cos^2 58° = 1$

Note: $\cos^2\alpha + \cos^2\beta = 1$

When, $\alpha + \beta = 90°$

228. (c) $\left(\frac{\sin\theta}{1+\cos\theta} + \frac{1+\cos\theta}{\sin\theta}\right)\left(\frac{1}{\tan\theta + \cot\theta}\right)$

$= \left(\frac{\sin^2\theta + 1 + \cos^2\theta + 2\cos\theta}{\sin\theta(1+\cos\theta)}\right)$

$(\sin\theta \cos\theta)$

$= \left(\frac{2+2\cos\theta}{\sin\theta(1+\cos\theta)}\right) \times \sin\theta \cos\theta$

$= 2\cos\theta$

229. (c) Put θ = 45°

Now,

$\frac{5}{1+\cot^2 45°} + \frac{3}{1+\tan^2 45°} + 2\cos^2 45°$

$= \frac{5}{2} + \frac{3}{2} + 2 \times \frac{1}{2} = 4 + 1 = 5$

Alternate Method:

$\frac{5}{1+\cot^2\theta} + \frac{3}{1+\tan^2\theta} + 2\cos^2\theta$

$= \frac{5}{\text{cosec}^2\theta} + \frac{3}{\sec^2\theta} + 2\cos^2\theta$

$= 5\sin^2\theta + 3\cos^2\theta + 2\cos^2\theta$

$= 5(\sin^2\theta + \cos^2\theta) = 5$

230. (a) $\frac{1+\sin A}{\cos A} + \frac{\cos A}{1+\sin A}$

$= \frac{1+\sin^2 A + 2\sin A + \cos^2 A}{\cos A(1+\sin A)}$

$= \frac{2(1+\sin A)}{\cos A(1+\sin A)}$

$= 2\sec A$

231. (a) $\frac{\sin A}{1+\cos A} + \frac{1+\cos A}{\sin A}$

$= \frac{\sin^2 A + (1+\cos A)^2}{\sin A(1+\cos A)}$

$= \frac{\sin^2 A + 1 + 2\cos A + \cos^2 A}{\sin A(1+\cos A)}$

$= \frac{2(1+\cos A)}{\sin A(1+\cos A)} = 2\,\text{cosec}A$

Alternate Method:

Put A = 30°

$\frac{\sin 30°}{1+\cos 30°} + \frac{1+\cos 30°}{\sin 30°}$

$= \frac{\frac{1}{2}}{1+\frac{\sqrt{3}}{2}} + \frac{1+\frac{\sqrt{3}}{2}}{\frac{1}{2}}$

$= \frac{\frac{1}{2}\times 2}{\sqrt{3}+2} + \frac{2+\sqrt{3}}{1} = \frac{1}{2+\sqrt{3}} + 2 + \sqrt{3}$

$= 2 - \sqrt{3} + 2 + \sqrt{3} = 4$

Go through options,

$2\,\text{cosec}\theta = 2\,\text{cosec}30° = 2 \times 2 = 4$

232. (d) $\frac{\cos A}{1+\sin A} + \frac{1+\sin A}{\cos A} + 2$

$= \frac{\cos^2 A + (1+\sin A)^2}{\cos A(1+\sin A)} + 2$

$= \frac{\cos^2 A + 1 + \sin^2 A + 2\sin A}{\cos A(1+\sin A)} + 2$

$= \frac{2(1+\sin A)}{\cos A(1+\sin A)} + 2 = 2(\sec A + 1)$

233. (a) $3(\sec^2 A - 1)(\text{cosec}A - 1)(\text{cosec}A + 1) + 4(\text{cosec}^2 A - \cot^2 A)$

$= 3\tan^2 A(\text{cosec}^2 A - 1) + 4 \times 1$

$= 3\tan^2 A \times \cot^2 A + 4 = 3 + 4 = 7$

$\left[\begin{array}{l}\because \sec^2 A - 1 = \tan^2 A \\ \text{cosec}^2 A - \cot^2 A = 1 \\ \tan^2 A \times \cot^2 A = 1\end{array}\right]$

Alternate Method:-

Put A = 45°

$3(\sec^2 45° - 1)(\text{cosec}45° - 1)(\text{cosec}45° + 1) + 4(\text{cosec}^2 45° - \cot^2 45°)$

$= 3(2-1)(\sqrt{2}-1)(\sqrt{2}+1) + 4(2 - 1) = 3 \times 1 + 4 = 7$

234. (a) $(1 + \tan^2\theta) + \left[1 + \frac{1}{\tan^2\theta}\right] = k$

put, θ = 45°

$\Rightarrow (1 + \tan^2 45°) + \left[1 + \frac{1}{\tan^2 45°}\right] = k$

$\Rightarrow (1 + 1) + \left[1 + \frac{1}{1}\right] = k$

$\Rightarrow 4 = k$

$\Rightarrow \sqrt{k} = \sqrt{4} = 2$

Go through option.

$\because \text{cosec}\theta \sec\theta$

$= \text{cosec}45° \sec 45° = \sqrt{2} \times \sqrt{2} = 2$

235. (d) $\left(\frac{1}{\sin\theta}+\frac{1}{\tan\theta}\right)\left(\frac{1}{\sin\theta}-\frac{1}{\tan\theta}\right)$

$= \frac{1}{\sin^2\theta}-\frac{1}{\tan^2\theta}$

$\Rightarrow \text{cosec}^2\theta - \cot^2\theta = 1$

236. (c) $(\sin\theta + \text{cosec}\theta)^2 + (\cos\theta + \sec\theta)^2$

$= \sin^2\theta + \text{cosec}^2\theta + 2 + \cos^2\theta + \sec^2\theta + 2$

$= 1 + 1 + \tan^2\theta + 1 + \cot^2\theta + 4$

$= 7 + \tan^2\theta + \cot^2\theta$

Alternate Method:

$(\sin\theta + \text{cosec}\theta)^2 + (\cos\theta + \sec\theta)^2$

Put $\theta = 45°$

$\left(\frac{1}{\sqrt{2}}+\sqrt{2}\right)^2+\left(\frac{1}{\sqrt{2}}+\sqrt{2}\right)^2=\left(\frac{3}{\sqrt{2}}\right)^2+\left(\frac{3}{\sqrt{2}}\right)^2$

$= \frac{9}{2}+\frac{9}{2} = 9$

Go through options,

$7 + \tan^2\theta + \cot^2\theta = 7 + \tan^2 45° + \cot^2 45° = 7 + 1 + 1 = 9$

237. (c) $\sin^2\theta \cos^2\theta = \frac{2}{9}$

$\therefore \text{cosec}^2\theta + \sec^2\theta = \frac{1}{\sin^2\theta}+\frac{1}{\cos^2\theta}$

$= \frac{1}{\sin^2\theta\cos^2\theta} = \frac{9}{2}$

Alternate Method:

$\sin^2\theta \times \cos^2\theta = \frac{2}{3}\times\frac{1}{3}$

$\therefore \text{cosec}^2\theta + \sec^2\theta = \frac{3}{2}+3 = \frac{9}{2}$

238. (a) Given, $\sec\theta - \cos\theta = 14$

$14\sec\theta = x$

Now,

$\sec\theta - \frac{1}{\sec\theta} = 14$

$\Rightarrow \sec^2\theta - 1 = 14\sec\theta$

$\Rightarrow x = \tan^2\theta$ ($\because \sec^2\theta - 1 = \tan^2\theta$)

239. (c) $\frac{\cos^4\theta - \sin^4\theta}{\sin^2\theta}$

$= \frac{(\cos^2\theta - \sin^2\theta)(\cos^2\theta + \sin^2\theta)}{\sin^2\theta}$

$= \frac{\cos^2\theta - \sin^2\theta}{\sin^2\theta}$

$= \frac{\cos^2\theta}{\sin^2\theta} - \frac{\sin^2\theta}{\sin^2\theta} = \cot^2\theta - 1$

Alternate Method:

Put $\theta = 30°$

$\frac{\cos^4 30° - \sin^4 30°}{\sin^2 30°} = \frac{\frac{9}{16}-\frac{1}{16}}{\frac{1}{4}}$

$= \frac{8}{16}\times 4 = 2$

Go through options,

$\cot^2 30° - 1 = 3 - 1 = 2$

240. (b) Given

$\cos^4\alpha - \sin^4\alpha = \frac{5}{6}$

$\Rightarrow (\cos^2\alpha - \sin^2\alpha)(\cos^2\alpha + \sin^2\alpha) = \frac{5}{6}$

$\Rightarrow (\cos^2\alpha - \sin^2\alpha)(1) = \frac{5}{6}$

$\Rightarrow \cos^2\alpha - (1 - \cos^2\alpha) = \frac{5}{6}$

$\Rightarrow 2\cos^2\alpha - 1 = \frac{5}{6}$

241. (c) $\frac{(1+\tan^2 A)}{\text{cosec}^2 A.\tan A} = \frac{\sec^2 A}{\text{cosec}^2 A \times \tan A}$

$= \frac{\frac{1}{\cos^2 A}}{\frac{1}{\sin^2 A}\times\tan A} = \frac{\sin^2 A}{\cos^2 A\times\tan A}$

$= \frac{\tan^2 A}{\tan A} = \tan A$

Alternate Method:

Put $\theta = 60°$

$\Rightarrow \frac{1+\tan^2 60°}{\text{cosec}^2 60°\times\tan 60°}$

$= \frac{1+3}{\left(\frac{2}{\sqrt{3}}\right)^2\times\sqrt{3}} = \frac{4}{\frac{4}{3}\times\sqrt{3}} = \frac{4\times 3}{4\sqrt{3}} = \sqrt{3}$

Go through options,

$\tan 60° = \sqrt{3}$

242. (b) A.T.Q,

$\frac{\cos\theta}{\sec\theta - 1}+\frac{\cos\theta}{\sec\theta + 1}$

$= \cos\theta\left[\frac{\sec\theta + 1 + \sec\theta - 1}{(\sec^2\theta - 1)}\right]$

$= \frac{\cos\theta\times 2\sec\theta}{\tan^2\theta} = 2\cot^2\theta$

243. (a) Given, $\frac{\cos\theta}{(1+\sin\theta)}+\frac{\cos\theta}{(1-\sin\theta)} = 4$

$\Rightarrow \frac{\cos\theta[1-\sin\theta+1+\sin\theta]}{\cos^2\theta} = 4$

$\Rightarrow \frac{2\cos\theta}{\cos^2\theta} = 4 \Rightarrow \sec^2\theta = \sec 60°$

$\Rightarrow \theta = 60°$

244. (a) $\frac{3-\sin^2 A+\cos^2 A}{2+2\cos^2 A}$

$= \frac{2+1-\sin^2 A+\cos^2 A}{2+2\cos^2 A}$

$= \frac{2+2\cos^2 A}{2+2\cos^2 A} = 1$

Alternate Method:

Put $A = 0°$

$\Rightarrow \frac{3-\sin^2 0°+\cos^2 0°}{2+2\cos^2 0°} = \frac{3+1}{2+2} = \frac{4}{4} = 1$

245. (a) $\frac{\sin^4 A+\cos^4 A}{1-2\sin^2 A\cos^2 A}$

$= \frac{1-2\sin^2 A\cos^2 A}{1-2\sin^2 A\cos^2 A} = 1$

Alternate Method:

Let, $A = 90°$

$\frac{\sin^4 A+\cos^4 A}{1-2\sin^2 A\cos^2 A} = \frac{(1)^4+0}{1-0} = 1$

246. (d) Given, $\sin\theta\cos\theta = \frac{1}{\sqrt{3}}$

$\therefore \sin^4\theta + \cos^4\theta = 1 - 2\sin^2\theta\cos^2\theta$

$= 1 - 2\times\left(\frac{1}{\sqrt{3}}\right)^2 = \frac{1}{3}$

247. (c) $\frac{1+\sin^4\theta+\cos^4\theta}{\cos^2\theta+\sin^4\theta}$

$= \frac{1+1-2\sin^2\theta\cos^2\theta}{1-\sin^2\theta+\sin^4\theta}$

$= \frac{2-2\sin^2\theta\cos^2\theta}{1-\sin^2\theta(1-\sin^2\theta)}$

$= \frac{2(1-\sin^2\theta\cos^2\theta)}{1-\sin^2\theta\cos^2\theta} = 2$

Alternate Method:

Put $\theta = 0 \Rightarrow \frac{1+0+1}{1+0} = \frac{2}{1} = 2$

248. (d) $\sin^4\theta + \cos^4\theta = 1 - 2\sin^2\theta\cos^2\theta$

$= 1 - 2\sin^2\theta(1 - \sin^2\theta)$

$= 1 - 2\sin^2\theta + 2\sin^4\theta$

$= 2\sin^4\theta - 2\sin^2\theta + 1$

249. (d) $2[\tan^2 A.\cos^2 A + \cos^2 A]$

$= 2\cos^2 A(\tan^2 A + 1)$

$= 2\cos^2 A \times \sec^2 A = 2$

Alternate Method:

Put $A = 45°$,

$\Rightarrow 2(\tan^2 45° \times \cos^2 45° + \cos^2 45°)$

$= 2\left(1\times\frac{1}{2}+\frac{1}{2}\right) = 2$

250. (d) $\tan\theta = \frac{8}{19}$

$\sec^2\theta = 1 + \tan^2\theta$

$= 1+\left(\frac{8}{19}\right)^2 = 1+\frac{64}{361} = 1\frac{64}{361}$

251. (b) Given, $\text{cosec}\theta = \frac{29}{22}$

$\because \cot^2\theta = \text{cosec}^2\theta - 1$

$= \frac{841}{484} - 1 = \frac{357}{484}$

252. (d) Given, $\cot\theta = \frac{9}{17}$

We know,

$\text{cosec}^2\theta = 1 + \cot^2\theta$

$= 1+\frac{81}{289} = 1\frac{81}{289}$

253. (a) $\frac{\sin^3 A - \cos^3 A}{\sin A - \cos A}$

$= \frac{(\sin A - \cos A)(\sin^2 A + \cos^2 A + \sin A\cos A)}{\sin A - \cos A}$

$= 1 + \sin A\cos A$

254. (d) $a^3 + b^3 = (a + b)(a^2 + b^2 - ab)$

$\therefore \frac{\sin^3 A + \cos^3 A}{\sin A + \cos A}$

$= \frac{(\sin A+\cos A)(\sin^2 A+\cos^2 A-\sin A\times\cos A)}{(\sin A+\cos A)}$

$= 1 - \sin A\cos A$

255. (b) $\frac{\cot^3 A - 1}{\cot A - 1}$

$= \frac{(\cot A - 1)(\cot^2 A + 1 + \cot A)}{(\cot A - 1)}$

$= \cot^2 A + 1 + \cot A$

$= \cot^2 A + \text{cosec}^2 A - \cot^2 A + \cot A$

$= \text{cosec}^2 A + \cot A$

256. (c) $\frac{\sin^3\alpha + \cos^3\alpha}{\sin\alpha + \cos\alpha}$

$= \frac{(\sin\alpha+\cos\alpha)(\sin^2\alpha+\cos^2\alpha-\sin\alpha.\cos\alpha)}{(\sin\alpha+\cos\alpha)}$

$= (1 - \sin\alpha.\cos\alpha)$

257. (a) $(a^3 + b^3) = (a + b)(a^2 + b^2 - ab)$
Now,
$(\sin A + \cos A)(1 - \sin A \cos A)$
$= (\sin A + \cos A)(\sin^2 A + \cos^2 A - \sin A \cos A)$ $[\because \sin^2 A + \cos^2 A = 1]$
$= \sin^3 A + \cos^3 A$

258. (b) $\sin^6\theta + \cos^6\theta = \frac{1}{3}$

$\Rightarrow 1 - 3\sin^2\theta\cos^2\theta = \frac{1}{3}$

$\Rightarrow 3\sin^2\theta\cos^2\theta = 1 - \frac{1}{3} = \frac{2}{3}$

$\Rightarrow \sin\theta\cos\theta = \frac{\sqrt{2}}{3}$

259. (b) $\frac{1}{\sin\theta} - \frac{\cot^2\theta}{1 + \text{cosec}\theta}$

$= \frac{1}{\sin\theta} - \frac{(\text{cosec}\,\theta - 1)(\text{cosec}\,\theta + 1)}{(\text{cosec}\,\theta + 1)}$

$= \text{cosec}\theta - \text{cosec}\theta + 1 = 1$

Alternate Method:
Put $\theta = 45°$,

$\Rightarrow \frac{1}{\sin 45°} - \frac{\cot^2 45°}{1 + \text{cosec}\,45°} = \sqrt{2} - \frac{1}{1+\sqrt{2}}$

$= \frac{\sqrt{2}+2-1}{\sqrt{2}+1} = \frac{\sqrt{2}+1}{\sqrt{2}+1} = 1$

260. (b) $(1 + \tan^2 A)(1 + \cot^2 A)$
$= \sec^2 A.\text{cosec}^2 A$

$= \frac{1}{\cos^2 A} \times \frac{1}{\sin^2 A} = \frac{1}{\sin^2 A(1 - \sin^2 A)}$

261. (c) $17\cos^2\theta + 10\sin^2\theta + \frac{7}{\text{cosec}^2\theta}$
$= 17\cos^2\theta + 10\sin^2\theta + 7\sin^2\theta$
$= 17\cos^2\theta + 17\sin^2\theta = 17(\cos^2 + \sin^2\theta)$
$= 17 \times 1 = 17$

262. (a) $\frac{1 + \tan^2 A}{\tan^2 A} = \frac{1}{\tan^2 A} + 1 = \cot^2 A + 1 = \text{cosec}^2 A \neq \sin^2 A$

263. (d) $\tan^2\theta + \tan^4\theta$
$= \tan^2\theta(1 + \tan^2\theta)$
$= \tan^2\theta.\sec^2\theta = (\sec^2\theta - 1)\sec^2\theta$
$= \sec^4\theta - \sec^2\theta$

Alternate Method:
Put $\theta = 45°$
$\tan^2 45° + \tan^4 45° = 1 + 1 = 2$
Go through options,
$\sec^4\theta - \sec^2\theta = \sec^4 45° - \sec^2 45°$

$= (\sqrt{2})^4 - (\sqrt{2})^2 = 4 - 2 = 2$

264. (c) $\frac{\cos A}{1 - \sin A} + \frac{\cos A}{1 + \sin A} = 4$

$\Rightarrow \frac{\cos A + \cos A \sin A + \cos A - \cos A \sin A}{1 - \sin^2 A} = 4$

$\Rightarrow \frac{\cos A \times 2}{1 - \sin^2 A} = 4 \Rightarrow \frac{2}{\cos A} = 4$

$\Rightarrow \cos A = \frac{1}{2} \Rightarrow A = 60°$

265. (d) $3\sin^2\theta + 5\cos^2\theta = 4$

$\Rightarrow 3 + 2\cos^2\theta = 4 \Rightarrow \cos^2\theta = \frac{1}{2}$

$\Rightarrow \cos\theta = \frac{1}{\sqrt{2}} = \cos 45°$

$\theta = 45°$
$\therefore \tan 45° = 1$

266. (c) $a = x\cos\theta + y\sin\theta$
$b = x\sin\theta - y\cos\theta$
$a^2 + b^2 = x^2\cos^2\theta + y^2\sin^2\theta + 2xy\sin\theta\cos\theta + x^2\sin^2\theta + y^2\cos^2\theta - 2xy\sin\theta\cos\theta = x^2 + y^2$

Alternate Method:-
Let $\theta = 45°$

$a = \frac{x}{\sqrt{2}} + \frac{y}{\sqrt{2}} = \frac{x+y}{\sqrt{2}}$

$b = \frac{x}{\sqrt{2}} - \frac{y}{\sqrt{2}} = \frac{x-y}{\sqrt{2}}$

$\therefore \quad a^2 + b^2 = \frac{1}{2}[(x + y)^2 + (x - y)^2]$

$= \frac{1}{2} \times 2(x^2 + y^2) = x^2 + y^2$

267. (a) $(\sec\phi - \tan\phi)^2 (1 + \sin\phi)^2 \div \cos^2\phi$
Put, $\phi = 45°$
$(\sec 45° - \tan 45°)^2 (1 + \sin 45°)^2 \div \cos^2 45°$

$= (\sqrt{2} - 1)^2 \left(1 + \frac{1}{\sqrt{2}}\right)^2 \div \left(\frac{1}{\sqrt{2}}\right)^2$

$= (\sqrt{2} - 1)^2 \left(\frac{\sqrt{2}+1}{\sqrt{2}}\right)^2 \div \frac{1}{2}$

$= \frac{(9-8)^2}{2} \times 2 = \frac{1}{2} \times 2 = 1$

268. (b) Given,
$p\cos A = 2q\sin A$ .....(i)
and $2p\,\text{cosec}\,A - q\sec A = 3$ .....(ii)
Let, $A = 45°$
By equation (i)

$p \times \frac{1}{\sqrt{2}} = 2q \times \frac{1}{\sqrt{2}}$

$\Rightarrow p = 2q$
By equation (ii)

$2p \times \sqrt{2} - q\sqrt{2} = 3$

$\Rightarrow 4q\sqrt{2} - \sqrt{2}\,q = 3$

$\Rightarrow 3\sqrt{2}\,q = 3$

$\Rightarrow q = \frac{1}{\sqrt{2}} \Rightarrow p = \sqrt{2}$

$\therefore p^2 + 4q^2 = 2 + 4 \times \frac{1}{2} = 4$

Alternate Method:-
$p\cos A = 2q\sin A$
$\Rightarrow p = 2q\tan A$ ....(i)
Also,
$2p\,\text{cosec}\,A - q\sec A = 3$
$\Rightarrow 4q\tan A \times \text{cosec}\,A - q\sec A = 3$
$\Rightarrow 4q\sec A - q\sec A = 3$
$\Rightarrow q = \cos A$ ....(ii)
$\Rightarrow p = 2\sin A$
$\therefore p^2 + 4q^2 = 4\sin^2 A + 4\cos^2 A$
$= 4(\sin^2 A + \cos^2 A) = 4$

269. (b) $x = r\cos(t)$
$y = r\sin(t)\cos(u)$
$z = r\sin(t)\sin(u)$
Let, $t = 45°$, $u = 30°$

$\Rightarrow x = \frac{r}{\sqrt{2}}$, $y = \frac{r}{\sqrt{2}} \times \frac{\sqrt{3}}{2} = \frac{\sqrt{3}r}{2\sqrt{2}}$

and, $z = \frac{r}{\sqrt{2}} \times \frac{1}{2} = \frac{r}{2\sqrt{2}}$

$\therefore x^2 + y^2 + z^2 = \frac{r^2}{2} + \frac{3r^2}{8} + \frac{r^2}{8}$

$= \frac{(4+3+1)r^2}{8} = r^2$

Alternate Method:
$x^2 + y^2 + z^2 = r^2\cos^2 t + r^2\sin^2 t\cos^2 u + r^2\sin^2 t\sin^2 u$
$= r^2[\cos^2 t + \sin^2 t\cos^2 u + \sin^2 t\sin^2 u]$
$= r^2[\cos^2 t + \sin^2 t(\cos^2 u + \sin^2 u)]$
$= r^2[\cos^2 t + \sin^2 t] = r^2$

270. (c) $\frac{1}{\text{cosec}\,\theta - \cot\theta} - \frac{1}{\sin\theta}$

$= \frac{\sin\theta}{1 - \cos\theta} - \frac{1}{\sin\theta}$

$= \frac{\sin^2\theta - 1 + \cos\theta}{\sin\theta(1 - \cos\theta)}$

$= \frac{-\cos^2\theta + \cos\theta}{\sin\theta(1 - \cos\theta)} = \frac{\cos\theta(-\cos + 1)}{\sin\theta(1 - \cos\theta)}$

$= \cot\theta$.

Alternate Method:

$\frac{1}{\text{cosec}\,\theta - \cot\theta} - \frac{1}{\sin\theta}$

Put, $\theta = 30°$

$\frac{1}{\text{cosec}\,30° - \cot 30°} - \frac{1}{\sin 30°} = \frac{1}{2 - \sqrt{3}} - 2$

$= 2 + \sqrt{3} - 2 = \sqrt{3}$

$\therefore \cot\theta = \cot 30° = \sqrt{3}$

271. (b) $(\sin A + \text{cosec}A)^2 + (\cos A + \sec A)^2$
$= \sin^2 A + \text{cosec}^2 A + 2 \times \sin A \times \text{cosec}A + \cos^2 A + \sec^2 A + 2 \times \cos A \times \sec A$
$= 1 + 2 + 2 + \tan^2 A + 1 + \cot^2 A + 1$
$= 7 + \tan^2 A + \cot^2 A$

Alternate Method:
By value putting $A = 45°$
$(\sin A + \text{cosec}\,A)^2 + (\cos A + \sec A)^2$
$= (\sin 45° + \text{cosec}\,45°)^2 + (\cos 45° + \sec 45°)^2$

$= \left(\frac{1}{\sqrt{2}} + \sqrt{2}\right)^2 + \left(\frac{1}{\sqrt{2}} + \sqrt{2}\right)^2$

$= \left(\frac{3}{\sqrt{2}}\right)^2 + \left(\frac{3}{\sqrt{2}}\right)^2$

$= \frac{9}{2} + \frac{9}{2} = \frac{18}{2} = 9$

Now, check option (b)
$7 + \tan^2 A + \cot^2 A = 7 + \tan^2 45° + \cot^2 45° = 7 + 1 + 1 = 9$
Hence, $A = 45°$ satisfy so, option 'b' will be the answer.

272. (d) $\frac{1 - \cot^2\theta}{\tan^2\theta - 1}$

$= \frac{1 - \frac{\cos^2\theta}{\sin^2\theta}}{\frac{\sin^2\theta}{\cos^2\theta} - 1} = \frac{\frac{\sin^2\theta - \cos^2\theta}{\sin^2\theta}}{\frac{\sin^2\theta - \cos^2\theta}{\cos^2\theta}} = \frac{\cos^2\theta}{\sin^2\theta}$

$= \cot^2\theta$

Alternate Method:

Put $\theta = 30°$

$\therefore \frac{1-\cot^2\theta}{\tan^2\theta-1} = \frac{1-(\cot 30°)^2}{(\tan 30°)^2-1}$

$= \frac{1-3}{\frac{1}{3}-1} = \frac{-2}{-2/3} = 3 = \cot^2 30° = \cot^2\theta$

273. (c) $\frac{3\sin\theta-\cos\theta}{\cos\theta+\sin\theta} = 1$

On dividing Nr and Dn by $\sin\theta$

$\Rightarrow \left(\frac{\frac{3\sin\theta-\cos\theta}{\sin\theta}}{\frac{\cos\theta+\sin\theta}{\sin\theta}}\right) = 1$

$\Rightarrow \frac{3-\cot\theta}{\cot\theta+1} = 1$

$\Rightarrow 3 - \cot\theta = \cot\theta + 1 \Rightarrow \cot\theta = 1$

274. (b) $\frac{\sin x-\cos x}{\sin x+\cos x} = \frac{2}{5}$

On dividing numerator and denominator by $\sin x$

$\Rightarrow \frac{\frac{\sin x}{\sin x}-\frac{\cos x}{\sin x}}{\frac{\sin x}{\sin x}+\frac{\cos x}{\sin x}} = \frac{2}{5}$

$\Rightarrow \frac{1-\cot x}{1+\cot x} = \frac{2}{5}$

$\Rightarrow 5 - 5\cot x = 2 + 2\cot x$

$\Rightarrow 7\cot x = 3 \Rightarrow \cot x = \frac{3}{7}$

$\therefore \frac{1+\cot^2 x}{1-\cot^2 x} = \frac{1+\left(\frac{3}{7}\right)^2}{1-\left(\frac{3}{7}\right)^2}$

$= \frac{1+\frac{9}{49}}{1-\frac{9}{49}} = \frac{\frac{58}{49}}{\frac{40}{49}} = \frac{58}{40} = 1.45$

275. (a) $(\sin\theta + \operatorname{cosec}\theta)^2 + (\cos\theta + \sec\theta)^2 = k + \tan^2\theta + \cot^2\theta$

$\Rightarrow \sin^2\theta + \operatorname{cosec}^2\theta + 2 + \cos^2\theta + \sec^2\theta + 2 = k + \tan^2\theta + \cot^2\theta$

$\Rightarrow 1 + 2 + 1 + \cot^2\theta + 1 + \tan^2\theta + 2 = k + \tan^2\theta + \cot^2\theta$

$\Rightarrow 1 + 2 + 2 + 1 + 1 = k$

$\Rightarrow k = 7$

Alternate Method:-

Put $\theta = 45°$

$\left(\frac{1}{\sqrt{2}}+\sqrt{2}\right)^2 + \left(\frac{1}{\sqrt{2}}+\sqrt{2}\right)^2 = k + 1 + 1$

$\Rightarrow \frac{9}{2}+\frac{9}{2} = k + 2$

$\Rightarrow 9 = k + 2 \Rightarrow k = 7$

276. (c) $1 + \sin^2\theta = a\cos^2\theta$

divide by $\cos^2\theta$ both side,

$\sec^2\theta + \tan^2\theta = a$

277. (c) $m + 5 = \sec^2 A \operatorname{cosec}^2 A (\sin^2 A) \cos^2 A + 5$

$\boxed{1-\cos^2 A = \sin^2 A}$

$\boxed{1-\sin^2 A = \cos^2 A}$

$\Rightarrow m + 5 = 1 + 5$

$\therefore m = 1$

278. (a) $\frac{x}{p} = \sec A.\cos B$ .....(i)

$\frac{y}{q} = \sec A.\sin B$ .....(ii)

$\frac{z}{r} = \tan A$ .....(iii)

On squaring both sides in all three equation

$\frac{x^2}{p^2} = \sec^2 A.\cos^2 B$ .....(iv)

$\frac{y^2}{q^2} = \sec^2 A.\sin^2 B$ .....(v)

$\frac{z^2}{r^2} = \tan^2 A$ .....(vi)

Now, (iv) + (v) – (vi)

$\frac{x^2}{p^2} + \frac{y^2}{q^2} - \frac{z^2}{r^2}$

$= \frac{\cos^2 B}{\cos^2 A} + \frac{\sin^2 B}{\cos^2 A} - \frac{\sin^2 A}{\cos^2 A}$

$= \frac{1-\sin^2 A}{\cos^2 A} = \frac{\cos^2 A}{\cos^2 A} = 1$

Alternate Method:-

Let, A = B = 45°

$\Rightarrow x = p, y = q, z = r$

$\therefore \frac{x^2}{p^2}+\frac{y^2}{q^2} - \frac{z^2}{r^2} = 1 + 1 - 1 = 1$

279. (b) $x = a\sec\theta\cos\phi, y = b\sec\theta\sin\phi$ and $z = c\tan\theta$

Let, $\theta = 45°, \phi = 30°$

$\Rightarrow x = a \times \sqrt{2} \times \frac{\sqrt{3}}{2} = \frac{a\sqrt{6}}{2} = a\frac{\sqrt{3}}{\sqrt{2}}$

$\Rightarrow y = b \times \sqrt{2} \times \frac{1}{2} = \frac{b}{\sqrt{2}}$

and $z = c$

$\therefore \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2}$

$= \frac{a^2\times\frac{3}{2}}{a^2} + \frac{\frac{b^2}{2}}{b^2} - \frac{c^2}{c^2} = \frac{3}{2} + \frac{1}{2} - 1 = 1$

Alternate Method:

$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = \frac{a^2\sec^2\theta\cos^2\phi}{a^2} + \frac{b^2\sec^2\theta\sin^2\phi}{b^2} - \frac{c^2\tan^2\theta}{c^2}$

$= \sec^2\theta(\cos^2\phi + \sin^2\phi) - \tan^2\theta$

$= \sec^2\theta - \tan^2\theta = 1$

280. (c) $\tan\theta = \frac{5}{8}$

$\therefore \frac{(1+\cos\theta)(1-\cos\theta)}{(1+\sin\theta)(1-\sin\theta)} = \frac{1-\cos^2\theta}{1-\sin^2\theta}$

$= \frac{\sin^2\theta}{\cos^2\theta} = \tan^2\theta = \left(\frac{5}{8}\right)^2 = \frac{25}{64}$

281. (c) $\frac{3\cos^3\theta - 2\cos\theta}{\sin\theta - 3\sin^3\theta}$

$= \frac{\cos\theta(3\cos^2\theta - 2)}{\sin\theta(1-3\sin^2\theta)}$

$= \frac{\cos\theta(1-3\sin^2\theta)}{\sin\theta(1-3\sin^2\theta)} = \cot\theta$

282. (d) $\frac{3(\sin^4 z - \cos^4 z + 1)}{\sin^2 z}$

$= \frac{3(\sin^2 z - \cos^2 z + 1)}{\sin^2 z}$

We know, $(\cos^2 z = 1 - \sin^2 z)$

$= \frac{3(2\sin^2 z)}{\sin^2 z} = 3 \times 2 = 6$

Alternate Method:

Put $z = 90°$

$\frac{3(\sin^4 90° - \cos^4 90° + 1)}{\sin^2 90°} = \frac{3(1-0+1)}{1} = 6$

283. (b) $\left(\frac{\operatorname{cosec} A}{\operatorname{cosec} A - 1} + \frac{\operatorname{cosec} A}{\operatorname{cosec} A + 1}\right)$

$\left(\frac{\sec A}{\sec A - 1} + \frac{\sec A}{\sec A + 1}\right)^{-1}$

$= \left(\frac{\operatorname{cosec}^2 A + \operatorname{cosec} A + \operatorname{cosec}^2 A - \operatorname{cosec} A}{\cot^2 A}\right)$

$\left(\frac{\sec^2 A + \sec^2 A}{\tan^2 A}\right)^{-1}$

$= \frac{2\operatorname{cosec}^2 A}{\cot^2 A} \times \frac{\tan^2 A}{2\sec^2 A} = \tan^2 A$

Put $A = 60°$

$\therefore \tan^2 60° = 3$

284. (a) $\tan(t) + \cot(t) = 1$

$\Rightarrow \frac{\sin t}{\cos t} + \frac{\cos t}{\sin t} = 1 \Rightarrow \frac{1}{\sin t.\cos t} = 1$

$\Rightarrow \sin t.\cos t = 1$

$\therefore (\sin t + \cos t)^2 = 1 + 2\sin t.\cos t$

$\Rightarrow (\sin t + \cos t)^2 = 1 + 2 \times 1 = 3$

$\Rightarrow \left(\frac{1}{\sin t + \cot t}\right) = \frac{1}{\sqrt{3}}$

285. (d) $(\operatorname{cosec}\theta - \sin\theta)(\sec\theta - \cos\theta)(\tan\theta + \cot\theta)$

$= \left(\frac{1}{\sin\theta} - \frac{\sin\theta}{1}\right)\left(\frac{1}{\cos\theta} - \frac{\cos\theta}{1}\right)\sec\theta\times\operatorname{cosec}\theta$

$\therefore [\sin^2\theta + \cos^2\theta = 1$ and $\tan\theta + \cot\theta$

$= \sec\theta \times \operatorname{cosec}\theta]$

$= \left(\frac{1-\sin^2\theta}{\sin\theta}\right)\left(\frac{1-\cos^2\theta}{\cos\theta}\right)(\sec\theta\times\operatorname{cosec}\theta)$

$= \frac{\cos^2\theta}{\sin\theta} \times \frac{\sin^2\theta}{\cos\theta} \times \sec\theta \times \operatorname{cosec}\theta = 1$

Alternate method:

Put $\theta = 45°$

$(\operatorname{cosec}45° - \sin45°)(\sec45° - \cos45°)(\tan45° + \cot45°)$

$= \left(\sqrt{2} - \frac{1}{\sqrt{2}}\right)\left(\sqrt{2} - \frac{1}{\sqrt{2}}\right)(1+1)$

$= \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} \times 2 = \frac{1}{2} \times 2 = 1$

286. (a) $p - q = \sin\theta + \tan\theta - \tan\theta + \sin\theta = 2\sin\theta$ ... (i)

$p + q = \sin\theta + \tan\theta + \tan\theta - \sin\theta = 2\tan\theta$ ... (ii)

$\Rightarrow pq = \tan^2\theta - \sin^2\theta = \sin^2\theta[\sec^2\theta - 1]$

$\Rightarrow pq = \sin^2\theta.\tan^2\theta$

$\Rightarrow \sqrt{pq} = \sin\theta.\tan\theta$

Multiply both eq.

$\Rightarrow p^2 - q^2 = 4 \times \sin\theta.\tan\theta$

$\Rightarrow p^2 - q^2 = 4\sqrt{pq}$

287. (c) $P\cos\alpha = 3$, $4\tan\alpha = Q$,

$\Rightarrow P \times \frac{1}{\sec\alpha} = 3$

$\sec\alpha = \frac{P}{3}$, $\tan\alpha = \frac{Q}{4}$

$\because \sec^2\alpha - \tan^2\alpha = 1$

$\Rightarrow \frac{P^2}{9} - \frac{Q^2}{16} = 1$

Alternate Method:

Put $\alpha = 45°$,

Then, $P = 3\sqrt{2}$ and $Q = 4$

option 'c' satisfy

288. (a) $(\cos x - \sin x)^2 + (\cos x + \sin x)^2$

$= \cos^2 x + \sin^2 x - 2\cos x \sin x$

$+ \cos^2 x + \sin^2 x + 2\cos x \sin x$

$= 2(\cos^2 x + \sin^2 x) = 2 \times 1 = 2$

289. (b) $\frac{1+\sin t}{4-4\sin t} - \frac{1-\sin t}{4+4\sin t}$

$= \frac{1}{4}\left(\frac{1+\sin t}{1-\sin t} - \frac{1-\sin t}{1+\sin t}\right)$

$= \frac{1}{4}\left(\frac{(1+\sin t)^2 - (1-\sin t)^2}{1-\sin^2 t}\right)$

$= \frac{1}{4} \times \frac{4\sin t}{\cos^2 t} = \tan t \sec t$

290. (d) $r = 15(\sin\theta + \cos\theta)$

$s = 16(\sin\theta - \cos\theta)$

$\Rightarrow r^2 = 15^2(\sin\theta + \cos\theta)^2$

$\Rightarrow s^2 = 16^2(\sin\theta - \cos\theta)^2$

ATQ,

$\therefore \frac{r^2}{15^2} + \frac{s^2}{16^2}$

$= \frac{15^2(\sin\theta + \cos\theta)^2}{15^2} + \frac{16^2(\sin\theta - \cos\theta)^2}{16^2}$

$= (\sin\theta + \cos\theta)^2 + (\sin\theta - \cos\theta)^2$

$= 2(\sin^2\theta + \cos^2\theta) = 2$

Alternate Method:

$r = 15(\sin\theta + \cos\theta)$, $s = 16(\sin\theta - \cos\theta)$

Put $\theta = 90°$

$r = 15$, $s = 16$

$\therefore \frac{r^2}{15^2} + \frac{s^2}{16^2} = 1 + 1 = 2$

291. (b) Given, $\sin m + \sin n = p$ ....(i)

$\cos m + \cos n = q$ ....(ii)

square Eq^n (i) & (ii) then add both Eq^n

$\Rightarrow 1 + 1 + 2(\sin m \sin n + \cos m \cos n) = p^2 + q^2$

$\Rightarrow 2 + 2(\sin m \sin n + \cos m \cos n) = p^2 + q^2$

$\Rightarrow \frac{p^2 + q^2 - 2}{2} = \sin m \sin n + \cos m \cos n$

292. (d) $\frac{(1+\cos\theta)^2 + \sin^2\theta}{(\operatorname{cosec}^2\theta - 1) \times \sin^2\theta}$

$= \frac{1 + \cos^2\theta + 2\cos\theta + \sin^2\theta}{\cot^2\theta \times \sin^2\theta}$

$= \frac{2 + 2\cos\theta}{\frac{\cos^2\theta}{\sin^2\theta} \times \sin^2\theta} = \frac{2(1+\cos\theta)}{\cos^2\theta}$

$= 2\sec\theta(1 + \sec\theta)$

Alternate Method:

Put $\theta = 45°$

$\frac{(1+\cos 45°)^2 + \sin^2 45°}{(\operatorname{cosec}^2 45° - 1) \times \sin^2 45°}$

$= \frac{\left(1 + \frac{1}{\sqrt{2}}\right)^2 + \frac{1}{2}}{\frac{1}{2}} = \left(\frac{(\sqrt{2}+1)^2}{2} + \frac{1}{2}\right) \times 2$

$= 3 + 2\sqrt{2} + 1 = 4 + 2\sqrt{2}$

Go through option,

$2\sec\theta(1 + \sec\theta) = 2\sec 45°(1 + \sec 45°)$

$= 2\sqrt{2}(1 + \sqrt{2}) = 4 + 2\sqrt{2}$

293. (b) $\sqrt{\frac{1}{\cos^2\theta} + \frac{1}{\sin^2\theta}} \times \sqrt{\frac{\sin^2\theta}{\cos^2\theta} - \sin^2\theta}$

$= \frac{1}{\cos\theta\sin\theta} \times \sqrt{\sin^2\theta\left(\frac{1-\cos^2\theta}{\cos^2\theta}\right)}$

$= \frac{1}{\cos\theta\sin\theta} \times \sin\theta \times \frac{\sin\theta}{\cos\theta}$

$= \frac{\sin\theta}{\cos^2\theta} = \sin\theta \sec^2\theta$

Alternate Method:

$\sqrt{\sec^2\theta + \operatorname{cosec}^2\theta} \times \sqrt{\tan^2\theta - \sin^2\theta}$

put $\theta = 45°$,

$= \sqrt{\left(\sqrt{2}\right)^2 + \left(\sqrt{2}\right)^2} \times \sqrt{1 - \left(\frac{1}{\sqrt{2}}\right)^2}$

$= \sqrt{2+2} \times \sqrt{1 - \frac{1}{2}} = \sqrt{4} \times \frac{1}{\sqrt{2}}$

$= 2 \times \frac{1}{\sqrt{2}} = \sqrt{2}$

Check options, option (b)

$\sin\theta \cdot \sec^2\theta = \sin 45° \sec^2 45°$

$= \frac{1}{\sqrt{2}}\left(\sqrt{2}\right)^2 = \frac{1}{\sqrt{2}} \times 2 = \sqrt{2}$

294. (b)

$\frac{1}{\sin^4(90°-2\alpha)} + \frac{1}{\cos^2(90°-2\alpha) - 1}$

$= \frac{1}{\cos^4 2\alpha} + \frac{1}{\sin^2 2\alpha - 1}$

$= \frac{1}{\cos^4 2\alpha} - \frac{1}{\cos^2 2\alpha}$

$= \frac{1 - \cos^2 2\alpha}{\cos^4 2\alpha} = \frac{\sin^2 2\alpha}{\cos^4 2\alpha}$

$= \tan^2 2\alpha \sec^2 2\alpha$

295. (a) $(1 + \tan^2 A)\left(1 + \frac{1}{\tan^2 A}\right)$

$= (1 + \tan^2 A)(1 + \cot^2 A)$

$= \left(1 + \frac{\sin^2 A}{\cos^2 A}\right)\left(1 + \frac{\cos^2 A}{\sin^2 A}\right)$

$= \frac{1}{\cos^2 A} \times \frac{1}{\sin^2 A} = \frac{1}{(1 - \sin^2 A)(\sin^2 A)}$

$= \left(\frac{1}{(\sin^2 A - \sin^4 A)}\right)$

Alternate Method:

$(1 + \tan^2 A)\left(1 + \frac{1}{\tan^2 A}\right)$

Put, $A = 60°$

$\Rightarrow (1 + 3)\left(1 + \frac{1}{3}\right) = 4 \times \frac{4}{3} = \frac{16}{3}$

Option (a), $\left(\frac{1}{(\sin^2 A - \sin^4 A)}\right)$

$= \left(\frac{1}{(\sin^2 60° - \sin^4 60°)}\right)$

$= \frac{1}{\frac{3}{4} - \frac{9}{16}} = \frac{16}{3}$

296. (b) $\left\{\tan\frac{A}{2} + \operatorname{cosec}\frac{B+C}{2}\right\}$

$\left\{\tan\frac{A}{2} - \operatorname{cosec}\frac{B+C}{2}\right\}$

$= \tan^2\frac{A}{2} - \operatorname{cosec}^2\frac{B+C}{2}$

$= \sec^2\frac{A}{2} - 1 - \operatorname{cosec}^2\frac{B+C}{2}$

$= \sec^2\frac{A}{2} - 1 - \operatorname{cosec}^2\left\{90° - \frac{A}{2}\right\}$

$= \sec^2\frac{A}{2} - 1 - \sec^2\frac{A}{2} = -1$

Alternate Method:

$\left\{\tan\frac{A}{2} + \operatorname{cosec}\frac{B+C}{2}\right\}$

$\left\{\tan\frac{A}{2} - \operatorname{cosec}\frac{B+C}{2}\right\}$

Put $A = 90°$, $B + C = 90°$

$(\tan 45° + \operatorname{cosec} 45°)(\tan 45° - \operatorname{cosec} 45°)$

$= (1 + \sqrt{2})(1 - \sqrt{2}) = 1 - 2 = -1$

297. (c) $\tan^4 A + \tan^2 A$

$= \tan^2 A(1 + \tan^2 A)$

$= \tan^2 A \sec^2 A = (\sec^2 A - 1)\sec^2 A$

$= \sec^4 A - \sec^2 A$

Alternate Method:

Put $\theta = 45°$

$\tan^4 A + \tan^2 A = \tan^4 45° + \tan^2 45°$

$= 1 + 1 = 2$

Now, Check option (c)

$\sec^4 A - \sec^2 A = \sec^4 45° - \sec^2 45°$

$= \left(\sqrt{2}\right)^4 - \left(\sqrt{2}\right)^2 = 4 - 2 = 2$

option (c) satisfy the condition

298. (d)

$\operatorname{cosec}\theta\left[\frac{1+\cos\theta}{\sin\theta} + \frac{\sin\theta}{1+\cos\theta}\right] - 2\cot^2\theta$

$= \operatorname{cosec}\theta\left[\frac{1+\cos\theta}{\sin\theta} + \frac{\sin\theta}{1+\cos\theta} \times \frac{1-\cos\theta}{1-\cos\theta}\right] - 2\cot^2\theta$

$= \operatorname{cosec}\theta\left[\frac{1+\cos\theta}{\sin\theta} + \frac{\sin\theta(1-\cos\theta)}{\sin^2\theta}\right] - 2\cot^2\theta$

[$\because 1 - \cos^2\theta = \sin^2\theta$, $\operatorname{cosec}^2\theta - \cot^2\theta = 1$]

$= \operatorname{cosec}\theta\left[\frac{1+\cos\theta+1-\cos\theta}{\sin\theta}\right] - 2\cot^2\theta$

$= 2\operatorname{cosec}^2\theta - 2\cot^2\theta$

$= 2(\operatorname{cosec}^2\theta - \cot^2\theta) = 2 \times 1 = 2$

299. (a) $\sin A\left(1+\frac{\sin A}{\cos A}\right)+\cos A\left(1+\frac{\cos A}{\sin A}\right)$

$=\frac{\sin A}{\cos A}(\cos A+\sin A)+\frac{\cos A}{\sin A}(\sin A+\cos A)$

$=(\cos A+\sin A)\left(\frac{\sin A}{\cos A}+\frac{\cos A}{\sin A}\right)$

$=\frac{\cos A+\sin A}{\sin A\times\cos A}=\sec A+\operatorname{cosec} A$

Alternate Method:

Put A = 45°

$\Rightarrow \sin 45°\left(1+\frac{\sin 45°}{\cos 45°}\right)+\cos 45°\left(1+\frac{\cos 45°}{\sin 45°}\right)$

$=\frac{1}{\sqrt{2}}\times 2+\frac{1}{\sqrt{2}}\times 2=\sqrt{2}+\sqrt{2}=2\sqrt{2}$

Go through options,

$\sec A+\operatorname{cosec} A=\sec 45°+\operatorname{cosec} 45°$

$=\sqrt{2}+\sqrt{2}=2\sqrt{2}$

300. (d) $\tan\theta+\cot\theta=2$

$\Rightarrow \tan\theta+\frac{1}{\tan\theta}=2$

$\Rightarrow \tan^2\theta+1=2\tan\theta$

$\Rightarrow \sec^2\theta=2\tan\theta \Rightarrow 1=2\sin\theta\cos\theta$

$\Rightarrow \sin 2\theta=\sin 90°$

$\Rightarrow 2\theta=90°$

$\Rightarrow \theta=45°$

Alternate Method:

$\tan\theta+\cot\theta=2$

$\Rightarrow \tan\theta+\frac{1}{\tan\theta}=2$

$\Rightarrow \tan\theta=1=\tan 45°$

$\Rightarrow \theta=45°$

301. (b) $\tan A+\cot A=2$

When we put A = 45° then,

$1+1=2$

$\therefore \tan^2 A+\cot^2 A$

$=\tan^2 45°+\cot^2 45°$

$=1+1=2$

302. (b) $\tan A+\cot A=2$

Put, A = 45°

$\tan 45°+\cot 45°=2$ (Satisfied)

$\therefore 7\tan^3 45°-6\cot^3 45°+8\sec^2 45°$

$=7\times 1-6\times 1+8\times(\sqrt{2})^2$

$=7-6+16=17$

303. (a) $\tan\theta+\cot\theta=2$

Let, $\theta=45°$ satisfy the condition.

$\therefore \tan^2\theta+\cot^3\theta+6\tan^3\theta\cot^2\theta$

$=\tan^2 45°+\cot^3 45°+6\tan^3 45°\cot^2 45°=1+1+6=8$

304. (c) $\cot\theta+\tan\theta=2$

Let, $\theta=45°$ satisfy the condition

$\therefore \tan^{12}\theta+\cot^{12}\theta+2\tan^5\theta\cot^7\theta$

$=\tan^{12}45°+\cot^{12}45°+2\tan^5 45°\cot^7 45°=1+1+2=4$

305. (b) $\tan\theta+\cot\theta=2$

Put $\theta=45°$

$\therefore \tan^{53}\theta+\cot^{38}\theta=1+1=2$

Alternate Method:

$\tan\theta+\cot\theta=2 \Rightarrow \tan\theta+\frac{1}{\tan\theta}=2$

We know,

$x+\frac{1}{x}=2$

$\Rightarrow x=1$

Hence, $\tan^{53}\theta+\cot^{38}\theta=1+1=2$

306. (a) If $\sin\theta+\operatorname{cosec}\theta=2$

Then, $\theta=90°$

$\therefore \sin^5\theta+\operatorname{cosec}^5\theta=2$

Alternate Method:

$\sin\theta+\operatorname{cosec}\theta=2$

$\Rightarrow \sin\theta+\frac{1}{\sin\theta}=2$

$\Rightarrow \frac{\sin^2\theta+1}{\sin\theta}=2$

$\Rightarrow \sin^2\theta-2\sin\theta+1=0$

$\Rightarrow (\sin\theta-1)^2=0 \Rightarrow \sin\theta=1$

$\Rightarrow \theta=90°$

$\therefore \sin^5\theta+\operatorname{cosec}^5\theta$

$=\sin^5 90°+\operatorname{cosec}^5 90°$

$=1+1=2$

307. (c) Given, $\cos\theta+\sec\theta=2$

We know,

$\cos 0°=1$ and $\sec 0°=1$

put $\theta=0°$

$\sin^6\theta+\cos^6\theta=(\sin 0°)^6+(\cos 0°)^6$

$=0+1=1$

308. (b) $\tan\theta+\cot\theta=6$

$\Rightarrow \tan^2\theta+\cot^2\theta=34$

$\because \left[x+\frac{1}{x}=n; x^2+\frac{1}{x^2}=n^2-2\right]$

309. (d) $\because \sin\theta+\operatorname{cosec}\theta=\sqrt{5}$

$\Rightarrow \sin\theta+\frac{1}{\sin\theta}=\sqrt{5}$

[If $x+\frac{1}{x}=k$,

Then $x^3+\frac{1}{x^3}=(k^3-3k)$]

$\therefore \sin^3\theta+\frac{1}{\sin^3\theta}=(\sqrt{5})^3-3\sqrt{5}$

$=5\sqrt{5}-3\sqrt{5}$

$\therefore \sin^3\theta+\operatorname{cosec}^3\theta=2\sqrt{5}$

310. (a) $\cos\theta+\sec\theta=\sqrt{3}$

$\cos\theta+\frac{1}{\cos\theta}=\sqrt{3}$

Cube both side

$\cos^3\theta+\frac{1}{\cos^3\theta}=(\sqrt{3})^3-3\sqrt{3}$

$=3\sqrt{3}-3\sqrt{3}=0$

311. (b) $\sec\theta+\cos\theta=32$

$\Rightarrow (\sec\theta+\cos\theta)^2=(32)^2$

$\Rightarrow \sec^2\theta+\cos^2\theta+2\sec\theta\cos\theta=1024$

$\Rightarrow \sec^2\theta+\cos^2\theta=1024-2$

$=1022$

Alternate Method:

If $x+\frac{1}{x}=k$

Then, $x^2+\frac{1}{x^2}=k^2-2=32^2-2$

$=1024-2=1022$

312. (c) $3x=\sec A, \frac{3}{x}=\tan A$

$\therefore \sec^2 A-\tan^2 A=1$

$\Rightarrow (3x)^2-\left(\frac{3}{x}\right)^2=1$

$\Rightarrow 9x^2-\frac{9}{x^2}=1$

$\Rightarrow 9\left(x^2-\frac{1}{x^2}\right)=1$

313. (b) $\sin\alpha-\operatorname{cosec}\theta=6$

Let, $\sin\alpha=x$

$x-\frac{1}{x}=6$

$x+\frac{1}{x}=\sqrt{\left(x-\frac{1}{x}\right)^2+4}=\sqrt{36+4}=\sqrt{40}$

$\sin\alpha+\operatorname{cosec}\theta=\sqrt{40}$

314. (c) $\sec x-\cos x=4$

$\Rightarrow \sec x+\cos x=\sqrt{(\sec x-\cos x)^2+4}$

$=\sqrt{20}=2\sqrt{5}$

$\left[\text{Using, if } x-\frac{1}{x}=a, x+\frac{1}{x}=\sqrt{a^2+4}\right]$

315. (c) $\sin x+\operatorname{cosec} x=2$

$\Rightarrow \sin x+\frac{1}{\sin x}=2$

$\Rightarrow \sin x=1$

$\Rightarrow x=90°$

$\therefore \sin^7 x+\operatorname{cosec}^7 x$

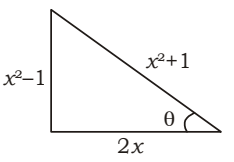
$=\sin^7 x+\frac{1}{\sin^7 x}=(1)^7+\frac{1}{(1)^7}=2$

316. (c) $\sec\theta+\tan\theta=x$

$\sec\theta-\tan\theta=\frac{1}{x}$

$2\tan\theta=x-\frac{1}{x}$

$\tan\theta=\frac{x^2-1}{2x}=\frac{P}{B}$



$\therefore H=\sqrt{P^2+B^2}=x^2+1$

$\therefore \sin\theta=\frac{P}{H}=\frac{x^2-1}{x^2+1}$

Note: $x^2-1, 2x, x^2+1$ are triplet

Where $x=2$

317. (c) If $\tan A+\sec A=k$

then, $\sec A-\tan A=\frac{1}{k}$

Now, $2\tan A=k-\frac{1}{k}$

$\tan A=\frac{k^2-1\to P}{2k\to B}$

$H=k^2+1$

$\Rightarrow \sin A=\frac{k^2-1}{k^2+1}$

318. (c) $\sec A - \tan A = p$ .....(i)

$\Rightarrow \sec A + \tan A = \frac{1}{p}$ .....(ii)

Adding equation (i) and (ii)

$\Rightarrow 2\sec A = p + \frac{1}{p}$

$\Rightarrow \sec A = \frac{p^2+1}{2p}$

319. (b) A.T.Q,

$\sec\theta + \tan\theta = a$ .....(i)

$\Rightarrow \sec\theta - \tan\theta = \frac{1}{a}$ .....(ii)

equ.[(i) + (ii)]

$2\sec\theta = a + \frac{1}{a} \Rightarrow \sec\theta = \frac{a^2+1}{2a}$

320. (c) $\sec\theta + \tan\theta = 5$

Then, $\sec\theta - \tan\theta = \frac{1}{5}$

Now, $\sec\theta = \frac{1}{2}\left(5+\frac{1}{5}\right)$

321. (b) $\sec\theta + \tan\theta = 5$

$\sec\theta - \tan\theta = \frac{1}{5}$

Now, $\sec\theta = \frac{1}{2}\left(5+\frac{1}{5}\right) = \frac{1}{2}\left(\frac{26}{5}\right)$

$= \frac{13}{5} \rightarrow$ Hypotenuse

$\frac{5}{} \rightarrow$ Base

By triplet = (5, 12, 13)

↓ ↓ ↓

B P H

$\sin\theta = \frac{12}{13}$

322. (a) Given, $\sec\theta + \tan\theta = 5$

We know, $\sec^2\theta - \tan^2\theta = 1$

$\Rightarrow (\sec\theta + \tan\theta)(\sec\theta - \tan\theta) = 1$

$\Rightarrow 5 \times (\sec\theta - \tan\theta) = 1$

$\Rightarrow \sec\theta - \tan\theta = \frac{1}{5}$

323. (b) Given,

$\sec\theta - \tan\theta = m$ .....(i)

We know,

$\sec^2\theta - \tan^2\theta = 1$

So, $\sec\theta + \tan\theta = \frac{1}{m}$ .....(ii)

Subtract Eq^n (i) from Eq^n (ii)

$\Rightarrow 2\tan\theta = \frac{1}{m} - m = \frac{1-m^2}{m}$

324. (d) Given,

$\sec\theta + \tan\theta = \sqrt{3}$ .....(i)

$\sec\theta - \tan\theta = \frac{1}{\sqrt{3}}$ .....(ii)

Adding Eq^n (i) + (ii)

$\Rightarrow 2\sec\theta = \sqrt{3} + \frac{1}{\sqrt{3}}$

$\Rightarrow \sec\theta = \frac{2}{\sqrt{3}} = \sec 30°$

$\Rightarrow \theta = 30°$

$\therefore \sin\theta + \cos\theta = \frac{1}{2} + \frac{\sqrt{3}}{2} = \left(\frac{\sqrt{3}+1}{2}\right)$

325. (d) Given,

$\text{Sec}^2\theta + \tan^2\theta = \frac{25}{18}$

We know, $\left(\sec^2\theta - \tan^2\theta = 1\right)$

$\therefore \sec^4\theta - \tan^4\theta$

$= \left(\text{Sec}^2\theta + \tan^2\theta\right)\left(\sec^2\theta - \tan^2\theta\right)$

$= \frac{25}{18} \times 1 = \frac{25}{18}$

326. (a) $\text{cosec}\theta + \cot\theta = \frac{3}{2}$ .....(i)

$\text{cosec}\theta - \cot\theta = \frac{2}{3}$ .....(ii)

On adding (i) and (ii), We get–

$2\text{cosec}\theta = \frac{2}{3} + \frac{3}{2} = \frac{4+9}{6} = \frac{13}{6}$

$\Rightarrow \text{cosec}\theta = \frac{13}{12}$

327. (a) We know that–

$\text{cosec}^2A - \cot^2A = 1$

$\Rightarrow (\text{cosec}A + \cot A)(\text{cosec}A - \cot A) = 1$

$\Rightarrow \text{cosec}A - \cot A = \frac{1}{\text{cosec}\,A + \cot A}$

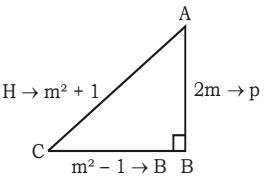
$\Rightarrow \text{cosec}A - \cot A = \frac{1}{m}$

328. (c) $\text{cosec}\theta + \cot\theta = m$ ...(i)

$\text{cosec}\theta - \cot\theta = \frac{1}{m}$ ....(ii)

On adding, $2\text{cosec}\theta = m + \frac{1}{m}$

$\Rightarrow \text{cosec}\theta = \frac{m^2+1}{2m}$



$\therefore \frac{m^2-1}{m^2+1} = \frac{B}{H} = \cos\theta$

Alternate Method:

Put $\theta = 45°$

$m = \text{cosec}\theta + \cot\theta = \sqrt{2}+1$

$\because \frac{m^2-1}{m^2+1} = \frac{(\sqrt{2}+1)^2-1}{(\sqrt{2}+1)^2+1} = \frac{3+2\sqrt{2}-1}{3+2\sqrt{2}+1}$

$= \frac{2+2\sqrt{2}}{4+2\sqrt{2}} = \frac{2(\sqrt{2}+1)}{2\sqrt{2}(\sqrt{2}+1)} = \frac{1}{\sqrt{2}}$

Go through options,

$\cos\theta = \cos 45° = \frac{1}{\sqrt{2}}$

329. (a) Given, $\text{cosec}A + \cot A = a\sqrt{b}$

$\Rightarrow \text{cosec}A = \frac{1}{2}\left(a\sqrt{b} + \frac{1}{a\sqrt{b}}\right) = \frac{a^2b+1}{2a\sqrt{b}}$

$\cot A = \frac{1}{2}\left(a\sqrt{b} - \frac{1}{a\sqrt{b}}\right) = \frac{a^2b-1}{2a\sqrt{b}}$

$\therefore \frac{\cot A}{\text{cosec}\,A} = \frac{\frac{a^2b-1}{2a\sqrt{b}}}{\frac{a^2b+1}{2a\sqrt{b}}}$

$\Rightarrow \cos A = \frac{a^2b-1}{a^2b+1}$

Alternate Method:

Put $A = 45°$

$\text{cosec}45° + \cot 45° = a\sqrt{b}$

$\Rightarrow \sqrt{2}+1 = a\sqrt{b}$

$\Rightarrow a^2b = 3 + 2\sqrt{2}$

$\therefore \frac{a^2b-1}{a^2b+1} = \frac{3+2\sqrt{2}-1}{3+2\sqrt{2}+1} = \frac{2+2\sqrt{2}}{4+2\sqrt{2}}$

$= \frac{2(\sqrt{2}+1)}{2\sqrt{2}(\sqrt{2}+1)} = \frac{1}{\sqrt{2}}$

Go through options,

$\cos A = \cos 45° = \frac{1}{\sqrt{2}}$

330. (b) $\sec x + \tan x = 5$

$\Rightarrow \sec x - \tan x = \frac{1}{5}$

$\text{cosec}\,y - \text{coty} = \frac{1}{3}$

$\Rightarrow \text{cosec}\,y + \text{coty} = 3$

ATQ,

$(\sec x + \text{cosec}\,y) - (\tan x - \text{coty})$

$= (\sec x + \text{cosec}\,y - \tan x + \text{coty})$

$= (\sec x - \tan x + \text{cosec}\,y + \cot y)$

$= \left(\frac{1}{5} + \frac{3}{1}\right) = \frac{16}{5} = 3.2$

331. (b) $2\sin^2\theta + 3\sin\theta - 2 = 0$

$\Rightarrow 2\sin^2\theta + 4\sin\theta - \sin\theta - 2 = 0$

$\Rightarrow 2\sin\theta(\sin\theta + 2) - 1(\sin\theta + 2) = 0$

$\Rightarrow (2\sin\theta - 1)(\sin\theta + 2) = 0$

$\sin\theta = -2$ **(not possible)**

so, $\sin\theta = \frac{1}{2}$

Then, $\theta = 30°$

Alternate Method:-

By option (b)

$2\sin^2 30° + 3\sin 30° - 2 = 0$

$\Rightarrow 2 \times \frac{1}{4} + 3 \times \frac{1}{2} - 2 = 0$

$\Rightarrow \left(\frac{1}{2} + \frac{3}{2}\right) - 2 = 0$

$\Rightarrow 2 - 2 = 0$

$\theta = 30°$ satisfy the above condition.

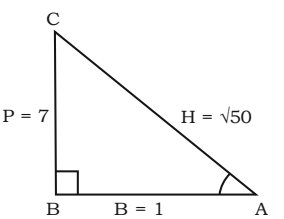
332. (c) $\tan^2A + 2\tan A - 63 = 0$

$\Rightarrow \tan^2A + 9\tan A - 7\tan A - 63 = 0$

$\Rightarrow \tan A(\tan A + 9) - 7(\tan A + 9) = 0$

$\Rightarrow (\tan A - 7)(\tan A + 9) = 0$

$\Rightarrow \tan A = \frac{7}{1} = \frac{P}{B}$



$\Rightarrow H = \sqrt{50}$

$\therefore 2\sin A + 5\cos A$

$= \left(2 \times \frac{7}{\sqrt{50}}\right) + \left(5 \times \frac{1}{\sqrt{50}}\right) = \frac{19}{\sqrt{50}}$

333. (c) $10\sin^2\theta + 6\cos^2\theta = 7$
$\Rightarrow 4\sin^2\theta + 6(\sin^2\theta + \cos^2\theta) = 7$
$\Rightarrow 4\sin^2\theta + 6 = 7 \Rightarrow 4\sin^2\theta = 1$
$\Rightarrow \sin\theta = \frac{1}{2} \Rightarrow \theta = 30°$
$\therefore \tan\theta = \tan30° = \frac{1}{\sqrt{3}}$

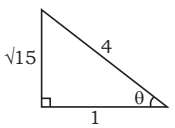
334. (a) $12\sin^2A + 2\cos^2A = 7$
$\Rightarrow 10\sin^2A + 2\sin^2A + 2\cos^2A = 7$
$\Rightarrow 10\sin^2A + 2 = 7 \Rightarrow 10\sin^2A = 5$
$\Rightarrow \sin^2A = \frac{1}{2} \Rightarrow \sin A = \frac{1}{\sqrt{2}}$
$\Rightarrow A = 45°, 135°$
$\therefore \cot A = \pm 1$

335. (c) $4\sin^2\theta - 4\sqrt{3}\sin\theta + 3 = 0$
$\Rightarrow 4\sin^2\theta - 2\sqrt{3}\sin\theta - 2\sqrt{3}\sin\theta + 3 = 0$
$\Rightarrow 2\sin\theta(2\sin\theta - \sqrt{3}) - \sqrt{3}(2\sin\theta - \sqrt{3}) = 0$
$\Rightarrow (2\sin\theta - \sqrt{3})^2 = 0$
$\Rightarrow 2\sin\theta = \sqrt{3}$
$\Rightarrow \sin\theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = 60°$

Alternate Method:
Put $\theta = 60°$
Only option (c) Satisfied.

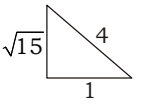
336. (c) $2\cos^2A + 3\cos A - 2 = 0$
$\Rightarrow 2\cos^2A + 4\cos A - \cos A - 2 = 0$
$\Rightarrow 2\cos A(\cos A + 2) - 1(\cos A + 2)$
$\Rightarrow (\cos A + 2)(2\cos A - 1)$
$\Rightarrow \cos A + 2 = 0, 2\cos A - 1 = 0$
$\cos A = -2$ (not possible)
So, we take $\cos A = \frac{1}{2} = \cos60°$
$\Rightarrow A = 60°$

337. (a) $4\sin^2\theta = 3(1 + \cos\theta)$
$\Rightarrow 4(1 - \cos^2\theta) = 3 + 3\cos\theta$
$\Rightarrow 4 - 4\cos^2\theta = 3 + 3\cos\theta$
$\Rightarrow 4\cos^2\theta + 3\cos\theta - 1 = 0$
$\Rightarrow 4\cos^2\theta + 4\cos\theta - \cos\theta - 1 = 0$
$\Rightarrow 4\cos\theta(\cos\theta + 1) - 1(\cos\theta + 1) = 0$
$\Rightarrow (4\cos\theta - 1)(\cos\theta + 1) = 0$
$\Rightarrow \cos\theta = \frac{1}{4}; -1$
we can't take -1 because cos between 0 to 90° is not–ve.



$\therefore \sqrt{15}\tan\theta + \frac{4}{\sqrt{15}}\sin\theta + 2\sec\theta$
$= \sqrt{15} \times \sqrt{15} + \frac{4}{\sqrt{15}} \times \frac{\sqrt{15}}{4} + 2 \times 4 = 24$

Alternate Method:
$4\sin^2\theta = 3(1 + \cos\theta)$
$4(1 - \cos^2\theta) = 3(1 + \cos\theta)$
$4(1 - \cos\theta)(1 + \cos\theta) = 3(1 + \cos\theta)$
$1 - \cos\theta = \frac{3}{4}$
$\cos\theta = \frac{1}{4}$



$\sqrt{15}\tan\theta + \frac{4}{\sqrt{15}}\sin\theta + 2\sec\theta$
$\sqrt{15} \times \sqrt{15} + \frac{4}{\sqrt{15}} \times \frac{\sqrt{15}}{4} + 2 \times \frac{4}{1}$
$= 24$

338. (a) $3\sec^4\theta + 8 = 10\sec^2\theta$
$\Rightarrow 3\sec^4\theta - 10\sec^2\theta + 8 = 0$
$\Rightarrow 3\sec^4\theta - 6\sec^2\theta - 4\sec^2\theta + 8 = 0$
$\Rightarrow 3\sec^2\theta(\sec^2\theta - 2) - 4(\sec^2\theta - 2) = 0$
$\Rightarrow (3\sec^2\theta - 4)(\sec^2\theta - 2) = 0$
$\Rightarrow 3\sec^2\theta - 4 = 0$
$\Rightarrow \sec\theta = \frac{2}{\sqrt{3}} \Rightarrow \theta = 30°$
and $\sec^2\theta - 2 = 0$
$\Rightarrow \sec^2\theta = 2 \Rightarrow \sec\theta = \sqrt{2} \Rightarrow \theta = 45°$
$\therefore \tan 30° = \frac{1}{\sqrt{3}}$; $\tan 45° = 1$

339. (c) $7\sin^2A + 3\cos^2A = 4$
$\Rightarrow 4\sin^2A + 3(\sin^2A + \cos^2A) = 4$
$\Rightarrow 4\sin^2A = 1$
$\Rightarrow \sin A = \frac{1}{2} \Rightarrow A = 30°$
$\therefore \cot A = \cot30° = \sqrt{3}$

340. (c) Given,
$2\operatorname{cosec}^2\theta + 3\cot^2\theta = 17$
$\Rightarrow 2\operatorname{cosec}^2\theta + 3\operatorname{cosec}^2\theta - 3 = 17$
$\Rightarrow 5\operatorname{cosec}^2\theta = 20$
$\Rightarrow \operatorname{cosec}\theta = \sqrt{4} = 2 = \operatorname{cosec}30°$
$\Rightarrow \theta = 30°$
Note: $\operatorname{cosec}^2\theta - \cot^2\theta = 1$

341. (d) $\sec^2\alpha + 4\cos^2\alpha = 4$
Put, $\theta = 45°$
$\Rightarrow \sec^2 45° + 4\cos^2 45° = 4$
$\Rightarrow (\sqrt{2})^2 + 4\left(\frac{1}{\sqrt{2}}\right)^2 = 4$
$\Rightarrow 2 + 4 \times \frac{1}{2} = 4$
$\Rightarrow 4 = 4$ [satisfy]

342. (a) $7\cos^2\theta + 5\sin^2\theta - 6 = 0$
$\Rightarrow 5(\sin^2\theta + \cos^2\theta) + 2\cos^2\theta - 6 = 0$
$\Rightarrow \cos\theta = \frac{1}{\sqrt{2}}$
$\Rightarrow \theta = 45°$
$\therefore 1 + \sqrt{\frac{\sec\theta + \tan\theta}{\sec\theta - \tan\theta}} = 1 + \sqrt{\frac{(\sec\theta + \tan\theta)^2}{\sec^2\theta - \tan^2\theta}}$
$= 1 + \sec\theta + \tan\theta = 1 + \sec45° + \tan45°$
$= 1 + \sqrt{2} + 1 = 2 + \sqrt{2}$

Alternate Method:
Let, $\theta = 45°$
$7\cos^2\theta + 5\sin^2\theta - 6 = 0$
$\Rightarrow 7 \times \frac{1}{2} + 5 \times \frac{1}{2} - 6 = 0$
$\Rightarrow 0 = 0$ (verified)
$\therefore 1 + \sqrt{\frac{\sec\theta + \tan\theta}{\sec\theta - \tan\theta}} = 1 + \sqrt{\frac{(\sec\theta + \tan\theta)^2}{\sec^2\theta - \tan^2\theta}}$
$= 1 + \sec\theta + \tan\theta = 1 + \sec45° + \tan45°$
$= 1 + \sqrt{2} + 1 = 2 + \sqrt{2}$

343. (d) $\sqrt{2}\sec^2\theta - 4\sec\theta + 2\sqrt{2} = 0$
$\Rightarrow \sqrt{2}\sec^2\theta - 2\sec\theta - 2\sec\theta + 2\sqrt{2} = 0$
$\Rightarrow \sqrt{2}\sec\theta(\sec\theta - \sqrt{2}) - 2(\sec\theta - \sqrt{2}) = 0$
$\Rightarrow \sec\theta = \sqrt{2} \Rightarrow \theta = 45°$
$\therefore \sin^2\theta + \tan^2\theta = \sin^2 45° + \tan^2 45°$
$= \frac{1}{2} + 1 = \frac{3}{2}$

Alternate Method:
$\sqrt{2}\sec^2\theta - 4\sec\theta + 2\sqrt{2} = 0$
Put $\theta = 45°$
$\Rightarrow \sqrt{2} \times (\sqrt{2})^2 - 4 \times \sqrt{2} + 2\sqrt{2} = 0$
$\Rightarrow 2\sqrt{2} - 4\sqrt{2} + 2\sqrt{2} = 0$
$\Rightarrow 4\sqrt{2} - 4\sqrt{2} = 0$ (Satisfied)
$\therefore \sin^2\theta + \tan^2\theta = \sin^2 45° + \tan^2 45°$
$= \frac{1}{2} + 1 = \frac{3}{2}$

344. (c) $\sin^2\theta - \cos^2\theta - 3\sin\theta + 2 = 0$
$\Rightarrow \sin^2\theta - (1 - \sin^2\theta) - 3\sin\theta + 2 = 0$
$\Rightarrow \sin^2\theta - 1 + \sin^2\theta - 3\sin\theta + 2 = 0$
$\Rightarrow 2\sin^2\theta - 3\sin\theta + 1 = 0$
$\Rightarrow 2\sin^2\theta - 2\sin\theta - \sin\theta + 1 = 0$
$\Rightarrow 2\sin\theta(\sin\theta - 1) - 1(\sin\theta - 1) = 0$
$\Rightarrow (2\sin\theta - 1)(\sin\theta - 1) = 0$
$\Rightarrow \sin\theta - 1 = 0 \Rightarrow \sin\theta = 1$
$\Rightarrow \theta = 90°$ [Not possible because $0° < \theta < 90°$]
$\Rightarrow 2\sin\theta - 1 = 0 \Rightarrow \sin\theta = \frac{1}{2}$
$\Rightarrow \theta = 30°$
$\therefore 1 + \sec\theta + \tan\theta = 1 + \sec30° + \tan30°$
$= 1 + \left(\frac{2}{\sqrt{3}}\right) + \left(\frac{1}{\sqrt{3}}\right) = \frac{(\sqrt{3} + 3)}{\sqrt{3}}$
$= 1 + \sqrt{3}$

345. (a) $6\cos^2\theta + \cos\theta - 2 = 0$
$\Rightarrow 6\cos^2\theta + 4\cos\theta - 3\cos\theta - 2 = 0$
$\Rightarrow 2\cos\theta(3\cos\theta + 2) - 1(3\cos\theta + 2) = 0$
$2\cos\theta = 1 \mid 3\cos\theta = -2$
$\cos\theta = \frac{1}{2} \mid \cos\theta = \frac{-2}{3}$
$\theta = 60°$ ($\because 0° < \theta < 90°$)
$\therefore (\operatorname{cosec}\theta + \cot\theta + \tan\theta)$
$= \frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}} + \sqrt{3}$
$= \frac{6}{\sqrt{3}} = 2\sqrt{3}$

346. (c) $8\cos^2\theta - 2\cos\theta - 3 = 0$
$\Rightarrow 8\cos^2\theta - 6\cos\theta + 4\cos\theta - 3 = 0$
$\Rightarrow 2\cos\theta(4\cos\theta - 3) + 1(4\cos\theta - 3) = 0$
$\Rightarrow (4\cos\theta - 3)(2\cos\theta + 1) = 0$
$\Rightarrow \cos\theta = \frac{3}{4}, -\frac{1}{2}$
$\Rightarrow \cos\theta = \frac{3}{4} \rightarrow B$
$P = \sqrt{4^2 - 3^2} = \sqrt{7}$
$\therefore 3\tan\theta + \operatorname{cosec}\theta + \cot\theta$
$= 3 \times \frac{\sqrt{7}}{3} + \frac{4}{\sqrt{7}} + \frac{3}{\sqrt{7}} = \frac{14}{\sqrt{7}} = 2\sqrt{7}$

347. (a) $7\sin^2\theta + 4\cos^2\theta = 5$

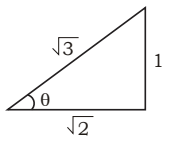
$\Rightarrow 3\sin^2\theta + 4\sin^2\theta + 4\cos^2\theta = 5$

$\Rightarrow 3\sin^2\theta + 4(\sin^2\theta + \cos^2\theta) = 5$

$\Rightarrow 3\sin^2\theta + 4 = 5 \Rightarrow 3\sin^2\theta = 1$

$\Rightarrow \sin\theta = \frac{1}{\sqrt{3}} \begin{matrix}\rightarrow P\\ \rightarrow H\end{matrix}$

$B = \sqrt{(\sqrt{3})^2 - 1^2} = \sqrt{2}$



$\therefore \frac{\sqrt{3}\sec\theta + \tan\theta}{\sqrt{2}\cot\theta - \sqrt{3}\cos\theta}$

$= \frac{\sqrt{3}\times\frac{\sqrt{3}}{\sqrt{2}} + \frac{1}{\sqrt{2}}}{\sqrt{2}\times\frac{\sqrt{2}}{1} - \sqrt{3}\times\frac{\sqrt{2}}{\sqrt{3}}} = \frac{\frac{4}{\sqrt{2}}}{\frac{2-\sqrt{2}}{1}}$

$= \frac{4}{\sqrt{2}\times\sqrt{2}(\sqrt{2}-1)}$

$= \frac{2(\sqrt{2}+1)}{(\sqrt{2}-1)(\sqrt{2}+1)} = 2(\sqrt{2}+1)$

348. (a) $\sin^2\theta - \cos^2\theta - 3\sin\theta + 2 = 0$

$\Rightarrow \sin^2\theta - 1 + \sin^2\theta - 3\sin\theta + 2 = 0$

$\Rightarrow 2\sin^2\theta - 3\sin\theta + 1 = 0$

$\Rightarrow \sin\theta = 1, \frac{1}{2}$

$0° < \theta < 90°$

$\sin\theta = \frac{1}{2} \Rightarrow \theta = 30°$

$\therefore \frac{1}{\sqrt{\sec\theta - \tan\theta}} = \frac{1}{\sqrt{\frac{2}{\sqrt{3}} - \frac{1}{\sqrt{3}}}} = \sqrt[4]{3}$

349. (d) $\sec^2\theta + (1-\sqrt{3})\tan\theta - (1+\sqrt{3}) = 0$

$\Rightarrow 1 + \tan^2\theta + (1-\sqrt{3})\tan\theta = 1 + \sqrt{3}$

$\Rightarrow \tan^2\theta + (1-\sqrt{3})\tan\theta - \sqrt{3} = 0$

$\Rightarrow \tan\theta = -1, \sqrt{3}$

Since θ is acute

$\Rightarrow \tan\theta = \sqrt{3} = \tan 60°$

$\Rightarrow \theta = 60°$

Alternate Method:

$\sec^2\theta + (1-\sqrt{3})\tan\theta - (1+\sqrt{3}) = 0$

by options

Put $\theta = 60°$

$4 + (1-\sqrt{3})\sqrt{3} - (1+\sqrt{3})$

$= 4 + \sqrt{3} - 3 - 1 - \sqrt{3} = 0$

350. (c) $4 - 2\sin^2\theta - 5\cos\theta = 0$

$\Rightarrow 4 - 2(1 - \cos^2\theta) - 5\cos\theta = 0$

$\Rightarrow 4 - 2 + 2\cos^2\theta - 5\cos\theta = 0$

$\Rightarrow 2\cos^2\theta - 5\cos\theta + 2 = 0$

$\Rightarrow 2\cos^2\theta - 4\cos\theta - \cos\theta + 2 = 0$

$\Rightarrow (2\cos\theta - 1)(\cos\theta - 2) = 0$

$\Rightarrow 2\cos\theta - 1 = 0$

$\cos\theta = \frac{1}{2} \Rightarrow \theta = 60°$

$\therefore \cos\theta + \tan\theta = \cos 60° + \tan 60°$

$= \frac{1}{2} + \sqrt{3} = \frac{1+2\sqrt{3}}{2}$

Alternate Method:-

$4 - 2\sin^2\theta - 5\cos\theta = 0$

Put $\theta = 60°$, $4 - 2\times\frac{3}{4} - 5\times\frac{1}{2}$

$= 4 - \frac{3}{2} - \frac{5}{2} = 4 - 4 = 0$

Go through options,

$\therefore \cos\theta + \tan\theta = \cos 60° + \tan 60°$

$= \frac{1}{2} + \sqrt{3} = \frac{1+2\sqrt{3}}{2}$

351. (d) $\frac{\sin^2\theta - 3\sin\theta + 2}{\cos^2\theta} = 1$

$\Rightarrow \text{Sin}^2\theta - 2\text{Sin}\theta - \text{Sin}\theta + 2 = \text{Cos}^2\theta$

$\Rightarrow (\text{Sin}\theta - 1)(\text{Sin}\theta - 2) = (1 - \text{Sin}\theta)(1 + \text{Sin}\theta)$

$\Rightarrow -\text{Sin}\theta + 2 = 1 + \text{Sin}\theta$

$\Rightarrow 2\text{Sin}\theta = 1$

$\Rightarrow \text{Sin}\theta = \frac{1}{2}$

$\Rightarrow \theta = 30°$

$\therefore \text{Cos}2\theta - \text{Sin}3\theta + \text{Cosec}2\theta$

$= \frac{1}{2} - 1 + \frac{2}{\sqrt{3}}$

$= \frac{2}{\sqrt{3}}\times\frac{\sqrt{3}}{\sqrt{3}} - \frac{1}{2} = \frac{2\sqrt{3}}{3} - \frac{1}{2} = \frac{4\sqrt{3}-3}{6}$

352. (d) $\frac{\sin^2\theta}{\cos^2\theta - 3\cos\theta + 2} = 1$

$\Rightarrow 1 - \cos^2\theta = \cos^2\theta - 3\cos\theta + 2$

$\Rightarrow 2\cos^2\theta - 3\cos\theta + 1 = 0$

$\Rightarrow 2\cos^2\theta - 2\cos\theta - \cos\theta + 1 = 0$

$\Rightarrow \cos\theta = 1$ and $\frac{1}{2}$ (since θ lies in 1st quadrant)

$\Rightarrow \cos\theta = \frac{1}{2} \Rightarrow \theta = 60°$

$\therefore \frac{\tan^2\frac{\theta}{2} + \sin^2\frac{\theta}{2}}{\tan\theta + \sin\theta} = \frac{\tan^2 30° + \sin^2 30°}{\tan 60° + \sin 60°}$

$= \frac{\left(\frac{1}{\sqrt{3}}\right)^2 + \left(\frac{1}{2}\right)^2}{\sqrt{3} + \frac{\sqrt{3}}{2}}$

$= \frac{\frac{1}{3}+\frac{1}{4}}{\frac{3\sqrt{3}}{2}} = \frac{\frac{7}{12}}{\frac{3\sqrt{3}}{2}} = \frac{7}{18\sqrt{3}}\times\frac{\sqrt{3}}{\sqrt{3}} = \frac{7\sqrt{3}}{54}$

353. (b) $\sin\theta\cos\theta = 2\cos^3\theta - \frac{1}{4}\cos\theta$

$\Rightarrow \sin\theta = 2\cos^2\theta - \frac{1}{4}$

$\Rightarrow \sin\theta = 2(1 - \sin^2\theta) - \frac{1}{4}$

$\Rightarrow 4\sin\theta = 8(1 - \sin^2\theta) - 1$

$\Rightarrow 8\sin^2\theta + 4\sin\theta - 7 = 0$

$\sin\theta = \frac{-4 \pm \sqrt{4^2 - 4\times 8\times(-7)}}{2\times 8}$

$= \frac{-4 \pm \sqrt{16 + 224}}{16}$

$= \frac{-4 \pm \sqrt{240}}{16} = \frac{-4 \pm \sqrt{15\times 16}}{16}$

$= \frac{+\sqrt{15}+1}{4}, \frac{\sqrt{15}-1}{4}$

$\because$ Angle is acute

So, $\sin\theta = \frac{\sqrt{15}-1}{4}$

354. (d) $[(\sin\theta\tan\theta + \cos\theta)^2 - 1]$

$= [\sin^2\theta\tan^2\theta + \cos^2\theta + 2\sin\theta.\tan\theta\cos\theta - 1]$

$= [\sin^2\theta\tan^2\theta + 2\sin\theta\tan\theta\cos\theta - \sin^2\theta]$

$= [\sin^2\theta\tan^2\theta + 2\sin^2\theta - \sin^2\theta]$

$= \sin^2\theta[\tan^2\theta + 1] = \sin^2\theta.\sec^2\theta$

$= \frac{\sin^2\theta}{\cos^2\theta} = \tan^2\theta$

Alternate Method:

put $\theta = 60°$

$[(\sin\theta.\tan\theta + \cos\theta)^2 - 1]$

$= \left[\left(\frac{\sqrt{3}}{2}\times\sqrt{3} + \frac{1}{2}\right)^2 - 1\right]$

$= [(2)^2 - 1] = 3$

from option (d)

$\tan^2\theta = \tan^2 60° = 3$ (verified)

355. (c) $(\text{cosec}\theta - \sin\theta)(\sec\theta - \cos\theta)(\tan\theta + \cot\theta)$

$= \left(\frac{1}{\sin\theta} - \sin\theta\right)\left(\frac{1}{\cos\theta} - \cos\theta\right)\left(\frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta}\right)$

$= \left(\frac{1-\sin^2\theta}{\sin\theta}\right)\left(\frac{1-\cos^2\theta}{\cos\theta}\right)\left(\frac{\sin^2\theta+\cos^2\theta}{\sin\theta\cos\theta}\right)$

$= \left(\frac{\cos^2\theta}{\sin\theta}\right)\left(\frac{\sin^2\theta}{\cos\theta}\right)\left(\frac{1}{\sin\theta\cos\theta}\right) = 1$

Alternate Method:-

Put $\theta = 45°$

$(\text{cosec}\theta - \sin\theta)(\sec\theta - \cos\theta)(\tan\theta + \cot\theta)$

$= (\text{cosec}45° - \sin45°)(\sec45° - \cos45°)(\tan45° + \cot45°)$

$= \left(\sqrt{2} - \frac{1}{\sqrt{2}}\right)\left(\sqrt{2} - \frac{1}{\sqrt{2}}\right)(1 + 1)$

$= \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} \times 2 = 1$

356. (b) $(\tan\theta + \cot\theta)(\sec\theta + \tan\theta)(1 - \sin\theta)$

$= \left(\frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta}\right)\left(\frac{1}{\cos\theta} + \frac{\sin\theta}{\cos\theta}\right)(1 - \sin\theta)$

$= \frac{1}{\sin\theta\cos\theta} \times \frac{(1+\sin\theta)(1-\sin\theta)}{\cos\theta}$

$= \frac{\cos^2\theta}{\sin\theta\cos^2\theta} = \frac{1}{\sin\theta} = \text{cosec}\theta$

Alternate Method:-

Put $\theta = 60°$

$(\tan60° + \cot60°)(\sec60° + \tan60°)(1 - \sin60°)$

$= \left(\sqrt{3} + \frac{1}{\sqrt{3}}\right)(2 + \sqrt{3})\left(1 - \frac{\sqrt{3}}{2}\right)$

$= \frac{4}{\sqrt{3}} \times (2 + \sqrt{3}) \times \frac{(2-\sqrt{3})}{2}$

$= \frac{4}{\sqrt{3}} \times \frac{1}{2} = \frac{2}{\sqrt{3}}$

Go through options,

$\text{cosec}60° = \frac{2}{\sqrt{3}}$

357. (d) $(\sin^6A + \cos^6A + 3\sin^2A\cos^2A) - (\sin^4A - \cos^4A + 1) \times \text{cosec}^2A$

We know that,

$(\sin^6A + \cos^6A = 1 - 3\sin^2A\cos^2A)$

$\Rightarrow (1 - 3\sin^2A\cos^2A + 3\sin^2A\cos^2A) - (\sin^2A - \cos^2A + 1) \times \text{cosec}^2A$

$= 1 - 2\sin^2A\,\text{cosec}^2A = -1$

Alternate Method:-

Put $A = 90° \Rightarrow 1 - (1 + 1) = -1$

358. (d) $4(\sin^6A + \cos^6A) - 6(\sin^4A + \cos^4A) + 8$

$= 4(1 - 3\sin^2A\cos^2A) - 6(1 - 2\sin^2A\cos^2A) + 8$

$= 4 - 12\sin^2A\cos^2A - 6 + 12\sin^2A\cos^2A + 8 = 6$

Alternate Method:

Put $A = 0°$

$\Rightarrow 4(\sin^60° + \cos^60°) - (\sin^40° + \cos^40°) + 8$

$= 4(0 + 1) - 6(0 + 1) + 8$

$= 4 - 6 + 8 = 6$

359. (a) $12(\sin^4\theta + \cos^4\theta) + 18(\sin^6\theta + \cos^6\theta) + 78\sin^2\theta\cos^2\theta$

$[\because \sin^4 + \cos^4\theta = 1 - 2\sin^2\theta.\cos^2\theta$

$\sin^6\theta + \cos^6\theta = 1 - 3\sin^2\theta.\cos^2\theta]$

So, $12(1 - 2\sin^2\theta.\cos^2\theta) + 18(1 - 3\sin^2\theta.\cos^2\theta) + 78\sin^2\theta.\cos^2\theta$

$= 12 - 24\sin^2\theta.\cos^2\theta + 18 - 54\sin^2\theta.\cos^2\theta + 78\sin^2\theta.\cos^2\theta$

$= 30 - 78\sin^2\theta.\cos^2\theta + 78\sin^2\theta.\cos^2\theta = 30$

Alternate Method:

Put $\theta = 0°$

$\Rightarrow 12(\sin^40° + \cos^40°) + 18(\sin^60° + \cos^60°) + 78\sin^20°\cos^20°$

$= 12 + 18 = 30$

360. (a) $5(\sin^4\theta + \cos^4\theta) + 3(\sin^6\theta + \cos^6\theta) + 19\sin^2\theta\cos^2\theta$

$= 5[(1)^2 - 2\sin^2\theta\cos^2\theta] + 3[1 - 3\sin^2\theta\cos^2\theta] + 19\sin^2\theta\cos^2\theta$

$= 5 - 10\sin^2\theta\cos^2\theta + 3 - 9\sin^2\theta\cos^2\theta + 19\cos^2\theta\sin^2\theta = 8$

Alternate Method:

Put $\theta = 90°$

$\Rightarrow 5(\sin^490° + \cos^490°) + 3(\sin^690° + \cos^690°) + 19\sin^290°\cos^290°$

$= 5 + 3 = 8$

361. (a) $(\cos^6\theta + \sin^6\theta - 1)(\tan^2\theta + \cot^2\theta + 2) + 3$

$= (1 - 3\sin^2\theta\cos^2\theta - 1)(1 + \tan^2\theta + 1 + \cot^2\theta) + 3$

$= (-3\sin^2\theta\cos^2\theta) \times (\sec^2\theta + \text{cosec}^2\theta) + 3$

$= -3\sin^2\theta\cos^2\theta \times \dfrac{1}{\sin^2\theta\cos^2\theta} + 3$

$= -3 + 3 = 0$

Alternate Method:

Put $\theta = 45°$,

$(\cos^645° + \sin^645° - 1)(\tan^245° + \cot^245° + 2) + 3$

$= \left[\left(\dfrac{1}{\sqrt{2}}\right)^6 + \left(\dfrac{1}{\sqrt{2}}\right)^6 - 1\right](1^2 + 1^2 + 2) + 3$

$= \left[\dfrac{1}{8} + \dfrac{1}{8} - 1\right](1 + 1 + 2) + 3$

$= \left[\dfrac{1+1-8}{8}\right] \times 4 + 3$

$= -\dfrac{3}{4} \times 4 + 3 = -3 + 3 = 0$

362. (a) $p\sin A - \cos A = 1$

Put $A = 90°$

$\Rightarrow p\sin90° - \cos90° = 1$

$\Rightarrow p \times 1 - 0 = 1$

$\Rightarrow p = 1$

Now,

$p^2 - (1 + p^2)\cos A$

$= 1 - (1 + 1^2)\cos90° = 1$

$- 2 \times 0 = 1 - 0 = 1$

363. (a) $\sin A + \cos A = \dfrac{1}{2\sqrt{2}}$

Squaring both sides,

$\sin^2A + \cos^2A + 2\sin A\cos A = \dfrac{1}{8}$

$\Rightarrow 2\sin A\cos A = \dfrac{1}{8} - 1 = \dfrac{-7}{8}$

$\Rightarrow 4\sin^2A\cos^2A = \dfrac{49}{64}$

$\therefore \sin^4A + \cos^4A = (\sin^2A + \cos^2A)^2$

$- 2\sin^2A\cos^2A$

$= 1 - 2\sin^2A\cos^2A$

$= 1 - \dfrac{49}{128} = \dfrac{79}{128}$

364. (d) $\sin A + \cos A = \sqrt{3}$

Squaring both side:-

$\Rightarrow 1 + 2\sin A\cos A = 3$

$\Rightarrow \sin A\cos A = 1$ .....(i)

$\therefore \tan A + \cot A + 2\sin A\cos A$

$= \dfrac{\sin A}{\cos A} + \dfrac{\cos A}{\sin A} + 2\sin A\cos A$

$= \dfrac{\sin^2A + \cos^2A}{\sin A\cos A} + 2\sin A\cos A$

$= \dfrac{1}{\sin A\cos A} + 2\sin A\cos A$

From equation (i) $= 1 + 2 = 3$

365. (c) $\sin A + \cos A = \dfrac{4}{3}$

Squaring both sides,

$\Rightarrow \sin^2A + \cos^2A + 2\sin A\cos A = \dfrac{16}{9}$

$\Rightarrow 1 + 2\sin A.\cos A = \dfrac{16}{9}$

$(\because \sin^2A + \cos^2A = 1)$

$\Rightarrow \sin A.\cos A = \dfrac{7}{18}$

$\therefore \tan A + \cot A = \dfrac{\sin A}{\cos A} + \dfrac{\cos A}{\sin A}$

$= \dfrac{\sin^2A + \cos^2A}{\cos A.\sin A} = \dfrac{1}{\sin A.\cos A}$

$= \dfrac{1}{\frac{7}{18}} = \dfrac{18}{7}$

366. (c) $\sin A - \cos A = \dfrac{\sqrt{3} - \sqrt{2}}{2}$

Squaring on both sides,

$\sin^2A + \cos^2A - 2\sin A\cos A = \dfrac{5 - 2\sqrt{6}}{4}$

$\Rightarrow 1 - \left(\dfrac{5 - 2\sqrt{6}}{4}\right) = 2\sin A.\cos A$

$\Rightarrow \dfrac{-1 + 2\sqrt{6}}{4} = 2\sin A.\cos A$

$\Rightarrow \sin A.\cos A = \dfrac{2\sqrt{6} - 1}{8}$

367. (c) Given, $(\sin\alpha + \cos\alpha) = \dfrac{2}{\sqrt{3}}$

Squaring both sides

$\Rightarrow \sin^2\alpha + \cos^2\alpha + 2\sin\alpha.\cos\alpha = \dfrac{4}{3}$

$\Rightarrow 1 + 2\sin\alpha.\cos\alpha = \dfrac{4}{3}$

$\Rightarrow 2\sin\alpha.\cos\alpha = \dfrac{1}{3} \Rightarrow \sin\alpha.\cos\alpha = \dfrac{1}{6}$

$\therefore \tan\alpha + \cot\alpha$

$= \dfrac{1}{\sin\alpha.\cos\alpha} = \dfrac{1}{\frac{1}{6}} = 6$

368. (b) $\sin\theta + \cos\theta = \dfrac{\sqrt{11}}{3}$

Squaring both sides

$\Rightarrow 1 + 2\sin\theta\cos\theta = \dfrac{11}{9}$

$\Rightarrow 2\sin\theta.\cos\theta = \dfrac{2}{9}$

$\Rightarrow \sin\theta.\cos\theta = \dfrac{1}{9}$

We know that,

$\tan\theta + \cot\theta = \dfrac{1}{\sin\theta.\cos\theta} = \dfrac{1}{\frac{1}{9}} = 9$

369. (a) $\tan\theta + \cot\theta$

$= \dfrac{\sin\theta}{\cos\theta} + \dfrac{\cos\theta}{\sin\theta} = \dfrac{1}{\sin\theta\cos\theta}$

Now,

$\sin\theta + \cos\theta = \dfrac{\sqrt{3} - 1}{2\sqrt{2}}$ .....(i)

squaring both sides in (i), we get

$\Rightarrow 1 + 2\sin\theta\cos\theta = \dfrac{4 - 2\sqrt{3}}{8}$

$\Rightarrow 2\sin\theta\cos\theta = \dfrac{4 - 2\sqrt{3} - 8}{8}$

$\Rightarrow 2\sin\theta\cos\theta = -\dfrac{(4 + 2\sqrt{3})}{8}$

$\Rightarrow \sin\theta\cos\theta = -\dfrac{(2 + \sqrt{3})}{8}$

$\therefore \tan\theta + \cot\theta = \left[\dfrac{-8}{2 + \sqrt{3}}\right]$

$= -8(2 - \sqrt{3}) = 8(\sqrt{3} - 2)$

370. (b) Given, $\sin t + \cos t = \frac{4}{5}$

Squaring both sides

$\Rightarrow (\sin t + \cos t)^2 = \left(\frac{4}{5}\right)^2$

$\Rightarrow \sin^2 t + \cos^2 t + 2\sin t.\cos t = \frac{16}{25}$

$\Rightarrow 1 + 2\sin t.\cos t = \frac{16}{25}$

$\Rightarrow 2\sin t.\cos t = \frac{16}{25} - 1$

$\Rightarrow \sin t.\cos t = \frac{-9}{50}$

371. (c) $f(p) = \sin^{p+2} x + \cos^{p+2} x$

$\therefore 6f(2) - 4f(4) + 10f(0) = 6(\sin^{2+2} x + \cos^4 x) - 4(\sin^6 x + \cos^6 x) + 10$

$= 6(1 - 2\sin^2 x \cos^2 x) - 4(1 - 3\sin^2 x \cos^2 x) + 10$

$= 6 - 12\sin^2 x \cos^2 x - 4 + 12\sin^2 x \cos x + 10 = 12$

372. (d) $x^2 - y^2 = (\tan A + \sin A)^2 - (\tan A - \sin A)^2$

$= 4\tan A \sin A$

$= 4 \times \frac{\sin A}{\cos A} \times \sin A = 4\frac{\sin^2 A}{\cos A}$

By options,

$4\sqrt{xy} = 4 \times \sqrt{\tan^2 A - \sin^2 A} = 4 \times \sqrt{\frac{\sin^2 A}{\cos^2 A} - \sin^2 A}$

$= 4 \times \sqrt{\frac{\sin^2 A(1 - \cos^2 A)}{\cos^2 A}}$

$= 4 \times \sqrt{\frac{\sin^4 A}{\cos^2 A}} = 4\frac{\sin^2 A}{\cos A}$

Alternate Method:

Put $A = 45°$

$x = 1 + \frac{1}{\sqrt{2}}, y = 1 - \frac{1}{\sqrt{2}}$

$x^2 - y^2 = 2\sqrt{2}$

By options,

$4\sqrt{xy} = 4 \times \sqrt{1 - \frac{1}{2}} = 4 \times \frac{1}{\sqrt{2}}$

$= 2\sqrt{2}$

373. (c) $\sin x + \sin y = a$

$\cos x + \cos y = b$

square both sides

$\sin^2 x + \sin^2 y + 2\sin x \sin y = a^2$ .....(i)

$\cos^2 x + \cos^2 y + 2\cos x \cos y = b^2$ .....(ii)

Adding (i) & (ii)

$a^2 + b^2 = \sin^2 x + \cos^2 x + \sin^2 y + \cos^2 y + 2[\sin x \sin y + \cos x \cos y]$

$\Rightarrow a^2 + b^2 = 1 + 1 + 2[\sin x \sin y + \cos x \cos y]$

$\Rightarrow \frac{a^2 + b^2 - 2}{2} = \sin x \sin y + \cos x \cos y$

Alternate Method:-

Put $x = y = 45°$

$\sin x + \sin y = a$

$\Rightarrow \sin 45° + \sin 45° = a \Rightarrow a = \sqrt{2}$

$\cos x + \cos y = b$

$\Rightarrow \cos 45° + \cos 45° = b \Rightarrow b = \sqrt{2}$

$\therefore \sin x \sin y + \cos x \cos y = \cos(x - y)$

$= \cos(45° - 45°) = \cos 0° = 1$

So, when we put value of $a = \sqrt{2}$ and $b = \sqrt{2}$ in option (c).

$\frac{a^2 + b^2 - 2}{2} = \frac{2 + 2 - 2}{2} = 1$

374. (a) $\cot^2\theta = 1 - e^2$

$\Rightarrow \operatorname{cosec}^2\theta - 1 = 1 - e^2$

$\Rightarrow \operatorname{cosec}^2\theta = 2 - e^2$

$\Rightarrow \operatorname{cosec}\theta = \sqrt{2 - e^2}$

$\therefore \operatorname{cosec}\theta + \cot^3\theta \sec\theta$

$= \frac{1}{\sin\theta} + \frac{\cos^2\theta}{\sin^3\theta}$

$= \frac{\cos^2\theta + \sin^2\theta}{\sin^3\theta} = \operatorname{cosec}^3\theta$

$= \left(\sqrt{2 - e^2}\right)^3 = (2 - e^2)^{\frac{3}{2}}$

375. (c) $\sec\theta + \tan^3\theta \operatorname{cosec}\theta$

$\Rightarrow \sec\theta + \tan^2\theta \sec\theta \Rightarrow \sec\theta(1 + \tan^2\theta)$

$\Rightarrow \sec^3\theta = ?$

Now,

$\tan^2\theta = 1 - a^2 \Rightarrow \sec^2\theta - 1 = 1 - a^2$

$\sec^2\theta = 2 - a^2$

$\sec^3\theta = (2 - a^2)^{\frac{3}{2}}$

Alternate Method:

$\tan^2\theta = 1 - a^2$

Put $\theta = 60° \Rightarrow 3 = 1 - a^2 \Rightarrow a^2 = -2$

$\therefore \sec\theta + \tan^3\theta \times \operatorname{cosec}\theta = \sec 60° + \tan^3 60° \operatorname{cosec} 60°$

$= 2 + 3\sqrt{3} \times \frac{2}{\sqrt{3}} = 2 + 6 = 8$

Now go through options,

$(2 - a^2)^{\frac{3}{2}} = (2 + 2)^{\frac{3}{2}} = 4^{\frac{3}{2}}$

$= (2^2)^{\frac{3}{2}} = 2^3 = 8$

376. (d) $\tan^2\alpha = 3 + Q^2$

$\Rightarrow \sec^2\alpha - 1 = 3 + Q^2$

$\Rightarrow \sec^2\alpha = 4 + Q^2$

$\Rightarrow \sec\alpha = (4 + Q^2)^{\frac{1}{2}}$

$\therefore \sec\alpha + \tan^3\alpha \operatorname{cosec}\alpha$

$= \sec\alpha + \tan^2\alpha \sec\alpha$

$= \sec\alpha[1 + \tan^2\alpha] = \sec^3\alpha = (4 + Q^2)^{\frac{3}{2}}$

Alternate Method:

$\tan^2\alpha = 3 + Q^2$

Put $\alpha = 45°$

$\tan^2 45° = 3 + Q^2$

$\Rightarrow Q^2 = -2$

$\therefore \sec 45° + \tan^3 45° \operatorname{cosec} 45°$

$= \sqrt{2} + 1 \times \sqrt{2} = 2\sqrt{2}$

Go through options,

$(4 + Q^2)^{\frac{3}{2}} = [4 + (-2)]^{\frac{3}{2}} = (2)^{\frac{3}{2}}$

$= 2\sqrt{2}$

377. (a) $(1 + \cot^2\theta)(1 + \tan^2\theta) \times (\sin\theta - \operatorname{cosec}\theta)(\cos\theta - \sec\theta)$

$= \operatorname{cosec}^2\theta \sec^2\theta$

$\times \left(\sin\theta - \frac{1}{\sin\theta}\right)\left(\cos\theta - \frac{1}{\cos\theta}\right)$

$= \frac{1}{\sin^2\theta} \times \frac{1}{\cos^2\theta} \times \frac{(\sin^2\theta - 1)}{\sin\theta} \times \frac{\cos^2\theta - 1}{\cos\theta}$

$= \frac{1}{\sin^2\theta\cos^2\theta} \times \frac{\cos^2\theta}{\sin\theta} \times \frac{\sin^2\theta}{\cos\theta}$

$= \sec\theta \operatorname{cosec}\theta$

Alternate Method:

$(1 + \cot^2\theta)(1 + \tan^2\theta)(\sin\theta - \operatorname{cosec}\theta)(\cos\theta - \sec\theta)$

Put $\theta = 45°$

$(1 + 1)(1 + 1)\left(\frac{1}{\sqrt{2}} - \sqrt{2}\right)\left(\frac{1}{\sqrt{2}} - \sqrt{2}\right)$

$= 4\left(\frac{1 - 2}{\sqrt{2}}\right)\left(\frac{1 - 2}{\sqrt{2}}\right) = 4\left(\frac{-1}{\sqrt{2}}\right)\left(\frac{-1}{\sqrt{2}}\right) = 2$

Go through options,

$\sec\theta \operatorname{cosec}\theta = \sec 45° \operatorname{cosec} 45°$

$= \sqrt{2} \times \sqrt{2} = 2$

378. (c) $x\sin^3\theta + y\cos^3\theta = \sin\theta\cos\theta$

$\Rightarrow x\sin^2\theta\sin\theta + y\cos^3\theta = \sin\theta\cos\theta$

$\Rightarrow y\cos\theta\sin^2\theta + y\cos^3\theta = \sin\theta\cos\theta$

$(\because x\sin\theta = y\cos\theta)$

$\Rightarrow y\cos\theta(\sin^2\theta + \cos^2\theta) = \sin\theta\cos\theta$

$y = \sin\theta$ .....(i)

$x\sin\theta = y\cos\theta$

$\Rightarrow x\sin\theta = \sin\theta\cos\theta$

$\Rightarrow x = \cos\theta$ .....(ii)

$\because x^2 + y^2 = \sin^2\theta + \cos^2\theta = 1$

Alternate Method:

Put $\theta = 45°$

$x\sin\theta = y\cos\theta \Rightarrow x = y$

$\therefore x\sin^3\theta + y\cos^3\theta = \sin\theta\cos\theta$

$\Rightarrow 2x\sin^3\theta = \sin^2\theta$

$\Rightarrow 2x\sin\theta = 1$

$\Rightarrow \frac{2x}{\sqrt{2}} = 1 \Rightarrow x = \frac{1}{\sqrt{2}} = y$

$\therefore x^2 + y^2 = \frac{1}{2} + \frac{1}{2} = 1$

SMART APPROACH:-

$x\sin^3\theta + y\cos^3\theta = \sin\theta.\cos\theta$

$\frac{x}{\cos\theta} \times \sin^2\theta + \frac{y}{\sin\theta} \times \cos^2\theta = 1$

So,

$\frac{x}{\cos\theta} = 1$ and $\frac{y}{\sin\theta} = 1$

$x = \cos\theta$ and $y = \sin\theta$

$x^2 + y^2 = \cos^2\theta + \sin^2\theta = 1$

379. (b) $a\sin X = b\cos X$ ... (i)

from (i) -

$b\cos X\sin^2 X + b\cos^3 X = \sin X\cos X$

$\Rightarrow b\cos X(\sin^2 X + \cos^2 X) = \sin X\cos X$

$\Rightarrow b = \sin X$

Similarly, $a = \cos X$

$\therefore a^2 + b^2 = \sin^2 X + \cos^2 X = 1$

Alternate Method:

$a\sin X = b\cos X$

Put, $X = 45°$

$\Rightarrow a = b$

$a\sin^3 X + b\cos^3 X = \sin X\cos X$

$\Rightarrow a \times \frac{1}{2\sqrt{2}} + b \times \frac{1}{2\sqrt{2}} = \frac{1}{2}$

$\Rightarrow a + b = \sqrt{2}$

$\Rightarrow a = \frac{1}{\sqrt{2}}, b = \frac{1}{\sqrt{2}}$

$\therefore a^2 + b^2 = \sin^2 X + \cos^2 X = 1$

SMART APPROACH:-

If $a\sin X = b\cos X$ and

$a\sin^3 X + b\cos^3 X = c\sin X\cos X$

Then, $a^2 + b^2 = c^2$

380. (d) $\left(\frac{\sin A}{1-\cos A}+\frac{1-\cos A}{\sin A}\right)\div\left(\frac{\cot^2 A}{1+\text{cosec } A}+1\right)$

$=\left(\frac{\sin A}{1-\cos A}\times\frac{1+\cos A}{1+\cos A}+\frac{1-\cos A}{\sin A}\right)$

$[\because 1-\cos^2 A=\sin^2 A]$

$=\left(\frac{\sin A(1+\cos A)}{\sin^2 A}+\frac{1-\cos A}{\sin A}\right)$

$=\frac{(1+\cos A)+(1-\cos A)}{\sin A}$

$=\frac{2}{\sin A}=2\text{cosec}A$ ....(i)

Now, $\left(\frac{\cot^2 A}{1+\text{cosec}A}+1\right)$

$(\because \text{cosec}^2 A-\cot^2 A=1)$

$=\frac{(\text{cosec } A-1)(\text{cosec } A+1)}{(1+\text{cosec } A)}+1$

$=\text{cosec}A-1+1=\text{cosec}A$ ....(ii)

$\therefore 2\text{ cosec}A\div\text{cosec}A=2$

Alternate Method:

Put $\theta=45°$

$\left(\frac{\sin 45°}{1-\cos 45°}+\frac{1-\cos 45°}{\sin 45°}\right)$

$\div\left(\frac{\cot^2 45°}{1+\text{cosec } 45°}+1\right)$

$=\left(\frac{\frac{1}{\sqrt2}}{1-\frac{1}{\sqrt2}}+\frac{1-\frac{1}{\sqrt2}}{\frac{1}{\sqrt2}}\right)\div\left(\frac{1}{1+\sqrt2}+1\right)$

$=\left(\frac{1}{\sqrt2}\times\frac{\sqrt2}{\sqrt2-1}+\frac{\sqrt2-1}{\sqrt2}\times\sqrt2\right)\div\left(\frac{1+1+\sqrt2}{\sqrt2+1}\right)$

$=(\sqrt2+1+\sqrt2-1)\div\left(\frac{2+\sqrt2}{\sqrt2+1}\right)=\frac{2\sqrt2}{\sqrt2}=2$

381. (b)

$\frac{\sec^2\theta(2+\tan^2\theta+\cot^2\theta)\div(\sin^2\theta-\tan^2\theta)}{(\text{cosec}^2\theta+\sec^2\theta)(1+\cot^2\theta)^2}$

$[\because \tan^2\theta-\sin^2\theta=\tan^2\theta\times\sin^2\theta]$

$1+\tan^2\theta=\sec^2\theta$

$1+\cot^2\theta=\text{cosec}^2\theta]$

$=\frac{\sec^2\theta(1+\tan^2\theta+1+\cot^2\theta)\div(\sin^2\theta\times\tan^2\theta)}{(\text{cosec}^2\theta+\sec^2\theta)(\text{cosec}^2)^2}$

$=\frac{\sec^2\theta(\sec^2\theta+\text{cosec}^2\theta)}{(\sin^2\theta\times\tan^2\theta)(\text{cosec}^2\theta+\sec^2\theta)(\text{cosec}^2\theta)^2}$

$=\frac{\sec^2\theta\times\sin^2\theta}{\tan^2\theta}$

$=\frac{\tan^2\theta}{\tan^2\theta}=1$

382. (d) $\frac{2+\tan^2 2A+\cot^2 2A}{\sec 2A.\text{cosec}2A}$

$=\frac{1+\tan^2 2A+1+\cot^2 2A}{\sec 2A.\text{ cosec}2A}$

$=\frac{\sec^2 2A+\text{cosec}^2 2A}{\sec 2A.\text{ cosec}2A}$

$=\frac{\sec^2 2A\times\text{cosec}^2 2A}{\sec 2A\times\text{cosec}2A}$

$=\sec 2A\text{ cosec}2A$

Note: $\sec^2\theta+\text{cosec}^2\theta=\sec^2\theta.\text{cosec}^2\theta$

383. (d)

$\left(\frac{\sin\theta-2\sin^3\theta}{2\cos^3\theta-\cos\theta}\right)^3\frac{1}{\tan\theta}-\sec^2\theta$

$=\left(\frac{\sin\theta}{\cos\theta}\frac{(1-2\sin^2\theta)}{(2\cos^2\theta-1)}\right)^3\times\frac{1}{\tan\theta}-\sec^2\theta$

$=(\tan\theta)^3\times\frac{1}{\tan\theta}-\sec^2\theta$

$=\tan^2\theta-\sec^2\theta=-1$

Note: $\cos 2\theta=1-2\sin^2\theta=2\cos^2\theta-1$

Alternate Method:

For $\theta=45°$, given value is not defined

$\therefore$ Put $\theta=30°$

$\left(\frac{\frac{1}{2}-2\times\frac{1}{8}}{2\times\left(\frac{\sqrt3}{2}\right)^3-\frac{\sqrt3}{2}}\right)^3\times\sqrt3-\left(\frac{2}{\sqrt3}\right)^2$

$=\left(\frac{\frac{1}{4}}{\frac{3\sqrt3-2\sqrt3}{4}}\right)^3\times\sqrt3-\frac{4}{3}$

$=\frac{1}{3\sqrt3}\times\sqrt3-\frac{4}{3}=\frac{1}{3}-\frac{4}{3}=-1$

384. (b)

$\frac{\sin A}{\cot A+\text{cosec } A}-\frac{\sin A}{\cot A-\text{cosec } A}-1$

$=\sin A\left(\frac{1}{\cot A+\text{cosec } A}-\frac{1}{\cot A-\text{cosec } A}\right)-1$

$=\sin A\left(\frac{\cot A-\text{cosec } A-\cot A-\text{cosec } A}{(\cot A+\text{cosec } A)(\cot A-\text{cosec } A)}\right)-1$

$=\sin A\left(\frac{-2\text{ cosec } A}{\cot^2 A-\text{cosec}^2 A}\right)-1$

$=\frac{-2\sin A\text{ cosec } A}{-(\text{cosec}^2 A-\cot^2 A)}-1=2-1=1$

Alternate Method:

Put $A=45°$

$\frac{\sin 45°}{\cot 45°+\text{cosec}45°}-\frac{\sin 45°}{\cot 45°-\text{cosec}45°}-1$

$=\frac{\frac{1}{\sqrt2}}{1+\sqrt2}-\frac{\frac{1}{\sqrt2}}{1-\sqrt2}-1$

$=\frac{1}{\sqrt2(\sqrt2+1)}+\frac{1}{\sqrt2(\sqrt2-1)}-1$

$=\frac{\sqrt2-1+\sqrt2+1}{\sqrt2}-1=\frac{2\sqrt2}{\sqrt2}-1=1$

385. (a) $\left[\frac{\tan^3\theta}{\sec^2\theta}+\frac{\cot^3\theta}{\text{cosec}^2\theta}+2\sin\theta\cos\theta\right]\div$

$(1+\text{cosec}^2\theta+\tan^2\theta)$

Consider, Numerator:-

$\frac{\frac{\sin^3\theta}{\cos^3\theta}}{\frac{1}{\cos^2\theta}}+\frac{\frac{\cos^3\theta}{\sin^3\theta}}{\frac{1}{\sin^2\theta}}+2\sin\theta\cos\theta$

$=\frac{\sin^3\theta}{\cos\theta}+\frac{\cos^3\theta}{\sin\theta}+2\sin\theta\cos\theta$

$=\frac{\sin^4\theta+\cos^4\theta+2\sin^2\theta\cos^2\theta}{\cos\theta\sin\theta}$

$=\frac{\left[\sin^2\theta+\cos^2\theta\right]^2}{\cos\theta\sin\theta}=\frac{1}{\cos\theta\sin\theta}$

Consider Denominator:

$1+\text{cosec}^2\theta+\tan^2\theta$

$=1+\frac{1}{\sin^2\theta}+\frac{\sin^2\theta}{\cos^2\theta}$

$=\frac{\sin^2\theta\cos^2\theta+\cos^2\theta+\sin^4\theta}{\sin^2\theta\cos^2\theta}$

$=\frac{\sin^2\theta(\cos^2\theta+\sin^2\theta)+\cos^2\theta}{\sin^2\theta\cos^2\theta}$

$=\frac{1}{\sin^2\theta\cos^2\theta}$

After combining,

$\frac{\frac{1}{\cos\theta\sin\theta}}{\frac{1}{\sin^2\theta\cos^2\theta}}=\frac{\sin^2\theta\cos^2\theta}{\sin\theta\cos\theta}=\sin\theta\cos\theta$

Alternate Method :-

Put $\theta=45°$

$\left(\frac{\tan^3 45°}{\sec^2 45°}+\frac{\cot^3 45°}{\text{cosec}^2 45°}+2\sin 45°\cos 45°\right)$

$\div(1+\text{cosec}^2 45°+\tan^2 45°)$

$=\left(\frac{1}{2}+\frac{1}{2}+\frac{2}{2}\right)\div(1+2+1)$

$=(1+1)\div 4=\frac{2}{4}=\frac{1}{2}$

Go through options,

$\sin\theta\cos\theta=\sin 45°\cos 45°=\frac{1}{2}$

386. (b)

$\frac{(1+\sec\theta\cos\text{ec}\theta)^2\times(\sec\theta-\tan\theta)^2(1+\sin\theta)}{(\sin\theta+\sec\theta)^2+(\cos\theta+\cos\text{ec}\theta)^2}$

$=\frac{\left(1+\frac{1}{\cos\theta\sin\theta}\right)^2\left(\frac{1}{\cos\theta}-\frac{\sin\theta}{\cos\theta}\right)^2(1+\sin\theta)}{\left(\sin\theta+\frac{1}{\cos\theta}\right)^2+\left(\cos\theta+\frac{1}{\sin\theta}\right)^2}$

$=\frac{\frac{(\cos\theta\sin\theta+1)^2}{\cos^2\theta\sin^2\theta}\left(\frac{1-\sin\theta}{\cos\theta}\right)^2(1+\sin\theta)}{\frac{(\cos\theta\sin\theta+1)^2}{\cos^2\theta}+\frac{(\cos\theta\sin\theta+1)^2}{\sin^2\theta}}$

$=\frac{\frac{(1-\sin^2\theta)(1-\sin\theta)}{\cos^2\theta\sin^2\theta\cos^2\theta}}{\frac{1}{\cos^2\theta\sin^2\theta}}=\frac{\cos^2\theta(1-\sin\theta)}{\cos^2\theta}$

$=1-\sin\theta$

Alternate Method:

$\frac{(1+\sec\theta\text{ cosec }\theta)^2(\sec\theta-\tan\theta)^2(1+\sin\theta)}{(\sin\theta+\sec\theta)^2+(\cos\theta+\text{cosec }\theta)^2}$

With the help of triplet 3, 4, 5 (P, B, H)

$\frac{\left(1+\frac{5}{4}\times\frac{5}{3}\right)^2\left(\frac{5}{4}-\frac{3}{4}\right)^2\left(1+\frac{3}{5}\right)}{\left(\frac{3}{5}+\frac{5}{4}\right)^2+\left(\frac{4}{5}+\frac{5}{3}\right)^2}$

$$= \frac{\left(1+\frac{25}{12}\right)^2 \times \frac{1}{4} \times \frac{8}{5}}{\left(\frac{12+25}{20}\right)^2 + \left(\frac{12+25}{15}\right)^2}$$

$$= \frac{\frac{1}{144} \times \frac{2}{5}}{\frac{1}{400} + \frac{1}{225}} = \frac{1}{72 \times 5} \times \frac{400 \times 225}{625}$$

$$= \frac{2}{5}$$

Go through option (b)

$\therefore$ $1 - \sin\theta = 1 - \frac{3}{5} = \frac{2}{5}$

387. (a) $\frac{\cot^3\theta}{\text{cosec}^2\theta} + \frac{\tan^3\theta}{\sec^2\theta} + 2\sin\theta\cos\theta$

$$= \frac{\cos^3\theta}{\sin\theta} + \frac{\sin^3\theta}{\cos\theta} + 2\sin\theta\cos\theta$$

$$= \frac{\cos^4\theta + \sin^4\theta + 2\cos^2\theta\sin^2\theta}{\sin\theta\cos\theta}$$

$$= \frac{(\cos^2\theta + \sin^2\theta)^2}{\sin\theta\cos\theta}$$

$$= \frac{1}{\sin\theta\cos\theta} = \text{cosec}\theta.\sec\theta$$

Alternate Method:-

$$\frac{\cot^3\theta}{\text{cosec}^2\theta} + \frac{\tan^3\theta}{\sec^2\theta} + 2\sin\theta\cos\theta$$

put $\theta = 30°$

$$= \frac{3\sqrt{3}}{4} + \frac{1\times 3}{3\sqrt{3}\times 4} + \frac{\sqrt{3}}{2} = \frac{9+1+6}{4\sqrt{3}} = \frac{4}{\sqrt{3}}$$

Go through options,

$$\text{cosec}30°.\sec 30° = 2 \times \frac{2}{\sqrt{3}} = \frac{4}{\sqrt{3}}$$

388. (c)

$$\sqrt{\sec^2\theta + \text{cosec}^2\theta}\left[\frac{\sin\theta(1+\cos\theta)}{1+\cos\theta-\sin^2\theta}\right]$$

$$= \frac{1}{\sin\theta\cos\theta}\left[\frac{\sin\theta(1+\cos\theta)}{1+\cos\theta-1+\cos^2\theta}\right]$$

$$= \frac{1}{\cos\theta}\left[\frac{\sin\theta(1+\cos\theta)}{\cos\theta(1+\cos\theta)}\right] = \sec^2\theta$$

Alternate Method:-

$$\sqrt{\sec^2\theta + \text{cosec}^2\theta}\left[\frac{\sin\theta(1+\cos\theta)}{1+\cos\theta-\sin^2\theta}\right]$$

put $\theta = 30°$

$$\sqrt{\left(\frac{4}{3}+4\right)}\left[\frac{\frac{1}{2}\left(1+\frac{\sqrt{3}}{2}\right)}{1+\frac{\sqrt{3}}{2}-\frac{1}{4}}\right]$$

$$= \frac{4}{\sqrt{3}}\left[\frac{\frac{1}{2}\left(1+\frac{\sqrt{3}}{2}\right)}{\frac{3}{4}+\frac{\sqrt{3}}{2}}\right] = \frac{4}{2\sqrt{3}}\left[\frac{1+\frac{\sqrt{3}}{2}}{\frac{\sqrt{3}}{2}\left(1+\frac{\sqrt{3}}{2}\right)}\right]$$

$$= \frac{4}{2\sqrt{3}} \times \frac{2}{\sqrt{3}} = \frac{4}{3} = \sec^2 30°$$

389. (a) $\frac{(2\sin A)(1+\sin A)}{1+\sin A+\cos A}$

$$= \frac{(2\sin A)(1+\sin A)(1+\sin A-\cos A)}{(1+\sin A+\cos A)(1+\sin A-\cos A)}$$

$$= \frac{(2\sin A)(1+\sin A)(1+\sin A-\cos A)}{(1+\sin A)^2-\cos^2 A}$$

$$= \frac{(2\sin A)(1+\sin A)(1+\sin A-\cos A)}{(1+\sin^2 A+2\sin A-\cos^2 A)}$$

$$= \frac{(2\sin A)(1+\sin A)(1+\sin A-\cos A)}{2\sin^2 A+2\sin A}$$

$$= \frac{(2\sin A)(1+\sin A)(1+\sin A-\cos A)}{2\sin A(\sin A+1)}$$

$= 1 + \sin A - \cos A$

Alternate Method:

$$\frac{(2\sin A)(1+\sin A)}{1+\sin A+\cos A}$$

Put $\theta = 90°$; $\frac{2\times 1(1+1)}{1+1+0} = \frac{2\times 2}{2} = 2$

Check option, option (a)

$1 + \sin A - \cos A = 1 + 1 - 0 = 2$

390. (d)

$$\frac{\sin\phi(1-\sin\phi)(\sin\phi+\cos\phi)(\sec\phi+\tan\phi)}{\sin\phi(1+\tan\phi)+\cos\phi(1+\cot\phi)}$$

$$= \frac{\sin\phi(1-\sin\phi)(\sin\phi+\cos\phi)\left(\frac{1}{\cos\phi}+\frac{\sin\phi}{\cos\phi}\right)}{\sin\phi\left(1+\frac{\sin\phi}{\cos\phi}\right)+\cos\phi\left(1+\frac{\cos\phi}{\sin\phi}\right)}$$

$$= \frac{\sin\phi(1-\sin\phi)(\sin\phi+\cos\phi)\left(\frac{(1+\sin\phi)}{\cos\phi}\right)}{\sin\phi\frac{(\cos\phi+\sin\phi)}{\cos\phi}+\cos\phi\left(\frac{\sin\phi+\cos\phi}{\sin\phi}\right)}$$

$$= \frac{\frac{\sin\phi(1-\sin^2\phi)}{\cos\phi}}{\frac{\sin^2\phi+\cos^2\phi}{\sin\phi\cos\phi}} = \frac{\frac{\sin\phi\cos^2\phi}{\cos\phi}}{\frac{1}{\sin\phi\cos\phi}}$$

$= \sin^2\phi\cos^2\phi$

Alternate Method:

$\phi = 45°$

$$\frac{\frac{1}{\sqrt{2}}\left(\frac{\sqrt{2}-1}{\sqrt{2}}\right)\times\sqrt{2}\times(\sqrt{2}+1)}{\frac{1}{\sqrt{2}}\times 2+\frac{1}{\sqrt{2}}\times 2} = \frac{1}{4}$$

Go through option $\sin^2\phi\cos^2\phi = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

Hence, option (d) satisfies.

391. (d)

$$\frac{1+\cos\theta-\sin^2\theta}{\sin\theta(1+\cos\theta)} \times \frac{\sqrt{\sec^2\theta+\text{cosec}^2\theta}}{\tan\theta+\cot\theta}$$

$$= \frac{\cos\theta(1+\cos\theta)}{\sin\theta(1+\cos\theta)} \times \frac{\sqrt{\frac{1}{\cos^2\theta}+\frac{1}{\sin^2\theta}}}{\frac{\sin\theta}{\cos\theta}+\frac{\cos\theta}{\sin\theta}}$$

$$= \cot\theta \times \frac{\frac{1}{\cos\theta\sin\theta}}{1} = \cot\theta$$

Alternate Method:-

Put $\theta = 30°$

$$\frac{1+\cos 30°-\sin^2 30°}{\sin 30°(1+\cos 30°)} \times \frac{\sqrt{\sec^2 30°+\text{cosec}^2 30°}}{\tan 30°+\cot 30°}$$

$$= \frac{1+\frac{\sqrt{3}}{2}-\frac{1}{4}}{\frac{1}{2}\left(1+\frac{\sqrt{3}}{2}\right)} \times \frac{\sqrt{\frac{4}{3}+4}}{\frac{1}{\sqrt{3}}+\sqrt{3}}$$

$$= \frac{4+2\sqrt{3}-1}{4} \times \frac{4}{2+\sqrt{3}} \times \frac{\sqrt{\frac{16}{3}}}{\frac{4}{\sqrt{3}}}$$

$$= \frac{3+2\sqrt{3}}{2+\sqrt{3}} \times \frac{4}{\sqrt{3}} \times \frac{\sqrt{3}}{4} = \frac{\sqrt{3}(2+\sqrt{3})}{2+\sqrt{3}} = \sqrt{3}$$

Go through options,

$\cot 30° = \sqrt{3}$

392. (d) $\frac{(\tan^6\theta-\sec^6\theta)+3\sec^2\theta.\tan^2\theta}{\tan^2\theta+\cot^2\theta+2}$

$$= \frac{(\tan^2\theta-\sec^2\theta)[\tan^4\theta+\sec^4\theta+\sec^2\theta\tan^2\theta]+3\sec^2\theta\tan^2\theta}{(1+\tan^2\theta+1+\cot^2\theta)}$$

$$= \frac{-[1+3\sec^2\theta\tan^2\theta]+3\sec^2\theta.\tan^2\theta}{(1+\tan^2\theta)(1+\cot^2\theta)}$$

$$= \frac{-1}{\sec^2\theta.\text{cosec}^2\theta} = -\sin^2\theta.\cos^2\theta$$

Alternate Method:-

Put $\theta = 45°$

$$\frac{\tan^6 45°-\sec^6 45°+3\sec^2 45°\tan^2 45°}{\tan^2 45°+\cot^2 45°+2}$$

$$= \frac{1-8+3\times 2}{1+1+2} = \frac{1-8+6}{4} = \frac{-1}{4}$$

Go through options,

$$-\sin^2 45°\cos^2 45° = \frac{-1}{2}\times\frac{1}{2} = \frac{-1}{4}$$

393. (a) $\frac{\sec^6\theta-\tan^6\theta-3\sec^2\theta\tan^2\theta+1}{\cos^4\theta-\sin^4\theta+2\sin^2\theta+2}$

$$= \frac{\left(\sec^2\theta-\tan^2\theta\right)^3+1}{\left(\cos^2\theta+\sin^2\theta\right)\left(\cos^2\theta-\sin^2\theta\right)+2\sin^2\theta+2}$$

$$= \frac{1+1}{\cos^2\theta-\sin^2\theta+2\sin^2\theta+2}$$

$$= \frac{2}{\cos^2\theta+\sin^2\theta+2} = \frac{2}{1+2} = \frac{2}{3}$$

Alternate Method:

Put $\theta = 0°$,

$$= \frac{1-0-0+1}{1-0+0+2} = \frac{2}{3}$$

394. (b)

$$\frac{\cos^4\theta-\sin^4\theta+2\sin^2\theta+3}{(\text{cosec}\,\theta+\cot\theta+1)(\text{cosec}\,\theta-\cot\theta+1)-2}$$

$$= \frac{(\cos^2\theta-\sin^2\theta)(\cos^2\theta+\sin^2\theta)+2\sin^2\theta+3}{(\text{cosec}\,\theta+1)^2-\cot^2\theta-2}$$

$$= \frac{\cos^2\theta-\sin^2\theta+2\sin^2\theta+3}{\text{cosec}^2\theta+1+2\text{cosec}\,\theta-\cot^2\theta-2}$$

$$= \frac{\cos^2\theta+\sin^2\theta+3}{2\text{cosec}\,\theta}$$

$$= \frac{1+3}{2\text{cosec}\,\theta} = \frac{4}{2\text{cosec}\,\theta} = 2\sin\theta$$

Alternate Method:-

Put $\theta = 60°$

$$\frac{\cos^4 60°-\sin^4 60°+2\sin^2 60°+3}{(\text{cosec}60°+\cot 60°+1)(\text{cosec}60°-\cot 60°+1)-2}$$

$$= \frac{\left(\frac{1}{16}-\frac{9}{16}\right)+2\times\frac{3}{4}+3}{\left(\frac{2}{\sqrt{3}}+\frac{1}{\sqrt{3}}+1\right)\left(\frac{2}{\sqrt{3}}-\frac{1}{\sqrt{3}}+1\right)-2}$$

$$= \frac{\frac{-1}{2}+\frac{3}{2}+3}{(\sqrt{3}+1)\frac{(\sqrt{3}+1)}{\sqrt{3}}-2}$$

$$= \frac{4\times\sqrt{3}}{4+2\sqrt{3}-2\sqrt{3}} = \frac{4\sqrt{3}}{4} = \sqrt{3}$$

Go through options,

$2\sin\theta = 2 \times \sin60° = 2 \times \frac{\sqrt{3}}{2} = \sqrt{3}$

395. (d)

$$\frac{(1-\sin\theta+\cos\theta)^2(1-\cos\theta)\sec^3\theta\,\text{cosec}^2\theta}{(\sec\theta-\tan\theta)\tan\theta+\cot\theta}$$

$$[1+\sin^2\theta-2\sin\theta+\cos^2\theta+2\cos\theta$$

$$= \frac{(1-\sin\theta)]\frac{1-\cos\theta}{\cos^3\theta}\frac{1}{\sin^2\theta}}{\left(\frac{1}{\cos\theta}-\frac{\sin\theta}{\cos\theta}\right)\left(\frac{\sin\theta}{\cos\theta}+\frac{\cos\theta}{\sin\theta}\right)}$$

$$[2(1-\sin\theta)+2\cos\theta(1-\sin\theta)]$$

$$= \frac{\frac{(1-\cos\theta)}{\cos^3\theta\sin^2\theta}}{\frac{(1-\sin\theta)}{\cos\theta}\left(\frac{1}{\cos\theta\sin\theta}\right)}$$

$$= \frac{\frac{2(1-\sin\theta)(1+\cos\theta)(1-\cos\theta)}{\cos^3\theta\sin^2\theta}}{\frac{(1-\sin\theta)}{\cos^2\theta\sin^2\theta}}$$

$$= 2\frac{\sin\theta}{\cos\theta} = 2\tan\theta$$

Alternate Method:-

Put $\theta = 45°$

$$\frac{(1-\sin45°+\cos45°)^2(1-\cos45°)\sec^3 45°\,\text{cosec}^2 45°}{(\sec45°-\tan45°)(\tan45°+\cot45°)}$$

$$= \frac{1\times\left(1-\frac{1}{\sqrt{2}}\right)\times2\sqrt{2}\times2}{(\sqrt{2}-1)\times2} = \frac{(\sqrt{2}-1)\times4\sqrt{2}}{\sqrt{2}\times2(\sqrt{2}-1)} = 2$$

Go through options,

$2\tan45° = 2 \times 1 = 2$

396. (b)

$$= \frac{(1-2\sin^2\theta.\cos^2\theta)(1+\cot\theta)\cos\theta}{(\sin^4\theta+\cos^4\theta)(1+\tan\theta)\,\text{cosec}\,\theta}-1$$

$$= \frac{(1-2\sin^2\theta.\cos^2\theta)(1+\tan\theta)\cos\theta}{\tan\theta(1-2\sin^2\theta.\cos^2\theta)(1+\tan\theta)\,\text{cosec}\theta}-1$$

$$= \frac{\sin\theta.\cos\theta\times\cos\theta}{\sin\theta}-1$$

$= -\sin^2\theta$

use $[\sin^4\theta + \cos^4\theta = 1 - 2\sin^2\theta.\cos^2\theta]$

Alternate Method:-

Put $\theta = 45°$

$$\frac{(1-2\sin^2 45°\cos^2 45°)(\cot45°+1)\cos45°}{(\sin^4 45°+\cos^4 45°)(1+\tan45°)\text{cosec}\,45°}-1$$

$$= \frac{\left(1-2\times\frac{1}{2}\times\frac{1}{2}\right)(2)\times\frac{1}{\sqrt{2}}}{\left(\frac{1}{4}+\frac{1}{4}\right)\times2\times\sqrt{2}}-1$$

$$= \frac{\left(1-\frac{1}{2}\right)\times\sqrt{2}}{\frac{1}{2}\times2\sqrt{2}}-1 \quad = \frac{\frac{1}{\sqrt{2}}}{\sqrt{2}}-1 = \frac{1}{2}-1 = \frac{-1}{2}$$

Go through options,

$-\sin^2 45° = \frac{-1}{2}$

397. (b) $\cot^4\theta - \cot^2\theta = 1$

$\Rightarrow \cot^4\theta = 1 + \cot^2\theta$

$\Rightarrow \cot^4\theta = \text{cosec}^2\theta$

$$\Rightarrow \frac{\cos^4\theta}{\sin^4\theta} = \frac{1}{\text{Sin}^2\theta}$$

$\Rightarrow \cos^4\theta = \sin^2\theta$ .....(i)

$\therefore \cos^4\theta + \cos^2\theta$

$= \sin^2\theta + \cos^2\theta = 1$ (From equation (i))

398. (b) We know, $\cot^2 A + 1 = \text{cosec}^2 A$

Given, $\cot^4 A + \cot^2 A = 1$

$\Rightarrow \cot^2 A (1 + \cot^2 A) = 1$

$\Rightarrow \cot^2 A = \sin^2 A$

$\Rightarrow \cos^2 A = \sin^4 A$ ..... (i)

$\therefore (\cos^2 A)^2 - 3\cos^2 A$

$= (1 - \sin^2 A)^2 - 3\cos^2 A$

$= (1 + \sin^4 A - 2\sin^2 A - 3\cos^2 A$

$= 1 + \cos^2 A - 2\sin^2 A - 3\cos^2 A$

$= 1 - 2(\sin^2 A + \cos^2 A) = 1 - 2 = -1$

399. (c) $\sin A + \sin^2 A = 1$

$\Rightarrow \sin A = 1 - \sin^2 A$

$\Rightarrow \sin A = \cos^2 A$

and $\cos^6 A = \sin^3 A$

$\cos^4 A = \sin^2 A$

$\therefore \cos^4 A + \cos^6 A = \sin^2 A + \sin^3 A$

$= \sin A (\sin A + \sin^2 A)$

$= \sin A \times 1 = \sin A$

400. (b) Given,

$\sin A + \sin^2 A = 1$

$\Rightarrow \sin A = 1 - \sin^2 A \Rightarrow \sin A = \cos^2 A$

$\therefore \cos^2 A + \cos^4 A = \cos^2 A + \sin^2 A = 1$

401. (a) $\sin x + \sin^2 x = 1$,

$\Rightarrow \sin x = 1 - \sin^2 x$

$\Rightarrow \sin x = \cos^2 x$

$\therefore \cos^8 x + 2\cos^6 x + \cos^4 x$

$= (\cos^4 x + \cos^2 x)^2 = (\sin^2 x + \cos^2 x)^2$

$= (1)^2 = 1$

402. (d) $\sin^4 x + \sin^2 x = 1$

$\Rightarrow (\sin^2 x \sin^2 x) = 1 - \sin^2 x$

$$\Rightarrow \sin^2 x = \frac{\cos^2 x}{\sin^2 x}$$

$\Rightarrow \sin^2 x = \cot^2 x$

$\therefore \cot^4 x + \cot^2 x = \sin^4 x + \sin^2 x = 1$

403. (d) $\cos A + \cos^2 A = 1$

$\Rightarrow \cos A = 1 - \cos^2 A$

$\Rightarrow \cos A = \sin^2 A$

Squaring on both side

$\Rightarrow \cos^2 A = \sin^4 A$

$\therefore \sin^2 A + \sin^4 A = \sin^2 A + \cos^2 A = 1$

404. (b) Given, $\cos A + \cos^2 A = 1$

$\Rightarrow 1 - \cos^2 A = \cos A$

$\Rightarrow \sin^2 A = \cos A$

$\therefore \sin^4 A + \sin^6 A$

$= \cos^2 A + \cos^3 A = \cos A(\cos A + \cos^2 A)$

$= \cos A \times 1 = \cos A$

405. (d) $\tan^4\theta + \tan^2\theta = 1$

$\Rightarrow \tan^2\theta (\tan^2\theta + 1) = 1$

$\Rightarrow \tan^2\theta \times \sec^2\theta = 1$

$$\Rightarrow \frac{\sin^2\theta}{\cos^2\theta}\times\frac{1}{\cos^2\theta} = 1$$

$\Rightarrow \sin^2\theta = \cos^4\theta$

$\therefore 11(\cos^4\theta + \cos^2\theta)$

$= 11(\sin^2\theta + \cos^2\theta) = 11 \times 1 = 11$

406. (a) Given,

$a\sin^{12}A + b\sin^{10}A + c\sin^8 A + \sin^6 A = 1$ ....(i)

$\cos A = \sin^2 A$

squaring both sides:

$\Rightarrow \cos^2 A = \sin^4 A \Rightarrow 1 - \sin^2 A = \sin^4 A$

$\Rightarrow 1 = \sin^4 A + \sin^2 A$

cubing both sides:

$\Rightarrow 1 = (\sin^4 A + \sin^2 A)^3$

$\Rightarrow 1 = \sin^{12} A + \sin^6 A + 3\sin^4 A \sin^2 A (\sin^4 A + \sin^2 A)$

$\Rightarrow 1 = \sin^{12} A + \sin^6 A + 3\sin^{10} A + 3\sin^8 A$

$\Rightarrow \sin^{12} A + 3\sin^{10} A + 3\sin^8 A + \sin^6 A = 1$ .....(ii)

On comparing (i) and (ii), we get a =1, b = 3 and c = 3

$\therefore$ Hence, a + b + c = 7

407. (a) Given,

$\sin^2\theta = \cos^3\theta$

Squaring both sides

$\sin^4\theta = \cos^6\theta$ .....(i)

$$\cot^2\theta - \cot^6\theta = \cot^2\theta - \frac{\cos^6\theta}{\sin^6\theta}$$

From equation (i),

$$= \cot^2\theta - \frac{\sin^4\theta}{\sin^6\theta} = \cot^2\theta - \frac{1}{\sin^2\theta}$$

$= \cot^2\theta - \text{cosec}^2\theta$

$= -\left(\text{cosec}^2\theta - \cot^2\theta\right) = -1$

408. (a) $$\sqrt{\frac{1+\sin\theta}{1-\sin\theta}} = \sqrt{\frac{(1+\sin\theta)^2}{1-\sin^2\theta}}$$

$$= \frac{1+\sin\theta}{\cos\theta} = \sec\theta + \tan\theta$$

409. (d) $$\frac{\sqrt{1-\sin A}}{\sqrt{1+\sin A}}$$

$$= \frac{\sqrt{(1-\sin A)\times(1-\sin A)}}{\sqrt{(1-\sin^2 A)}} = \frac{\sqrt{(1-\sin A)^2}}{\cos A}$$

$$= \frac{1-\sin A}{\cos A} = \sec A - \tan A$$

410. (a) Put $\theta = 30°$

$$\frac{(\sin30°+\cos30°-1)}{(\sin30°-\cos30°+1)}\times\frac{1+\sin30°}{\cos30°}$$

$$= \frac{\left(\frac{1}{2}+\frac{\sqrt{3}}{2}-1\right)}{\left(\frac{1}{2}-\frac{\sqrt{3}}{2}+1\right)}\times\frac{1+\frac{1}{2}}{\frac{\sqrt{3}}{2}}$$

$$= \frac{\sqrt{3}-1}{3-\sqrt{3}}\times\sqrt{3} = \frac{\sqrt{3}-1}{3-\sqrt{3}}\times\frac{3+\sqrt{3}}{3+\sqrt{3}}\times\sqrt{3}$$

$$= \frac{3\sqrt{3}+3-3-\sqrt{3}}{6}\times\sqrt{3} = \frac{2\sqrt{3}\times\sqrt{3}}{6} = 1$$

Alternate Method:

$$\frac{\sin\theta+\cos\theta-1}{\sin\theta-\cos\theta+1}\times\sqrt{\frac{1+\sin\theta}{1-\sin\theta}}$$

With the help of triplet $\underset{\downarrow}{3}, \underset{\downarrow}{4}, \underset{\downarrow}{5}$

P B H

$$\left(\frac{\frac{3}{5}+\frac{4}{5}-1}{\frac{3}{5}-\frac{4}{5}+1}\right)\times\sqrt{\frac{1+\frac{3}{5}}{1-\frac{3}{5}}} = \frac{2}{5}\times\frac{5}{4}\times2 = 1$$

411. (d) $\sqrt{\frac{\text{cosec}A+1}{\text{cosec}A-1}}+\sqrt{\frac{\text{cosec}A-1}{\text{cosec}A+1}}$

By rationalizing

$=\frac{\text{cosec}A+1}{\cot A}+\frac{\text{cosec}A-1}{\cot A}$

$=\frac{\text{cosec}A+1+\text{cosec}A-1}{\cot A}=\frac{2\text{cosec}A}{\cot A}$

$=2\times\frac{1}{\sin A}\times\frac{\sin A}{\cos A}=2\sec A$

Alternate Method:

$\sqrt{\frac{\text{cosec}A+1}{\text{cosec}A-1}}+\sqrt{\frac{\text{cosec}A-1}{\text{cosec}A+1}}$

Put A = 45°

$\sqrt{\frac{\sqrt{2}+1}{\sqrt{2}-1}}+\sqrt{\frac{\sqrt{2}-1}{\sqrt{2}+1}}$

$=\sqrt{2}+1+\sqrt{2}-1=2\sqrt{2}$

Option (d), $2\sec A=2\sqrt{2}$

412. (d) $\sqrt{\frac{1+\cos p}{1-\cos p}}$

$=\sqrt{\frac{(1+\cos p)\times(1+\cos p)}{(1-\cos p)\times(1+\cos p)}}$

$=\sqrt{\frac{(1+\cos p)^2}{1-\cos^2 p}}=\sqrt{\frac{(1+\cos p)^2}{\sin^2 p}}$

$=\frac{1+\cos p}{\sin p}=\frac{1}{\sin p}+\frac{\cos p}{\sin p}$

$=\text{cosec}p+\cot p$

413. (a) $\sqrt{\frac{1-\cos A}{1+\cos A}}+\sqrt{\frac{1+\cos A}{1-\cos A}}$

$=\sqrt{\frac{1-\cos^2 A}{(1+\cos A)^2}}+\sqrt{\frac{(1+\cos A)^2}{1-\cos^2 A}}$

$=\frac{\sin A}{1+\cos A}+\frac{1+\cos A}{\sin A}$

$=\frac{\sin^2 A+1+\cos^2 A+2\cos A}{\sin A(1+\cos A)}$

$=\frac{2(1+\cos A)}{\sin A(1+\cos A)}=2\,\text{cosec}A$

Alternate Method:

Put A = 60°

$\sqrt{\frac{1-\cos A}{1+\cos A}}+\sqrt{\frac{1+\cos A}{1-\cos A}}$

$=\sqrt{\frac{1-\frac{1}{2}}{1+\frac{1}{2}}}+\sqrt{\frac{1+\frac{1}{2}}{1-\frac{1}{2}}}$

$=\sqrt{\frac{1}{2}\times\frac{2}{3}}+\sqrt{\frac{3}{2}\times 2}$

$=\frac{1}{\sqrt{3}}+\sqrt{3}=\frac{4}{\sqrt{3}}$

$\therefore$ 2 cosecA = 2 × cosec60°

$=2\times\frac{2}{\sqrt{3}}=\frac{4}{\sqrt{3}}$

414. (b) $\sqrt{\frac{\sec\theta+1}{\sec\theta-1}}+\sqrt{\frac{\sec\theta-1}{\sec\theta+1}}$

Rationalising denominator,

$\frac{\sec\theta+1}{\tan\theta}+\frac{\sec\theta-1}{\tan\theta}=\frac{2\sec\theta}{\tan\theta}$

$=\frac{\frac{2}{\cos\theta}}{\frac{\sin\theta}{\cos\theta}}=2\text{cosec}\theta$

Alternate Method:

Put, θ = 60°

$\sqrt{\frac{\sec\theta+1}{\sec\theta-1}}+\sqrt{\frac{\sec\theta-1}{\sec\theta+1}}=\sqrt{\frac{2+1}{2-1}}+\sqrt{\frac{2-1}{2+1}}$

$=\sqrt{3}+\frac{1}{\sqrt{3}}=\frac{4}{\sqrt{3}}$

By options,

$2\text{cosec}\theta=2\text{cosec}60°=2\times\frac{2}{\sqrt{3}}=\frac{4}{\sqrt{3}}$

415. (b) Given

$\left\{\left(\frac{\sec\theta-1}{\sec\theta+1}\right)^n\right\}=(\text{cosec}\,\theta-\cot\theta)$

R.H.S = cosecθ − cotθ

$=\frac{1}{\sin\theta}-\frac{\cos\theta}{\sin\theta}$

$=\frac{1-\cos\theta}{\sin\theta}$ ..... (i)

$=\frac{1-\cos\theta}{\sin\theta}\times\frac{1+\cos\theta}{1+\cos\theta}$

$=\frac{1-\cos^2\theta}{\sin\theta(1+\cos\theta)}=\frac{\sin^2\theta}{\sin\theta(1+\cos\theta)}$

$=\frac{\sin\theta}{1+\cos\theta}$ ..... (ii)

Multiplying (i) and (ii), we get–

$\frac{1-\cos\theta}{1+\cos\theta}=(\text{cosec}\,\theta-\cot\theta)^2$

$\frac{\sec\theta-1}{\sec\theta+1}=(\text{cosec}\,\theta-\cot\theta)^2$

$\Rightarrow\left(\frac{\sec\theta-1}{\sec\theta+1}\right)^{\frac{1}{2}}=\text{cosec}\,\theta-\cot\theta$

Hence, $n=\frac{1}{2}=0.5$

Alternate Method:

Put θ = 45°

$\left\{\left(\frac{\sec\theta-1}{\sec\theta+1}\right)\right\}^n=\text{cosec}\theta-\cot\theta$

$\left(\frac{\sqrt{2}-1}{\sqrt{2}+1}\right)^n=(\sqrt{2}-1)$

By rationalizing we get

$\Rightarrow[(\sqrt{2}-1)(\sqrt{2}-1)]^n=(\sqrt{2}-1)$

$\Rightarrow[(\sqrt{2}-1)^2]^n=(\sqrt{2}-1)$

$\Rightarrow 2n=1$

$\Rightarrow n=\frac{1}{2}=0.5$

416. (d) $\sqrt{\frac{1-\tan A}{1+\tan A}}$

$=\sqrt{\frac{1-\frac{\sin A}{\cos A}}{1+\frac{\sin A}{\cos A}}}=\sqrt{\frac{\cos A-\sin A}{\cos A+\sin A}}$

$=\sqrt{\frac{\cos^2 A+\sin^2 A-2\sin A\cos A}{\cos^2 A-\sin^2 A}}$

$=\sqrt{\frac{1-2\sin A\cos A}{\cos^2 A-\sin^2 A}}=\sqrt{\frac{1-\sin 2A}{\cos 2A}}$

417. (c) $\left[\frac{\tan\theta+\sec\theta-1}{\tan\theta-\sec\theta+1}\right]$

[$\because \sec^2\theta-\tan^2\theta=1$]

$=\frac{[\tan\theta+\sec\theta-(\sec^2\theta-\tan^2\theta)]}{(\tan\theta-\sec\theta+1)}$

$=(\tan\theta+\sec\theta)\left[\frac{1-\sec\theta+\tan\theta}{\tan\theta-\sec\theta+1}\right]$

$=\tan\theta+\sec\theta=\frac{1+\sin\theta}{\cos\theta}$

Alternate Method:

Put θ = 60°

$\frac{\tan 60°+\sec 60°-1}{\tan 60°-\sec 60°+1}=\frac{\sqrt{3}+2-1}{\sqrt{3}-2+1}$

$=\frac{\sqrt{3}+1}{\sqrt{3}-1}$

By rationalizing we get

$=\frac{(\sqrt{3}+1)^2}{2}=\frac{4+2\sqrt{3}}{2}=2+\sqrt{3}$

Go through options,

$\frac{1+\sin 60°}{\cos 60°}=\frac{1+\frac{\sqrt{3}}{2}}{\frac{1}{2}}=2+\sqrt{3}$

418. (d) $\frac{\tan\theta-\sec\theta+1}{\tan\theta+\sec\theta-1}$

Put θ = 45°

$\Rightarrow\frac{1-\sqrt{2}+1}{1+\sqrt{2}-1}=\frac{\sqrt{2}(\sqrt{2}-1)}{\sqrt{2}}$

$=(\sqrt{2}-1)$

By option,

$\frac{\cos\theta}{1+\sin\theta}=\frac{\frac{1}{\sqrt{2}}}{1+\frac{1}{\sqrt{2}}}=\frac{1}{\sqrt{2}+1}=(\sqrt{2}-1)$

Alternate Method:

$\frac{\tan\theta-\sec\theta+1}{\tan\theta+\sec\theta-1}$

$=\frac{\tan\theta-\sec\theta+(\sec^2\theta-\tan^2\theta)}{\tan\theta+\sec\theta-1}$

$=\frac{(\sec\theta-\tan\theta)[-1+\sec\theta+\tan\theta]}{(\tan\theta+\sec\theta-1)}$

$=\sec\theta-\tan\theta=\frac{1}{\cos\theta}-\frac{\sin\theta}{\cos\theta}$

$=\frac{1-\sin\theta}{\cos\theta}=\frac{\cos\theta}{1+\sin\theta}$

419. (d) $\frac{\cot\theta+\operatorname{cosec}\theta-1}{\cot\theta-\operatorname{cosec}+1}$

put triplet = (3, 4, 5)

$\Rightarrow \frac{\frac{4}{3}+\frac{5}{3}-\frac{1}{1}}{\frac{4}{3}-\frac{5}{3}+\frac{1}{1}} = \frac{\frac{6}{3}}{\frac{2}{3}} = 3$

option (d), (cosecθ + cotθ)

$= \frac{5}{3}+\frac{4}{3} = \frac{9}{3} = 3$

Alternate Method:

$\frac{\cot\theta+\operatorname{cosec}\theta-1}{\cot\theta-\operatorname{cosec}\theta+1}$

Put θ = 45°

$\Rightarrow \frac{1+\sqrt{2}-1}{1-\sqrt{2}+1} = \frac{\sqrt{2}}{2-\sqrt{2}} = \frac{\sqrt{2}(2+\sqrt{2})}{4-2}$

$= \frac{2+\sqrt{2}}{\sqrt{2}} = \sqrt{2}+1$

Go through options,

cosecθ + cotθ = cosec45° + cot45°

$= \sqrt{2}+1$

420. (a) $\frac{\sin A}{1+\cos A} \times \frac{1-\cos A}{1-\cos A}$

$= \frac{\sin(1-\cos A)}{\sin^2 A} = \frac{1-\cos A}{\sin A}$

421. (a) $p = \frac{\sin A}{1+\cos A}$

$\Rightarrow p = \frac{\sin A}{1+\cos A} \times \frac{1-\cos A}{1-\cos A}$

$\Rightarrow p = \frac{\sin A(1-\cos A)}{(1-\cos^2 A)} = \frac{\sin A(1-\cos A)}{\sin^2 A}$

$\Rightarrow p = \frac{1-\cos A}{\sin A} \Rightarrow \frac{1}{p} = \frac{\sin A}{1-\cos A}$

422. (d) $\frac{\sin A}{1+\cos A} + \cot A$

Let, θ = 30°

$\Rightarrow \frac{\frac{1}{2}}{1+\frac{\sqrt{3}}{2}} + \sqrt{3} = \frac{1}{2+\sqrt{3}} + \sqrt{3}$

$= \frac{1+2\sqrt{3}+3}{2+\sqrt{3}} = \frac{4+2\sqrt{3}}{2+\sqrt{3}}$

$= \frac{2\left(2+\sqrt{3}\right)}{2+\sqrt{3}} = 2$

From options (d) cosec30° = 2

Alternate Method:-

$\frac{\sin A}{1+\cos A} + \frac{\cos A}{\sin A}$

$= \frac{\sin^2 A+\cos A+\cos^2 A}{\sin A(1+\cos A)}$

$= \frac{1+\cos A}{\sin A(1+\cos A)}$

$= \frac{1}{\sin A} = \operatorname{cosec} A.$

423. (d) $\frac{1+\sin\theta-\cos\theta}{1+\sin\theta+\cos\theta}$

Put θ = 60°

$\frac{1+\sin 60°-\cos 60°}{1+\sin 60°+\cos 60°}$

$= \frac{1+\frac{\sqrt{3}}{2}-\frac{1}{2}}{1+\frac{\sqrt{3}}{2}+\frac{1}{2}} = \frac{\frac{2+\sqrt{3}-1}{2}}{\frac{3+\sqrt{3}}{2}}$

$= \frac{1+\sqrt{3}}{\sqrt{3}(1+\sqrt{3})} = \frac{1}{\sqrt{3}}$

Now put the value of θ = 60° in option (d)

$\therefore \tan\frac{\theta}{2} = \tan 30° = \frac{1}{\sqrt{3}}$ (satisfied)

424. (c) $\frac{\sin\theta+\cos\theta-1}{\sin\theta-\cos\theta+1} \times \frac{\tan^2\theta(\operatorname{cosec}^2\theta-1)}{\sec\theta-\tan\theta}$

Put, θ = 45°

$\frac{\sin 45°+\cos 45°-1}{\sin 45°-\cos 45°+1} \times \frac{\tan^2 45°(\operatorname{cosec}^2 45°-1)}{\sec 45°-\tan 45°}$

$= \frac{\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{2}}-1}{1} \times \frac{1((\sqrt{2})^2-1)}{\sqrt{2}-1}$

$= \sqrt{2}-1 \times \frac{1}{\sqrt{2}-1} = 1$

425. (d) $\frac{1+\sin\theta}{\cos\theta}$

By rationalizing we get,

$\Rightarrow \frac{1+\sin\theta}{\cos\theta} \times \frac{(\cos\theta)}{(\cos\theta)} = \frac{\cos\theta}{1-\sin\theta}$

Alternate Method:

Put θ = 30°

$\frac{1+\sin 30°}{\cos 30°} = \frac{1+\frac{1}{2}}{\frac{\sqrt{3}}{2}} = \frac{3}{\sqrt{3}} = \sqrt{3}$

Go through options,

$\frac{\cos 30°}{1-\sin 30°} = \frac{\frac{\sqrt{3}}{2}}{1-\frac{1}{2}} = \sqrt{3}$

426. (a) (1 + cotA – cosecA) (1 + tanA + secA) – 3(sin²A + cos²A)

Put A = 45°

$\Rightarrow (1+1-\sqrt{2})(1+1+\sqrt{2})-3$

$= (2-\sqrt{2})(2+\sqrt{2})-3$

$= (2)^2-(\sqrt{2})^2-3$

$= 4-2-3 = -1$

427. (a) $(\tan\theta+\sec\theta+1)(\cot\theta-\operatorname{cosec}\theta+1)$

$= \left(\frac{\sin\theta}{\cos\theta}+\frac{1}{\cos\theta}+\frac{1}{1}\right)\left(\frac{\cos\theta}{\sin\theta}-\frac{1}{\sin\theta}+\frac{1}{1}\right)$

$= \left(\frac{\sin\theta+1+\cos\theta}{\cos\theta}\right)\left(\frac{\cos\theta-1+\sin\theta}{\sin\theta}\right)$

$= \frac{(\sin\theta+\cos\theta)^2-(1)^2}{\sin\theta\times\cos\theta} \quad \therefore\left[(a+b)(a-b)=a^2-b^2\right]$

$= \frac{\sin^2\theta+\cos^2\theta+2\sin\theta\cos\theta-1}{\sin\theta\times\cos\theta}$

$= \frac{1+2\sin\theta\cos\theta-1}{\sin\theta.\cos\theta} \quad \therefore\left[\sin^2\theta+\cos^2\theta=1\right]$

$= 2$

Alternate Method:

Put θ = 45°

(tan45° + sec45° + 1)(cot45° – cosec45° + 1)

$= (1+\sqrt{2}+1)(1-\sqrt{2}+1)$

$= (2+\sqrt{2})(2-\sqrt{2})$

$\therefore [a^2-b^2 = (a+b)(a-b)]$

$= 4-2 = 2$

428. (d) (1 + tanθ–secθ)(1 + cotθ + cosecθ)

$= \left(1+\frac{\sin\theta}{\cos\theta}-\frac{1}{\cos\theta}\right)\left(1+\frac{\cos\theta}{\sin\theta}+\frac{1}{\sin\theta}\right)$

$= \left(\frac{\cos\theta+\sin\theta-1}{\cos\theta}\right)\left(\frac{\sin\theta+\cos\theta+1}{\sin\theta}\right)$

$= \frac{1+2\sin\theta\ \cos\theta-1}{\cos\theta\sin\theta} = 2$

Alternate Method:

(1 + tanθ – secθ)(1 + cotθ + cosecθ)

Put θ = 45°

$\Rightarrow (1+1-\sqrt{2})(1+1+\sqrt{2})$

$= (2-\sqrt{2})(2+\sqrt{2}) = 4-2 = 2$

429. (a) (1 + cotA – cosecA) (1 + tanA + secA) –1

$= \left(1+\frac{\cos A}{\sin A}-\frac{1}{\sin A}\right)\left(1+\frac{\sin A}{\cos A}+\frac{1}{\cos A}\right)-1$

$= \left\{\frac{(\sin A+\cos A)^2-1^2}{\sin A.\cos A}\right\}-1$

$= \left\{\frac{\sin^2 A+\cos^2 A+2\sin A\cos A-1}{\sin A.\cos A}\right\}-1$

$[\sin^2 A+\cos^2 A = 1]$

$= \left\{\frac{1+2\sin A\cos A-1}{\sin A.\cos A}\right\}-1$

$= \left\{\frac{2\sin A\cos A}{\sin A.\cos A}\right\}-1$

$= 2-1 = 1$

Alternate Method:

Put A = 45°

(1 + cot45° – cosec45°) (1 + tan45° + sec45°) – 1

$= (1+1-\sqrt{2})(1+1+\sqrt{2})-1$

$= (2-\sqrt{2})(2+\sqrt{2})-1$

$= 2-1 = 1$

430. (b) (cosecA + cotA+1)(cosecA – cotA +1) –2cosec A

= (cosec A + cot A +1) (cosec A + 1 – cot A) – 2 cosec A

= (cosec A + 1)² – cot²A – 2 cosecA

= cosec²A + 1 + 2cosecA – cot²A – 2cosecA

= cosec² A – cot² A + 1

= 1 + 1 = 2

Alternate Method:

Put θ = 45°

(cosec45° + cot45° + 1) (cosec45° – cot45° + 1) – 2 cosec45°

$= (\sqrt{2}+2)(\sqrt{2}-1+1)-2\sqrt{2}$

$= (\sqrt{2}+2)\times\sqrt{2}-2\sqrt{2}$

$= 2+2\sqrt{2}-2\sqrt{2} = 2$

431. (c) $(1 + \cot A + \tan A)(\sin A - \cos A)\left(\frac{\sin A \cos A}{\sin^3 A - \cos^3 A}\right)$

$= \left(1 + \frac{\cos A}{\sin A} + \frac{\sin A}{\cos A}\right)(\sin A - \cos A) \times \left(\frac{\sin A \cos A}{(\sin A - \cos A)(\sin^2 A + \cos^2 A + \sin A \cos A)}\right)$

$= (\cos A \sin A + \cos^2 A + \sin^2 A)(\sin A - \cos A) \times \frac{1}{(\sin A - \cos A)(1 + \sin A \cos A)}$

$= \frac{(1 + \cos A \sin A)}{1 + \sin A \cos A} = 1$

432. (c) We know,

$2\sin^2 A = 1 - \cos 2A$

$2\cos^2 A = 1 + \cos 2A$

On solving option (c) we get

$= \frac{\sqrt{1 - \cos A}}{\sqrt{1 + \cos A}} = \frac{\sqrt{2\sin^2 \frac{A}{2}}}{\sqrt{2\cos^2 \frac{A}{2}}} = \tan \frac{A}{2}$

433. (a) $\sqrt{\frac{1 - \sin 3\theta}{1 + \sin 3\theta}}$

By rationalization

$= \sqrt{\frac{(1 - \sin 3\theta)^2}{1 - \sin^2 3\theta}} = \frac{1 - \sin 3\theta}{\cos 3\theta}$

$= \sec 3\theta - \tan 3\theta$

Alternate Method:

$\frac{\sqrt{1 - \sin 3\theta}}{\sqrt{1 + \sin 3\theta}}$

Put $\theta = 10°$, $\frac{\sqrt{1 - \sin 30°}}{\sqrt{1 + \sin 30°}} = \frac{\sqrt{\frac{1}{2}}}{\sqrt{\frac{3}{2}}} = \frac{1}{\sqrt{3}}$

Using option (a), $\sec 3\theta - \tan 3\theta$

$= \sec 30° - \tan 30°$

$= \frac{2}{\sqrt{3}} - \frac{1}{\sqrt{3}} = \frac{1}{\sqrt{3}}$ (Satisfied)

434. (d) $\tan \frac{\theta}{2} = \sqrt{\frac{2\sin^2 \frac{A}{2}}{2\cos^2 \frac{A}{2}}} = \sqrt{\frac{1 - \cos A}{1 + \cos A}}$

$= \frac{\sin \theta}{1 + \cos \theta}$

Alternate Method:

Put $\theta = 60°$, $\tan = \frac{60°}{2} = \tan 30° = \frac{1}{\sqrt{3}}$

Go through option

(d) $\frac{\sin \theta}{1 + \cos \theta} = \frac{\sin 60°}{1 + \cos 60°}$

$= \frac{\frac{\sqrt{3}}{2}}{1 + \frac{1}{2}} = \frac{\frac{\sqrt{3}}{2}}{\frac{3}{2}} = \frac{1}{\sqrt{3}}$

435. (c) $\frac{\tan A - \sin A}{\tan A + \sin A}$

$\therefore \tan A = \frac{\sin A}{\cos A} = \sin A \sec A$

$= \frac{\sin A \sec A - \sin A}{\sin A \sec A + \sin A}$

$= \frac{\sin A(\sec A - 1)}{\sin A(\sec A + 1)} = \frac{(\sec A - 1)}{(\sec A + 1)}$

436. (a) $\frac{1 - \cos \theta}{\sin \theta} = \frac{1}{5}$

$\Rightarrow \frac{1}{\sin \theta} - \frac{\cos \theta}{\sin \theta} = \frac{1}{5}$

$\Rightarrow \operatorname{cosec}\theta - \cot\theta = \frac{1}{5}$

$\therefore \frac{1 + \cos \theta}{\sin \theta} = \operatorname{cosec}\theta + \cot\theta = 5$

Alternate Method:

$\frac{1 - \cos \theta}{\sin \theta} = \frac{1}{5}$

$\frac{1 - \cos \theta}{\sin \theta} \times \frac{\sin \theta}{\sin \theta} = \frac{1}{5}$

$\frac{\sin \theta}{1 + \cos \theta} = \frac{1}{5}$

$\frac{1 + \cos \theta}{\sin \theta} = \frac{5}{1}$

437. (b) $\frac{\cos A}{1 + \tan A} - \frac{\sin A}{1 + \cot A}$

$= \frac{\cos A}{1 + \frac{\sin A}{\cos A}} - \frac{\sin A}{1 + \frac{\cos A}{\sin A}}$

$= \frac{\cos^2 A}{\cos A + \sin A} - \frac{\sin^2 A}{\sin A + \cos A}$

$= \frac{\cos^2 A - \sin^2 A}{\cos A + \sin A} = \cos A - \sin A$

Alternate Method:

$\frac{\cos A}{1 + \tan A} - \frac{\sin A}{1 + \cot A}$

Put $\theta = 45°$,

$\Rightarrow \frac{\cos 45°}{1 + \tan 45°} - \frac{\sin 45°}{1 + \cot 45°} = 0$

Go through options,

$\cos A - \sin A = \cos 45° - \sin 45°$

$= \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} = 0$

438. (d) $\frac{\cos A}{1 - \tan A} + \frac{\sin A}{1 - \cot A}$

$= \frac{\cos^2 A}{\cos A - \sin A} + \frac{\sin^2 A}{\sin A - \cos A}$

$= \frac{\cos^2 A - \sin^2 A}{\cos A - \sin A}$

$= \cos A + \sin A$

439. (c) $\frac{\cos A}{1 - \tan A} + \frac{\sin A}{1 - \cot A} - \sin A$

$= \frac{\cos A}{1 - \frac{\sin A}{\cos A}} + \frac{\sin A}{1 - \frac{\cos A}{\sin A}} - \sin A$

$= \frac{\cos^2 A}{\cos A - \sin A} + \frac{\sin^2 A}{\sin A - \cos A} - \sin A$

$= \frac{\cos^2 A}{\cos A - \sin A} - \frac{\sin^2 A}{\cos A - \sin A} - \sin A$

$= \frac{\cos^2 A - \sin^2 A}{\cos A - \sin A} - \sin A$

$= \cos A + \sin A - \sin A = \cos A$

Alternate Method:-

$\frac{\cos A}{1 - \tan A} + \frac{\sin A}{1 - \cot A} - \sin A$

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$\Rightarrow \frac{\frac{4}{5}}{1 - \frac{3}{4}} + \frac{\frac{3}{5}}{1 - \frac{4}{3}} - \frac{3}{5} \Rightarrow \frac{16}{5} - \frac{9}{5} - \frac{3}{5} = \frac{4}{5}$

Go through option

$\cos A = \frac{4}{5}$

440. (b) $1 + \sin\theta = m \cos\theta$

$\Rightarrow m = \frac{1 + \sin \theta}{\cos \theta} = \sqrt{\frac{(1 + \sin \theta)^2}{1 - \sin^2 \theta}}$

$= \sqrt{\frac{(1 + \sin \theta)(1 + \sin \theta)}{(1 - \sin \theta)(1 + \sin \theta)}} = \sqrt{\frac{1 + \sin \theta}{1 - \sin \theta}}$

$\Rightarrow \frac{m^2}{1} = \frac{1 + \sin \theta}{1 - \sin \theta}$

Using Componendo and dividendo,

$\frac{m^2 + 1}{m^2 - 1} = \frac{1}{\sin \theta} \Rightarrow \sin\theta = \frac{m^2 - 1}{m^2 + 1}$

Alternate Method:

$1 + \sin\theta = m \cos\theta$

Put $\theta = 45°$

$\Rightarrow 1 + \frac{1}{\sqrt{2}} = \frac{m}{\sqrt{2}} \Rightarrow \sqrt{2} + 1 = m$

Go through Options,

$\frac{m^2 - 1}{m^2 + 1} = \frac{(\sqrt{2} + 1)^2 - 1}{(\sqrt{2} + 1)^2 + 1} = \frac{3 + 2\sqrt{2} - 1}{3 + 2\sqrt{2} + 1}$

$= \frac{2 + 2\sqrt{2}}{4 + 2\sqrt{2}} = \frac{1}{\sqrt{2}}$

441. (d) $1 + \sqrt{\frac{\cot \theta + \cos \theta}{\cot \theta - \cos \theta}}$, $0° < \theta < 90°$

$= 1 + \sqrt{\frac{\frac{\cos \theta}{\sin \theta} + \cos \theta}{\frac{\cos \theta}{\sin \theta} - \cos \theta}}$

$= 1 + \sqrt{\frac{\frac{\cos \theta}{\sin \theta}(1 + \sin \theta)}{\frac{\cos \theta}{\sin \theta}(1 - \sin \theta)}} = 1 + \frac{\sqrt{1 + \sin \theta}}{\sqrt{1 - \sin \theta}}$

Rationalise, we get

$= 1 + \frac{\sqrt{1 + \sin \theta}}{\sqrt{1 - \sin \theta}} \times \frac{\sqrt{1 + \sin \theta}}{\sqrt{1 + \sin \theta}}$

$= 1 + \frac{\sqrt{(1 + \sin \theta)^2}}{\sqrt{(1 - \sin^2 \theta)}}$

$= 1 + \frac{1 + \sin \theta}{\cos \theta} = 1 + \frac{1}{\cos \theta} + \frac{\sin \theta}{\cos \theta}$

$= 1 + \sec\theta + \tan\theta$

Alternate Method:

Put $\theta = 45°$,

$1 + \sqrt{\frac{\cot 45° + \cos 45°}{\cot 45° - \cos 45°}}$

$= 1 + \sqrt{\frac{1 + \frac{1}{\sqrt{2}}}{1 - \frac{1}{\sqrt{2}}}} = 1 + \sqrt{\frac{\sqrt{2} + 1}{\sqrt{2} - 1}}$

$= 1 + \sqrt{2} + 1 = 2 + \sqrt{2}$

Go through options,

$1 + \sec\theta + \tan\theta = 1 + \sec 45° + \tan 45°$

$= 1 + \sqrt{2} + 1 = 2 + \sqrt{2}$

442. (c) $2-\sqrt{\frac{\cot\theta+\cos\theta}{\cot\theta-\cos\theta}}$

$= 2-\sqrt{\frac{\cos\theta(1+\sin\theta)}{\cos\theta(1-\sin\theta)}}$

$= 2-\sqrt{\left[\frac{1+\sin\theta}{\cos\theta}\right]^2}$

$= 2 - \sec\theta - \tan\theta$

Alternate Method:

Put $\theta = 45°$,

$2-\sqrt{\frac{\cot45°+\cos45°}{\cot45°-\cos45°}}$

$= 2-\sqrt{\frac{1+\frac{1}{\sqrt2}}{1-\frac{1}{\sqrt2}}} = 2-\sqrt{\frac{\sqrt2+1}{\sqrt2-1}} = 2-(\sqrt2+1)$

$= 2-\sqrt2-1 = 1-\sqrt2$

Go through options,

$2 - \sec\theta - \tan\theta = 2-\sqrt2-1 = 1-\sqrt2$

443. (d) $1 + 2\tan^2\theta + 2\sin\theta\sec^2\theta$

$= 1+\frac{2\sin^2\theta}{\cos^2\theta}+\frac{2\sin\theta}{\cos^2\theta}$

$= \frac{\cos^2\theta+2\sin^2\theta+2\sin\theta}{\cos^2\theta}$

$= \frac{1+\sin^2\theta+2\sin\theta}{\cos^2\theta}$

$= \frac{(\sin\theta+1)^2}{(1-\sin^2\theta)} = \frac{1+\sin\theta}{1-\sin\theta}$

Alternate Method:-

$1 + 2\tan^2\theta + 2\sin\theta\sec^2\theta$

Put $\theta = 30°$,

$1+\frac{2}{3}+\frac{2}{2}\times\frac{4}{3} = \frac{9}{3} = 3$

from given options

$\frac{1+\sin30°}{1-\sin30°} = \frac{1+\frac12}{1-\frac12} = \frac{3/2}{1/2} = 3$

444. (c) $1+ 2\tan^2\theta + 2\sin\theta\sec^2\theta = \frac{a}{b}$

$\Rightarrow 1 + 2\frac{\sin^2\theta}{\cos^2\theta}+\frac{2\sin\theta}{\cos^2\theta} = \frac{a}{b}$

$\Rightarrow \frac{\cos^2\theta+2\sin^2\theta+2\sin\theta}{\cos^2\theta} = \frac{a}{b}$

$\Rightarrow \frac{\cos^2\theta+\sin^2\theta+\sin^2\theta+2\sin\theta}{\cos^2\theta} = \frac{a}{b}$

$\Rightarrow \frac{1+\sin^2\theta+2\sin\theta}{\cos^2\theta} = \frac{a}{b}$

$\Rightarrow \frac{(1+\sin\theta)^2}{1-\sin^2\theta} = \frac{a}{b} \Rightarrow \frac{1+\text{Sin}\,\theta}{1-\text{Sin}\,\theta} = \frac{a}{b}$

$\therefore \frac{a+b}{a-b} = \frac{1}{\sin\theta} = \text{cosec}\,\theta$

Alternate Method:-

Put $\theta = 60°$

$1 + 2\tan^2 60° + 2\sin60°\sec^2 60° = \frac{a}{b}$

$\Rightarrow 1 + 6 + 2\times\frac{\sqrt3}{2}\times4 = \frac{a}{b}$

$\Rightarrow \frac{7+4\sqrt3}{1} = \frac{a}{b}$

$\frac{a+b}{a-b} = \frac{7+4\sqrt3+1}{7+4\sqrt3-1} = \frac{8+4\sqrt3}{6+4\sqrt3} = \frac{4(2+\sqrt3)}{2\sqrt3(\sqrt3+2)}$

$= \frac{2}{\sqrt3} = \text{cosec}60°$

445. (d) $\left(\frac{1+\sec^2A}{1+\cos^2A}\right)\left(\frac{1+\sin^2A}{1+\text{cosec}^2A}\right)$

$= \left(\frac{1+\sec^2A}{1+\frac{1}{\sec^2A}}\right)\times\left(\frac{1+\sin^2A}{1+\frac{1}{\sin^2A}}\right)$

$= \left[\frac{(1+\sec^2A)}{(\sec^2A+1)}\times\sec^2A\right]$

$\times\left[\frac{(1+\sin^2A)}{(\sin^2A+1)}\times\sin^2A\right]$

$= \sin^2A\sec^2A = \tan^2A$

Alternate Method:

Put $\theta = 45°$,

$\left(\frac{1+\sec^2 45°}{1+\cos^2 45°}\right)\left(\frac{1+\sin^2 45°}{1+\text{cosec}^2 45°}\right)$

$= \left(\frac{1+2}{1+\frac12}\right)\left(\frac{1+\frac12}{1+2}\right) = \left(\frac{3\times2}{3}\right)\left(\frac{3}{2\times3}\right) = 1$

Go through options,

$\tan^2\theta = \tan^2 45° = 1$

446. (c) $\frac{\sin\theta+\cos\theta}{\sin\theta-\cos\theta}+\frac{\sin\theta-\cos\theta}{\sin\theta+\cos\theta}$

We know,

$\frac{a+b}{a-b}+\frac{a-b}{a+b} = \frac{2(a^2+b^2)}{a^2-b^2}$

$\Rightarrow \frac{2(\sin^2\theta+\cos^2\theta)}{\sin^2\theta-\cos^2\theta} = \frac{2}{\sin^2\theta-\cos^2\theta}$

447. (a) $(2\cos^2\theta-1)\left[\frac{1+\tan\theta}{1-\tan\theta}+\frac{1-\tan\theta}{1+\tan\theta}\right]$

We know,

$\frac{a+b}{a-b}+\frac{a-b}{a+b} = \frac{2(a^2+b^2)}{a^2-b^2}$

$= (2\cos^2\theta-1)\left[\frac{2(1+\tan^2\theta)}{(1-\tan^2\theta)}\right]$

$= \cos2\theta\times2\times\frac{1}{\cos2\theta} = 2$

$\left[\begin{array}{l}2\cos^2\theta-1=\cos2\theta\\ \frac{1-\tan^2\theta}{1+\tan^2\theta}=\cos2\theta\end{array}\right]$

Alternate Method:

$(2\cos^2\theta-1)\left[\frac{1+\tan\theta}{1-\tan\theta}+\frac{1-\tan\theta}{1+\tan\theta}\right]$

Put $\theta = 30°$

$\left(2\times\frac34-1\right)\left\{\frac{1+\frac{1}{\sqrt3}}{1-\frac{1}{\sqrt3}}+\frac{1-\frac{1}{\sqrt3}}{1+\frac{1}{\sqrt3}}\right\}$

$= \left(\frac32-1\right)\left(\frac{\sqrt3+1}{\sqrt3-1}+\frac{\sqrt3-1}{\sqrt3+1}\right)$

$= \frac12\times\left(\frac{(\sqrt3+1)^2+(\sqrt3-1)^2}{2}\right)$

$= \frac{4+2\sqrt3+4-2\sqrt3}{4} = 2$

448. (b) $\frac{1+\tan A}{\text{cosec}A}+\frac{1+\cot A}{\sec A}$

$= \frac{1+\frac{\sin A}{\cos A}}{\frac{1}{\sin A}}+\frac{1+\frac{\cos A}{\sin A}}{\frac{1}{\cos A}}$

$= \frac{\cos A+\sin A}{\cos A}\times\sin A+\frac{\sin A+\cos A}{\sin A}\times\cos A$

$= (\cos A+\sin A)\left(\frac{\sin A}{\cos A}+\frac{\cos A}{\sin A}\right)$

$= \frac{\cos A+\sin A}{\sin A\ \cos A} = \frac{1}{\sin A}+\frac{1}{\cos A}$

$= \text{cosec}A + \sec A$

Alternate Method:

$\frac{1+\tan A}{\text{cosec}A}+\frac{1+\cot A}{\sec A}$

Put $A = 45°$

$\Rightarrow \frac{1+1}{\sqrt2}+\frac{1+1}{\sqrt2} \Rightarrow \frac{2}{\sqrt2}+\frac{2}{\sqrt2} = \frac{4}{\sqrt2} = 2\sqrt2$

$(\sec A + \text{cosec}A) = 2\sqrt2$

Hence, Option (b) satisfied

449. (c) $\text{cosec}^4A\ (1 - \cos^4A) - 2\cot^2A - 1$

$= \frac{\text{cosec}^2A}{\sin^2A}\ (1 - \cos^2A)\ (1 + \cos^2A) - 2\cot^2A - 1$

$= \text{cosec}^2A\ (1 + \cos^2A) - 2\cot^2A - 1$

$= \text{cosec}^2A + \cot^2A - 2\cot^2A - 1$

$= \text{cosec}^2A - \cot^2A - 1$

$= 1 - 1 = 0$

Alternate Method:-

Put $A = 45°$

$\text{cosec}^4 45°(1 - \cos^4 45°) - 2\cot^2 45° - 1$

$= 4\left(1-\frac14\right) - 2 - 1$

$= 4\times\frac34 - 3 = 3 - 3 = 0$

SMART APPROACH:-

$\text{cosec}^4A\ (1 - \cos^4A) - 2\cot^2A - 1$
$= \text{cosec}^4A - \cot^4A - 2\cot^2A - 1$
$= \text{cosec}^2A + \cot^2A - 2\cot^2A - 1$
$= \text{cosec}^2A - \cot^2A - 1$
$= 1 - 1 = 0$

450. (b) $\sin\theta - \cos\theta = \frac15$

$\therefore \sin\theta + \cos\theta$

$= \sqrt{2-\left(\frac15\right)^2} = \sqrt{2-\frac{1}{25}} = \sqrt{\frac{49}{25}} = \frac75$

451. (c) If, $\cos\theta - \sin\theta = \sqrt2\sin\theta$

Comparing with $a\cos\theta - b\sin\theta = c$

$a = 1, b = 1, c = \sqrt2\sin\theta$

$\cos\theta + \sin\theta = \sqrt{a^2+b^2-c^2}$

$= \sqrt{1+1-2\sin^2\theta} = \sqrt2\cos\theta$

452. (c) $4\cos\theta + 3\sin\theta = x$

squaring both sides–

$16\cos^2\theta + 9\sin^2\theta + 24\sin\theta\cos\theta = x^2$ .....(i)

Again,

$4\sin\theta - 3\cos\theta = y$

squaring both sides–

$16\sin^2\theta + 9\cos^2\theta - 24\sin\theta\cos\theta = y^2$ .....(ii)

Adding equation (i) and (ii), we get–

$x^2 + y^2 = 25\sin^2\theta + 25\cos^2\theta$

$= 25(\sin^2\theta + \cos^2\theta) = 25(1) = 25$

SMART APPROACH:-
Special Case:
If $A\cos\theta + B\sin\theta = x$
and $A\sin\theta - B\cos\theta = y$
then –
$x^2 + y^2 = A^2 + B^2$
Given,
$4\cos\theta + 3\sin\theta = x$
$4\sin\theta - 3\sin\theta = y$
$\therefore x^2 + y^2 = 4^2 + 3^2$
$= 16 + 9 = 25$

453. (d) $\sin x + \cos x = \sqrt{2}\sin x$
$\sin x - \cos x = c$

$$\begin{bmatrix} a\sin x + b\cos x = p \\ b\sin x + a\cos x = c \\ a^2 + b^2 = p^2 + c^2 \end{bmatrix}$$

$\Rightarrow (1)^2 + (1)^2 = (\sqrt{2}\sin x)^2 + c^2$
$\Rightarrow 2 = 2\sin^2 x + c^2$
$\Rightarrow c^2 = 2 - 2\sin^2 x$
$\Rightarrow c^2 = 2(1 - \sin^2 x)$
$\Rightarrow c^2 = 2\cos^2 x$
$\therefore c = \sqrt{2}\cos x$

454. (d) $\sin\theta + \cos\theta = \frac{\sqrt{11}}{3}$
Comparing with $a\sin\theta + b\cos\theta = c$
$a = 1, b = 1, c = \frac{\sqrt{11}}{3}$
$\cos\theta - \sin\theta = \sqrt{a^2 + b^2 - c^2}$
$= \sqrt{1 + 1 - \frac{11}{9}} = \sqrt{\frac{7}{9}} = \frac{\sqrt{7}}{3}$

455. (a) Given, $\sin\theta + \cos\theta = \frac{1}{29}$
Squaring both side
$\Rightarrow 1 + 2\sin\theta.\cos\theta = \frac{1}{841}$
$\Rightarrow 2\sin\theta.\cos\theta = \frac{-840}{841}$
We know that,
$(\sin\theta - \cos\theta)^2 = (\sin\theta + \cos\theta)^2 - 4\sin\theta.\cos\theta$
$= \frac{1}{841} + \frac{1680}{841} = \sqrt{\frac{1681}{841}} = \frac{41}{29}$
$\therefore \frac{\sin\theta + \cos\theta}{\sin\theta - \cos\theta} = \frac{\frac{1}{29}}{\frac{41}{29}} = \frac{1}{41}$

Alternate Method:
$\sin\theta + \cos\theta = \frac{1}{29}$
$\sin\theta - \cos\theta = \sqrt{2 - \left(\frac{1}{29}\right)^2} = \sqrt{2 - \frac{1}{841}}$
$= \sqrt{\frac{1682 - 1}{841}} = \sqrt{\frac{1681}{841}} = \frac{41}{29}$
$\therefore \frac{\sin\theta + \cos\theta}{\sin\theta - \cos\theta} = \frac{\frac{1}{29}}{\frac{41}{29}} = \frac{1}{41}$

456. (a) $3\sin A + 4\cos A = 5$
$\Rightarrow \frac{3}{5}\sin A + \frac{4}{5}\cos A = 1$
Comparing by $\sin^2\theta + \cos^2\theta = 1$
$\sin A = \frac{3}{5}, \cos A = \frac{4}{5}$
$\therefore \tan A = \frac{\sin A}{\cos A} = \frac{3}{4}$

457. (a) $a\cot\theta + b\,\text{cosec}\theta = p$ .....(i)
$b\cot\theta + a\,\text{cosec}\theta = q$ .....(ii)
Squaring both (i) and (ii) then subtract
$a^2(\cot^2\theta - \text{cosec}^2\theta) + b^2(\text{cosec}^2\theta - \cot^2\theta) = p^2 - q^2$
$\Rightarrow -a^2 + b^2 = p^2 - q^2 \Rightarrow p^2 - q^2 = b^2 - a^2$

SMART APPROACH:-
$a\cot\theta + b\,\text{cosec}\theta = p$
$b\cot\theta + a\,\text{cosec}\theta = q$
Let $\theta = 30°$
Then,
$\sqrt{3}a + 2b = p$
$2a + \sqrt{3}b = q$
$p - q = (\sqrt{3} - 2)a + (2 - \sqrt{3})b$
$p + q = (\sqrt{3} + 2)a + (2 + \sqrt{3})b$
$\Rightarrow p^2 - q^2 = -a^2 + b^2 = b^2 - a^2$

458. (c) Given,
$\sqrt{3}\cos\theta + \sin\theta = 1$
from options:-
(b) $\sqrt{3}\cos 60° + \sin 60° = \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2}$
$= \sqrt{3} \neq 1$
(c) $\sqrt{3}\cos 90° + \sin 90° = 1.$
Which satisfied.

459. (b) $\sin^2 t = \frac{1}{2}(\tan t)$
Put $t = 45°$
$\Rightarrow \sin^2 45° = \frac{1}{2}\tan t \Rightarrow \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2} \times 1$
$\Rightarrow \frac{1}{2} = \frac{1}{2}$ (satisfied)

460. (d) $(1 + \sin^4 A - \cos^4 A)\,\text{cosec}^2 A$
Let, $A = 45°$
$\Rightarrow (1 + \sin^4 45° - \cos^4 45°)\,\text{cosec}^2 45°$
$= \left[1 + \left(\frac{1}{\sqrt{2}}\right)^4 - \left(\frac{1}{\sqrt{2}}\right)^4\right](\sqrt{2})^2 = 2$

Alternate Method:
$(1 + \sin^4 A - \cos^4 A)\,\text{cosec}^2 A$
$= \{(1 + (\sin^2 A - \cos^2 A)(\sin^2 A + \cos^2 A)\}\,\text{cosec}^2 A$
$= (1 + (\sin^2 A - \cos^2 A)\,\text{cosec}^2 A$
$= (1 - \cos^2 A + \sin^2 A)\,\text{cosec}^2 A$
$= 2\sin^2 A\,\text{cosec}^2 A = 2$

461. (d) $(\cot\theta + \tan\theta)(\text{cosec}\theta - \sin\theta)(\cos\theta - \sec\theta)$
Put $\theta = 45°$
$(\cot 45° + \tan 45°)(\text{cosec}45° - \sin 45°)(\cos 45° - \sec 45°)$
$= 2 \times \left(\sqrt{2} - \frac{1}{\sqrt{2}}\right) \times \left(\frac{1}{\sqrt{2}} - \sqrt{2}\right)$
$= 2 \times \frac{1}{\sqrt{2}} \times \left(-\frac{1}{\sqrt{2}}\right) = -1$

462. (a) $2\tan^2 A + 4\cos^4 A = 3$
Possible value of $A = 45°$
$\Rightarrow 2 + 4 \times \left(\frac{1}{\sqrt{2}}\right)^4 = 3$
$\Rightarrow 2 + 1 = 3 \Rightarrow 3 = 3$

463. (d) $m = a\sec A$ and $y = b\tan A$
Put, $A = 45°$
$m = a\sqrt{2}, y = b$
$\therefore b^2m^2 - a^2y^2 + \frac{a^2y^2}{b^2m^2} + \cos^2 A$
$= b^2 \times 2a^2 - a^2b^2 + \frac{a^2b^2}{b^2 \times 2a^2} + \cos^2 45°$
$= a^2b^2 + \frac{1}{2} + \frac{1}{2} = a^2b^2 + 1$

464. (d) $m = \sec\theta - \tan\theta$, $n = \text{cosec}\theta + \cot\theta$
put $\theta = 45°$
$m = \sqrt{2} - 1, n = \sqrt{2} + 1$
$\therefore m + n\,(m - 1)$
$= m - n + mn$
$= \sqrt{2} - 1 - \sqrt{2} - 1 + [(\sqrt{2} - 1) \times \sqrt{2} + 1]$
$= -2 + 1 = -1$

465. (b) $\frac{1 - \sin 2t}{1 + \sin 2t} \times \frac{\cos t + \sin t}{\cos t - \sin t}$
Put, $t = 30°$
$\Rightarrow \frac{1 - \sin 60°}{1 + \sin 60°} \times \frac{\cos 30° + \sin 30°}{\cos 30° - \sin 30°}$
$= \frac{1 - \frac{\sqrt{3}}{2}}{1 + \frac{\sqrt{3}}{2}} \times \frac{\frac{\sqrt{3}}{2} + \frac{1}{2}}{\frac{\sqrt{3}}{2} - \frac{1}{2}} = \frac{2 - \sqrt{3}}{2 + \sqrt{3}} \times \frac{\sqrt{3} + 1}{\sqrt{3} - 1}$
Rationalize the denominators,
$= (7 - 4\sqrt{3}) \times \frac{(4 + 2\sqrt{3})}{2}$
$= (7 - 4\sqrt{3}) \times (2 + \sqrt{3})$
$= 14 + 7\sqrt{3} - 8\sqrt{3} - 12 = 2 - \sqrt{3}$
From options (b)
$\frac{1 - \tan 30°}{1 + \tan 30°} = \frac{1 - \frac{1}{\sqrt{3}}}{1 + \frac{1}{\sqrt{3}}} = \frac{\sqrt{3} - 1}{\sqrt{3} + 1}$
$= 2 - \sqrt{3}$
Hence, satisfies.

Alternate Method:-
$\frac{1 - \sin 2t}{1 + \sin 2t} \times \frac{(\cos t + \sin t)}{(\cos t - \sin t)}$
$= \frac{1 - 2\sin t\cos t}{1 + 2\sin t\cos t} \times \frac{\cos t + \sin t}{\cos t - \sin t}$
$= \frac{(\cos t - \sin t)^2}{(\cos t + \sin t)^2} \times \frac{\cos t + \sin t}{\cos t - \sin t}$
$= \frac{\cos t - \sin t}{\cos t + \sin t} = \frac{1 - \tan t}{1 + \tan t}$

466. (a) $\tan A = a \tan B$ and $\sin A = b \sin B$

Let $A = 30°$ and $B = 60°$

$\tan 30° = a \tan 60° \Rightarrow \frac{1}{\sqrt{3}} = a\sqrt{3}$

$\Rightarrow a = \frac{1}{3}$

$\sin 30° = b \sin 60° \Rightarrow \frac{1}{2} = b\ \frac{\sqrt{3}}{2}$

$\Rightarrow b = \frac{1}{\sqrt{3}}$

So, $\cos^2 A = \cos^2 30° = \frac{3}{4}$

Go through options,

$$\frac{b^2-1}{a^2-1} = \frac{\left(\frac{1}{3}\right)-1}{\frac{1}{9}-1} = \frac{\frac{-2}{3}}{\frac{-8}{9}}$$

$$= \frac{-2}{3} \times \frac{9}{-8} = \frac{3}{4}$$

467. (d) $\cot\alpha = \frac{P}{Q}$

Put $\alpha = 45° \Rightarrow P = Q$

$$\therefore \frac{P\cos\alpha - Q\sin\alpha}{P\cos\alpha + Q\sin\alpha} - \frac{P^2-Q^2}{P^2+Q^2} + 3$$

$$= \frac{P\cos 45° - Q\sin 45°}{P\cos 45° + Q\sin 45°} - \frac{P^2-P^2}{P^2+P^2} + 3 = 3$$

468. (c) Put $A = 45°$

$x = \sin 45° + \cos 45° = \sqrt{2}$

$y = \sec A + \text{cosec}\, A = \sec 45° + \text{cosec}\, 45° = 2\sqrt{2}$

so option (c), $y + 2x = x^2y$

$\Rightarrow 2\sqrt{2} + 2\times\sqrt{2} = \left(\sqrt{2}\right)^2 \times 2\sqrt{2}$

$\Rightarrow 4\sqrt{2} = 4\sqrt{2}$

469. (c) $\frac{[(\sin x + \sin y)(\sin y - \sin x)]}{[(\cos x + \cos y)(\cos y - \cos x)]}$

Put $x = 90°$, $y = 0°$

$$\frac{[(\sin 90° + \sin 0°)(\sin 0° - \sin 90°)]}{[(\cos 90° + \cos 0°)(\cos 0° - \cos 90°)]}$$

$$= \frac{[(1+0)(0-1)]}{[(0+1)(1-0)]} = \frac{-1}{1} = -1$$

Alternate Method:

$$\frac{[(\sin x + \sin y)(\sin y - \sin x)]}{[(\cos x + \cos y)(\cos y - \cos x)]}$$

$$= \frac{-\sin^2 x + \sin x.\sin y + \sin^2 y - \sin x.\sin y}{\cos x.\cos y - \cos^2 x + \cos^2 y - \cos x.\cos y}$$

$$= \frac{\sin^2 y - \sin^2 x}{\cos^2 y - \cos^2 x}$$

$$= \frac{1-\cos^2 y - 1 + \cos^2 x}{\cos^2 y - \cos^2 x}$$

$$= \frac{\cos^2 x - \cos^2 y}{\cos^2 y - \cos^2 x} = -1$$

470. (c) Given,

$p = \cot A + \cos A$

$q = \cot A - \cos A$

Put $A = 45°$ we get,

$p = \frac{\sqrt{2}+1}{\sqrt{2}}$, $q = \frac{\sqrt{2}-1}{\sqrt{2}}$

Then go through options.

On checking option (c).

$$\text{LHS} = \sqrt{16pq} = \sqrt{16 \times \frac{\sqrt{2}+1}{\sqrt{2}} \times \frac{\sqrt{2}-1}{\sqrt{2}}}$$

$= \sqrt{8}$

$$\text{RHS} = p^2 - q^2 = \frac{3+2\sqrt{2}}{2} - \frac{3-2\sqrt{2}}{2}$$

$$= \frac{4\sqrt{2}}{2} = 2\sqrt{2} = \sqrt{8}$$

471. (d) $\left(\frac{\sin(x-y)}{\sin(x+y)}\right)\left(\frac{\tan x + \tan y}{\tan x - \tan y}\right)$

Put $x = 60°$, $y = 30°$

$$= \left(\frac{\sin 30°}{\sin 90°}\right)\left(\frac{\tan 60° + \tan 30°}{\tan 60° - \tan 30°}\right)$$

$$= \frac{\frac{1}{2}}{1} \times \frac{\sqrt{3} + \frac{1}{\sqrt{3}}}{\sqrt{3} - \frac{1}{\sqrt{3}}} = \frac{1}{2} \times \frac{4}{2} = 1$$

472. (c) $\text{cosec}\,(\alpha + \beta)$

$$= \frac{\sec\alpha.\sec\beta \times \text{cosec}\alpha.\text{cosec}\beta}{\sec\alpha.\text{cosec}\beta + \text{cosec}\alpha.\sec\beta}$$

Let $\alpha = 45°$ & $\beta = 30°$

$\because \text{cosec}\, 75°$

$$= \frac{\sec 45°.\sec 30° \times \text{cosec}45°.\text{cosec}30°}{\sec 45°.\text{cosec}30° + \text{cosec}45°.\sec 30°}$$

$$= \frac{\sqrt{2} \times \frac{2}{\sqrt{3}} \times \sqrt{2} \times 2}{\sqrt{2} \times 2 + \sqrt{2} \times \frac{2}{\sqrt{3}}} = \frac{8}{\sqrt{3}} \times \frac{\sqrt{3}}{2\sqrt{2}(\sqrt{3}+1)}$$

$$= \frac{4}{\sqrt{2}(\sqrt{3}+1)} \times \frac{\sqrt{3}-1}{\sqrt{3}-1}$$

$= \sqrt{2}(\sqrt{3}-1) = \sqrt{6} - \sqrt{2}$

SMART APPROACH:-

We know $\sin 75° = \frac{\sqrt{3}+1}{2\sqrt{2}}$

Then, $\text{cosec}\, 75° = \frac{2\sqrt{2}}{\sqrt{3}+1} \times \frac{\sqrt{3}-1}{\sqrt{3}-1}$

$$= \frac{2\sqrt{6} - 2\sqrt{2}}{2} = \sqrt{6} - \sqrt{2}$$

473. (d) If $\alpha + \beta = 45°$

then $(\tan\alpha + 1)(\tan\beta + 1) = 2$

$\Rightarrow 2x = 2 \Rightarrow x = 1$

474. (d) We know,

If $A + B = 45°$

then $(1 + \tan A)(1 + \tan B) = 2$

$\therefore (1 + \tan 10°)(1 + \tan 35°) = 2$

475. (c) $\sin^2\theta + \cos^2\theta = 1$

$\Rightarrow \theta$ have infinite values

476. (a) $a = \cot A + \cos A$ ... (i)

$b = \cot A - \cos A$ ... (ii)

Multiply Eq^n (i) × Eq^n (ii),

$ab = \cot^2 A - \cos^2 A$

Add Eq^n (i) + Eq^n (ii)

$a + b = 2\cot A$ ... (iii)

Subtract Eq^n (i) – Eq^n (ii)

$a - b = 2\cos A$ ... (iv)

Multiply Eq^n (iii) × Eq^n (iv)

$a^2 - b^2 = 4\cot A.\cos A$

$\therefore a^2 - b^2 = 4\sqrt{ab}$

$= 4\cot A.\cos A - 4 \times \sqrt{\cot^2 A \times \cos^2 A}$

$= 4\cot A.\cos A - 4\cot A.\cos A = 0$

Alternate Method:

$a = \cot A + \cos A$

$b = \cot A - \cos A$

Put value of $A = 90°$

$\Rightarrow a = \cot 90° + \cos 90° \Rightarrow a = 0$

$\Rightarrow b = \cot 90° - \cos 90° \Rightarrow b = 0$

$\therefore a^2 - b^2 - 4\sqrt{ab} = 0$

477. (b) $\cos^2\theta + \sin\theta + 1 = 0$

$1 - \sin^2\theta + \sin\theta + 1 = 0$

We know

$\sin\theta \le 1$

This is possible only when $\sin\theta = -1$

Then,

θ lies b/w $\left(\frac{5\pi}{4}, \frac{7\pi}{4}\right)$

478. (c) $0° < t < 90°$

$\Rightarrow \sin t < \cos t$ when $t < 45°$

e.g. $\sin 30° < \cos 30°$

$\Rightarrow \frac{1}{2} < \frac{\sqrt{3}}{2}$

479. (c) $2\left(\frac{\cos^2 x - \sec^2 x}{\tan^2 x}\right) = a + b\cos^2 x$

$$\Rightarrow 2\frac{\left(\cos^2 x - \frac{1}{\cos^2 x}\right)}{\frac{\sin^2 x}{\cos^2 x}} = a + b\cos^2 x$$

$$\Rightarrow \frac{2(\cos^4 x - 1)}{\sin^2 x} = a + b\cos^2 x$$

$$\Rightarrow \frac{2[(\cos^2 x - 1)(\cos^2 x + 1)]}{\sin^2 x}$$

$= a + b\cos^2 x$

$\Rightarrow -2(\cos^2 x + 1) = a + b\cos^2 x$

$\Rightarrow -(2\cos^2 x - 1) - 3 = a + b\cos^2 x$

$\Rightarrow -\cos^2 x - 3 = a + b\cos^2 x$

$\Rightarrow a = -3,\ b = -1$

Alternate Method:

Put $x = 30°$,

$$\Rightarrow 2\ \frac{\cos^2 30° - \sec^2 30°}{\tan^2 30°} = a + b\cos 60°$$

$$\Rightarrow 2\ \frac{\frac{3}{4} - \frac{4}{3}}{\frac{1}{3}} = a + \frac{b}{2}$$

$$\Rightarrow 2 \times \frac{9-16}{12} \times 3 = a + \frac{b}{2}$$

$$\Rightarrow -\frac{7}{2} = a + \frac{b}{2}$$

$$\Rightarrow -3 - \frac{1}{2} = a + \frac{b}{2}$$

$\therefore a = -3,\ b = -1$

480. (b) Given,

$$x = \frac{2\sin\theta}{1+\cos\theta+\sin\theta}$$

By rationalization

$$\frac{2\sin\theta}{1+\cos\theta+\sin\theta}\times\frac{1-\cos\theta+\sin\theta}{1-\cos\theta+\sin\theta}$$

$$=\frac{2\sin\theta(1-\cos\theta+\sin\theta)}{(1+\sin\theta)^2-(\cos\theta)^2}$$

$$=\frac{2\sin\theta(1-\cos\theta+\sin\theta)}{1+2\sin\theta+\sin^2\theta-\cos^2\theta}$$

$$=\frac{2\sin\theta(1-\cos\theta+\sin\theta)}{2\sin\theta+2\sin^2\theta}$$

$$=\frac{2\sin\theta(1-\cos\theta+\sin\theta)}{2\sin\theta(1+\sin\theta)}$$

$$=\frac{1-\cos\theta+\sin\theta}{1+\sin\theta}=x$$

Alternate Method:

Let, $\sin\theta=\frac{3}{5}$ Then, $\cos\theta=\frac{4}{5}$

$$x=\frac{2\sin\theta}{1+\cos\theta+\sin\theta}=\frac{2\times\frac{3}{5}}{1+\frac{4}{5}+\frac{3}{5}}$$

$$=\frac{\frac{6}{5}}{\frac{12}{5}}=\frac{1}{2}=\frac{1-\cos\theta+\sin\theta}{1+\sin\theta}$$

$$=\frac{1-\frac{4}{5}+\frac{3}{5}}{1+\frac{3}{5}}=\frac{\frac{4}{5}}{\frac{8}{5}}=\frac{1}{2}=x$$

481. (b) $a=\frac{2\sin x}{1+\sin x+\cos x}$,

$b=\frac{c}{1+\sin x}$

$a = b$

$$\Rightarrow \frac{2\sin x}{1+\sin x+\cos x}=\frac{c}{1+\sin x}$$

$$\Rightarrow c=\frac{2\sin x(1+\sin x)(1+\sin x-\cos x)}{(1+\sin x+\cos x)(1+\sin x-\cos x)}$$

$$\Rightarrow c=\frac{2\sin x(1+\sin x)(1+\sin x-\cos x)}{(1+\sin x)^2-\cos^2 x}$$

$$\Rightarrow c=\frac{2\sin x(1+\sin x)(1+\sin x-\cos x)}{1^2+\sin^2 x+2\sin x-\cos^2 x}$$

$$\Rightarrow c=\frac{2\sin x(1+\sin x)(1+\sin x-\cos x)}{2\sin^2 x+2\sin x}$$

$$\Rightarrow c=\frac{2\sin x(1+\sin x)(1+\sin x-\cos x)}{2\sin x\times(\sin x+1)}$$

$$\Rightarrow c=1+\sin x-\cos x$$

Alternate Method:

Put $x = 90°$

$a=\frac{2}{1+1}=1$, $b=\frac{c}{1+1}=\frac{c}{2}$

$a = b$ (Given) $\Rightarrow 1=\frac{c}{2}\Rightarrow c=2$

Go through options,

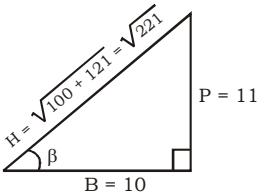
$1 + \sin x - \cos x = 1 + \sin 90° - \cos 90°$

$= 1 + 1 = 2$

482. (b) $\frac{\cos\alpha}{\sin\beta}=10$ & $\frac{\cos\alpha}{\cos\beta}=11$

On divinding, $\frac{\sin\beta}{\cos\beta}=\frac{11}{10}$

$$\Rightarrow \tan\beta=\frac{11}{10}$$



$$\cos^2\beta=\left(\frac{B}{H}\right)^2=\left(\frac{10}{\sqrt{221}}\right)^2=\frac{100}{221}$$

SMART APPROACH:-
$\cos^2\beta$ is always a perfect square and only option (b) is perfect square.

483. (b) Given,

$$\sin18°=\frac{\sqrt{5}-1}{4}$$

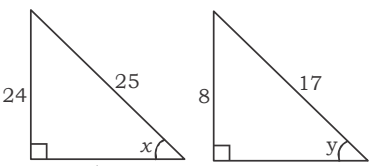
ATQ, $\text{cosec}18°=\frac{1}{\sin18°}=\frac{1}{\frac{\sqrt{5}-1}{4}}$

$$=\frac{4}{\sqrt{5}-1}$$

After rationalising

$$\frac{4}{\sqrt{5}-1}\times\frac{\sqrt{5}+1}{\sqrt{5}+1}=\frac{4(\sqrt{5}+1)}{4}=\sqrt{5}+1$$

484. (a)



$$\cos^{-1}\left(\frac{7}{25}\right)-\cos^{-1}\left(\frac{15}{17}\right)=?$$

Let, $\cos^{-1}\left(\frac{7}{25}\right)=x$ and $\cos^{-1}\left(\frac{15}{17}\right)=y$

$\because \sin(x-y)=\sin x\cos y-\cos x\sin y$

$$\Rightarrow \sin(x-y)=\frac{24}{25}\times\frac{15}{17}-\frac{7}{25}\times\frac{8}{17}$$

$$\Rightarrow \sin(x-y)=\frac{360-56}{425}$$

$$\Rightarrow \sin(x-y)=\frac{304}{425}$$

$$\Rightarrow x-y=\sin^{-1}\left(\frac{304}{425}\right)$$

485. (a) $\cos^4\theta-\sin^4\theta=\frac{4}{5}$

$$\Rightarrow (\cos^2\theta-\sin^2\theta)(\cos^2\theta+\sin^2\theta)=\frac{4}{5}$$

$$\Rightarrow (\cos^2\theta-\sin^2\theta)=\frac{4}{5}$$

$$\Rightarrow \cos2\theta=\frac{4}{5}$$

So, $\sin2\theta=\frac{3}{5}$ [3, 4, 5 are triplet]

$\therefore \sin4\theta = 2\times\sin2\theta\times\cos2\theta$

$$=2\times\frac{3}{5}\times\frac{4}{5}=\frac{24}{25}$$

486. (b) We know, $\cos2\theta = 2\cos^2\theta - 1$

$$\Rightarrow \cos^2\theta=\frac{\cos2\theta+1}{2}$$

Put $\theta = 15°$

$$\cos^2 15°=\frac{\cos30°+1}{2}=\frac{2+\sqrt{3}}{4}$$

487. (d) $\cos^2\theta-\sin^2\theta=\frac{1}{2}$

$[\because \cos^2\theta-\sin^2\theta=\cos2\theta]$

So, $\cos2\theta=\cos60°$

$\Rightarrow 2\theta = 60°$

$\Rightarrow \theta = 30°$

488. (d) Given, $\sin\theta=\frac{6}{10}$, $\cos\theta=\frac{8}{10}$

We know,

$\because \sin2\theta = 2\sin\theta\cos\theta$

$$=2\times\frac{6}{10}\times\frac{8}{10}=\frac{24}{25}$$

489. (a) Let, $x = 45°$

$$\tan22.5=\frac{45°}{2}=\frac{x}{2}$$

ATQ,

$$\tan\frac{x}{2}=\frac{1-\cos x}{\sin x}$$

$$\Rightarrow \tan\frac{45°}{2}=\frac{1-\cos45°}{\sin45°}$$

$$\Rightarrow \tan22.5°=\frac{1-\frac{1}{\sqrt{2}}}{\frac{1}{\sqrt{2}}}$$

$$\Rightarrow \tan22.5°=\frac{\sqrt{2}-1}{\sqrt{2}}\times\sqrt{2}=\sqrt{2}-1$$

490. (d) $4\tan\theta - 3 = 0$

$$\tan\theta=\frac{3}{4}\begin{matrix}\rightarrow P\\ \rightarrow B\end{matrix}\Rightarrow H=5$$

$$\therefore \frac{1-\cos2\theta}{1+\cos2\theta}=\frac{1-(\cos^2\theta-\sin^2\theta)}{1+(\cos^2\theta-\sin^2\theta)}$$

By triplet (3, 4, 5)

$$=\frac{1-\left[\left(\frac{4}{5}\right)^2-\left(\frac{3}{5}\right)^2\right]}{1+\left[\left(\frac{4}{5}\right)^2-\left(\frac{3}{5}\right)^2\right]}=\frac{1-\frac{7}{25}}{1+\frac{7}{25}}$$

$$=\frac{18}{32}=\frac{9}{16}$$

491. (c) $\left(\frac{2\cos^3\theta-\cos\theta}{\sin\theta-2\sin^3\theta}\right)^2+1$

$$=\left[\frac{\cos\theta(2\cos^2\theta-1)}{\sin\theta(1-2\sin^2\theta)}\right]^2+1$$

$[\cos2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta]$

$$=\left[\frac{\cos\theta\times\cos2\theta}{\sin\theta\times\cos2\theta}\right]^2+1$$

$$=\left(\frac{\cos\theta}{\sin\theta}\right)^2+1=1+\cot^2\theta=\text{cosec}^2\theta$$

492. (d) $\frac{2\tan A}{1+\tan^2 A}$

$= \dfrac{2\dfrac{\sin A}{\cos A}}{\dfrac{\cos^2 A + \sin^2 A}{\cos^2 A}} \quad \left[\because \sin 2A = \dfrac{2\tan A}{1+\tan^2 A}\right]$

$= 2\sin A\cos A = \sin 2A$

Alternate Method:

Put A = 45°

$\dfrac{2\tan 45°}{1+\tan^2 45°} = \dfrac{2}{1+1} = 1$

Go through options,

$\sin 2A = \sin(2 \times 45°) = \sin 90° = 1$

493. (a) $\dfrac{\sin\theta - 2\sin^3\theta}{2\cos^3\theta - \cos\theta}$

$= \dfrac{\sin\theta(1-2\sin^2\theta)}{\cos\theta(2\cos^2\theta - 1)} = \dfrac{\sin\theta}{\cos\theta} = \tan\theta$

494. (c) $\dfrac{2\cos^3\theta - \cos\theta}{\sin\theta - 2\sin^3\theta}$

$= \dfrac{\cos\theta\left(2\cos^2\theta - 1\right)}{\sin\theta\left(1 - 2\sin^2\theta\right)}$

$= \dfrac{\cos\theta \times \cos 2\theta}{\sin\theta \times \cos 2\theta} = \cot\theta$

Alternate Method:

Put $\theta = 30°$

$\dfrac{2\cos^3 30° - \cos 30°}{\sin 30° - 2\sin^3 30°} = \dfrac{2\times\dfrac{3\sqrt{3}}{8} - \dfrac{\sqrt{3}}{2}}{\dfrac{1}{2} - 2\times\dfrac{1}{8}}$

$= \dfrac{\dfrac{3\sqrt{3}}{4} - \dfrac{\sqrt{3}}{2}}{\dfrac{1}{2} - \dfrac{1}{4}} = \dfrac{3\sqrt{3} - 2\sqrt{3}}{4} \times \dfrac{4}{1} = \sqrt{3}$

Go through options,

$\cot\theta = \cot 30° = \sqrt{3}$

495. (a) $\sqrt{2+\sqrt{2+\sqrt{2+2\cos 8\theta}}}$

[$\because \cos 2\theta = 2\cos^2\theta - 1$

$\cos 8\theta = 2\cos^2 4\theta - 1$

$2\cos^2 4\theta = \cos 8\theta + 1$]

$= \sqrt{2+\sqrt{2+\sqrt{2(1+\cos 8\theta)}}}$

$= \sqrt{2+\sqrt{2+\sqrt{2\times 2\cos^2 4\theta}}}$

$= \sqrt{2+\sqrt{2+\sqrt{4\cos^2 4\theta}}}$

$= \sqrt{2+\sqrt{2+2\cos 4\theta}}$

$= \sqrt{2+\sqrt{2(1+\cos 4\theta)}}$

$= \sqrt{2+\sqrt{2\times 2\cos^2 2\theta}}$

$= \sqrt{2+2\cos 2\theta} = \sqrt{2(1+\cos 2\theta)}$

$= \sqrt{2\times 2\cos^2\theta} = 2\cos\theta$

496. (c) $\dfrac{\left(2\cos^3 A - \cos A\right)}{\left(\sin A - 2\sin^3 A\right)} = 1$

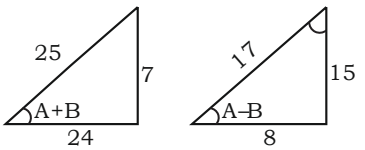
$\Rightarrow \dfrac{\cos A\left(2\cos^2 A - 1\right)}{\sin A\left(1 - 2\sin^2 A\right)} = 1$

$\Rightarrow \cot A \times \dfrac{\cos 2A}{\cos 2A} = 1$

$\Rightarrow \cot A = \cot 45°$

$\Rightarrow A = 45°$

497. (c) $\cos(A+B) = \dfrac{24}{25}, \sin(A-B) = \dfrac{15}{17}$



$\tan(2A) = \tan\left[(A+B)+(A-B)\right]$

$= \dfrac{\tan(A+B)+\tan(A-B)}{1-\tan(A+B)+\tan(A-B)}$

$= \dfrac{\dfrac{7}{24}+\dfrac{15}{8}}{1-\dfrac{7}{24}\times\dfrac{15}{8}} = \dfrac{\dfrac{52}{24}}{\dfrac{29}{64}} = \dfrac{416}{87}$

498. (b) $\sin 3A = 3\sin A - 4\sin^3 A$

499. (c) $\sin 3A = 3\sin A - 4\sin^3 A$

$\therefore 3\sin 15° - 4\sin^3 15° = \sin(3 \times 15°)$

$= \sin 45° = \dfrac{1}{\sqrt{2}}$

500. (d) Given, A = 60°, B = 30°

We know,

$\sin A\cos B + \cos A\sin B = \sin(A+B)$

$= \sin(60° + 30°) = \sin 90° = 1$

501. (c) $\cos(A+B) = \cos A.\cos B - \sin A.\sin B$

Let, A = 45°, B = 30°

$\therefore \cos 75° = \cos(45° + 30°)$

$= \cos 45°.\cos 30° - \sin 45°.\sin 30°$

$= \dfrac{1}{\sqrt{2}}\times\dfrac{\sqrt{3}}{2} - \dfrac{1}{\sqrt{2}}\times\dfrac{1}{2}$

$= \dfrac{1}{\sqrt{2}}\left(\dfrac{\sqrt{3}}{2} - \dfrac{1}{2}\right) = \dfrac{\sqrt{3}-1}{2\sqrt{2}}\times\dfrac{\sqrt{2}}{\sqrt{2}}$

$= \dfrac{\sqrt{6}-\sqrt{2}}{4}$

502. (d) Given, $\angle B = 90°$

Remaining $\angle A + \angle C = 90°$

$\sin(A + C) = \sin(90°) = 1$

503. (b) $\cos(x+y) - \sin(x-y) + \tan 2z$

$= \cos x\cos y - \sin x\sin y - \sin x\cos y + \cos x\sin y + \dfrac{2\tan z}{1-\tan^2 z}$

504. (b) $\tan(A-B) = \dfrac{\tan A - \tan B}{1+\tan A\tan B}$

$\therefore \tan(72° - 27°) = \dfrac{\tan 72° - \tan 27°}{1+\tan 72°.\tan 27°}$

$\Rightarrow 1 + \tan 72°.\tan 27° = \tan 72° - \tan 27°$

$\Rightarrow \tan 72° - \tan 27° - \tan 72°.\tan 27° = 1$

Note: In these type of questions whenever the difference of angle = 45° answer will be 1.

505. (b) $\sin(X-Y) = \sin X\cos Y - \cos X\sin Y$

$\therefore \dfrac{\sin(A-B)}{\cos A\cos B} + \dfrac{\sin(B-C)}{\cos B\cos C}$

$= \dfrac{\sin A\cos B - \cos A\sin B}{\cos A\cos B} + \dfrac{\sin B\cos C - \cos B\sin C}{\cos B\cos C}$

$= \dfrac{\sin A\cos B\cos C - \cos A\sin B\cos C + \sin B\cos C\cos A - \cos B\sin C\cos A}{\cos A\cos B\cos C}$

$= \dfrac{\cos B(\sin A\cos C - \sin C\cos A)}{\cos A\cos B\cos C}$

$= \dfrac{\sin(A-C)}{\cos A\cos C}$

Alternate Method:

Put B = C = 0°

ATQ,

$\dfrac{\sin A}{\cos A} = \tan A$

Only option (b) satisfied

506. (c) $\operatorname{cosec} 31° = \sec 59°$

$\therefore \left[\dfrac{\operatorname{cosec} 31° - \sec 59°}{(\sin 18° \div \cos 72°)} \times \dfrac{\tan 26°}{\cot 64°}\right] + \left[\dfrac{\tan 30° + \tan 15°}{(1-\tan 30°\tan 15°)}\right]$

$= 0 + \dfrac{\tan 30° + \tan 15°}{1-\tan 30°\times\tan 15°}$

$= \tan(30° + 15°) = \tan 45° = 1$

507. (c) $\dfrac{\cos 13° + \sin 13°}{\cos 13° - \sin 13°}$

Divide numerator and denominator by cos13°

$\Rightarrow \dfrac{\cos 13°(1+\tan 13°)}{\cos 13°(1-\tan 13°)}$

$= \dfrac{\tan 45° + \tan 13°}{1-\tan 45°\times\tan 13°}$

$= \tan(45° + 13°) = \tan 58°$

508. (a) $\dfrac{\tan 71° + \tan 19°}{1-\tan 71°.\tan 19°}$

$= \tan(71° + 19°)$

$= \tan 90° =$ Not defined

$\left[\tan(A+B) = \dfrac{\tan A + \tan B}{1-\tan A\times\tan B}\right]$

509. (c) Given,

$\tan(A-B) = \dfrac{\tan A - \tan B}{1+\tan A\tan B}$

Put A = 45° and B = 30°

$\tan(45° - 30°) = \dfrac{\tan 45° - \tan 30°}{1+\tan 45°\tan 30°}$

$= \dfrac{1-\dfrac{1}{\sqrt{3}}}{1+(1)\left(\dfrac{1}{\sqrt{3}}\right)} = \dfrac{\sqrt{3}-1}{\sqrt{3}}\times\dfrac{\sqrt{3}}{\sqrt{3}+1}$

$= \dfrac{\sqrt{3}-1}{\sqrt{3}+1}\times\dfrac{\sqrt{3}-1}{\sqrt{3}-1} = \dfrac{\left(\sqrt{3}-1\right)^2}{\left(\sqrt{3}\right)^2 - (1)^2}$

$= \dfrac{3-2\sqrt{3}+1}{3-1} = \dfrac{4-2\sqrt{3}}{2}$

$= \dfrac{2\left(2-\sqrt{3}\right)}{2} = 2-\sqrt{3}$

$\therefore \tan 15° = 2-\sqrt{3}$

510. (a) $\sin(A+B) = \cos(A+B)$

$\Rightarrow \dfrac{\sin(A+B)}{\cos(A+B)} = 1$

$\Rightarrow \tan(A+B) = 1$

$\Rightarrow \frac{\tan A + \tan B}{1 - \tan A \tan B} = 1$

$\Rightarrow$ tan A + tan B = 1 – tan A tan B

$\Rightarrow$ tan A + tan B tan A = 1 – tan B

$\Rightarrow$ tan A (1 + tan B) = 1 – tan B

$\Rightarrow \tan A = \frac{1 - \tan B}{1 + \tan B}$

Alternate Method:

sin(A + B) = cos(A + B)

Put, A = 0, B = 45°

Now,

tanA = tan0° = 0

Now check option

Put B = 45°

Which option given us zero

$\frac{1 - \tan B}{1 + \tan B} = \frac{1-1}{1+1} = 0$

Hence, option (a) is satisfy

511. (b) $\sin A = \frac{4}{5}, \quad \sin B = \frac{15}{17}$

$\Rightarrow \cos A = \frac{3}{5}, \quad \cos B = \frac{8}{17}$

$\therefore$ sin (A – B) = sin A cos B – cos A

$\sin B = \frac{4}{5} \times \frac{8}{17} - \frac{3}{5} \times \frac{15}{17}$

$= \frac{32}{85} - \frac{45}{85} = -\frac{13}{85}$

512. (c) $\frac{\cos x - \sqrt{3} \sin x}{2}$

$= \frac{1}{2}\cos x - \frac{\sqrt{3}}{2}\sin x$

= cos 60° cos x – sin 60° sin x

= cos (60° + x) $= \cos\left(\frac{\pi}{3} + x\right)$

513. (a) tanA –tanB – tanC = tanA tanB tanC

$\Rightarrow$ tanA–tanB=tanA tanB tanC + tanC

$\Rightarrow$ tanA – tanB = tanC (tanA tanB + 1)

$\Rightarrow \tan C = \frac{\tan A - \tan B}{\tan A \tan B + 1}$

$\Rightarrow$ tanC = tan(A – B)

$\Rightarrow$ C = A – B

$\Rightarrow$ A = B + C

514. (a) tan (α + β) = a, tan (α – β) = b

$\tan 2\alpha = \tan(\alpha + \beta + \alpha - \beta)$

Let α + β = a, α – β = b

$\therefore \tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$

$\tan 2\alpha = \frac{a + b}{1 - ab}$

Alternate Method:

tan(α + β) = a, tan(α – β) = b

Put α = 30°, β = 0

$\tan 30° = a \Rightarrow a = \frac{1}{\sqrt{3}}$

$\tan 30° = b \Rightarrow b = \frac{1}{\sqrt{3}}$

$\tan 2\alpha = \tan 60° = \sqrt{3}$

Go through options,

$\frac{a + b}{1 - ab} = \frac{\frac{1}{\sqrt{3}} + \frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}} \times \frac{1}{\sqrt{3}}} = \frac{\frac{2}{\sqrt{3}}}{1 - \frac{1}{3}} = \frac{2}{\sqrt{3}} \times \frac{3}{2}$

$= \sqrt{3}$

515. (c) $\tan A \tan B + \frac{\cos x}{\cos A \cos B} = 1$

$\Rightarrow \frac{\sin A \sin B}{\cos A \cos B} + \frac{\cos x}{\cos A \cos B} = 1$

$\Rightarrow$ sinA sinB + cos$x$ = cosA cosB

$\Rightarrow$ cos$x$ = cosA cosB – sinA sinB

$\Rightarrow$ cos$x$ = cos (A + B)

$\Rightarrow x$ = A + B

Alternate Method:

Put A = B = 45°

$\tan 45° \tan 45° + \frac{\cos x}{\cos 45° \cos 45°} = 1$

$\Rightarrow$ 1 + 2 cos$x$ = 1

$\Rightarrow$ cos$x$ = 0

$\Rightarrow x$ = 90° = A + B

516. (c) Given,

$\sin A = \frac{1}{3}$

$\therefore \cos A = \sqrt{1 - \sin^2 A} = \sqrt{1 - \left(\frac{1}{3}\right)^2} = \frac{2\sqrt{2}}{3}$

$\sin B = \frac{1}{5}$

$\therefore \cos B = \sqrt{1 - \sin^2 B} = \sqrt{1 - \left(\frac{1}{5}\right)^2} = \frac{2\sqrt{6}}{5}$

Since the angles given in the second quadrant.

So,

$\cos A = \frac{-2\sqrt{2}}{3} \Rightarrow \cos B = \frac{-2\sqrt{6}}{5}$

We know that–

cos(A – B) = cosAcosB + sinAsinB

$= \frac{-2\sqrt{2}}{3} \times \frac{-2\sqrt{6}}{5} + \frac{1}{3} \times \frac{1}{5}$

$= \frac{2\sqrt{2}}{3} \times \frac{2\sqrt{2} \times \sqrt{3}}{5} + \frac{1}{3} \times \frac{1}{5} = \frac{8\sqrt{3} + 1}{15}$

517. (a) We know that,

sin(A + B) = sinAcosB + cosAsinB

$\therefore \frac{\sin(A + B)}{\sin A \cos B}$

$= \frac{\sin A \cos B + \cos A \sin B}{\sin A \cos B}$

$= \frac{\sin A \cos B}{\sin A \cos B} + \frac{\cos A \sin B}{\sin A \cos B}$

= 1 + cotA tanB

Alternate Method:

Put A = 60°, B = 30°

$\frac{\sin 90°}{\sin 60° \cos 30°} = \frac{1}{\frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2}} = \frac{4}{3}$

Go through options,

1 + cotA tanB = 1 + cot60° tan30°

$= 1 + \frac{1}{\sqrt{3}} \times \frac{1}{\sqrt{3}} = 1 + \frac{1}{3} = \frac{4}{3}$

518. (a) $\because$ sinC + sinD

$= 2\sin\left(\frac{C + D}{2}\right).\cos\left(\frac{C - D}{2}\right)$

$\therefore$ sin75° + sin15° = 2sin45°.cos30°

$= 2 \times \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{\sqrt{2}} = \frac{\sqrt{6}}{2}$

519. (a) sin2x + 2sin4x + sin6x

= (sin6x + sin2x) + 2sin4x

$\boxed{\sin A + \sin B = 2\sin\left(\frac{A + B}{2}\right)\cos\left(\frac{A - B}{2}\right)}$

$= 2\sin\frac{6x + 2x}{2} \cos\frac{6x - 2x}{2} + 2\sin 4x$

= 2sin 4x cos 2x + 2sin 4x

= 2sin 4x (cos2x + 1)

[We know, cos 2x + 1 = 2cos^2x]

= 2 sin 4x × 2 cos^2x

= 4 cos^2x sin 4x

Alternate Method:

Put θ = 30°

sin60° + 2sin120° + sin180°

$= \frac{\sqrt{3}}{2} + 2 \times \frac{\sqrt{3}}{2} + 0 = \frac{3\sqrt{3}}{2}$

By option (A)

4cos²x.sin4x

$= 4 \times \frac{3}{4} \times \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{2}$

520. (a) By formula,

$\sin\alpha - \sin\beta = 2\cos\frac{\alpha + \beta}{2}\sin\frac{\alpha - \beta}{2}$

521. (a) $\frac{\sin 4\theta + \sin 2\theta}{\cos 4\theta + \cos 2\theta}$

Using the formula:

$\sin A + \sin B = 2\sin\frac{A + B}{2} \cos\frac{A - B}{2}$

$\cos A + \cos B = 2\cos\frac{A + B}{2} \cos\frac{A - B}{2}$

$\therefore \frac{\sin 4\theta + \sin 2\theta}{\cos 4\theta + \cos 2\theta} = \frac{2\sin 3\theta \cos\theta}{2\cos 3\theta \cos\theta}$

= tan3θ

522. (d) cos(36° + A) cos (36° – A) + cos (54° + A) cos (54° – A)

We know,

$\boxed{\cos(A + B).\cos(A - B) = \cos^2 A - \sin^2 B}$

= cos².36 – sin²A + cos²54 – sin²A

= cos²36 – sin²A + sin²36 – sin²A

= 1 – sin²A – sin²A

= cos²A – sin²A = cos2A

523. (b) cos75° cos15° = $\frac{1}{2}$

[cos(75° + 15°) + cos(75° – 15°)]

$= \frac{1}{2}$ [cos90° + cos60°]

$= \frac{1}{2}\left[\frac{1}{2}\right] = \frac{1}{4}$

524. (d) By option,

$\sin A \sin B = \frac{1}{2}$ {cos(A – B) – cos(A + B)}

By value putting, let A = 60°, B = 30°

LHS = sinA sinB = sin60° × sin30°

$= \frac{\sqrt{3}}{2} \times \frac{1}{2} = \frac{\sqrt{3}}{4}$

RHS = $\frac{1}{2}$ {cos(A – B) – cos(A + B)}}

= $\frac{1}{2}$ (cos30° – cos90°)

= $\frac{1}{2}\left\{\frac{\sqrt{3}}{2}-0\right\}$ = $\frac{\sqrt{3}}{4}$

525. (d) $\frac{\sin 8\theta\cos\theta-\sin 6\theta\cos 3\theta}{\cos 2\theta\cos\theta-\sin 3\theta\sin 4\theta}$

Multiplying and dividing by 2

$=\frac{\sin 8\theta\cos\theta-\sin 6\theta\cos 3\theta}{\cos 2\theta\cos\theta-\sin 3\theta\sin 4\theta}\times\frac{2}{2}$

$=\frac{2\sin 8\theta\cos\theta-2\sin 6\theta\cos 3\theta}{2\cos 2\theta\cos\theta-2\sin 3\theta\sin 4\theta}$

[We know,

2sinAcosB = sin(A + B) + sin(A –B)

2cosAcosB = cos(A + B) + cos(A –B)

2sinAsinB = cos(A – B) – cos(A + B)]

$=\frac{\sin 9\theta+\sin 7\theta-(\sin 9\theta+\cos 3\theta)}{\cos 3\theta+\cos\theta-(\cos\theta-\cos 7\theta)}$

$=\frac{\sin 7\theta-\sin 3\theta}{\cos 3\theta+\cos 7\theta}=\frac{2\cos 5\theta\sin 2\theta}{2\cos 5\theta\cos 2\theta}$

= tan2θ

[We know,

$\sin C-\sin D=2\cos\left(\frac{C+D}{2}\right)\sin\left(\frac{C-D}{2}\right)$

$\cos C+\cos D=2\cos\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)$]

526. (d) Max value of (asinθ + bcosθ)

= $\sqrt{a^2+b^2}$

Here, a = 2, b = 3

∴ Max value of (2sinθ + 3cosθ)

= $\sqrt{4+9}$ = $\sqrt{13}$

527. (a) Maximum value of a sinA + cosA

= $\sqrt{a^2+b^2}$

Maximum value of 7cosA + 24 sinA

= $\sqrt{7^2+24^2}$ = 25

∴ 7cosA + 24 sinA + 32 = 25 + 32 = 57

528. (a) 19 sinθ + 6cotθ sinθ

= 19 sinθ + 6 × $\frac{\cos\theta}{\sin\theta}$ × sinθ

= 19 sinθ + 6 cosθ

∴ Max value = $\sqrt{19^2+6^2}$ = $\sqrt{397}$

529. (d) Least value of acosec²θ + bsin²θ

= $2\sqrt{ab}$

Least value of 16 cosec² θ + 25 sin²θ

= $2\sqrt{16\times 25}$ = 2 × 4 × 5 = 40

530. (d) We have,

sinθ cosθ

Now, this can be written us

sinθ. cosθ = $\frac{1}{2}$ (2sinθ cosθ)

[∴ sin2θ = 2sinθ cosθ]

as, we know that

– 1 ≤ sin nx ≤ 1 n = 1, 2 .... A

Here, – 1 ≤ sin nx [for minimum]

Hence, minimum value of sin2θ is –1

Required least value is $\left(\frac{-1}{2}\right)$

531. (c) ($\sin^{12}\theta+\cos^{20}\theta$)

[We know, $\sin^2\theta+\cos^2\theta=1$

$\cos^2\theta=1-\sin^2\theta$]

= $(\sin^2\theta)^6+(\cos^2\theta)^{10}$

= $(\sin^2\theta)^6+(1-\sin^2\theta)^{10}$

The maximum value of $\sin^2\theta$ = 1

= $(1)^6+(1-1)^{10}$ =1 + 0 = 1

532. (b) $\sin^4\theta+\cos^4\theta$

= $(\sin^2\theta+\cos^2\theta)^2-2\sin^2\theta\cos^2\theta$

= $1-2\sin^2\theta\cos^2\theta$

⇒ $\sin^2\theta\cos^2\theta$ should be minimum which is '0'.

∴ max value = 1 – 2 × 0 = 1

Alternate Method:

Whenere the powers of sin & cos are even then then maximum value of $\sin^n\theta+\cos^n\theta$ is always 1.

SELECTION मंत्र #07



"How to handle failure?"

#Hacks:-

☞ **Accept & Analyze the Failure–**

- Accept that failure is part of the process, not the end.
- Analyze your mistakes**:** Was it lack of preparation, time management, revision, or exam pressure?
- Check your score vs. cut-off to identify weak areas.

☞ **Reset Your Mindset –**

- Take a short break to refresh your mind, but don't dwell on negativity.
- Watch success stories of aspirants who cleared after multiple attempts (e.g., SSC, UPSC, Banking).
- Remind yourself: "Many have faced setbacks but succeeded. I will too!"

☞ **Make a Better Study Plan–**

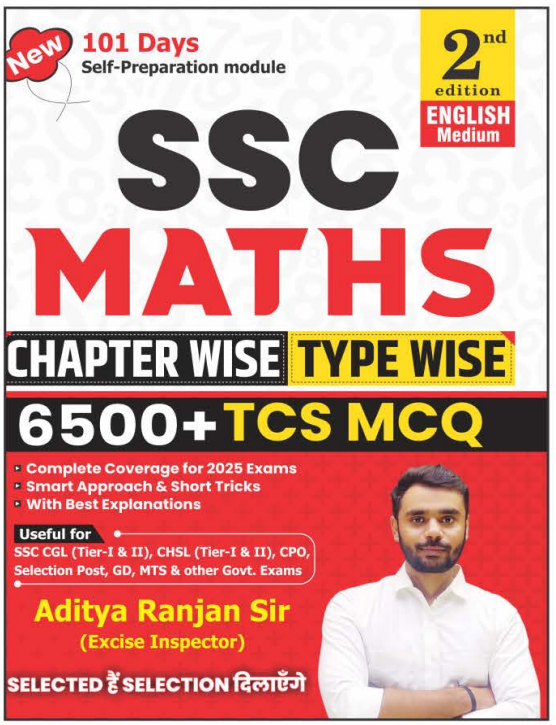
- Focus on weak areas: Improve sections where you lost marks.
- Revise more smartly: Use PYQs, mock tests, and short notes.
- Time Management: Improve speed and accuracy with timed practice.

☞ **Believe in Yourself–**

- Your experience from failure is an asset, not a burden.
- Success may take time, but with persistence and smart effort, you WILL crack it!
- **Tips:**Restart with full energy—your selection is just one right attempt away

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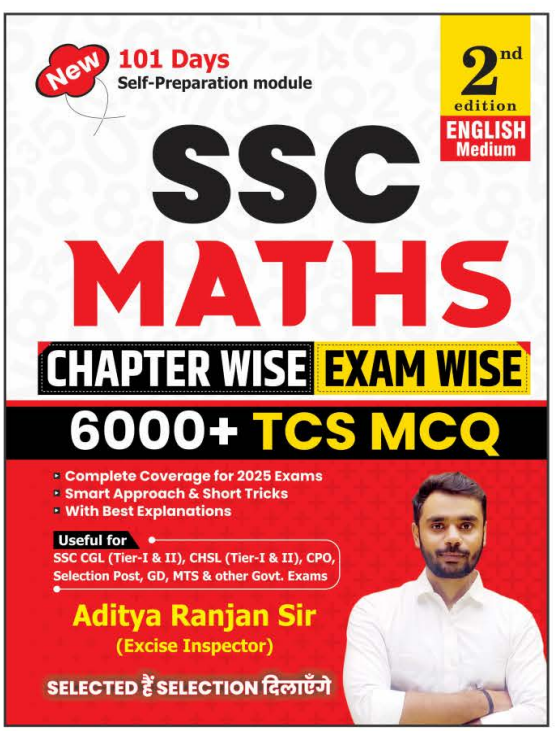
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